National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Interview - Individual Foods, First Day (DR1IFF_L)

Data File: DR1IFF_L.xpt

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Component Description

The objective of the dietary interview component is to obtain detailed dietary intake information from NHANES participants. The dietary intake data are used to estimate the types and amounts of foods and beverages (including all types of water) consumed during the 24-hour period prior to the interview (midnight to midnight), and to estimate intakes of energy, nutrients, and other food components from those foods and beverages. Following the dietary recall, participants are asked questions on salt use, whether the person's overall intake on the previous day was much more than usual, usual or much less than usual, and whether the participant is on any type of special diet. Questions on frequency of fish and shellfish consumed during the past 30 days are asked of participants 1 year or older, with the use of proxies for young children (see the Dietary Interview Procedure Manuals (cdc.gov) for more information on the proxy interview).

The dietary interview component, called What We Eat in America (WWEIA), is conducted as a partnership between the U.S. Department of Agriculture (USDA) and the U.S. Department of Health and Human Services (DHHS). Under this partnership, DHHS' National Center for Health Statistics (NCHS), Division of Health and Nutrition Examination Surveys is responsible for the survey sample design and all aspects of data collection and USDA's Food Surveys Research Group (FSRG) is responsible for the dietary data collection methodology, maintenance of the databases used to code and process the data, and data review and processing.

All NHANES participants are eligible for two 24-hour dietary recall interviews. Traditionally, the first dietary recall interview was collected in-person in the Mobile Examination Center (MEC) and the second interview was collected by telephone 3 to 10 days later. In August 2021-August 2023, the time participants spent in the MEC was limited to reduce risk of exposure to SARS-CoV-2. As a result, both dietary recall interviews were administered via telephone. Further information about changes in data collection are described in the Plan and Operations report of the NHANES August 2021-August 2023.

As in previous years, two types of dietary intake data are available for the August 2021-August 2023 survey cycle: Individual Foods files and Total Nutrient Intakes files.

What's New with the August 2021-August 2023 WWEIA Release:

In response to the COVID-19 pandemic, the mode of the first dietary recall changed from in-person during the MEC examination to telephone 3 to 7 days after the examination. Both dietary interviews were completed in English or Spanish. During the mobile examination, participants scheduled an appointment for the first telephone dietary interview and received a set of measuring guides to help estimate the amounts of foods and beverages consumed. **Appendix 1** provides a summary of changes among the 5 latest cycles of data collection. No new data variables were added in the August 2021-August 2023 WWEIA release.

Dietary Interview Data Files: Four data files were produced from the information collected in the dietary interviews: two Individual Foods files and two Total Nutrient Intakes files. Each

file includes one day of intake data. The number "1" or "2" in the file name identifies the day of the interview: 1 = first day, 2 = second day. File names are as follows:

File Names for Dietary Interview Data:

File	Day 1	Day 2
Individual Foods File	DR1IFF_L	DR2IFF_L
Total Nutrient Intakes File	DR1TOT_L	DR2TOT_L

The amounts in these files reflect only nutrients obtained from foods, beverages, and water, including tap and bottled water. They do not include nutrients obtained from dietary supplement intakes, antacids, or medications. Data on intake of dietary supplement use are available on the August 2021-August 2023 Dietary Data (cdc.gov) page.

Individual Foods Files (DR1IFF_L and DR2IFF_L): Detailed information about each food/beverage item (including the description, amount of, and nutrient content) reported by each participant is included in the Individual Foods files. The names for both Day 1 and Day 2 variables are listed in **Appendix 2**.

The Individual Foods files include, for each interview day, one record for each food/beverage consumed by a participant. Each record is uniquely numbered within a participant's set of records and contains the information listed below:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Whether the food/beverage was eaten in combination with other foods, such as in a sandwich;
- Time of eating occasion/when the food was eaten;
- Eating occasion name;
- Where the food/beverage was obtained;
- Whether the meal/snack was eaten at home or not;
- A USDA Food and Nutrient Database for Dietary Studies (FNDDS) code identifying the food/beverage;
- Amount of food/beverage consumed, in grams; and
- Food energy and 64 nutrients/food components (listed in **Appendix 3**) from each food/beverage as calculated using USDA's Food and Nutrient Database for Dietary Studies 2021-2023 (FNDDS 2021-2023).

Descriptions for the USDA FNDDS food codes are provided in the Food Code Description file (DRXFCD_L). The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code in the FNDDS 2021-2023. **Appendix 4** provides SAS code examples that may be used to link the food code description to the Individual Foods file.

Total Nutrient Intakes Files (DR1TOT_L and DR2TOT_L): For each participant, daily total energy and nutrient intakes from foods and beverages, and whether the amount of food consumed was usual, much more than usual, or much less than usual, are included in the Total Nutrient Intakes files. The Day 1 file also includes information on salt use in cooking and at the table; whether the participant is currently on any kind of diet to lose weight or for another health-related reason and, if so, the type of diet; and information on frequency of fish and shellfish consumption for participants aged 1 or older. The names for both Day 1 and Day 2 variables are listed in **Appendix 5**.

The Total Nutrient Intakes files provide a summary record of total nutrient intakes for each participant. Each total intake record contains the following information:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Type of salt used and how often added at the table and in food preparation (Day 1 file only);
- Use of salt at the table yesterday and the type of salt used;
- Whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet (Day 1 file only);
- Total number of foods and beverages including water reported for that participant for that day's intake;
- Daily aggregates of food energy and 64 nutrients/food components (listed in **Appendix 3**) from all foods/beverages as calculated using USDA's Food and Nutrient Database for Dietary Studies August 2021-August 2023 (FNDDS 2021- 2023)
- Whether the amount of food consumed was usual, much more than usual, or much less than usual;
- Total amount of tap and bottled water consumed (calculated as the sum of reports of water drunk by itself in the 24-hour recall) and the usual source of tap water; and
- Frequency of fish and shellfish consumption in the past 30 days (participants 1 year or older, Day 1 file only).

Eligible Sample

All participants in the August 2021-August 2023 sample who spoke English or Spanish were eligible. Participants aged 1 year or older were eligible for the frequency of fish and shellfish consumption questions following the 24-hour recall.

Protocol and Procedure

The examination protocol and data collection methods are fully documented in the NHANES dietary interviewer procedures manuals.

Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant's intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child's intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish.

During the MEC visit, participants were scheduled an appointment for the first dietary interview and were given a set of measuring guides (NHANES Measuring Guides for the Dietary Recall Interview) to use for reporting food amounts during the telephone interview. The measuring guides included measuring cups, spoons, a ruler, and a food model booklet, which contained two-dimensional drawings of glasses, mugs, bowls, mounds, circles, wedges, a shape chart and a chicken chart. The first telephone dietary interview was scheduled 3 to 7 days post MEC examination while the second interview was collected 3 to 10 days following the first dietary interview and were generally scheduled on a different day of the week as the first interview. In August 2021-August 2023, 204 participants were interviewed on the same day of the week for both day 1 and day 2 interviews due to their scheduling availability. Any participant who did not have a telephone was given a toll-free number to call so that the recall could be conducted.

What We Eat in America data were collected using USDA's dietary data collection instrument, the Automated Multiple Pass Method (AMPM), available at:

http://www.ars.usda.gov/nea/bhnrc/fsrg. The AMPM was designed to provide an efficient and accurate means of collecting intakes for large-scale national surveys. The AMPM is a fully computerized recall method that uses a 5-step interview outlined below:

1. **Quick List** - Participant recalls all foods and beverages consumed the day before the interview (midnight to midnight).
2. **Forgotten Foods** - Participant is asked about consumption of foods commonly forgotten during the Quick List step.
3. **Time and Occasion** - Time and eating occasion are collected for each food.
4. **Detail Cycle** - For each food, a detailed description, amount eaten, and additions to the food are collected. Eating occasions and times between eating occasions are reviewed to elicit forgotten foods.
5. **Final Probe** - Additional foods not remembered earlier are collected.

The AMPM includes an extensive compilation of standardized food-specific questions and possible response options. Routing of questions is based on previous responses. The AMPM is updated for each 2-year collection of WWEIA to reflect the changing food supply and to address research needs from the data user community. Additional information about the AMPM is provided in Raper et al. (Raper et al., 2004).

The AMPM was validated in a large study and shown to be an effective method for collecting accurate group energy intake of adults. Completed in 2004, this extensive research project included 524 healthy, weight-stable volunteers, aged 30-69 years. The accuracy of the AMPM was evaluated by comparing reported energy intake (EI) to total energy expenditure (TEE) using the doubly labeled water technique (Moshfegh et al., 2008). Among the findings were that EI compared to TEE was under-reported by 11% overall, by less than 3% for normal weight subjects with body mass index (BMI) < 25 and 16% for overweight subjects with BMI ≥ 25.

Additional studies provide evidence that the AMPM accurately measures group energy intake. Blanton (Blanton et al., 2006) reported that EI was not significantly different from TEE for a sample of 20 adult females. Rumpler (Rumpler et al., 2008) found that mean EIs were accurately reported for a sample of 12 adult males.

Additional evidence for the accuracy of AMPM has been provided by analysis of the 24-hour urinary sodium data collected in the AMPM Validation Study, which suggest the AMPM is a valid measure for estimating mean sodium intake in adults. Dietary sodium intake calculated from 24-hour recall data of 465 subjects collected via AMPM was compared with sodium values from 24-hour urine collections measured during the same 24-hour period. The AMPM-derived mean dietary sodium estimates reflected over 90% of the biomarker-based estimates (Rhodes et al., 2013).

For additional information about the dietary interview component and related survey protocols, please visit the August 2021-August 2023 Dietary Interviewer Procedures Manual page. A description of changes to the NHANES August 2021-August 2023 survey is provided in the Plan and Operations of the National Heath And Nutrition Examination Survey, August 2021-August 2023.

Quality Assurance & Quality Control

All dietary interviewers were required to complete an intensive one-week training course and to conduct supervised practice interviews before working independently in the field. Retraining sessions were conducted annually to reinforce the proper protocols and technique.

Interviewers were monitored throughout the data collection period. Monitoring consisted of the following:

- Reviews of audio recorded interviews were conducted for approximately 5% of each interviewer's work.

- Quality control of interviews, which were checked for completeness of the recalls, missing information, inconsistent reports, and unclear notes. Written notification and feedback were provided to the interviewers.

Data Processing and Editing

Interview data files were sent electronically from the field and were imported into Survey Net, a computer-assisted food coding and data management system developed by USDA (Raper et al., 2004).

USDA's Food and Nutrient Database for Dietary Studies (FNDDS) 2021-2023 was used to process the intakes reported by the August 2021-August 2023 sample. The FNDDS includes comprehensive information that can be used to code individual foods/beverages and portion sizes reported by participants and includes nutrient values for calculating nutrient intakes. FNDDS nutrient values as well as food codes and portion sizes are updated for every 2-year WWEIA, NHANES release cycle. The basis for the nutrient values as well as food codes and portion weights in FNDDS are detailed in the documentation for FNDDS 2021-2023 available at http://www.ars.usda.gov/nea/bhnrc/fsrg.

Coders were required to pass a certification test after the initial training. They were routinely monitored to ensure the quality and completeness of their work. Approximately 10 percent of the coder's work was randomly selected to be independently coded by another coder. Results from the two codings were compared and adjudicated, if necessary.

After intake data were coded, various types of reviews and quality assurance procedures were conducted to ensure the quality of the data. Examples of reviews include the following:

- Interviewers' and coders' questions and comments were reviewed to ensure that they have been addressed.
- Decisions made by coders about how to code new or unusual foods/beverages or quantities reported by participants were reviewed for accuracy, reasonableness, and consistency across intake data; items in question were resolved by coder supervisors.
- Specific data integrity checks for reasonableness, consistency, and logic were conducted.

Analytic Notes

Each Individual Foods file (Day 1 and Day 2) is comprised of food records. For most participants, there are multiple records in each file. For each Total Nutrient Intakes file (Day 1 and Day 2) there is one record for each participant. These files can be linked with other NHANES files by the respondent sequence number (SEQN).

Variable names: For data collected on both Day 1 and Day 2, variable names are differentiated by having the number "1" or "2" in the third position of the variable name to identify the collection day. For example, the USDA food code variable (in the Individual Foods File), which identifies the food reported by the participant, is named DR1IFDCD in the Day 1 file and DR2IFDCD in the Day 2 file. Appendices **2** and **5** list the Day 1 and Day 2 variable names for the Individual Foods file and the Total Nutrient Intakes file, respectively.

Names for the following variables are the same for both days in the Individual Foods file and the Total Nutrient Intakes file:

Variables with the Same Name for Both Days in the Dietary Interview Files

Day 1 and Day 2 variable name	Label
SEQN	Respondent sequence number
WTDRD1	Dietary day one sample weight
WTDR2D	Dietary two-day sample weight
DRABF	Breast-fed infant (either day)
DRDINT	Number of days of intake

Number of days of intake: A variable has been included to indicate the number of days of intake collected from each participant. The variable name is DRDINT. In the August 2021-August 2023 sample, 6,754 participants provided complete dietary intakes for Day 1. The 6,754 participants with a complete Day 1 intake include 21 participants who had an unreliable day 1 intake but reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall. Of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2.

Dietary recall status code: A status code (DR1DRSTZ or DR2DRSTZ) is used in both the Individual Foods and Total Nutrient Intakes files to indicate the quality and completeness of a survey participant's response to the dietary recall section. The codes are the following:

1 = Reliable and met the following minimum criteria:

- The first 4 steps of the 5-step AMPM completed.
- Food/beverages consumed for each reported eating occasion identified.

For individuals with a code 1, all relevant variables associated with the 24-hour dietary recall contain a value.

2 = Not reliable or did not meet the minimum criteria

Individuals with a code 2 have incomplete records. No data on total nutrient intakes and the total number of foods reported are provided for these cases. These individuals have no records in the Individual Foods files.

3 [Code 3 is not included in the current datasets. It was only used for data from the 1999-2000 survey cycle.]

4 = Reported consuming breast milk

For infants and children who consumed human milk, there is a record in the Individual Foods files for each report of human milk. However, because amounts of human milk intake are not quantified, these records contain missing values for the amount consumed and for the amounts of energy and nutrients from human milk. Also, records of human milk have a missing value for the food source variable (DR1FS, DR2FS) and the eaten at home variable (DR1_040Z, DR2_040Z) in the Individual Foods files. Records for any other foods and beverages consumed by breast-fed infants and children are included in the Individual Foods files along with their amounts and nutrient information. Because of the missing amount or quantity information for human milk, no total nutrient intakes (contained in the Total Nutrient Intakes files) were computed for participants with a code 4.

A variable that identifies breast-fed children, DRABF, is included. This variable has a code of 1 if a child consumed human milk in either intake day.

5 = Not done

This code is assigned when the dietary recall section of the interview did not take place due to various reasons (such as refusal, equipment failure, or unable to contact the participant). These individuals have no records in the Individual Foods files. These individuals have a record in the Total Nutrients file with values only for the following variables: the respondent sequence number (SEQN), the dietary recall status code (DR1DRSTZ or DR2DRSTZ) and for participants 1 year or older, the fish and shellfish questions in the DR1TOT_L file (DRD340, DRD350A-K, DRD350AQ-JQ, DRD360, DRD370A-V, and DRD370AQ-UQ).

Only codes 1 and 4 appear in the Individual Foods file.

Distinguishing Between Foods/Beverages and Dietary Supplements in NHANES: In August 2021-August 2023, there was no 24-hour dietary supplement use collection. The 30-day dietary supplement use collection changed from being collected during the home interview to collection after the first 24-hour dietary recall via telephone. All NHANES participants responding to the 24-hour dietary recall interview are eligible for the dietary supplement and non-prescription antacid use questions. Information is obtained on all vitamins, minerals, herbals, and other dietary supplements as well as non-prescription antacids that were consumed during a 30-day period, including the name and the amount of supplement or antacid taken.

Distinguishing between foods/beverages and supplements can be challenging. NCHS and FSRG review questionable items reported in the dietary supplement and dietary recall components to resolve disposition of these items into the appropriate component. Products that are labeled as a dietary supplement, that have a supplement facts panel on the label, and are in tablets, capsules, softgels, gelcaps, or other pill forms, are considered dietary supplements. Items that are powders or liquids can be hard to distinguish. General guidelines used state that if powders and liquid concentrates have product directions stating that they be added to a liquid, they are classified as beverages. Examples are teas and protein powders. An exception is made for fiber products, which are classified as dietary supplements. Along this same guideline, energy drinks are considered beverages, but "energy shot" type products are considered dietary supplements.

It is best to refer to the databases that detail every food/beverage and dietary supplement reported in NHANES to identify exact determination used. The databases are:

- 2021-2023 Food and Nutrient Database for Dietary Studies
- NHANES Dietary Supplement Database

Participants who reported consuming only water, no food or other beverages: Records are included in the Individual Foods file for participants who consumed only water. There are 5 such individuals in the August 2021-August 2023 datasets, 4 in the Day 1 data and 1 in the Day 2 data. Their dietary recall status variable for the day is coded as "1" (complete and reliable) in the Total Nutrients file and the total number of items is the number of times water was reported. Individuals with just water intake and no food intake will have zero energy intake for the day.

Participants who reported consuming no water, food or other beverages: There can be participants whose intakes are determined to be complete even though they reported no water, food, or other beverage records for the day. For such participants there are no records in the Individual Foods file, but their dietary recall status is coded as complete and reliable, and the Total Nutrients file will include records with zero values for all nutrients. In the August 2021-August 2023 datasets, there are 3 individuals in the day 1 data that reported no water, food, or other beverage records for the day.

Number of days between the intake day and the day of family interview: Each of the four intake files includes a variable (DR1DBIH for Day 1 files and DR2DBIH for Day 2 files) to indicate the number of days between the intake day (i.e., the period covered by the 24-hour recall) and the day that the family questionnaire was administered in the household. A positive value in DR1BHIH or DR2BHIH indicates the family interview occurred prior to the intake day. In the survey, most of the family interviews were done before the participant participated in the dietary interview. A value of "0" in DR1BHIH or DR2BHIH indicates the family interview

occurred on the same date as the intake day. A negative value (i.e., DR1BHIH<0 or DR2BHIH<0) means that the family interview occurred after the intake day.

Food source: The source from which each food/beverage was obtained (e.g., from a store, fast food restaurant, cafeteria) is identified by the variables DR1FS (day 1) and DR2FS (day 2) in the Individual Foods files.

The code descriptions for this variable are:

Code Descriptions for Source of Food Variable

Code	Description
1	Store grocery/supermarket
2	Restaurant with waiter/waitress
3	Restaurant fast food/Pizza
4	Bar/Tavern/Lounge
5	Restaurant, no additional information
6	Cafeteria NOT in a K-12 school
7	Cafeteria in a K-12 school
8	Child/Adult care center
9	Child/Adult home care
10	Soup kitchen/shelter/food pantry facility
11	Meals on Wheels Program
12	Community food program – other
13	Community program, no additional info
14	Vending machine
15	Common coffee pot or snack tray
16	From someone else/gift
17	Mail order purchase
18	Residential dining facility
19	Grown or caught by you or someone you know
20	Fish caught by you or someone you know
24	Sport, recreation, or entertainment
25	Street vendor, vending truck
26	Fundraiser sales
27	Store - convenience type
28	Store - no additional information
91	Other, specify

Eating occasion: The variables DR1_030Z and DR2_030Z are located in the Individual Foods file. The code descriptions for the eating occasion variables are shown in the table below.

Code Descriptions for Eating Occasion Variable

Code	Description
1	Breakfast
2	Lunch
3	Dinner
4	Supper
5	Brunch
6	Snack
7	Beverage/Drink
8	Feeding-infant only
9	Extended consumption
10	Desayuno
11	Almuerzo
12	Comida
13	Merienda
14	Cena
15	Entre comida
16	Botana
17	Bocadillo
18	Tentempie
19	Bebida
91	Other

Eating occasion was designated by the respondent. During the interview, a list of eating occasion names was available to the respondent for selection. However, eating occasion names were not defined for the respondent.

Foods and beverages coded as part of a combination: 41 percent of foods and beverages reported in the WWEIA, NHANES August 2021-August 2023 sample were identified as items consumed together as combinations. Items consumed as a combination were identified by one of sixteen unique "combination food types." Foods and beverages not coded in combination have the code "0" for the combination food type variable.

The combination types provide a linkage for:

- Foods or beverages with additions, such as cereal with milk, coffee with cream;
- Multi-component foods that have specific protocol for collection such as some salads and sandwiches; and
- Other combinations that do not have a unique code in the FNDDS.

Combination Type, Code, Examples, and Percent of Food and Beverages Reported by Type, August 2021-August 2023, Day 1

Combination Type	Code	Examples of Combination Type	% Items
Not in combination	0	NA	59
Beverage w/ additions	1	Coffee, tea with: milk, cream, sugar.	9
Cereal w/ additions	2	Cereals (ready-to-eat, cooked) with: milk, sugar, fruit, butter.	4
Bread/baked product w/additions	3	Breads, rolls, pancakes with: butter, jam, syrup, fruit. Cakes, pies with: ice cream, toppings. Crackers with: cheese, dip, peanut butter.	5
Salad	4	Components of salads that do not have a single code in FNDDS. It may also designate additional items to single code salads.	5
Sandwiches	5	Components of sandwiches that do not have a single code in FNDDS. It may also designate additional items added to single code sandwiches.	6
Soup	6	Soup with: crackers, croutons, cheese.	<1
Frozen meals	7	Components of a prepackaged frozen meal and additions to the meal.	<1
Ice cream/ frozen yogurt w/ additions	8	Ice cream with: syrup, nuts, toppings.	<1
Dried beans or Vegetable w/ additions	9	French fries, potatoes with: catsup, gravy, butter, toppings. Beans with: sauce, butter.	3
Fruit w/ additions	10	Fruit with: toppings, milk, honey. Components of fruit mixtures or salads that do not have a single code in FNDDS.	1
Tortilla products	11	Components of tacos and tortilla products that do not have a single code in FNDDS. It may also designate additional items to single code tacos or tortilla products.	2
Meat, Poultry, Fish	12	Meat, poultry, fish with: gravy, sauce, and condiments.	2
Lunchables®	13	Components of pre-packaged lunch kits.	<1
Chips w/ additions	14	Potato chips, corn chips with: dip, cheese, salsa.	1
Baby Toddler Food and Infant Formula	15	Baby Toddler cereal with: infant formula, fruit. Infant formula with: baby cereal.	<1
Other mixtures	90	Rice, pasta, spaghetti, eggs, other mixtures with: butter, gravy, sauce, condiments.	4

All items given a combination food type are given an additional variable to identify each of the items within the combination. This variable is the "combination food number" that is unique to the combination food type within the individual intake.

Variable Labels and Names for Combination Coding

Combination Coding	Variable Name, Day 1	Variable Name, Day 2
Combination food type	DR1CCMTX	DR2CCMTX
Combination food number	DR1CCMNM	DR2CCMNM

The What We Eat in America Food Categories, available on the FSRG website (http://www.ars.usda.gov/nea/bhnrc/fsrg), is a grouping scheme that combines foods and beverages together that have similar usage and nutrient content with the emphasis on

how they are commonly consumed in the American diet. There are more than 170 unique categories, and each is assigned a 4-digit number and description. The WWEIA Food Categories contain discrete food items and are not disaggregated (e.g., pizza vs. grain, cheese, tomatoes, etc.). Designed to be flexible, the categories can be combined as needed to address specific research questions. A new version of the WWEIA Food Categories is produced for each 2-year release cycle of WWEIA, NHANES.

Special diet: Information on whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet, was provided. The variable DRQSDIET identifies whether a participant was on a special diet. The variables DRQSDT1 through DRQSDT12 and DRQSDT91 identify the type of diet or diets that the participant was following. These variables can be found in the Total Nutrient Intakes file.

Sample weights for dietary intake data: The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, **the appropriate sample weights must be used.**

For the NHANES August 2021-August 2023 sample, there were 22,660 persons selected; of these 8,860 were considered participants to the MEC examination and data collection. A total of 6,754 MEC participants provided complete dietary intakes for Day 1, and of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2. The 6,754 participants who provided complete Day 1 intakes includes 21 participants who had an unreliable day 1 intake and reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall.

Most analyses of NHANES data use data collected in the MEC and the variable WTMEC2YR should be used for the sample weights. However, for the WWEIA dietary data, different sample weights are recommended for analysis. Because food intake can vary by weekdays and weekends, use of the MEC weights disproportionately represents intakes on weekends.

A set of weights (WTDRD1) is provided that should be used when an analysis uses the Day 1 dietary recall data (either alone or when Day 1 nutrient data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with Day 1 data. Day 1 weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These Day 1 weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using Day 1 data and Day 1 weights might be larger than standard errors for similar estimates based on MEC weights.

When analysis is based on both days of dietary intake, only 5,879 sample participants have complete data. The NHANES protocol requires an attempt to collect the second day of dietary data at least 3 days after the first day, but the actual number of days between the two interviews is variable. A set of adjusted weights, WTDR2D, is to be used when an analysis uses the smaller sample with completed Day 1 and Day 2 dietary data. This two-day weight was constructed for the 5,879 participants by taking the Day 1 weights (WTDRD1) and further adjusting for: (a) the additional non-response for the second recall; and (b) for the proportion of weekend (Friday through Sunday) and weekday (Monday through Thursday) combinations of Day 1 and Day 2 recalls.

NOTE: All sample weights are person-level weights, and each set of dietary weights should sum to the same overall population control total as the MEC weights (WTMEC2YR). For the August 2021-August 2023 cycle, the Day 1 dietary recall weights (WTDRD1) are appropriate to use in the analysis of the fish and shellfish consumption data (i.e., variables DRD340, DRD350A-K, DRD350AQ-JQ DRD360, DRD370A-V, and DRD370AQ-UQ) located in the Day 1 Total Nutrient Intake File (DR1TOT_L), irrespective of whether other dietary data are included in the analysis. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

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Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
English Instructions:	One of the key variables for the file. The primary key variables are SEQN and DR1ILINE.
Target:	Both males and females 0 YEARS - 150 YEARS

WTDRD1 - Dietary day one sample weight

Variable Name: WTDRD1

SAS Label: Dietary day one sample weight

English Text: Dietary day one sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5708.632624 to 408505.81364	Range of Values	100116	100116	
0	Day 1 dietary recall not done/incomplete	0	100116	
.	Missing	0	100116	

WTDR2D - Dietary two-day sample weight

Variable Name: WTDR2D

SAS Label: Dietary two-day sample weight

English Text: Dietary two-day sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6018.694738 to 759868.32478	Range of Values	87162	87162	
0	Day 2 dietary recall not done/incomplete	12954	100116	
.	Missing	0	100116	

DR1ILINE - Food/Individual component number

Variable Name: DR1ILINE

SAS Label: Food/Individual component number

English Text: Food/Individual component number

English Instructions: One of the key variables for the file. The primary key variables are SEQN and DR1ILINE.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 49	Range of Values	100116	100116	
.	Missing	0	100116	

DR1DRSTZ - Dietary recall status

Variable Name: DR1DRSTZ

SAS Label: Dietary recall status

English Text: Dietary recall status

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Reliable and met the minimum criteria	99345	99345	
2	Not reliable or not met the minimum criteria	0	99345	
4	Reported consuming breast-milk	771	100116	
5	Not done	0	100116	
.	Missing	0	100116	

DR1EXMER - Interviewer ID code

Variable Name: DR1EXMER

SAS Label: Interviewer ID code

English Text: Interviewer ID code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 96	Range of Values	100116	100116	
.	Missing	0	100116	

DRABF - Breast-fed infant (either day)

Variable Name: DRABF

SAS Label: Breast-fed infant (either day)

English Text: Indicates whether the sample person was an infant who was breast-fed on either of the two recall days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	771	771	
2	No	99076	99847	
.	Missing	269	100116	

DRDINT - Number of days of intake

Variable Name: DRDINT

SAS Label: Number of days of intake

English Text: Indicates whether the sample person has intake data for one or two days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Day 1 only	12954	12954	
2	Day 1 and day 2	87162	100116	
.	Missing	0	100116	

DR1DBIH - # of days b/w intake and HH interview

Variable Name: DR1DBIH

SAS Label: # of days b/w intake and HH interview

English Text: Number of days between intake day and the day of family questionnaire administered in the household.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
-30 to 149	Range of Values	94128	94128	
.	Missing	5988	100116	

DR1DAY - Intake day of the week

Variable Name: DR1DAY

SAS Label: Intake day of the week

English Text: Intake day of the week

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Sunday	16662	16662	
2	Monday	14772	31434	
3	Tuesday	17312	48746	
4	Wednesday	23200	71946	
5	Thursday	13842	85788	
6	Friday	6474	92262	
7	Saturday	7854	100116	
.	Missing	0	100116	

DR1LANG - Language respondent used mostly

Variable Name: DR1LANG

SAS Label: Language respondent used mostly

English Text: The respondent spoke mostly:

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	English	92150	92150	
2	Spanish	7315	99465	
3	English and Spanish	536	100001	
4	Other	24	100025	
.	Missing	91	100116	

DR1CCMNM - Combination food number

Variable Name: DR1CCMNM

SAS Label: Combination food number

English Text: Combination food number (sequential number)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 15	Range of Values	100116	100116	
.	Missing	0	100116	

DR1CCMTX - Combination food type

Variable Name: DR1CCMTX

SAS Label: Combination food type

English Text: Combination food type

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	Non-combination food	58813	58813	
1	Beverage w/ additions	8694	67507	
2	Cereal w/ additions	3523	71030	
3	Bread/baked products w/ additions	5250	76280	
4	Salad	4970	81250	
5	Sandwiches	6126	87376	
6	Soup	497	87873	
7	Frozen meals	46	87919	
8	Ice cream/frozen yogurt w/ additions	218	88137	
9	Dried beans and vegetable w/ additions	2562	90699	
10	Fruit w/ additions	672	91371	
11	Tortilla products	2180	93551	
12	Meat, poultry, fish	2044	95595	
13	Lunchables®	175	95770	
14	Chips w/ additions	514	96284	
15	Baby Toddler Food and Infant Formula	118	96402	
90	Other mixtures	3714	100116	
.	Missing	0	100116	

DR1_020 - Time of eating occasion (HH:MM)

Variable Name: DR1_020

SAS Label: Time of eating occasion (HH:MM)

English Text: What time did you begin to eat/drink the meal/food?

English Instructions: Format HHMM6.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 86340	Range of Values	100116	100116	
.	Missing	0	100116	

DR1_030Z - Name of eating occasion

Variable Name: DR1_030Z

SAS Label: Name of eating occasion

English Text: Name of eating occasion

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Breakfast	19146	19146	
2	Lunch	19521	38667	
3	Dinner	21058	59725	
4	Supper	4333	64058	
5	Brunch	859	64917	
6	Snack	17164	82081	
7	Drink	6656	88737	
8	Infant feeding	984	89721	
9	Extended consumption	4155	93876	
10	Desayano	1402	95278	
11	Almuerzo	1073	96351	
12	Comida	936	97287	
13	Merienda	660	97947	
14	Cena	1356	99303	
15	Entre comida	157	99460	
16	Botana	139	99599	
17	Bocadillo	169	99768	
18	Tentempie	33	99801	
19	Bebida	315	100116	
91	Other	0	100116	
99	Don't know	0	100116	
.	Missing	0	100116	

DR1FS - Source of food

Variable Name: DR1FS

SAS Label: Source of food

English Text: Where did you get (this/most of the ingredients for this) {FOODNAME}?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Store - grocery/supermarket	67563	67563	
2	Restaurant with waiter/waitress	5266	72829	
3	Restaurant fast food/pizza	6991	79820	
4	Bar/tavern/lounge	294	80114	
5	Restaurant no additional information	15	80129	
6	Cafeteria NOT in a K-12 school	620	80749	
7	Cafeteria in a K-12 school	1547	82296	
8	Child/Adult care center	242	82538	
9	Child/Adult home care	37	82575	
10	Soup kitchen/shelter/food pantry	76	82651	
11	Meals on Wheels	73	82724	
12	Community food program - other	195	82919	
13	Community program no additional information	8	82927	
14	Vending machine	165	83092	
15	Common coffee pot or snack tray	518	83610	
16	From someone else/gift	3634	87244	
17	Mail order purchase	1131	88375	
18	Residential dining facility	81	88456	
19	Grown or caught by you or someone you know	571	89027	
20	Fish caught by you or someone you know	8	89035	
24	Sport, recreation, or entertainment facility	346	89381	
25	Street vendor, vending truck	242	89623	
26	Fundraiser sales	61	89684	
27	Store - convenience type	2522	92206	
28	Store - no additional info	391	92597	
91	Other, specify	0	92597	
99	Don't know	187	92784	
.	Missing	7332	100116	

DR1_040Z - Did you eat this meal at home?

Variable Name: DR1_040Z

SAS Label: Did you eat this meal at home?

English Text: Did you eat this meal at home?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	75025	75025	
2	No	24759	99784	
7	Refused	0	99784	
9	Don't know	3	99787	
.	Missing	329	100116	

DR1IFDCD - USDA food code

Variable Name: DR1IFDCD

SAS Label: USDA food code

English Text: USDA food code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11000000 to 95342000	Range of Values	100116	100116	
.	Missing	0	100116	

DR1IGRMS - Grams

Variable Name: DR1IGRMS

SAS Label: Grams

English Text: Gram weight of the food/individual component

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.05 to 15360	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IKCAL - Energy (kcal)

Variable Name: DR1IKCAL

SAS Label: Energy (kcal)

English Text: Energy (kcal)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4575	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IPROT - Protein (gm)

Variable Name: DR1IPROT

SAS Label: Protein (gm)

English Text: Protein (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 335.48	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICARB - Carbohydrate (gm)

Variable Name: DR1ICARB

SAS Label: Carbohydrate (gm)

English Text: Carbohydrate (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 832.95	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ISUGR - Total sugars (gm)

Variable Name: DR1ISUGR

SAS Label: Total sugars (gm)

English Text: Total sugars (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 702.46	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IFIBE - Dietary fiber (gm)

Variable Name: DR1IFIBE

SAS Label: Dietary fiber (gm)

English Text: Dietary fiber (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 91.2	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ITFAT - Total fat (gm)

Variable Name: DR1ITFAT

SAS Label: Total fat (gm)

English Text: Total fat (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 261.46	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ISFAT - Total saturated fatty acids (gm)

Variable Name: DR1ISFAT

SAS Label: Total saturated fatty acids (gm)

English Text: Total saturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 118.608	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IMFAT - Total monounsaturated fatty acids (gm)

Variable Name: DR1IMFAT

SAS Label: Total monounsaturated fatty acids (gm)

English Text: Total monounsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 109.2	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IPFAT - Total polyunsaturated fatty acids (gm)

Variable Name: DR1IPFAT

SAS Label: Total polyunsaturated fatty acids (gm)

English Text: Total polyunsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 89.814	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICHOL - Cholesterol (mg)

Variable Name: DR1ICHOL

SAS Label: Cholesterol (mg)

English Text: Cholesterol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2269	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IATOC - Vitamin E as alpha-tocopherol (mg)

Variable Name: DR1IATOC

SAS Label: Vitamin E as alpha-tocopherol (mg)

English Text: Vitamin E as alpha-tocopherol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 116.74	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IATOA - Added alpha-tocopherol (Vitamin E) (mg)

Variable Name: DR1IATOA

SAS Label: Added alpha-tocopherol (Vitamin E) (mg)

English Text: Added alpha-tocopherol (Vitamin E) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 108.87	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IRET - Retinol (mcg)

Variable Name: DR1IRET

SAS Label: Retinol (mcg)

English Text: Retinol (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 38108	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVARA - Vitamin A, RAE (mcg)

Variable Name: DR1IVARA

SAS Label: Vitamin A, RAE (mcg)

English Text: Vitamin A as retinol activity equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 38190	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IACAR - Alpha-carotene (mcg)

Variable Name: DR1IACAR

SAS Label: Alpha-carotene (mcg)

English Text: Alpha-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30805	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IBCAR - Beta-carotene (mcg)

Variable Name: DR1IBCAR

SAS Label: Beta-carotene (mcg)

English Text: Beta-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 62410	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICRYP - Beta-cryptoxanthin (mcg)

Variable Name: DR1ICRYP

SAS Label: Beta-cryptoxanthin (mcg)

English Text: Beta-cryptoxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4920	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ILYCO - Lycopene (mcg)

Variable Name: DR1ILYCO

SAS Label: Lycopene (mcg)

English Text: Lycopene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 132561	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ILZ - Lutein + zeaxanthin (mcg)

Variable Name: DR1ILZ

SAS Label: Lutein + zeaxanthin (mcg)

English Text: Lutein + zeaxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 60565	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVB1 - Thiamin (Vitamin B1) (mg)

Variable Name: DR1IVB1

SAS Label: Thiamin (Vitamin B1) (mg)

English Text: Thiamin (Vitamin B1) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 7.286	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVB2 - Riboflavin (Vitamin B2) (mg)

Variable Name: DR1IVB2

SAS Label: Riboflavin (Vitamin B2) (mg)

English Text: Riboflavin (Vitamin B2) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 16.936	Range of Values	99787	99787	
.	Missing	329	100116	

DR1INIAC - Niacin (mg)

Variable Name: DR1INIAC

SAS Label: Niacin (mg)

English Text: Niacin (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 259.891	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVB6 - Vitamin B6 (mg)

Variable Name: DR1IVB6

SAS Label: Vitamin B6 (mg)

English Text: Vitamin B6 (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 25.958	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IFOLA - Total folate (mcg)

Variable Name: DR1IFOLA

SAS Label: Total folate (mcg)

English Text: Total folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1941	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IFA - Folic acid (mcg)

Variable Name: DR1IFA

SAS Label: Folic acid (mcg)

English Text: Folic acid (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1941	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IFF - Food folate (mcg)

Variable Name: DR1IFF

SAS Label: Food folate (mcg)

English Text: Food folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1588	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IFDFE - Folate, DFE (mcg)

Variable Name: DR1IFDFE

SAS Label: Folate, DFE (mcg)

English Text: Folate as dietary folate equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3301	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICHL - Total choline (mg)

Variable Name: DR1ICHL

SAS Label: Total choline (mg)

English Text: Total choline (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2080.9	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVB12 - Vitamin B12 (mcg)

Variable Name: DR1IVB12

SAS Label: Vitamin B12 (mcg)

English Text: Vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 409.38	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IB12A - Added vitamin B12 (mcg)

Variable Name: DR1IB12A

SAS Label: Added vitamin B12 (mcg)

English Text: Added vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 78.34	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVC - Vitamin C (mg)

Variable Name: DR1IVC

SAS Label: Vitamin C (mg)

English Text: Vitamin C (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 887.7	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVD - Vitamin D (D2 + D3) (mcg)

Variable Name: DR1IVD

SAS Label: Vitamin D (D2 + D3) (mcg)

English Text: Vitamin D (D2 + D3) (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 54.5	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IVK - Vitamin K (mcg)

Variable Name: DR1IVK

SAS Label: Vitamin K (mcg)

English Text: Vitamin K (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2098.8	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICALC - Calcium (mg)

Variable Name: DR1ICALC

SAS Label: Calcium (mg)

English Text: Calcium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5675	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IPHOS - Phosphorus (mg)

Variable Name: DR1IPHOS

SAS Label: Phosphorus (mg)

English Text: Phosphorus (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4859	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IMAGN - Magnesium (mg)

Variable Name: DR1IMAGN

SAS Label: Magnesium (mg)

English Text: Magnesium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1700	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IIRON - Iron (mg)

Variable Name: DR1IIRON

SAS Label: Iron (mg)

English Text: Iron (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 120.23	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IZINC - Zinc (mg)

Variable Name: DR1IZINC

SAS Label: Zinc (mg)

English Text: Zinc (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 63.75	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICOPP - Copper (mg)

Variable Name: DR1ICOPP

SAS Label: Copper (mg)

English Text: Copper (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 72.041	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ISODI - Sodium (mg)

Variable Name: DR1ISODI

SAS Label: Sodium (mg)

English Text: Sodium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 19050	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IPOTA - Potassium (mg)

Variable Name: DR1IPOTA

SAS Label: Potassium (mg)

English Text: Potassium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 12081	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ISELE - Selenium (mcg)

Variable Name: DR1ISELE

SAS Label: Selenium (mcg)

English Text: Selenium (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1054.4	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ICAFF - Caffeine (mg)

Variable Name: DR1ICAFF

SAS Label: Caffeine (mg)

English Text: Caffeine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2448	Range of Values	99787	99787	
.	Missing	329	100116	

DR1ITHEO - Theobromine (mg)

Variable Name: DR1ITHEO

SAS Label: Theobromine (mg)

English Text: Theobromine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1094	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IALCO - Alcohol (gm)

Variable Name: DR1IALCO

SAS Label: Alcohol (gm)

English Text: Alcohol (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 300.3	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IMOIS - Moisture (gm)

Variable Name: DR1IMOIS

SAS Label: Moisture (gm)

English Text: Moisture (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 15344.64	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS040 - SFA 4:0 (Butanoic) (gm)

Variable Name: DR1IS040

SAS Label: SFA 4:0 (Butanoic) (gm)

English Text: SFA 4:0 (Butanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4.008	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS060 - SFA 6:0 (Hexanoic) (gm)

Variable Name: DR1IS060

SAS Label: SFA 6:0 (Hexanoic) (gm)

English Text: SFA 6:0 (Hexanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.02	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS080 - SFA 8:0 (Octanoic) (gm)

Variable Name: DR1IS080

SAS Label: SFA 8:0 (Octanoic) (gm)

English Text: SFA 8:0 (Octanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.792	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS100 - SFA 10:0 (Decanoic) (gm)

Variable Name: DR1IS100

SAS Label: SFA 10:0 (Decanoic) (gm)

English Text: SFA 10:0 (Decanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4.6	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS120 - SFA 12:0 (Dodecanoic) (gm)

Variable Name: DR1IS120

SAS Label: SFA 12:0 (Dodecanoic) (gm)

English Text: SFA 12:0 (Dodecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 31.05	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS140 - SFA 14:0 (Tetradecanoic) (gm)

Variable Name: DR1IS140

SAS Label: SFA 14:0 (Tetradecanoic) (gm)

English Text: SFA 14:0 (Tetradecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 16.9	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS160 - SFA 16:0 (Hexadecanoic) (gm)

Variable Name: DR1IS160

SAS Label: SFA 16:0 (Hexadecanoic) (gm)

English Text: SFA 16:0 (Hexadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 58.256	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IS180 - SFA 18:0 (Octadecanoic) (gm)

Variable Name: DR1IS180

SAS Label: SFA 18:0 (Octadecanoic) (gm)

English Text: SFA 18:0 (Octadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 27.168	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IM161 - MFA 16:1 (Hexadecenoic) (gm)

Variable Name: DR1IM161

SAS Label: MFA 16:1 (Hexadecenoic) (gm)

English Text: MFA 16:1 (Hexadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 16.797	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IM181 - MFA 18:1 (Octadecenoic) (gm)

Variable Name: DR1IM181

SAS Label: MFA 18:1 (Octadecenoic) (gm)

English Text: MFA 18:1 (Octadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 107.013	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IM201 - MFA 20:1 (Eicosenoic) (gm)

Variable Name: DR1IM201

SAS Label: MFA 20:1 (Eicosenoic) (gm)

English Text: MFA 20:1 (Eicosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 12.41	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IM221 - MFA 22:1 (Docosenoic) (gm)

Variable Name: DR1IM221

SAS Label: MFA 22:1 (Docosenoic) (gm)

English Text: MFA 22:1 (Docosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 17.575	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP182 - PFA 18:2 (Octadecadienoic) (gm)

Variable Name: DR1IP182

SAS Label: PFA 18:2 (Octadecadienoic) (gm)

English Text: PFA 18:2 (Octadecadienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 89.536	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP183 - PFA 18:3 (Octadecatrienoic) (gm)

Variable Name: DR1IP183

SAS Label: PFA 18:3 (Octadecatrienoic) (gm)

English Text: PFA 18:3 (Octadecatrienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 14.943	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP184 - PFA 18:4 (Octadecatetraenoic) (gm)

Variable Name: DR1IP184

SAS Label: PFA 18:4 (Octadecatetraenoic) (gm)

English Text: PFA 18:4 (Octadecatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 0.605	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP204 - PFA 20:4 (Eicosatetraenoic) (gm)

Variable Name: DR1IP204

SAS Label: PFA 20:4 (Eicosatetraenoic) (gm)

English Text: PFA 20:4 (Eicosatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.519	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP205 - PFA 20:5 (Eicosapentaenoic) (gm)

Variable Name: DR1IP205

SAS Label: PFA 20:5 (Eicosapentaenoic) (gm)

English Text: PFA 20:5 (Eicosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2.868	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP225 - PFA 22:5 (Docosapentaenoic) (gm)

Variable Name: DR1IP225

SAS Label: PFA 22:5 (Docosapentaenoic) (gm)

English Text: PFA 22:5 (Docosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.099	Range of Values	99787	99787	
.	Missing	329	100116	

DR1IP226 - PFA 22:6 (Docosahexaenoic) (gm)

Variable Name: DR1IP226

SAS Label: PFA 22:6 (Docosahexaenoic) (gm)

English Text: PFA 22:6 (Docosahexaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.256	Range of Values	99787	99787	
.	Missing	329	100116	

Appendix 1. Changes between WWEIA survey cycles 2013-2014 thru the August 2021-August 2023 sample

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Administration of 24-hour dietary recall	Same as 2003-2012: Day 1 in-person in MEC, Day 2 by telephone	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Day 1 by telephone, Day 2 by telephone
Language, Day 1	Same as 1999-2012: Interpreters assisted as needed	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Language, Day 2	Same as 2011-2012: English, Spanish, or Asian languages	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Person who did not provide reliable intake for Day 1 but did for Day 2	Same as 2003-2012: Neither intake was included in the release datasets	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	The single reliable intake was included in the release datasets as Day 1 intake (n=21)
Food source (where food was obtained)	Codes 8 and 9 revised descriptions.	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014
Combination food types	Same as 2003-2012: Values for 15 combination types	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	Values for 16 combination types; added “Baby Toddler Food and Infant Formula”
Number of intakes that include only water consumption for the day	6 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (all in Day 2 data), records are included in Individual Foods file.	2 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (1 in Day 1, 4 in Day 2 data), records are included in Individual Foods file.	5 intakes (4 in Day 1, 1 in Day 2), records are included in Individual Foods file.
Number of intakes that include no water or food consumption for the day	1 intake in Day 2 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	2 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.	3 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.
Tap water source	Same as 2003-2012: Reported in 6 categories.	Same as 2003-2004	Responses combined into 4 categories	Same as 2017-2018	Same as 2017-2018
Modification codes: DR1MC	All remaining modification	No modification	No modification	No modification	No modification

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Day 2 Modification codes: DR2MC Modification Code Description file: DRXMCD	codes deleted; new food codes addressing modifications added in FNDDS 2013-2014	codes	codes	codes	codes
Salt use at the table and in cooking or preparing foods and type	Same as 2003-2004	Same as 2003-2004	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Salt used at the table yesterday and type	Question asked about salt use at the table yesterday and kind of salt to coincide with 24-hour recall.	Same as 2013-2014	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Main respondent and person whom helped in responding for the interview	Same as 2003-2012	Response category changed	Same as 2015-2016	Same as 2015-2016	Same as 2015-2016

Appendix 2. Variables in the Individual Foods Files (DR1IFF_L and DR2IFF_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1ILINE	DR2ILINE	Food/Individual component number
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1CCMNM	DR2CCMNM	Combination food number
DR1CCMTX	DR2CCMTX	Combination food type
DR1_020	DR2_020	Time of eating occasion (HH:MM)
DR1_030Z	DR2_030Z	Name of eating occasion
DR1FS	DR2FS	Source of food
DR1_040Z	DR2_040Z	Did you eat this meal at home?
DR1IFDCD	DR2IFDCD	USDA food code
DR1IGRMS	DR2IGRMS	Grams
DR1IKCAL	DR2IKCAL	Energy (kcal)
DR1IPROT	DR2IPROT	Protein (gm)
DR1ICARB	DR2ICARB	Carbohydrate (gm)
DR1ISUGR	DR2ISUGR	Total sugars (gm)
DR1IFIBE	DR2IFIBE	Dietary fiber (gm)
DR1ITFAT	DR2ITFAT	Total fat (gm)
DR1ISFAT	DR2ISFAT	Total saturated fatty acids (gm)
DR1IMFAT	DR2IMFAT	Total monounsaturated fatty acids (gm)
DR1IPFAT	DR2IPFAT	Total polyunsaturated fatty acids (gm)
DR1ICHOL	DR2ICHOL	Cholesterol (mg)
DR1IATOC	DR2IATOC	Vitamin E as alpha-tocopherol (mg)
DR1IATOA	DR2IATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1IRET	DR2IRET	Retinol (mcg)
DR1IVARA	DR2IVARA	Vitamin A, RAE (mcg)
DR1IACAR	DR2IACAR	Alpha-carotene (mcg)
DR1IBCAR	DR2IBCAR	Beta-carotene (mcg)
DR1ICRYP	DR2ICRYP	Beta-cryptoxanthin (mcg)
DR1ILYCO	DR2ILYCO	Lycopene (mcg)
DR1ILZ	DR2ILZ	Lutein + zeaxanthin (mcg)
DR1IVB1	DR2IVB1	Thiamin (Vitamin B1) (mg)

Day1 Name	Day2 Name	Variable Label
DR1IVB2	DR2IVB2	Riboflavin (Vitamin B2) (mg)
DR1INIAC	DR2INIAC	Niacin (mg)
DR1IVB6	DR2IVB6	Vitamin B6 (mg)
DR1IFOLA	DR2IFOLA	Total folate (mcg)
DR1IFA	DR2IFA	Folic acid (mcg)
DR1IFF	DR2IFF	Food folate (mcg)
DR1IFDFE	DR2IFDFE	Folate, DFE (mcg)
DR1ICHL	DR2ICHL	Total choline (mg)
DR1IVB12	DR2IVB12	Vitamin B12 (mcg)
DR1IB12A	DR2IB12A	Added vitamin B12 (mcg)
DR1IVC	DR2IVC	Vitamin C (mg)
DR1IVD	DR2IVD	Vitamin D (D2 + D3) (mcg)
DR1IVK	DR2IVK	Vitamin K (mcg)
DR1ICALC	DR2ICALC	Calcium (mg)
DR1IPHOS	DR2IPHOS	Phosphorus (mg)
DR1IMAGN	DR2IMAGN	Magnesium (mg)
DR1IIRON	DR2IIRON	Iron (mg)
DR1IZINC	DR2IZINC	Zinc (mg)
DR1ICOPP	DR2ICOPP	Copper (mg)
DR1ISODI	DR2ISODI	Sodium (mg)
DR1IPOTA	DR2IPOTA	Potassium (mg)
DR1ISELE	DR2ISELE	Selenium (mcg)
DR1ICAFF	DR2ICAFF	Caffeine (mg)
DR1ITHEO	DR2ITHEO	Theobromine (mg)
DR1IALCO	DR2IALCO	Alcohol (gm)
DR1IMOIS	DR2IMOIS	Moisture (gm)
DR1IS040	DR2IS040	SFA 4:0 (Butanoic) (gm)
DR1IS060	DR2IS060	SFA 6:0 (Hexanoic) (gm)
DR1IS080	DR2IS080	SFA 8:0 (Octanoic) (gm)
DR1IS100	DR2IS100	SFA 10:0 (Decanoic) (gm)
DR1IS120	DR2IS120	SFA 12:0 (Dodecanoic) (gm)
DR1IS140	DR2IS140	SFA 14:0 (Tetradecanoic) (gm)
DR1IS160	DR2IS160	SFA 16:0 (Hexadecanoic) (gm)
DR1IS180	DR2IS180	SFA 18:0 (Octadecanoic) (gm)
DR1IM161	DR2IM161	MFA 16:1 (Hexadecenoic) (gm)
DR1IM181	DR2IM181	MFA 18:1 (Octadecenoic) (gm)
DR1IM201	DR2IM201	MFA 20:1 (Eicosenoic) (gm)
DR1IM221	DR2IM221	MFA 22:1 (Docosenoic) (gm)
DR1IP182	DR2IP182	PFA 18:2 (Octadecadienoic) (gm)

Day1 Name	Day2 Name	Variable Label
DR1IP183	DR2IP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1IP184	DR2IP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1IP204	DR2IP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1IP205	DR2IP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1IP225	DR2IP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1IP226	DR2IP226	PFA 22:6 (Docosahexaenoic) (gm)

Appendix 3. List of Nutrients/Food Components (Unit)

Energy and Macronutrients

Food energy (kcal)
Protein (gm)
Carbohydrate (gm)
Fat, total (gm)
Alcohol (gm)

Sugars, total (gm)
Dietary fiber, total (gm)
Water (moisture) (gm)*

Saturated fatty acids, total (gm)
Monounsaturated fatty acids, total (gm)
Polyunsaturated fatty acids, total (gm)
Cholesterol (mg)

Individual fatty acids:
- 4:0 (gm)
- 6:0 (gm)
- 8:0 (gm)
- 10:0 (gm)
- 12:0 (gm)
- 14:0 (gm)
- 16:0 (gm)
- 18:0 (gm)
- 16:1 (gm)
- 18:1 (gm)
- 20:1 (gm)
- 22:1 (gm)
- 18:2 (gm)
- 18:3 (gm)
- 18:4 (gm)
- 20:4 (gm)
- 20:5 n-3 (gm)
- 22:5 n-3 (gm)
- 22:6 n-3 (gm)

Vitamins, Minerals, and Other Components

Vitamin A as retinol activity equivalents (mcg)
Retinol (mcg)
Carotenoids:
- Carotene, alpha (mcg)
- Carotene, beta (mcg)
- Cryptoxanthin, beta (mcg)
- Lycopene (mcg)
- Lutein + zeaxanthin (mcg)

Vitamin E as alpha-tocopherol (mg)
- **Added vitamin E as alpha-tocopherol (mg)

Vitamin D (D2 + D3) (mcg)
Vitamin K as phylloquinone (mcg)
Vitamin C (mg)
Thiamin (mg)
Riboflavin (mg)
Niacin (mg)
Vitamin B-6 (mg)
Folate, total (mcg)
- Folate as dietary folate equivalents (mcg)

Folic acid (mcg)
Food folate (mcg)
Choline, total (mg)
Vitamin B-12 (mcg)
***Added vitamin B-12 (mcg)

Calcium (mg)
Iron (mg)
Magnesium (mg)
Phosphorus (mg)
Potassium (mg)
Sodium (mg)
Zinc (mg)
Copper (mg)
Selenium (mcg)
Caffeine (mg)
Theobromine (mg)

** Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)*

*** Represents a synthetic subcomponent of vitamin E and is included in the vitamin E value.*

**** Represents a fortified subcomponent of vitamin B12 and is included in the vitamin B12 value.*

Appendix 4. Adding Food Code Descriptions to Your Files

One supporting file is included with the Individual Foods files: the Food Code Description file (DRXFCD_L).

The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code included in the Individual Foods files.

The Food Code Description file (DRXFCD_L) contains three variables:

DRXFDCD a numeric value corresponding to DR1IFDCD in the file DR1IFF_L or DR2IFDCD in the file DR2IFF_L;

DRXFCSD a short description (up to 60 characters) of the food code;

DRXFDLD a long description (up to 200 characters) of the food code.

The following SQL code is an example of appending the shorter food code description (here renamed DR1IFCSD) to one of the Individual Foods files using PROC SQL from SAS®. Other SQL implementations may be different.

```
proc sql;
  create table DR1IFF_L_PLUS as
  select iff.*, desc.DRXFCSD as DR1FCSD, desc.DRXFCLD as DR1FCLD
  from NHANES.DR1IFF_L iff
  left join NHANES.DRXFCD_L desc
  on iff.DR1IFDCD = desc.DRXFDCD
  order by SEQN, DR1ILINE;
quit;
```

SAS® users may wish to use Proc Format to assign labels to the food codes. The following example generates and saves a picture format for food codes and a separate format for each food code that includes both the food code itself and the short food code description. It is assumed that the user has stored the Individual Foods files and the Food Code Description file in a library called NHANES and wishes to store the formats there as well.

```
options fmtsearch = (NHANES);

proc format library = library;
  picture foodcode
  low - high = '000-00000';
quit;

data tmp;
  set NHANES.DRXFCD_L;
  length cfoodcode $9 label $72;
  cfoodcode = put(DRXFDCD, foodcode.);
  label = cfoodcode || '  ' || DRXFCSD;
run;

data fmt (keep = fmtname start label);
  set tmp;
  retain fmtname 'DRXFDCD';
  rename DRXFDCD = start;
run;
```

```
proc format cntlin = fmt library = library;
run;
```

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Appendix 5. Variables in the Total Nutrients Files (DR1TOT_L and DR2TOT_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1MRESP	DR2MRESP	Main respondent for this interview
DR1HELP	DR2HELP	Helped in responding for this interview
DBQ095Z	N/A	Type of table salt used
DBD100	N/A	How often add salt to food at table
DRQSPREP	N/A	Salt used in preparation?
DR1STY	DR2STY	Salt used at table yesterday?
DR1SKY	DR2SKY	Type of salt used yesterday
DRQSDIET	N/A	On special diet?
DRQSDT1	N/A	Weight loss/Low calorie diet
DRQSDT2	N/A	Low fat/Low cholesterol diet
DRQSDT3	N/A	Low salt/Low sodium diet
DRQSDT4	N/A	Sugar free/Low sugar diet
DRQSDT5	N/A	Low fiber diet
DRQSDT6	N/A	High fiber diet
DRQSDT7	N/A	Diabetic diet
DRQSDT8	N/A	Weight gain/Muscle building diet
DRQSDT9	N/A	Low carbohydrate diet
DRQSDT10	N/A	High protein diet
DRQSDT11	N/A	Gluten-free/Celiac diet
DRQSDT12	N/A	Renal/Kidney diet
DRQSDT91	N/A	Other special diet
DR1TNUMF	DR2TNUMF	Number of foods/beverages reported
DR1TKCAL	DR2TKCAL	Energy (kcal)
DR1TPROT	DR2TPROT	Protein (gm)
DR1TCARB	DR2TCARB	Carbohydrate (gm)
DR1TSUGR	DR2TSUGR	Total sugars (gm)
DR1TFIBE	DR2TFIBE	Dietary fiber (gm)
DR1TTFAT	DR2TTFAT	Total fat (gm)
DR1TSFAT	DR2TSFAT	Total saturated fatty acids (gm)

Day1 Name	Day2 Name	Variable Label
DR1TMFAT	DR2TMFAT	Total monounsaturated fatty acids (gm)
DR1TPFAT	DR2TPFAT	Total polyunsaturated fatty acids (gm)
DR1TCHOL	DR2TCHOL	Cholesterol (mg)
DR1TATOC	DR2TATOC	Vitamin E as alpha-tocopherol (mg)
DR1TATOA	DR2TATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1TRET	DR2TRET	Retinol (mcg)
DR1TVARA	DR2TVARA	Vitamin A, RAE (mcg)
DR1TACAR	DR2TACAR	Alpha-carotene (mcg)
DR1TBCAR	DR2TBCAR	Beta-carotene (mcg)
DR1TCRYP	DR2TCRYP	Beta-cryptoxanthin (mcg)
DR1TLYCO	DR2TLYCO	Lycopene (mcg)
DR1TLZ	DR2TLZ	Lutein + zeaxanthin (mcg)
DR1TVB1	DR2TVB1	Thiamin (Vitamin B1) (mg)
DR1TVB2	DR2TVB2	Riboflavin (Vitamin B2) (mg)
DR1TNIAC	DR2TNIAC	Niacin (mg)
DR1TVB6	DR2TVB6	Vitamin B6 (mg)
DR1TFOLA	DR2TFOLA	Total folate (mcg)
DR1TFA	DR2TFA	Folic acid (mcg)
DR1TFF	DR2TFF	Food folate (mcg)
DR1TFDFE	DR2TFDFE	Folate, DFE (mcg)
DR1TCHL	DR2TCHL	Total choline (mg)
DR1TVB12	DR2TVB12	Vitamin B12 (mcg)
DR1TB12A	DR2TB12A	Added vitamin B12 (mcg)
DR1TVC	DR2TVC	Vitamin C (mg)
DR1TVD	DR2TVD	Vitamin D (D2 + D3) (mcg)
DR1TVK	DR2TVK	Vitamin K (mcg)
DR1TCALC	DR2TCALC	Calcium (mg)
DR1TPHOS	DR2TPHOS	Phosphorus (mg)
DR1TMAGN	DR2TMAGN	Magnesium (mg)
DR1TIRON	DR2TIRON	Iron (mg)
DR1TZINC	DR2TZINC	Zinc (mg)
DR1TCOPP	DR2TCOPP	Copper (mg)
DR1TSODI	DR2TSODI	Sodium (mg)
DR1TPOTA	DR2TPOTA	Potassium (mg)
DR1TSELE	DR2TSELE	Selenium (mcg)
DR1TCAFF	DR2TCAFF	Caffeine (mg)
DR1TTHEO	DR2TTHEO	Theobromine (mg)
DR1TALCO	DR2TALCO	Alcohol (gm)
DR1TMOIS	DR2TMOIS	Moisture (gm)

Day1 Name	Day2 Name	Variable Label
DR1TS040	DR2TS040	SFA 4:0 (Butanoic) (gm)
DR1TS060	DR2TS060	SFA 6:0 (Hexanoic) (gm)
DR1TS080	DR2TS080	SFA 8:0 (Octanoic) (gm)
DR1TS100	DR2TS100	SFA 10:0 (Decanoic) (gm)
DR1TS120	DR2TS120	SFA 12:0 (Dodecanoic) (gm)
DR1TS140	DR2TS140	SFA 14:0 (Tetradecanoic) (gm)
DR1TS160	DR2TS160	SFA 16:0 (Hexadecanoic) (gm)
DR1TS180	DR2TS180	SFA 18:0 (Octadecanoic) (gm)
DR1TM161	DR2TM161	MFA 16:1 (Hexadecenoic) (gm)
DR1TM181	DR2TM181	MFA 18:1 (Octadecenoic) (gm)
DR1TM201	DR2TM201	MFA 20:1 (Eicosenoic) (gm)
DR1TM221	DR2TM221	MFA 22:1 (Docosenoic) (gm)
DR1TP182	DR2TP182	PFA 18:2 (Octadecadienoic) (gm)
DR1TP183	DR2TP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1TP184	DR2TP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1TP204	DR2TP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1TP205	DR2TP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1TP225	DR2TP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1TP226	DR2TP226	PFA 22:6 (Docosahexaenoic) (gm)
DR1_300	DR2_300	Compare food consumed yesterday to usual
DR1_320Z	DR2_320Z	Total plain water drank yesterday (gm)
DR1_330Z	DR2_330Z	Total tap water drank yesterday (gm)
DR1BWATZ	DR2BWATZ	Total bottled water drank yesterday (gm)
DR1TWSZ	DR2TWSZ	Tap water source
DRD340	N/A	Shellfish eaten during past 30 days
DRD350A	N/A	Clams eaten during past 30 days
DRD350AQ	N/A	# of times clams eaten in past 30 days
DRD350B	N/A	Crabs eaten during past 30 days
DRD350BQ	N/A	# of times crabs eaten in past 30 days
DRD350C	N/A	Crayfish eaten during past 30 days
DRD350CQ	N/A	# of times crayfish eaten past 30 days
DRD350D	N/A	Lobsters eaten during past 30 days
DRD350DQ	N/A	# of times lobsters eaten past 30 days
DRD350E	N/A	Mussels eaten during past 30 days
DRD350EQ	N/A	# of times mussels eaten in past 30 days
DRD350F	N/A	Oysters eaten during past 30 days
DRD350FQ	N/A	# of times oysters eaten in past 30 days
DRD350G	N/A	Scallops eaten during past 30 days
DRD350GQ	N/A	# of times scallops eaten past 30 days

Day1 Name	Day2 Name	Variable Label
DRD350H	N/A	Shrimp eaten during past 30 days
DRD350HQ	N/A	# of times shrimp eaten in past 30 days
DRD350I	N/A	Other shellfish eaten past 30 days
DRD350IQ	N/A	# of times other shellfish eaten
DRD350J	N/A	Other unknown shellfish eaten past 30 d
DRD350JQ	N/A	# of times other unknown shellfish eaten
DRD350K	N/A	Refused on shellfish eaten past 30 days
DRD360	N/A	Fish eaten during past 30 days
DRD370A	N/A	Breaded fish products eaten past 30 days
DRD370AQ	N/A	# of times breaded fish products eaten
DRD370B	N/A	Tuna eaten during past 30 days
DRD370BQ	N/A	# of times tuna eaten in past 30 days
DRD370C	N/A	Bass eaten during past 30 days
DRD370CQ	N/A	# of times bass eaten in past 30 days
DRD370D	N/A	Catfish eaten during past 30 days
DRD370DQ	N/A	# of times catfish eaten in past 30 days
DRD370E	N/A	Cod eaten during past 30 days
DRD370EQ	N/A	# of times cod eaten in past 30 days
DRD370F	N/A	Flatfish eaten during past 30 days
DRD370FQ	N/A	# of times flatfish eaten past 30 days
DRD370G	N/A	Haddock eaten during past 30 days
DRD370GQ	N/A	# of times haddock eaten in past 30 days
DRD370H	N/A	Mackerel eaten during past 30 days
DRD370HQ	N/A	# of times mackerel eaten past 30 days
DRD370I	N/A	Perch eaten during past 30 days
DRD370IQ	N/A	# of times perch eaten in past 30 days
DRD370J	N/A	Pike eaten during past 30 days
DRD370JQ	N/A	# of times pike eaten in past 30 days
DRD370K	N/A	Pollock eaten during past 30 days
DRD370KQ	N/A	# of times pollock eaten in past 30 days
DRD370L	N/A	Porgy eaten during past 30 days
DRD370LQ	N/A	# of times porgy eaten in past 30 days
DRD370M	N/A	Salmon eaten during past 30 days
DRD370MQ	N/A	# of times salmon eaten in past 30 days
DRD370N	N/A	Sardines eaten during past 30 days
DRD370NQ	N/A	# of times sardines eaten past 30 days
DRD370O	N/A	Sea bass eaten during past 30 days
DRD370OQ	N/A	# of times sea bass eaten past 30 days
DRD370P	N/A	Shark eaten during past 30 days

Day1 Name	Day2 Name	Variable Label
DRD370PQ	N/A	# of times shark eaten in past 30 days
DRD370Q	N/A	Swordfish eaten during past 30 days
DRD370QQ	N/A	# of times swordfish eaten past 30 days
DRD370R	N/A	Trout eaten during past 30 days
DRD370RQ	N/A	# of times trout eaten in past 30 days
DRD370S	N/A	Walleye eaten during past 30 days
DRD370SQ	N/A	# of times walleye eaten in past 30 days
DRD370T	N/A	Other fish eaten during past 30 days
DRD370TQ	N/A	# of times other fish eaten past 30 days
DRD370U	N/A	Other unknown fish eaten in past 30 days
DRD370UQ	N/A	# of times other unknown fish eaten
DRD370V	N/A	Refused on fish eaten past 30 days

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Interview - Total Nutrient Intakes, First Day (DR1TOT_L)

Data File: DR1TOT_L.xpt

First Published: October 2024

Last Revised: NA

Component Description

The objective of the dietary interview component is to obtain detailed dietary intake information from NHANES participants. The dietary intake data are used to estimate the types and amounts of foods and beverages (including all types of water) consumed during the 24-hour period prior to the interview (midnight to midnight), and to estimate intakes of energy, nutrients, and other food components from those foods and beverages. Following the dietary recall, participants are asked questions on salt use, whether the person's overall intake on the previous day was much more than usual, usual or much less than usual, and whether the participant is on any type of special diet. Questions on frequency of fish and shellfish consumed during the past 30 days are asked of participants 1 year or older, with the use of proxies for young children (see the Dietary Interview Procedure Manuals (cdc.gov) for more information on the proxy interview).

The dietary interview component, called What We Eat in America (WWEIA), is conducted as a partnership between the U.S. Department of Agriculture (USDA) and the U.S. Department of Health and Human Services (DHHS). Under this partnership, DHHS' National Center for Health Statistics (NCHS), Division of Health and Nutrition Examination Surveys is responsible for the survey sample design and all aspects of data collection and USDA's Food Surveys Research Group (FSRG) is responsible for the dietary data collection methodology, maintenance of the databases used to code and process the data, and data review and processing.

All NHANES participants are eligible for two 24-hour dietary recall interviews. Traditionally, the first dietary recall interview was collected in-person in the Mobile Examination Center (MEC) and the second interview was collected by telephone 3 to 10 days later. In August 2021-August 2023, the time participants spent in the MEC was limited to reduce risk of exposure to SARS-CoV-2. As a result, both dietary recall interviews were administered via telephone. Further information about changes in data collection are described in the Plan and Operations report of the NHANES August 2021-August 2023.

As in previous years, two types of dietary intake data are available for the August 2021-August 2023 survey cycle: Individual Foods files and Total Nutrient Intakes files.

What's New with the August 2021-August 2023 WWEIA Release:

In response to the COVID-19 pandemic, the mode of the first dietary recall changed from in-person during the MEC examination to telephone 3 to 7 days after the examination. Both dietary interviews were completed in English or Spanish. During the mobile examination, participants scheduled an appointment for the first telephone dietary interview and received a set of measuring guides to help estimate the amounts of foods and beverages consumed. **Appendix 1** provides a summary of changes among the 5 latest cycles of data collection. No new data variables were added in the August 2021-August 2023 WWEIA release.

Dietary Interview Data Files: Four data files were produced from the information collected in the dietary interviews: two Individual Foods files and two Total Nutrient Intakes files. Each

file includes one day of intake data. The number “1” or “2” in the file name identifies the day of the interview: 1 = first day, 2 = second day. File names are as follows:

File Names for Dietary Interview Data:

File	Day 1	Day 2
Individual Foods File	DR1IFF_L	DR2IFF_L
Total Nutrient Intakes File	DR1TOT_L	DR2TOT_L

The amounts in these files reflect only nutrients obtained from foods, beverages, and water, including tap and bottled water. They do not include nutrients obtained from dietary supplement intakes, antacids, or medications. Data on intake of dietary supplement use are available on the August 2021-August 2023 Dietary Data (cdc.gov) page.

Individual Foods Files (DR1IFF_L and DR2IFF_L): Detailed information about each food/beverage item (including the description, amount of, and nutrient content) reported by each participant is included in the Individual Foods files. The names for both Day 1 and Day 2 variables are listed in **Appendix 2**.

The Individual Foods files include, for each interview day, one record for each food/beverage consumed by a participant. Each record is uniquely numbered within a participant’s set of records and contains the information listed below:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Whether the food/beverage was eaten in combination with other foods, such as in a sandwich;
- Time of eating occasion/when the food was eaten;
- Eating occasion name;
- Where the food/beverage was obtained;
- Whether the meal/snack was eaten at home or not;
- A USDA Food and Nutrient Database for Dietary Studies (FNDDS) code identifying the food/beverage;
- Amount of food/beverage consumed, in grams; and
- Food energy and 64 nutrients/food components (listed in **Appendix 3**) from each food/beverage as calculated using USDA's Food and Nutrient Database for Dietary Studies 2021-2023 (FNDDS 2021-2023).

Descriptions for the USDA FNDDS food codes are provided in the Food Code Description file (DRXFCD_L). The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code in the FNDDS 2021-2023. **Appendix 4** provides SAS code examples that may be used to link the food code description to the Individual Foods file.

Total Nutrient Intakes Files (DR1TOT_L and DR2TOT_L): For each participant, daily total energy and nutrient intakes from foods and beverages, and whether the amount of food consumed was usual, much more than usual, or much less than usual, are included in the Total Nutrient Intakes files. The Day 1 file also includes information on salt use in cooking and at the table; whether the participant is currently on any kind of diet to lose weight or for another health-related reason and, if so, the type of diet; and information on frequency of fish and shellfish consumption for participants aged 1 or older. The names for both Day 1 and Day 2 variables are listed in **Appendix 5**.

The Total Nutrient Intakes files provide a summary record of total nutrient intakes for each participant. Each total intake record contains the following information:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Type of salt used and how often added at the table and in food preparation (Day 1 file only);
- Use of salt at the table yesterday and the type of salt used;
- Whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet (Day 1 file only);
- Total number of foods and beverages including water reported for that participant for that day's intake;
- Daily aggregates of food energy and 64 nutrients/food components (listed in **Appendix 3**) from all foods/beverages as calculated using USDA's Food and Nutrient Database for Dietary Studies August 2021-August 2023 (FNDDS 2021- 2023)
- Whether the amount of food consumed was usual, much more than usual, or much less than usual;
- Total amount of tap and bottled water consumed (calculated as the sum of reports of water drunk by itself in the 24-hour recall) and the usual source of tap water; and
- Frequency of fish and shellfish consumption in the past 30 days (participants 1 year or older, Day 1 file only).

Eligible Sample

All participants in the August 2021-August 2023 sample who spoke English or Spanish were eligible. Participants aged 1 year or older were eligible for the frequency of fish and shellfish consumption questions following the 24-hour recall.

Protocol and Procedure

The examination protocol and data collection methods are fully documented in the NHANES dietary interviewer procedures manuals.

Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant’s intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child’s intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish.

During the MEC visit, participants were scheduled an appointment for the first dietary interview and were given a set of measuring guides (NHANES Measuring Guides for the Dietary Recall Interview) to use for reporting food amounts during the telephone interview. The measuring guides included measuring cups, spoons, a ruler, and a food model booklet, which contained two-dimensional drawings of glasses, mugs, bowls, mounds, circles, wedges, a shape chart and a chicken chart. The first telephone dietary interview was scheduled 3 to 7 days post MEC examination while the second interview was collected 3 to 10 days following the first dietary interview and were generally scheduled on a different day of the week as the first interview. In August 2021-August 2023, 204 participants were interviewed on the same day of the week for both day 1 and day 2 interviews due to their scheduling availability. Any participant who did not have a telephone was given a toll-free number to call so that the recall could be conducted.

What We Eat in America data were collected using USDA's dietary data collection instrument, the Automated Multiple Pass Method (AMPM), available at:

http://www.ars.usda.gov/nea/bhnrc/fsrg. The AMPM was designed to provide an efficient and accurate means of collecting intakes for large-scale national surveys. The AMPM is a fully computerized recall method that uses a 5-step interview outlined below:

1. **Quick List** - Participant recalls all foods and beverages consumed the day before the interview (midnight to midnight).
2. **Forgotten Foods** - Participant is asked about consumption of foods commonly forgotten during the Quick List step.
3. **Time and Occasion** - Time and eating occasion are collected for each food.
4. **Detail Cycle** - For each food, a detailed description, amount eaten, and additions to the food are collected. Eating occasions and times between eating occasions are reviewed to elicit forgotten foods.
5. **Final Probe** - Additional foods not remembered earlier are collected.

The AMPM includes an extensive compilation of standardized food-specific questions and possible response options. Routing of questions is based on previous responses. The AMPM is updated for each 2-year collection of WWEIA to reflect the changing food supply and to address research needs from the data user community. Additional information about the AMPM is provided in Raper et al. (Raper et al., 2004).

The AMPM was validated in a large study and shown to be an effective method for collecting accurate group energy intake of adults. Completed in 2004, this extensive research project included 524 healthy, weight-stable volunteers, aged 30-69 years. The accuracy of the AMPM was evaluated by comparing reported energy intake (EI) to total energy expenditure (TEE) using the doubly labeled water technique (Moshfegh et al., 2008). Among the findings were that EI compared to TEE was under-reported by 11% overall, by less than 3% for normal weight subjects with body mass index (BMI) < 25 and 16% for overweight subjects with BMI ≥ 25.

Additional studies provide evidence that the AMPM accurately measures group energy intake. Blanton (Blanton et al., 2006) reported that EI was not significantly different from TEE for a sample of 20 adult females. Rumpler (Rumpler et al., 2008) found that mean EIs were accurately reported for a sample of 12 adult males.

Additional evidence for the accuracy of AMPM has been provided by analysis of the 24-hour urinary sodium data collected in the AMPM Validation Study, which suggest the AMPM is a valid measure for estimating mean sodium intake in adults. Dietary sodium intake calculated from 24-hour recall data of 465 subjects collected via AMPM was compared with sodium values from 24-hour urine collections measured during the same 24-hour period. The AMPM-derived mean dietary sodium estimates reflected over 90% of the biomarker-based estimates (Rhodes et al., 2013).

For additional information about the dietary interview component and related survey protocols, please visit the August 2021-August 2023 Dietary Interviewer Procedures Manual page. A description of changes to the NHANES August 2021-August 2023 survey is provided in the Plan and Operations of the National Heath And Nutrition Examination Survey, August 2021-August 2023.

Quality Assurance & Quality Control

All dietary interviewers were required to complete an intensive one-week training course and to conduct supervised practice interviews before working independently in the field. Retraining sessions were conducted annually to reinforce the proper protocols and technique.

Interviewers were monitored throughout the data collection period. Monitoring consisted of the following:

- Reviews of audio recorded interviews were conducted for approximately 5% of each interviewer's work.

- Quality control of interviews, which were checked for completeness of the recalls, missing information, inconsistent reports, and unclear notes. Written notification and feedback were provided to the interviewers.

Data Processing and Editing

Interview data files were sent electronically from the field and were imported into Survey Net, a computer-assisted food coding and data management system developed by USDA (Raper et al., 2004).

USDA's Food and Nutrient Database for Dietary Studies (FNDDS) 2021-2023 was used to process the intakes reported by the August 2021-August 2023 sample. The FNDDS includes comprehensive information that can be used to code individual foods/beverages and portion sizes reported by participants and includes nutrient values for calculating nutrient intakes. FNDDS nutrient values as well as food codes and portion sizes are updated for every 2-year WWEIA, NHANES release cycle. The basis for the nutrient values as well as food codes and portion weights in FNDDS are detailed in the documentation for FNDDS 2021-2023 available at http://www.ars.usda.gov/nea/bhnrc/fsrg.

Coders were required to pass a certification test after the initial training. They were routinely monitored to ensure the quality and completeness of their work. Approximately 10 percent of the coder's work was randomly selected to be independently coded by another coder. Results from the two codings were compared and adjudicated, if necessary.

After intake data were coded, various types of reviews and quality assurance procedures were conducted to ensure the quality of the data. Examples of reviews include the following:

- Interviewers' and coders' questions and comments were reviewed to ensure that they have been addressed.
- Decisions made by coders about how to code new or unusual foods/beverages or quantities reported by participants were reviewed for accuracy, reasonableness, and consistency across intake data; items in question were resolved by coder supervisors.
- Specific data integrity checks for reasonableness, consistency, and logic were conducted.

Analytic Notes

Each Individual Foods file (Day 1 and Day 2) is comprised of food records. For most participants, there are multiple records in each file. For each Total Nutrient Intakes file (Day 1 and Day 2) there is one record for each participant. These files can be linked with other NHANES files by the respondent sequence number (SEQN).

Variable names: For data collected on both Day 1 and Day 2, variable names are differentiated by having the number "1" or "2" in the third position of the variable name to identify the collection day. For example, the USDA food code variable (in the Individual Foods File), which identifies the food reported by the participant, is named DR1IFDCD in the Day 1 file and DR2IFDCD in the Day 2 file. Appendices **2** and **5** list the Day 1 and Day 2 variable names for the Individual Foods file and the Total Nutrient Intakes file, respectively.

Names for the following variables are the same for both days in the Individual Foods file and the Total Nutrient Intakes file:

Variables with the Same Name for Both Days in the Dietary Interview Files

Day 1 and Day 2 variable name	Label
SEQN	Respondent sequence number
WTDRD1	Dietary day one sample weight
WTDR2D	Dietary two-day sample weight
DRABF	Breast-fed infant (either day)
DRDINT	Number of days of intake

Number of days of intake: A variable has been included to indicate the number of days of intake collected from each participant. The variable name is DRDINT. In the August 2021-August 2023 sample, 6,754 participants provided complete dietary intakes for Day 1. The 6,754 participants with a complete Day 1 intake include 21 participants who had an unreliable day 1 intake but reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall. Of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2.

Dietary recall status code: A status code (DR1DRSTZ or DR2DRSTZ) is used in both the Individual Foods and Total Nutrient Intakes files to indicate the quality and completeness of a survey participant's response to the dietary recall section. The codes are the following:

1 = Reliable and met the following minimum criteria:

- The first 4 steps of the 5-step AMPM completed.
- Food/beverages consumed for each reported eating occasion identified.

For individuals with a code 1, all relevant variables associated with the 24-hour dietary recall contain a value.

2 = Not reliable or did not meet the minimum criteria

Individuals with a code 2 have incomplete records. No data on total nutrient intakes and the total number of foods reported are provided for these cases. These individuals have no records in the Individual Foods files.

3 [Code 3 is not included in the current datasets. It was only used for data from the 1999-2000 survey cycle.]

4 = Reported consuming breast milk

For infants and children who consumed human milk, there is a record in the Individual Foods files for each report of human milk. However, because amounts of human milk intake are not quantified, these records contain missing values for the amount consumed and for the amounts of energy and nutrients from human milk. Also, records of human milk have a missing value for the food source variable (DR1FS, DR2FS) and the eaten at home variable (DR1_040Z, DR2_040Z) in the Individual Foods files. Records for any other foods and beverages consumed by breast-fed infants and children are included in the Individual Foods files along with their amounts and nutrient information. Because of the missing amount or quantity information for human milk, no total nutrient intakes (contained in the Total Nutrient Intakes files) were computed for participants with a code 4.

A variable that identifies breast-fed children, DRABF, is included. This variable has a code of 1 if a child consumed human milk in either intake day.

5 = Not done

This code is assigned when the dietary recall section of the interview did not take place due to various reasons (such as refusal, equipment failure, or unable to contact the participant). These individuals have no records in the Individual Foods files. These individuals have a record in the Total Nutrients file with values only for the following variables: the respondent sequence number (SEQN), the dietary recall status code (DR1DRSTZ or DR2DRSTZ) and for participants 1 year or older, the fish and shellfish questions in the DR1TOT_L file (DRD340, DRD350A-K, DRD350AQ-JQ, DRD360, DRD370A-V, and DRD370AQ-UQ).

Only codes 1 and 4 appear in the Individual Foods file.

Distinguishing Between Foods/Beverages and Dietary Supplements in NHANES: In August 2021-August 2023, there was no 24-hour dietary supplement use collection. The 30-day dietary supplement use collection changed from being collected during the home interview to collection after the first 24-hour dietary recall via telephone. All NHANES participants responding to the 24-hour dietary recall interview are eligible for the dietary supplement and non-prescription antacid use questions. Information is obtained on all vitamins, minerals, herbals, and other dietary supplements as well as non-prescription antacids that were consumed during a 30-day period, including the name and the amount of supplement or antacid taken.

Distinguishing between foods/beverages and supplements can be challenging. NCHS and FSRG review questionable items reported in the dietary supplement and dietary recall components to resolve disposition of these items into the appropriate component. Products that are labeled as a dietary supplement, that have a supplement facts panel on the label, and are in tablets, capsules, softgels, gelcaps, or other pill forms, are considered dietary supplements. Items that are powders or liquids can be hard to distinguish. General guidelines used state that if powders and liquid concentrates have product directions stating that they be added to a liquid, they are classified as beverages. Examples are teas and protein powders. An exception is made for fiber products, which are classified as dietary supplements. Along this same guideline, energy drinks are considered beverages, but "energy shot" type products are considered dietary supplements.

It is best to refer to the databases that detail every food/beverage and dietary supplement reported in NHANES to identify exact determination used. The databases are:

- 2021-2023 Food and Nutrient Database for Dietary Studies
- NHANES Dietary Supplement Database

Participants who reported consuming only water, no food or other beverages: Records are included in the Individual Foods file for participants who consumed only water. There are 5 such individuals in the August 2021-August 2023 datasets, 4 in the Day 1 data and 1 in the Day 2 data. Their dietary recall status variable for the day is coded as "1" (complete and reliable) in the Total Nutrients file and the total number of items is the number of times water was reported. Individuals with just water intake and no food intake will have zero energy intake for the day.

Participants who reported consuming no water, food or other beverages: There can be participants whose intakes are determined to be complete even though they reported no water, food, or other beverage records for the day. For such participants there are no records in the Individual Foods file, but their dietary recall status is coded as complete and reliable, and the Total Nutrients file will include records with zero values for all nutrients. In the August 2021-August 2023 datasets, there are 3 individuals in the day 1 data that reported no water, food, or other beverage records for the day.

Number of days between the intake day and the day of family interview: Each of the four intake files includes a variable (DR1DBIH for Day 1 files and DR2DBIH for Day 2 files) to indicate the number of days between the intake day (i.e., the period covered by the 24-hour recall) and the day that the family questionnaire was administered in the household. A positive value in DR1BHIH or DR2BHIH indicates the family interview occurred prior to the intake day. In the survey, most of the family interviews were done before the participant participated in the dietary interview. A value of "0" in DR1BHIH or DR2BHIH indicates the family interview

occurred on the same date as the intake day. A negative value (i.e., DR1BHIH<0 or DR2BHIH<0) means that the family interview occurred after the intake day.

Food source: The source from which each food/beverage was obtained (e.g., from a store, fast food restaurant, cafeteria) is identified by the variables DR1FS (day 1) and DR2FS (day 2) in the Individual Foods files.

The code descriptions for this variable are:

Code Descriptions for Source of Food Variable

Code	Description
1	Store grocery/supermarket
2	Restaurant with waiter/waitress
3	Restaurant fast food/Pizza
4	Bar/Tavern/Lounge
5	Restaurant, no additional information
6	Cafeteria NOT in a K-12 school
7	Cafeteria in a K-12 school
8	Child/Adult care center
9	Child/Adult home care
10	Soup kitchen/shelter/food pantry facility
11	Meals on Wheels Program
12	Community food program – other
13	Community program, no additional info
14	Vending machine
15	Common coffee pot or snack tray
16	From someone else/gift
17	Mail order purchase
18	Residential dining facility
19	Grown or caught by you or someone you know
20	Fish caught by you or someone you know
24	Sport, recreation, or entertainment
25	Street vendor, vending truck
26	Fundraiser sales
27	Store - convenience type
28	Store - no additional information
91	Other, specify

Eating occasion: The variables DR1_030Z and DR2_030Z are located in the Individual Foods file. The code descriptions for the eating occasion variables are shown in the table below.

Code Descriptions for Eating Occasion Variable

Code	Description
1	Breakfast
2	Lunch
3	Dinner
4	Supper
5	Brunch
6	Snack
7	Beverage/Drink
8	Feeding-infant only
9	Extended consumption
10	Desayuno
11	Almuerzo
12	Comida
13	Merienda
14	Cena
15	Entre comida
16	Botana
17	Bocadillo
18	Tentempie
19	Bebida
91	Other

Eating occasion was designated by the respondent. During the interview, a list of eating occasion names was available to the respondent for selection. However, eating occasion names were not defined for the respondent.

Foods and beverages coded as part of a combination: 41 percent of foods and beverages reported in the WWEIA, NHANES August 2021-August 2023 sample were identified as items consumed together as combinations. Items consumed as a combination were identified by one of sixteen unique "combination food types." Foods and beverages not coded in combination have the code "0" for the combination food type variable.

The combination types provide a linkage for:

- Foods or beverages with additions, such as cereal with milk, coffee with cream;
- Multi-component foods that have specific protocol for collection such as some salads and sandwiches; and
- Other combinations that do not have a unique code in the FNDDS.

Combination Type, Code, Examples, and Percent of Food and Beverages Reported by Type, August 2021-August 2023, Day 1

Combination Type	Code	Examples of Combination Type	% Items
Not in combination	0	NA	59
Beverage w/ additions	1	Coffee, tea with: milk, cream, sugar.	9
Cereal w/ additions	2	Cereals (ready-to-eat, cooked) with: milk, sugar, fruit, butter.	4
Bread/baked product w/additions	3	Breads, rolls, pancakes with: butter, jam, syrup, fruit. Cakes, pies with: ice cream, toppings. Crackers with: cheese, dip, peanut butter.	5
Salad	4	Components of salads that do not have a single code in FNDDS. It may also designate additional items to single code salads.	5
Sandwiches	5	Components of sandwiches that do not have a single code in FNDDS. It may also designate additional items added to single code sandwiches.	6
Soup	6	Soup with: crackers, croutons, cheese.	<1
Frozen meals	7	Components of a prepackaged frozen meal and additions to the meal.	<1
Ice cream/ frozen yogurt w/ additions	8	Ice cream with: syrup, nuts, toppings.	<1
Dried beans or Vegetable w/ additions	9	French fries, potatoes with: catsup, gravy, butter, toppings. Beans with: sauce, butter.	3
Fruit w/ additions	10	Fruit with: toppings, milk, honey. Components of fruit mixtures or salads that do not have a single code in FNDDS.	1
Tortilla products	11	Components of tacos and tortilla products that do not have a single code in FNDDS. It may also designate additional items to single code tacos or tortilla products.	2
Meat, Poultry, Fish	12	Meat, poultry, fish with: gravy, sauce, and condiments.	2
Lunchables®	13	Components of pre-packaged lunch kits.	<1
Chips w/ additions	14	Potato chips, corn chips with: dip, cheese, salsa.	1
Baby Toddler Food and Infant Formula	15	Baby Toddler cereal with: infant formula, fruit. Infant formula with: baby cereal.	<1
Other mixtures	90	Rice, pasta, spaghetti, eggs, other mixtures with: butter, gravy, sauce, condiments.	4

All items given a combination food type are given an additional variable to identify each of the items within the combination. This variable is the "combination food number" that is unique to the combination food type within the individual intake.

Variable Labels and Names for Combination Coding

Combination Coding	Variable Name, Day 1	Variable Name, Day 2
Combination food type	DR1CCMTX	DR2CCMTX
Combination food number	DR1CCMNM	DR2CCMNM

The What We Eat in America Food Categories, available on the FSRG website (http://www.ars.usda.gov/nea/bhnrc/fsrg), is a grouping scheme that combines foods and beverages together that have similar usage and nutrient content with the emphasis on

how they are commonly consumed in the American diet. There are more than 170 unique categories, and each is assigned a 4-digit number and description. The WWEIA Food Categories contain discrete food items and are not disaggregated (e.g., pizza vs. grain, cheese, tomatoes, etc.). Designed to be flexible, the categories can be combined as needed to address specific research questions. A new version of the WWEIA Food Categories is produced for each 2-year release cycle of WWEIA, NHANES.

Special diet: Information on whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet, was provided. The variable DRQSDIET identifies whether a participant was on a special diet. The variables DRQSDT1 through DRQSDT12 and DRQSDT91 identify the type of diet or diets that the participant was following. These variables can be found in the Total Nutrient Intakes file.

Sample weights for dietary intake data: The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, **the appropriate sample weights must be used.**

For the NHANES August 2021-August 2023 sample, there were 22,660 persons selected; of these 8,860 were considered participants to the MEC examination and data collection. A total of 6,754 MEC participants provided complete dietary intakes for Day 1, and of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2. The 6,754 participants who provided complete Day 1 intakes includes 21 participants who had an unreliable day 1 intake and reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall.

Most analyses of NHANES data use data collected in the MEC and the variable WTMEC2YR should be used for the sample weights. However, for the WWEIA dietary data, different sample weights are recommended for analysis. Because food intake can vary by weekdays and weekends, use of the MEC weights disproportionately represents intakes on weekends.

A set of weights (WTDRD1) is provided that should be used when an analysis uses the Day 1 dietary recall data (either alone or when Day 1 nutrient data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with Day 1 data. Day 1 weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These Day 1 weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using Day 1 data and Day 1 weights might be larger than standard errors for similar estimates based on MEC weights.

When analysis is based on both days of dietary intake, only 5,879 sample participants have complete data. The NHANES protocol requires an attempt to collect the second day of dietary data at least 3 days after the first day, but the actual number of days between the two interviews is variable. A set of adjusted weights, WTDR2D, is to be used when an analysis uses the smaller sample with completed Day 1 and Day 2 dietary data. This two-day weight was constructed for the 5,879 participants by taking the Day 1 weights (WTDRD1) and further adjusting for: (a) the additional non-response for the second recall; and (b) for the proportion of weekend (Friday through Sunday) and weekday (Monday through Thursday) combinations of Day 1 and Day 2 recalls.

NOTE: All sample weights are person-level weights, and each set of dietary weights should sum to the same overall population control total as the MEC weights (WTMEC2YR). For the August 2021-August 2023 cycle, the Day 1 dietary recall weights (WTDRD1) are appropriate to use in the analysis of the fish and shellfish consumption data (i.e., variables DRD340, DRD350A-K, DRD350AQ-JQ DRD360, DRD370A-V, and DRD370AQ-UQ) located in the Day 1 Total Nutrient Intake File (DR1TOT_L), irrespective of whether other dietary data are included in the analysis. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

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Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

WTDRD1 - Dietary day one sample weight

Variable Name: WTDRD1

SAS Label: Dietary day one sample weight

English Text: Dietary day one sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5708.632624 to 408505.81364	Range of Values	6754	6754	
0	Day 1 dietary recall not done/incomplete	2106	8860	
.	Missing	0	8860	

WTDR2D - Dietary two-day sample weight

Variable Name: WTDR2D

SAS Label: Dietary two-day sample weight

English Text: Dietary two-day sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6018.694738 to 759868.32478	Range of Values	5879	5879	
0	Day 2 dietary recall not done/incomplete	875	6754	
.	Missing	2106	8860	

DR1DRSTZ - Dietary recall status

Variable Name: DR1DRSTZ

SAS Label: Dietary recall status

English Text: Dietary recall status

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Reliable and met the minimum criteria	6694	6694	
2	Not reliable or not met the minimum criteria	43	6737	
4	Reported consuming breast-milk	60	6797	
5	Not done	2063	8860	
.	Missing	0	8860	

DR1EXMER - Interviewer ID code

Variable Name: DR1EXMER

SAS Label: Interviewer ID code

English Text: Interviewer ID code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 96	Range of Values	6797	6797	
.	Missing	2063	8860	

DRABF - Breast-fed infant (either day)

Variable Name: DRABF

SAS Label: Breast-fed infant (either day)

English Text: Indicates whether the sample person was an infant who was breast-fed on either of the two recall days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	60	60	
2	No	6674	6734	
.	Missing	2126	8860	

DRDINT - Number of days of intake

Variable Name: DRDINT

SAS Label: Number of days of intake

English Text: Indicates whether the sample person has intake data for one or two days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Day 1 only	875	875	
2	Day 1 and day 2	5879	6754	
.	Missing	2106	8860	

DR1DBIH - # of days b/w intake and HH interview

Variable Name: DR1DBIH

SAS Label: # of days b/w intake and HH interview

English Text: Number of days between intake day and the day of family questionnaire administered in the household.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
-30 to 149	Range of Values	6375	6375	
.	Missing	2485	8860	

DR1DAY - Intake day of the week

Variable Name: DR1DAY

SAS Label: Intake day of the week

English Text: Intake day of the week

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Sunday	1134	1134	
2	Monday	991	2125	
3	Tuesday	1167	3292	
4	Wednesday	1569	4861	
5	Thursday	945	5806	
6	Friday	441	6247	
7	Saturday	550	6797	
.	Missing	2063	8860	

DR1LANG - Language respondent used mostly

Variable Name: DR1LANG

SAS Label: Language respondent used mostly

English Text: The respondent spoke mostly:

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	English	6245	6245	
2	Spanish	519	6764	
3	English and Spanish	34	6798	
4	Other	4	6802	
.	Missing	2058	8860	

DR1MRESP - Main respondent for this interview

Variable Name: DR1MRESP

SAS Label: Main respondent for this interview

English Text: Who was the main respondent for this interview?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	SP	5680	5680	
2	Mother of SP	876	6556	
3	Father of SP	133	6689	
5	Spouse of SP	10	6699	
6	Child of SP	14	6713	
7	Grandparent of SP	20	6733	
8	Friend, partner, non-relative	0	6733	
9	Translator, not a HH member	0	6733	
10	Child care provider, caretaker	7	6740	
11	Other relative	7	6747	
77	Refused	0	6747	
99	Don't know	0	6747	
.	Missing	2113	8860	

DR1HELP - Helped in responding for this interview

Variable Name: DR1HELP

SAS Label: Helped in responding for this interview

English Text: Who helped in responding for this interview

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	SP	310	310	
4	Parent of SP	441	751	
5	Spouse of SP	177	928	
6	Child of SP	24	952	
7	Grandparent of SP	16	968	
8	Friend, partner, non-relative	13	981	
9	Translator, not a HH member	2	983	
10	Child care provider, caretaker	2	985	
11	Other relative	14	999	
12	No One	5742	6741	
77	Refused	0	6741	
99	Don't know	6	6747	
.	Missing	2113	8860	

DBQ095Z - Type of table salt used

Variable Name: DBQ095Z

SAS Label: Type of table salt used

English Text: What type of salt {do you/does SP} usually add to {your/his/her/SP's} food at the table? Would you say . . .

English Instructions: CAPI INSTRUCTION: IF SP AGE <= 5, DISPLAY "DO YOU" FOR FIRST DISPLAY AND {SP'S} FOR SECOND DISPLAY.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Ordinary salt [includes regular iodized salt, sea salt and seasoning salts made with regular salt]	4556	4556	
2	Lite salt	124	4680	
3	Salt substitute	62	4742	
4	Doesn't use or add salt products at the table	2002	6744	DRQSPREP
91	Other	0	6744	
99	Don't know	53	6797	DRQSPREP
.	Missing	2063	8860	

DBD100 - How often add salt to food at table

Variable Name: DBD100

SAS Label: How often add salt to food at table

English Text: How often {do you/does SP} add this salt to {your/his/her/SP's} food at the table? Would you say . . .

English Instructions: CAPI INSTRUCTION: IF SP AGE <= 5, DISPLAY "DO YOU" FOR FIRST DISPLAY AND {SP'S} FOR SECOND DISPLAY.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Rarely	2406	2406	
2	Occasionally	1562	3968	
3	Very often	764	4732	
7	Refused	0	4732	
9	Don't know	10	4742	
.	Missing	4118	8860	

DRQSPREP - Salt used in preparation?

Variable Name: DRQSPREP

SAS Label: Salt used in preparation?

English Text: How often is ordinary salt or seasoned salt added in cooking or preparing foods in your household? Is it never, rarely, occasionally, or very often?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Never	549	549	
2	Rarely	1228	1777	
3	Occasionally	2216	3993	
4	Very often	2717	6710	
9	Don't know	87	6797	
.	Missing	2063	8860	

DR1STY - Salt used at table yesterday?

Variable Name: DR1STY

SAS Label: Salt used at table yesterday?

English Text: Did {you/SP} add any salt to {your/her/his} food at the table yesterday? Salt includes ordinary or sea salt, lite salt, or a salt substitute.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	1197	1197	
2	No	5555	6752	DRQSDIET
9	Don't know	45	6797	DRQSDIET
.	Missing	2063	8860	

DR1SKY - Type of salt used yesterday

Variable Name: DR1SKY

SAS Label: Type of salt used yesterday

English Text: What type of salt was it? (Was it ordinary or sea salt, lite salt, or a salt substitute?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Ordinary, sea, seasoned, or other flavored salt	1138	1138	
2	Lite salt	35	1173	
3	Salt substitute	18	1191	
91	Other	0	1191	
99	Don't know	6	1197	
.	Missing	7663	8860	

DRQSDIET - On special diet?

Variable Name: DRQSDIET

SAS Label: On special diet?

English Text: Are you currently on any kind of diet, either to lose weight or for some other health-related reason?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	859	859	
2	No	5899	6758	DR1TNUMF
9	Don't know	39	6797	DR1TNUMF
.	Missing	2063	8860	

DRQSDT1 - Weight loss/Low calorie diet

Variable Name: DRQSDT1

SAS Label: Weight loss/Low calorie diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Weight loss or low calorie diets	447	447	
.	Missing	8413	8860	

DRQSDT2 - Low fat/Low cholesterol diet

Variable Name: DRQSDT2

SAS Label: Low fat/Low cholesterol diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2	Low fat or low cholesterol diet	70	70	
.	Missing	8790	8860	

DRQSDT3 - Low salt/Low sodium diet

Variable Name: DRQSDT3

SAS Label: Low salt/Low sodium diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3	Low salt or low sodium diet (including diet to lower blood pressure or hypertension)	96	96	
.	Missing	8764	8860	

DRQSDT4 - Sugar free/Low sugar diet

Variable Name: DRQSDT4

SAS Label: Sugar free/Low sugar diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4	Sugar free or low sugar diet	44	44	
.	Missing	8816	8860	

DRQSDT5 - Low fiber diet

Variable Name: DRQSDT5

SAS Label: Low fiber diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5	Low fiber or low residue diet	2	2	
.	Missing	8858	8860	

DRQSDT6 - High fiber diet

Variable Name: DRQSDT6

SAS Label: High fiber diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6	High fiber or high residue diet	7	7	
.	Missing	8853	8860	

DRQSDT7 - Diabetic diet

Variable Name: DRQSDT7

SAS Label: Diabetic diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
7	Diabetic diet (including gestational diabetic diets)	93	93	
.	Missing	8767	8860	

DRQSDT8 - Weight gain/Muscle building diet

Variable Name: DRQSDT8

SAS Label: Weight gain/Muscle building diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
8	Weight gain/Muscle building diet	27	27	
.	Missing	8833	8860	

DRQSDT9 - Low carbohydrate diet

Variable Name: DRQSDT9

SAS Label: Low carbohydrate diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
9	Low carbohydrate diet	70	70	
.	Missing	8790	8860	

DRQSDT10 - High protein diet

Variable Name: DRQSDT10

SAS Label: High protein diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10	High protein diet	33	33	
.	Missing	8827	8860	

DRQSDT11 - Gluten-free/Celiac diet

Variable Name: DRQSDT11

SAS Label: Gluten-free/Celiac diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11	Gluten-free/Celiac diet	32	32	
.	Missing	8828	8860	

DRQSDT12 - Renal/Kidney diet

Variable Name: DRQSDT12

SAS Label: Renal/Kidney diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
12	Renal/Kidney diet	13	13	
.	Missing	8847	8860	

DRQSDT91 - Other special diet

Variable Name: DRQSDT91

SAS Label: Other special diet

English Text: What kind of diet are you on? (Is it a weight loss or low calorie diet: low fat or cholesterol diet; low salt or sodium diet; sugar free or low sugar diet; low fiber diet; high fiber diet; diabetic diet; or another type of diet?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
91	Other special diet	148	148	
.	Missing	8712	8860	

DR1TNUMF - Number of foods/beverages reported

Variable Name: DR1TNUMF

SAS Label: Number of foods/beverages reported

English Text: Total number of foods/beverages reported in the individual foods file

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 49	Range of Values	6754	6754	
.	Missing	2106	8860	

DR1TKCAL - Energy (kcal)

Variable Name: DR1TKCAL

SAS Label: Energy (kcal)

English Text: Energy (kcal)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 10446	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TPROT - Protein (gm)

Variable Name: DR1TPROT

SAS Label: Protein (gm)

English Text: Protein (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 459.15	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCARB - Carbohydrate (gm)

Variable Name: DR1TCARB

SAS Label: Carbohydrate (gm)

English Text: Carbohydrate (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1300.94	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TSUGR - Total sugars (gm)

Variable Name: DR1TSUGR

SAS Label: Total sugars (gm)

English Text: Total sugars (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 953.9	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TFIBE - Dietary fiber (gm)

Variable Name: DR1TFIBE

SAS Label: Dietary fiber (gm)

English Text: Dietary fiber (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 127.3	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TTFAT - Total fat (gm)

Variable Name: DR1TTFAT

SAS Label: Total fat (gm)

English Text: Total fat (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 505.62	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TSFAT - Total saturated fatty acids (gm)

Variable Name: DR1TSFAT

SAS Label: Total saturated fatty acids (gm)

English Text: Total saturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 208.842	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TMFAT - Total monounsaturated fatty acids (gm)

Variable Name: DR1TMFAT

SAS Label: Total monounsaturated fatty acids (gm)

English Text: Total monounsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 205.377	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TPFAT - Total polyunsaturated fatty acids (gm)

Variable Name: DR1TPFAT

SAS Label: Total polyunsaturated fatty acids (gm)

English Text: Total polyunsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 193.753	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCHOL - Cholesterol (mg)

Variable Name: DR1TCHOL

SAS Label: Cholesterol (mg)

English Text: Cholesterol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3598	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TATOC - Vitamin E as alpha-tocopherol (mg)

Variable Name: DR1TATOC

SAS Label: Vitamin E as alpha-tocopherol (mg)

English Text: Vitamin E as alpha-tocopherol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 122.97	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TATOA - Added alpha-tocopherol (Vitamin E) (mg)

Variable Name: DR1TATOA

SAS Label: Added alpha-tocopherol (Vitamin E) (mg)

English Text: Added alpha-tocopherol (Vitamin E) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 109.19	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TRET - Retinol (mcg)

Variable Name: DR1TRET

SAS Label: Retinol (mcg)

English Text: Retinol (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 38879	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVARA - Vitamin A, RAE (mcg)

Variable Name: DR1TVARA

SAS Label: Vitamin A, RAE (mcg)

English Text: Vitamin A as retinol activity equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 39008	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TACAR - Alpha-carotene (mcg)

Variable Name: DR1TACAR

SAS Label: Alpha-carotene (mcg)

English Text: Alpha-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30805	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TBCAR - Beta-carotene (mcg)

Variable Name: DR1TBCAR

SAS Label: Beta-carotene (mcg)

English Text: Beta-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 62861	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCRYP - Beta-cryptoxanthin (mcg)

Variable Name: DR1TCRYP

SAS Label: Beta-cryptoxanthin (mcg)

English Text: Beta-cryptoxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5268	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TLYCO - Lycopene (mcg)

Variable Name: DR1TLYCO

SAS Label: Lycopene (mcg)

English Text: Lycopene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 133549	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TLZ - Lutein + zeaxanthin (mcg)

Variable Name: DR1TLZ

SAS Label: Lutein + zeaxanthin (mcg)

English Text: Lutein + zeaxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 61219	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVB1 - Thiamin (Vitamin B1) (mg)

Variable Name: DR1TVB1

SAS Label: Thiamin (Vitamin B1) (mg)

English Text: Thiamin (Vitamin B1) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 18.153	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVB2 - Riboflavin (Vitamin B2) (mg)

Variable Name: DR1TVB2

SAS Label: Riboflavin (Vitamin B2) (mg)

English Text: Riboflavin (Vitamin B2) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 22.32	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TNIAC - Niacin (mg)

Variable Name: DR1TNIAC

SAS Label: Niacin (mg)

English Text: Niacin (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 283.721	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVB6 - Vitamin B6 (mg)

Variable Name: DR1TVB6

SAS Label: Vitamin B6 (mg)

English Text: Vitamin B6 (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 27.574	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TFOLA - Total folate (mcg)

Variable Name: DR1TFOLA

SAS Label: Total folate (mcg)

English Text: Total folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2538	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TFA - Folic acid (mcg)

Variable Name: DR1TFA

SAS Label: Folic acid (mcg)

English Text: Folic acid (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2254	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TFF - Food folate (mcg)

Variable Name: DR1TFF

SAS Label: Food folate (mcg)

English Text: Food folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2064	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TFDFE - Folate, DFE (mcg)

Variable Name: DR1TFDFE

SAS Label: Folate, DFE (mcg)

English Text: Folate as dietary folate equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4027	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCHL - Total choline (mg)

Variable Name: DR1TCHL

SAS Label: Total choline (mg)

English Text: Total choline (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2403.5	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVB12 - Vitamin B12 (mcg)

Variable Name: DR1TVB12

SAS Label: Vitamin B12 (mcg)

English Text: Vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 413.18	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TB12A - Added vitamin B12 (mcg)

Variable Name: DR1TB12A

SAS Label: Added vitamin B12 (mcg)

English Text: Added vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 78.34	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVC - Vitamin C (mg)

Variable Name: DR1TVC

SAS Label: Vitamin C (mg)

English Text: Vitamin C (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 975.4	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVD - Vitamin D (D2 + D3) (mcg)

Variable Name: DR1TVD

SAS Label: Vitamin D (D2 + D3) (mcg)

English Text: Vitamin D (D2 + D3) (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 79.2	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TVK - Vitamin K (mcg)

Variable Name: DR1TVK

SAS Label: Vitamin K (mcg)

English Text: Vitamin K (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2263.9	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCALC - Calcium (mg)

Variable Name: DR1TCALC

SAS Label: Calcium (mg)

English Text: Calcium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 9266	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TPHOS - Phosphorus (mg)

Variable Name: DR1TPHOS

SAS Label: Phosphorus (mg)

English Text: Phosphorus (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 7956	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TMAGN - Magnesium (mg)

Variable Name: DR1TMAGN

SAS Label: Magnesium (mg)

English Text: Magnesium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2234	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TIRON - Iron (mg)

Variable Name: DR1TIRON

SAS Label: Iron (mg)

English Text: Iron (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 146.47	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TZINC - Zinc (mg)

Variable Name: DR1TZINC

SAS Label: Zinc (mg)

English Text: Zinc (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 80.63	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCOPP - Copper (mg)

Variable Name: DR1TCOPP

SAS Label: Copper (mg)

English Text: Copper (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 74.122	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TSODI - Sodium (mg)

Variable Name: DR1TSODI

SAS Label: Sodium (mg)

English Text: Sodium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 20006	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TPOTA - Potassium (mg)

Variable Name: DR1TPOTA

SAS Label: Potassium (mg)

English Text: Potassium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 14296	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TSELE - Selenium (mcg)

Variable Name: DR1TSELE

SAS Label: Selenium (mcg)

English Text: Selenium (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1154	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TCAFF - Caffeine (mg)

Variable Name: DR1TCAFF

SAS Label: Caffeine (mg)

English Text: Caffeine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2460	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TTHEO - Theobromine (mg)

Variable Name: DR1TTHEO

SAS Label: Theobromine (mg)

English Text: Theobromine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1687	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TALCO - Alcohol (gm)

Variable Name: DR1TALCO

SAS Label: Alcohol (gm)

English Text: Alcohol (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 448.1	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TMOIS - Moisture (gm)

Variable Name: DR1TMOIS

SAS Label: Moisture (gm)

English Text: Moisture (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 29329.36	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS040 - SFA 4:0 (Butanoic) (gm)

Variable Name: DR1TS040

SAS Label: SFA 4:0 (Butanoic) (gm)

English Text: SFA 4:0 (Butanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4.958	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS060 - SFA 6:0 (Hexanoic) (gm)

Variable Name: DR1TS060

SAS Label: SFA 6:0 (Hexanoic) (gm)

English Text: SFA 6:0 (Hexanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.552	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS080 - SFA 8:0 (Octanoic) (gm)

Variable Name: DR1TS080

SAS Label: SFA 8:0 (Octanoic) (gm)

English Text: SFA 8:0 (Octanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 11.91	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS100 - SFA 10:0 (Decanoic) (gm)

Variable Name: DR1TS100

SAS Label: SFA 10:0 (Decanoic) (gm)

English Text: SFA 10:0 (Decanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 6.6	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS120 - SFA 12:0 (Dodecanoic) (gm)

Variable Name: DR1TS120

SAS Label: SFA 12:0 (Dodecanoic) (gm)

English Text: SFA 12:0 (Dodecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 32.281	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS140 - SFA 14:0 (Tetradecanoic) (gm)

Variable Name: DR1TS140

SAS Label: SFA 14:0 (Tetradecanoic) (gm)

English Text: SFA 14:0 (Tetradecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 22.271	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS160 - SFA 16:0 (Hexadecanoic) (gm)

Variable Name: DR1TS160

SAS Label: SFA 16:0 (Hexadecanoic) (gm)

English Text: SFA 16:0 (Hexadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 108.107	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TS180 - SFA 18:0 (Octadecanoic) (gm)

Variable Name: DR1TS180

SAS Label: SFA 18:0 (Octadecanoic) (gm)

English Text: SFA 18:0 (Octadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 49.46	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TM161 - MFA 16:1 (Hexadecenoic) (gm)

Variable Name: DR1TM161

SAS Label: MFA 16:1 (Hexadecenoic) (gm)

English Text: MFA 16:1 (Hexadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 19.038	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TM181 - MFA 18:1 (Octadecenoic) (gm)

Variable Name: DR1TM181

SAS Label: MFA 18:1 (Octadecenoic) (gm)

English Text: MFA 18:1 (Octadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 199.904	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TM201 - MFA 20:1 (Eicosenoic) (gm)

Variable Name: DR1TM201

SAS Label: MFA 20:1 (Eicosenoic) (gm)

English Text: MFA 20:1 (Eicosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 12.786	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TM221 - MFA 22:1 (Docosenoic) (gm)

Variable Name: DR1TM221

SAS Label: MFA 22:1 (Docosenoic) (gm)

English Text: MFA 22:1 (Docosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 17.596	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP182 - PFA 18:2 (Octadecadienoic) (gm)

Variable Name: DR1TP182

SAS Label: PFA 18:2 (Octadecadienoic) (gm)

English Text: PFA 18:2 (Octadecadienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 171.424	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP183 - PFA 18:3 (Octadecatrienoic) (gm)

Variable Name: DR1TP183

SAS Label: PFA 18:3 (Octadecatrienoic) (gm)

English Text: PFA 18:3 (Octadecatrienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 21.585	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP184 - PFA 18:4 (Octadecatetraenoic) (gm)

Variable Name: DR1TP184

SAS Label: PFA 18:4 (Octadecatetraenoic) (gm)

English Text: PFA 18:4 (Octadecatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 0.61	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP204 - PFA 20:4 (Eicosatetraenoic) (gm)

Variable Name: DR1TP204

SAS Label: PFA 20:4 (Eicosatetraenoic) (gm)

English Text: PFA 20:4 (Eicosatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.577	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP205 - PFA 20:5 (Eicosapentaenoic) (gm)

Variable Name: DR1TP205

SAS Label: PFA 20:5 (Eicosapentaenoic) (gm)

English Text: PFA 20:5 (Eicosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.012	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP225 - PFA 22:5 (Docosapentaenoic) (gm)

Variable Name: DR1TP225

SAS Label: PFA 22:5 (Docosapentaenoic) (gm)

English Text: PFA 22:5 (Docosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.337	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1TP226 - PFA 22:6 (Docosahexaenoic) (gm)

Variable Name: DR1TP226

SAS Label: PFA 22:6 (Docosahexaenoic) (gm)

English Text: PFA 22:6 (Docosahexaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.962	Range of Values	6694	6694	
.	Missing	2166	8860	

DR1_300 - Compare food consumed yesterday to usual

Variable Name: DR1_300

SAS Label: Compare food consumed yesterday to usual

English Text: Was the amount of food that {you/NAME} ate yesterday much more than usual, usual, or much less than usual?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Much more than usual	363	363	
2	Usual	5559	5922	
3	Much less than usual	814	6736	
7	Refused	0	6736	
9	Don't know	61	6797	
.	Missing	2063	8860	

DR1_320Z - Total plain water drank yesterday (gm)

Variable Name: DR1_320Z

SAS Label: Total plain water drank yesterday (gm)

English Text: Total plain water drank yesterday - including plain tap water, water from a drinking fountain, water from a water cooler, bottled water, and spring water.

English Instructions: Calculated from water consumption records reported as part of the 24-hour dietary recall interview.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 26879.61	Range of Values	6754	6754	
.	Missing	2106	8860	

DR1_330Z - Total tap water drank yesterday (gm)

Variable Name: DR1_330Z

SAS Label: Total tap water drank yesterday (gm)

English Text: Total tap water drank yesterday - including filtered tap water and water from a drinking fountain.

English Instructions: Calculated from tap water consumption records reported as part of the 24-hour dietary recall interview.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 26879.61	Range of Values	6754	6754	
.	Missing	2106	8860	

DR1BWATZ - Total bottled water drank yesterday (gm)

Variable Name: DR1BWATZ

SAS Label: Total bottled water drank yesterday (gm)

English Text: Total bottled water drank yesterday (gm)

English Instructions: Calculated from bottle water consumption records reported as part of the 24-hour dietary recall interview.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 9570	Range of Values	6754	6754	
.	Missing	2106	8860	

DR1TWSZ - Tap water source

Variable Name: DR1TWSZ

SAS Label: Tap water source

English Text: When you drink tap water, what is the main source of the tap water?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Community supply	4470	4470	
4	Don't drink tap water	1345	5815	
91	Other	769	6584	
99	Don't know	213	6797	
.	Missing	2063	8860	

DRD340 - Shellfish eaten during past 30 days

Variable Name: DRD340

SAS Label: Shellfish eaten during past 30 days

English Text: Please look at this list of shellfish. During the past 30 days did you eat any types of shellfish listed on this card? Include any foods that had shellfish in them such as sandwiches, soups, or salads.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	3033	3033	
2	No	3607	6640	DRD360
7	Refused	29	6669	DRD360
9	Don't know	25	6694	DRD360
.	Missing	2166	8860	

DRD350A - Clams eaten during past 30 days

Variable Name: DRD350A

SAS Label: Clams eaten during past 30 days

English Text: Clams eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	256	256	
2	No	2777	3033	DRD350B
.	Missing	5827	8860	

DRD350AQ - # of times clams eaten in past 30 days

Variable Name: DRD350AQ

SAS Label: # of times clams eaten in past 30 days

English Text: Number of times clams were eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 8	Range of Values	256	256	
.	Missing	8604	8860	

DRD350B - Crabs eaten during past 30 days

Variable Name: DRD350B

SAS Label: Crabs eaten during past 30 days

English Text: Crabs eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	548	548	
2	No	2485	3033	DRD350C
.	Missing	5827	8860	

DRD350BQ - # of times crabs eaten in past 30 days

Variable Name: DRD350BQ

SAS Label: # of times crabs eaten in past 30 days

English Text: Number of times crab was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 10	Range of Values	548	548	
.	Missing	8312	8860	

DRD350C - Crayfish eaten during past 30 days

Variable Name: DRD350C

SAS Label: Crayfish eaten during past 30 days

English Text: Crayfish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	62	62	
2	No	2971	3033	DRD350D
.	Missing	5827	8860	

DRD350CQ - # of times crayfish eaten past 30 days

Variable Name: DRD350CQ

SAS Label: # of times crayfish eaten past 30 days

English Text: Number of times crayfish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 15	Range of Values	62	62	
.	Missing	8798	8860	

DRD350D - Lobsters eaten during past 30 days

Variable Name: DRD350D

SAS Label: Lobsters eaten during past 30 days

English Text: Lobsters eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	313	313	
2	No	2720	3033	DRD350E
.	Missing	5827	8860	

DRD350DQ - # of times lobsters eaten past 30 days

Variable Name: DRD350DQ

SAS Label: # of times lobsters eaten past 30 days

English Text: Number of times lobster was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 6	Range of Values	313	313	
.	Missing	8547	8860	

DRD350E - Mussels eaten during past 30 days

Variable Name: DRD350E

SAS Label: Mussels eaten during past 30 days

English Text: Mussels eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	217	217	
2	No	2816	3033	DRD350F
.	Missing	5827	8860	

DRD350EQ - # of times mussels eaten in past 30 days

Variable Name: DRD350EQ

SAS Label: # of times mussels eaten in past 30 days

English Text: Number of times mussels were eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 5	Range of Values	217	217	
.	Missing	8643	8860	

DRD350F - Oysters eaten during past 30 days

Variable Name: DRD350F

SAS Label: Oysters eaten during past 30 days

English Text: Oysters eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	230	230	
2	No	2803	3033	DRD350G
.	Missing	5827	8860	

DRD350FQ - # of times oysters eaten in past 30 days

Variable Name: DRD350FQ

SAS Label: # of times oysters eaten in past 30 days

English Text: Number of times oysters were eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 9	Range of Values	230	230	
.	Missing	8630	8860	

DRD350G - Scallops eaten during past 30 days

Variable Name: DRD350G

SAS Label: Scallops eaten during past 30 days

English Text: Scallops eaten during the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	358	358	
2	No	2675	3033	DRD350H
.	Missing	5827	8860	

DRD350GQ - # of times scallops eaten past 30 days

Variable Name: DRD350GQ

SAS Label: # of times scallops eaten past 30 days

English Text: Number of times scallops were eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 12	Range of Values	357	357	
.	Missing	8503	8860	

DRD350H - Shrimp eaten during past 30 days

Variable Name: DRD350H

SAS Label: Shrimp eaten during past 30 days

English Text: Shrimp eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	2661	2661	
2	No	372	3033	DRD350I
.	Missing	5827	8860	

DRD350HQ - # of times shrimp eaten in past 30 days

Variable Name: DRD350HQ

SAS Label: # of times shrimp eaten in past 30 days

English Text: Number of times shrimp was eaten in the last 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 22	Range of Values	2660	2660	
.	Missing	6200	8860	

DRD350I - Other shellfish eaten past 30 days

Variable Name: DRD350I

SAS Label: Other shellfish eaten past 30 days

English Text: Other shellfish (ex. octopus, squid) eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	223	223	
2	No	2809	3032	DRD350J
.	Missing	5828	8860	

DRD350IQ - # of times other shellfish eaten

Variable Name: DRD350IQ

SAS Label: # of times other shellfish eaten

English Text: Number of times other shellfish (ex. octopus, squid) was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 16	Range of Values	223	223	
.	Missing	8637	8860	

DRD350J - Other unknown shellfish eaten past 30 d

Variable Name: DRD350J

SAS Label: Other unknown shellfish eaten past 30 d

English Text: Other unknown shellfish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	15	15	
2	No	3017	3032	DRD350K
.	Missing	5828	8860	

DRD350JQ - # of times other unknown shellfish eaten

Variable Name: DRD350JQ

SAS Label: # of times other unknown shellfish eaten

English Text: Number of times other unknown shellfish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 4	Range of Values	15	15	
.	Missing	8845	8860	

DRD350K - Refused on shellfish eaten past 30 days

Variable Name: DRD350K

SAS Label: Refused on shellfish eaten past 30 days

English Text: Refused to give detailed information on shellfish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	0	0	
2	No	3031	3031	
.	Missing	5829	8860	

DRD360 - Fish eaten during past 30 days

Variable Name: DRD360

SAS Label: Fish eaten during past 30 days

English Text: Please look at this list of fish. During the past 30 days did you eat any types of fish listed on this card? Include any foods that had fish in them such as sandwiches, soups, or salads.

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	4175	4175	
2	No	2467	6642	End of Section
7	Refused	29	6671	End of Section
9	Don't know	24	6695	End of Section
.	Missing	2165	8860	

DRD370A - Breaded fish products eaten past 30 days

Variable Name: DRD370A

SAS Label: Breaded fish products eaten past 30 days

English Text: Breaded fish products eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	628	628	
2	No	3547	4175	DRD370B
.	Missing	4685	8860	

DRD370AQ - # of times breaded fish products eaten

Variable Name: DRD370AQ

SAS Label: # of times breaded fish products eaten

English Text: Number of times breaded fish products were eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 20	Range of Values	628	628	
.	Missing	8232	8860	

DRD370B - Tuna eaten during past 30 days

Variable Name: DRD370B

SAS Label: Tuna eaten during past 30 days

English Text: Tuna eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	1864	1864	
2	No	2311	4175	DRD370C
.	Missing	4685	8860	

DRD370BQ - # of times tuna eaten in past 30 days

Variable Name: DRD370BQ

SAS Label: # of times tuna eaten in past 30 days

English Text: Number of times tuna was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 82	Range of Values	1864	1864	
.	Missing	6996	8860	

DRD370C - Bass eaten during past 30 days

Variable Name: DRD370C

SAS Label: Bass eaten during past 30 days

English Text: Bass eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	60	60	
2	No	4115	4175	DRD370D
.	Missing	4685	8860	

DRD370CQ - # of times bass eaten in past 30 days

Variable Name: DRD370CQ

SAS Label: # of times bass eaten in past 30 days

English Text: Number of times bass was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 6	Range of Values	60	60	
.	Missing	8800	8860	

DRD370D - Catfish eaten during past 30 days

Variable Name: DRD370D

SAS Label: Catfish eaten during past 30 days

English Text: Catfish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	347	347	
2	No	3828	4175	DRD370E
.	Missing	4685	8860	

DRD370DQ - # of times catfish eaten in past 30 days

Variable Name: DRD370DQ

SAS Label: # of times catfish eaten in past 30 days

English Text: Number of times catfish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 15	Range of Values	347	347	
.	Missing	8513	8860	

DRD370E - Cod eaten during past 30 days

Variable Name: DRD370E

SAS Label: Cod eaten during past 30 days

English Text: Cod eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	701	701	
2	No	3474	4175	DRD370F
.	Missing	4685	8860	

DRD370EQ - # of times cod eaten in past 30 days

Variable Name: DRD370EQ

SAS Label: # of times cod eaten in past 30 days

English Text: Number of times cod was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 29	Range of Values	701	701	
.	Missing	8159	8860	

DRD370F - Flatfish eaten during past 30 days

Variable Name: DRD370F

SAS Label: Flatfish eaten during past 30 days

English Text: Flatfish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	134	134	
2	No	4041	4175	DRD370G
.	Missing	4685	8860	

DRD370FQ - # of times flatfish eaten past 30 days

Variable Name: DRD370FQ

SAS Label: # of times flatfish eaten past 30 days

English Text: Number of times flatfish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 24	Range of Values	134	134	
.	Missing	8726	8860	

DRD370G - Haddock eaten during past 30 days

Variable Name: DRD370G

SAS Label: Haddock eaten during past 30 days

English Text: Haddock eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	223	223	
2	No	3952	4175	DRD370H
.	Missing	4685	8860	

DRD370GQ - # of times haddock eaten in past 30 days

Variable Name: DRD370GQ

SAS Label: # of times haddock eaten in past 30 days

English Text: Number of times haddock was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 22	Range of Values	223	223	
.	Missing	8637	8860	

DRD370H - Mackerel eaten during past 30 days

Variable Name: DRD370H

SAS Label: Mackerel eaten during past 30 days

English Text: Mackerel eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	85	85	
2	No	4090	4175	DRD370I
.	Missing	4685	8860	

DRD370HQ - # of times mackerel eaten past 30 days

Variable Name: DRD370HQ

SAS Label: # of times mackerel eaten past 30 days

English Text: Number of times mackerel was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 10	Range of Values	85	85	
.	Missing	8775	8860	

DRD370I - Perch eaten during past 30 days

Variable Name: DRD370I

SAS Label: Perch eaten during past 30 days

English Text: Perch eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	76	76	
2	No	4099	4175	DRD370J
.	Missing	4685	8860	

DRD370IQ - # of times perch eaten in past 30 days

Variable Name: DRD370IQ

SAS Label: # of times perch eaten in past 30 days

English Text: Number of times perch was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 6	Range of Values	76	76	
.	Missing	8784	8860	

DRD370J - Pike eaten during past 30 days

Variable Name: DRD370J

SAS Label: Pike eaten during past 30 days

English Text: Pike eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	12	12	
2	No	4163	4175	DRD370K
.	Missing	4685	8860	

DRD370JQ - # of times pike eaten in past 30 days

Variable Name: DRD370JQ

SAS Label: # of times pike eaten in past 30 days

English Text: Number of times pike was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 6	Range of Values	12	12	
.	Missing	8848	8860	

DRD370K - Pollock eaten during past 30 days

Variable Name: DRD370K

SAS Label: Pollock eaten during past 30 days

English Text: Pollock eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	172	172	
2	No	4003	4175	DRD370L
.	Missing	4685	8860	

DRD370KQ - # of times pollock eaten in past 30 days

Variable Name: DRD370KQ

SAS Label: # of times pollock eaten in past 30 days

English Text: Number of times pollock was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 15	Range of Values	172	172	
.	Missing	8688	8860	

DRD370L - Porgy eaten during past 30 days

Variable Name: DRD370L

SAS Label: Porgy eaten during past 30 days

English Text: Porgy eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	18	18	
2	No	4157	4175	DRD370M
.	Missing	4685	8860	

DRD370LQ - # of times porgy eaten in past 30 days

Variable Name: DRD370LQ

SAS Label: # of times porgy eaten in past 30 days

English Text: Number of times porgy was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 7	Range of Values	18	18	
.	Missing	8842	8860	

DRD370M - Salmon eaten during past 30 days

Variable Name: DRD370M

SAS Label: Salmon eaten during past 30 days

English Text: Salmon eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	2037	2037	
2	No	2138	4175	DRD370N
.	Missing	4685	8860	

DRD370MQ - # of times salmon eaten in past 30 days

Variable Name: DRD370MQ

SAS Label: # of times salmon eaten in past 30 days

English Text: Number of times salmon was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 30	Range of Values	2037	2037	
.	Missing	6823	8860	

DRD370N - Sardines eaten during past 30 days

Variable Name: DRD370N

SAS Label: Sardines eaten during past 30 days

English Text: Sardines eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	213	213	
2	No	3962	4175	DRD370O
.	Missing	4685	8860	

DRD370NQ - # of times sardines eaten past 30 days

Variable Name: DRD370NQ

SAS Label: # of times sardines eaten past 30 days

English Text: Number of times sardines were eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 30	Range of Values	213	213	
.	Missing	8647	8860	

DRD370O - Sea bass eaten during past 30 days

Variable Name: DRD370O

SAS Label: Sea bass eaten during past 30 days

English Text: Sea bass eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	83	83	
2	No	4092	4175	DRD370P
.	Missing	4685	8860	

DRD370OQ - # of times sea bass eaten past 30 days

Variable Name: DRD370OQ

SAS Label: # of times sea bass eaten past 30 days

English Text: Number of times sea bass was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 8	Range of Values	83	83	
.	Missing	8777	8860	

DRD370P - Shark eaten during past 30 days

Variable Name: DRD370P

SAS Label: Shark eaten during past 30 days

English Text: Shark eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	4	4	
2	No	4171	4175	DRD370Q
.	Missing	4685	8860	

DRD370PQ - # of times shark eaten in past 30 days

Variable Name: DRD370PQ

SAS Label: # of times shark eaten in past 30 days

English Text: Number of times shark was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 3	Range of Values	4	4	
.	Missing	8856	8860	

DRD370Q - Swordfish eaten during past 30 days

Variable Name: DRD370Q

SAS Label: Swordfish eaten during past 30 days

English Text: Swordfish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	45	45	
2	No	4130	4175	DRD370R
.	Missing	4685	8860	

DRD370QQ - # of times swordfish eaten past 30 days

Variable Name: DRD370QQ

SAS Label: # of times swordfish eaten past 30 days

English Text: Number of times swordfish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 3	Range of Values	45	45	
.	Missing	8815	8860	

DRD370R - Trout eaten during past 30 days

Variable Name: DRD370R

SAS Label: Trout eaten during past 30 days

English Text: Trout eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	153	153	
2	No	4022	4175	DRD370S
.	Missing	4685	8860	

DRD370RQ - # of times trout eaten in past 30 days

Variable Name: DRD370RQ

SAS Label: # of times trout eaten in past 30 days

English Text: Number of times trout was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 10	Range of Values	153	153	
.	Missing	8707	8860	

DRD370S - Walleye eaten during past 30 days

Variable Name: DRD370S

SAS Label: Walleye eaten during past 30 days

English Text: Walleye eaten during the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	96	96	
2	No	4079	4175	DRD370T
.	Missing	4685	8860	

DRD370SQ - # of times walleye eaten in past 30 days

Variable Name: DRD370SQ

SAS Label: # of times walleye eaten in past 30 days

English Text: Number of times walleye was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 6	Range of Values	96	96	
.	Missing	8764	8860	

DRD370T - Other fish eaten during past 30 days

Variable Name: DRD370T

SAS Label: Other fish eaten during past 30 days

English Text: Other type of fish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	982	982	
2	No	3193	4175	DRD370U
.	Missing	4685	8860	

DRD370TQ - # of times other fish eaten past 30 days

Variable Name: DRD370TQ

SAS Label: # of times other fish eaten past 30 days

English Text: Number of times other type of fish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 30	Range of Values	982	982	
.	Missing	7878	8860	

DRD370U - Other unknown fish eaten in past 30 days

Variable Name: DRD370U

SAS Label: Other unknown fish eaten in past 30 days

English Text: Other unknown type eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	124	124	
2	No	4051	4175	DRD370V
.	Missing	4685	8860	

DRD370UQ - # of times other unknown fish eaten

Variable Name: DRD370UQ

SAS Label: # of times other unknown fish eaten

English Text: Number of times other unknown type of fish was eaten in the past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 6	Range of Values	124	124	
.	Missing	8736	8860	

DRD370V - Refused on fish eaten past 30 days

Variable Name: DRD370V

SAS Label: Refused on fish eaten past 30 days

English Text: Refused to give detailed information on fish eaten during past 30 days

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	3	3	
2	No	4170	4173	
.	Missing	4687	8860	

Appendix 1. Changes between WWEIA survey cycles 2013-2014 thru the August 2021-August 2023 sample

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Administration of 24-hour dietary recall	Same as 2003-2012: Day 1 in-person in MEC, Day 2 by telephone	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Day 1 by telephone, Day 2 by telephone
Language, Day 1	Same as 1999-2012: Interpreters assisted as needed	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Language, Day 2	Same as 2011-2012: English, Spanish, or Asian languages	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Person who did not provide reliable intake for Day 1 but did for Day 2	Same as 2003-2012: Neither intake was included in the release datasets	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	The single reliable intake was included in the release datasets as Day 1 intake (n=21)
Food source (where food was obtained)	Codes 8 and 9 revised descriptions.	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014
Combination food types	Same as 2003-2012: Values for 15 combination types	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	Values for 16 combination types; added "Baby Toddler Food and Infant Formula"
Number of intakes that include only water consumption for the day	6 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (all in Day 2 data), records are included in Individual Foods file.	2 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (1 in Day 1, 4 in Day 2 data), records are included in Individual Foods file.	5 intakes (4 in Day 1, 1 in Day 2), records are included in Individual Foods file.
Number of intakes that include no water or food consumption for the day	1 intake in Day 2 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	2 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.	3 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.
Tap water source	Same as 2003-2012: Reported in 6 categories.	Same as 2003-2004	Responses combined into 4 categories	Same as 2017-2018	Same as 2017-2018
Modification codes: DR1MC	All remaining modification	No modification	No modification	No modification	No modification

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Day 2 Modification codes: DR2MC Modification Code Description file: DRXMCD	codes deleted; new food codes addressing modifications added in FNDDS 2013-2014	codes	codes	codes	codes
Salt use at the table and in cooking or preparing foods and type	Same as 2003-2004	Same as 2003-2004	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Salt used at the table yesterday and type	Question asked about salt use at the table yesterday and kind of salt to coincide with 24-hour recall.	Same as 2013-2014	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Main respondent and person whom helped in responding for the interview	Same as 2003-2012	Response category changed	Same as 2015-2016	Same as 2015-2016	Same as 2015-2016

Appendix 2. Variables in the Individual Foods Files (DR1IFF_L and DR2IFF_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1ILINE	DR2ILINE	Food/Individual component number
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1CCMNM	DR2CCMNM	Combination food number
DR1CCMTX	DR2CCMTX	Combination food type
DR1_020	DR2_020	Time of eating occasion (HH:MM)
DR1_030Z	DR2_030Z	Name of eating occasion
DR1FS	DR2FS	Source of food
DR1_040Z	DR2_040Z	Did you eat this meal at home?
DR1IFDCD	DR2IFDCD	USDA food code
DR1IGRMS	DR2IGRMS	Grams
DR1IKCAL	DR2IKCAL	Energy (kcal)
DR1IPROT	DR2IPROT	Protein (gm)
DR1ICARB	DR2ICARB	Carbohydrate (gm)
DR1ISUGR	DR2ISUGR	Total sugars (gm)
DR1IFIBE	DR2IFIBE	Dietary fiber (gm)
DR1ITFAT	DR2ITFAT	Total fat (gm)
DR1ISFAT	DR2ISFAT	Total saturated fatty acids (gm)
DR1IMFAT	DR2IMFAT	Total monounsaturated fatty acids (gm)
DR1IPFAT	DR2IPFAT	Total polyunsaturated fatty acids (gm)
DR1ICHOL	DR2ICHOL	Cholesterol (mg)
DR1IATOC	DR2IATOC	Vitamin E as alpha-tocopherol (mg)
DR1IATOA	DR2IATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1IRET	DR2IRET	Retinol (mcg)
DR1IVARA	DR2IVARA	Vitamin A, RAE (mcg)
DR1IACAR	DR2IACAR	Alpha-carotene (mcg)
DR1IBCAR	DR2IBCAR	Beta-carotene (mcg)
DR1ICRYP	DR2ICRYP	Beta-cryptoxanthin (mcg)
DR1ILYCO	DR2ILYCO	Lycopene (mcg)
DR1ILZ	DR2ILZ	Lutein + zeaxanthin (mcg)
DR1IVB1	DR2IVB1	Thiamin (Vitamin B1) (mg)

Day1 Name	Day2 Name	Variable Label
DR1IVB2	DR2IVB2	Riboflavin (Vitamin B2) (mg)
DR1INIAC	DR2INIAC	Niacin (mg)
DR1IVB6	DR2IVB6	Vitamin B6 (mg)
DR1IFOLA	DR2IFOLA	Total folate (mcg)
DR1IFA	DR2IFA	Folic acid (mcg)
DR1IFF	DR2IFF	Food folate (mcg)
DR1IFDFE	DR2IFDFE	Folate, DFE (mcg)
DR1ICHL	DR2ICHL	Total choline (mg)
DR1IVB12	DR2IVB12	Vitamin B12 (mcg)
DR1IB12A	DR2IB12A	Added vitamin B12 (mcg)
DR1IVC	DR2IVC	Vitamin C (mg)
DR1IVD	DR2IVD	Vitamin D (D2 + D3) (mcg)
DR1IVK	DR2IVK	Vitamin K (mcg)
DR1ICALC	DR2ICALC	Calcium (mg)
DR1IPHOS	DR2IPHOS	Phosphorus (mg)
DR1IMAGN	DR2IMAGN	Magnesium (mg)
DR1IIRON	DR2IIRON	Iron (mg)
DR1IZINC	DR2IZINC	Zinc (mg)
DR1ICOPP	DR2ICOPP	Copper (mg)
DR1ISODI	DR2ISODI	Sodium (mg)
DR1IPOTA	DR2IPOTA	Potassium (mg)
DR1ISELE	DR2ISELE	Selenium (mcg)
DR1ICAFF	DR2ICAFF	Caffeine (mg)
DR1ITHEO	DR2ITHEO	Theobromine (mg)
DR1IALCO	DR2IALCO	Alcohol (gm)
DR1IMOIS	DR2IMOIS	Moisture (gm)
DR1IS040	DR2IS040	SFA 4:0 (Butanoic) (gm)
DR1IS060	DR2IS060	SFA 6:0 (Hexanoic) (gm)
DR1IS080	DR2IS080	SFA 8:0 (Octanoic) (gm)
DR1IS100	DR2IS100	SFA 10:0 (Decanoic) (gm)
DR1IS120	DR2IS120	SFA 12:0 (Dodecanoic) (gm)
DR1IS140	DR2IS140	SFA 14:0 (Tetradecanoic) (gm)
DR1IS160	DR2IS160	SFA 16:0 (Hexadecanoic) (gm)
DR1IS180	DR2IS180	SFA 18:0 (Octadecanoic) (gm)
DR1IM161	DR2IM161	MFA 16:1 (Hexadecenoic) (gm)
DR1IM181	DR2IM181	MFA 18:1 (Octadecenoic) (gm)
DR1IM201	DR2IM201	MFA 20:1 (Eicosenoic) (gm)
DR1IM221	DR2IM221	MFA 22:1 (Docosenoic) (gm)
DR1IP182	DR2IP182	PFA 18:2 (Octadecadienoic) (gm)

Day1 Name	Day2 Name	Variable Label
DR1IP183	DR2IP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1IP184	DR2IP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1IP204	DR2IP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1IP205	DR2IP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1IP225	DR2IP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1IP226	DR2IP226	PFA 22:6 (Docosahexaenoic) (gm)

Appendix 3. List of Nutrients/Food Components (Unit)

Energy and Macronutrients

Food energy (kcal)
Protein (gm)
Carbohydrate (gm)
Fat, total (gm)
Alcohol (gm)

Sugars, total (gm)
Dietary fiber, total (gm)
Water (moisture) (gm)*

Saturated fatty acids, total (gm)
Monounsaturated fatty acids, total (gm)
Polyunsaturated fatty acids, total (gm)
Cholesterol (mg)

Individual fatty acids:
- 4:0 (gm)
- 6:0 (gm)
- 8:0 (gm)
- 10:0 (gm)
- 12:0 (gm)
- 14:0 (gm)
- 16:0 (gm)
- 18:0 (gm)
- 16:1 (gm)
- 18:1 (gm)
- 20:1 (gm)
- 22:1 (gm)
- 18:2 (gm)
- 18:3 (gm)
- 18:4 (gm)
- 20:4 (gm)
- 20:5 n-3 (gm)
- 22:5 n-3 (gm)
- 22:6 n-3 (gm)

Vitamins, Minerals, and Other Components

Vitamin A as retinol activity equivalents (mcg)
Retinol (mcg)
Carotenoids:
- Carotene, alpha (mcg)
- Carotene, beta (mcg)
- Cryptoxanthin, beta (mcg)
- Lycopene (mcg)
- Lutein + zeaxanthin (mcg)

Vitamin E as alpha-tocopherol (mg)
- **Added vitamin E as alpha-tocopherol (mg)

Vitamin D (D2 + D3) (mcg)
Vitamin K as phylloquinone (mcg)
Vitamin C (mg)
Thiamin (mg)
Riboflavin (mg)
Niacin (mg)
Vitamin B-6 (mg)
Folate, total (mcg)
- Folate as dietary folate equivalents (mcg)

Folic acid (mcg)
Food folate (mcg)
Choline, total (mg)
Vitamin B-12 (mcg)
***Added vitamin B-12 (mcg)

Calcium (mg)
Iron (mg)
Magnesium (mg)
Phosphorus (mg)
Potassium (mg)
Sodium (mg)
Zinc (mg)
Copper (mg)
Selenium (mcg)
Caffeine (mg)
Theobromine (mg)

** Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)*

*** Represents a synthetic subcomponent of vitamin E and is included in the vitamin E value.*

**** Represents a fortified subcomponent of vitamin B12 and is included in the vitamin B12 value.*

Appendix 4. Adding Food Code Descriptions to Your Files

One supporting file is included with the Individual Foods files: the Food Code Description file (DRXFCD_L).

The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code included in the Individual Foods files.

The Food Code Description file (DRXFCD_L) contains three variables:

DRXFDCD a numeric value corresponding to DR1IFDCD in the file DR1IFF_L or DR2IFDCD in the file DR2IFF_L;

DRXFCSD a short description (up to 60 characters) of the food code;

DRXFDLD a long description (up to 200 characters) of the food code.

The following SQL code is an example of appending the shorter food code description (here renamed DR1IFCSD) to one of the Individual Foods files using PROC SQL from SAS®. Other SQL implementations may be different.

```
proc sql;
  create table DR1IFF_L_PLUS as
  select iff.*, desc.DRXFCSD as DR1FCSD, desc.DRXFCLD as DR1FCLD
  from NHANES.DR1IFF_L iff
  left join NHANES.DRXFCD_L desc
  on iff.DR1IFDCD = desc.DRXFDCD
  order by SEQN, DR1ILINE;
quit;
```

SAS® users may wish to use Proc Format to assign labels to the food codes. The following example generates and saves a picture format for food codes and a separate format for each food code that includes both the food code itself and the short food code description. It is assumed that the user has stored the Individual Foods files and the Food Code Description file in a library called NHANES and wishes to store the formats there as well.

```
options fmtsearch = (NHANES);

proc format library = library;
  picture foodcode
  low - high = '000-00000';
quit;

data tmp;
  set NHANES.DRXFCD_L;
  length cfoodcode $9 label $72;
  cfoodcode = put(DRXFDCD, foodcode.);
  label = cfoodcode || '  ' || DRXFCSD;
run;

data fmt (keep = fmtname start label);
  set tmp;
  retain fmtname 'DRXFDCD';
  rename DRXFDCD = start;
run;
```

```
proc format cntlin = fmt library = library;
run;
```

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Appendix 5. Variables in the Total Nutrients Files (DR1TOT_L and DR2TOT_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1MRESP	DR2MRESP	Main respondent for this interview
DR1HELP	DR2HELP	Helped in responding for this interview
DBQ095Z	N/A	Type of table salt used
DBD100	N/A	How often add salt to food at table
DRQSPREP	N/A	Salt used in preparation?
DR1STY	DR2STY	Salt used at table yesterday?
DR1SKY	DR2SKY	Type of salt used yesterday
DRQSDIET	N/A	On special diet?
DRQSDT1	N/A	Weight loss/Low calorie diet
DRQSDT2	N/A	Low fat/Low cholesterol diet
DRQSDT3	N/A	Low salt/Low sodium diet
DRQSDT4	N/A	Sugar free/Low sugar diet
DRQSDT5	N/A	Low fiber diet
DRQSDT6	N/A	High fiber diet
DRQSDT7	N/A	Diabetic diet
DRQSDT8	N/A	Weight gain/Muscle building diet
DRQSDT9	N/A	Low carbohydrate diet
DRQSDT10	N/A	High protein diet
DRQSDT11	N/A	Gluten-free/Celiac diet
DRQSDT12	N/A	Renal/Kidney diet
DRQSDT91	N/A	Other special diet
DR1TNUMF	DR2TNUMF	Number of foods/beverages reported
DR1TKCAL	DR2TKCAL	Energy (kcal)
DR1TPROT	DR2TPROT	Protein (gm)
DR1TCARB	DR2TCARB	Carbohydrate (gm)
DR1TSUGR	DR2TSUGR	Total sugars (gm)
DR1TFIBE	DR2TFIBE	Dietary fiber (gm)
DR1TTFAT	DR2TTFAT	Total fat (gm)
DR1TSFAT	DR2TSFAT	Total saturated fatty acids (gm)

Day1 Name	Day2 Name	Variable Label
DR1TMFAT	DR2TMFAT	Total monounsaturated fatty acids (gm)
DR1TPFAT	DR2TPFAT	Total polyunsaturated fatty acids (gm)
DR1TCHOL	DR2TCHOL	Cholesterol (mg)
DR1TATOC	DR2TATOC	Vitamin E as alpha-tocopherol (mg)
DR1TATOA	DR2TATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1TRET	DR2TRET	Retinol (mcg)
DR1TVARA	DR2TVARA	Vitamin A, RAE (mcg)
DR1TACAR	DR2TACAR	Alpha-carotene (mcg)
DR1TBCAR	DR2TBCAR	Beta-carotene (mcg)
DR1TCRYP	DR2TCRYP	Beta-cryptoxanthin (mcg)
DR1TLYCO	DR2TLYCO	Lycopene (mcg)
DR1TLZ	DR2TLZ	Lutein + zeaxanthin (mcg)
DR1TVB1	DR2TVB1	Thiamin (Vitamin B1) (mg)
DR1TVB2	DR2TVB2	Riboflavin (Vitamin B2) (mg)
DR1TNIAC	DR2TNIAC	Niacin (mg)
DR1TVB6	DR2TVB6	Vitamin B6 (mg)
DR1TFOLA	DR2TFOLA	Total folate (mcg)
DR1TFA	DR2TFA	Folic acid (mcg)
DR1TFF	DR2TFF	Food folate (mcg)
DR1TFDFE	DR2TFDFE	Folate, DFE (mcg)
DR1TCHL	DR2TCHL	Total choline (mg)
DR1TVB12	DR2TVB12	Vitamin B12 (mcg)
DR1TB12A	DR2TB12A	Added vitamin B12 (mcg)
DR1TVC	DR2TVC	Vitamin C (mg)
DR1TVD	DR2TVD	Vitamin D (D2 + D3) (mcg)
DR1TVK	DR2TVK	Vitamin K (mcg)
DR1TCALC	DR2TCALC	Calcium (mg)
DR1TPHOS	DR2TPHOS	Phosphorus (mg)
DR1TMAGN	DR2TMAGN	Magnesium (mg)
DR1TIRON	DR2TIRON	Iron (mg)
DR1TZINC	DR2TZINC	Zinc (mg)
DR1TCOPP	DR2TCOPP	Copper (mg)
DR1TSODI	DR2TSODI	Sodium (mg)
DR1TPOTA	DR2TPOTA	Potassium (mg)
DR1TSELE	DR2TSELE	Selenium (mcg)
DR1TCAFF	DR2TCAFF	Caffeine (mg)
DR1TTHEO	DR2TTHEO	Theobromine (mg)
DR1TALCO	DR2TALCO	Alcohol (gm)
DR1TMOIS	DR2TMOIS	Moisture (gm)

Day1 Name	Day2 Name	Variable Label
DR1TS040	DR2TS040	SFA 4:0 (Butanoic) (gm)
DR1TS060	DR2TS060	SFA 6:0 (Hexanoic) (gm)
DR1TS080	DR2TS080	SFA 8:0 (Octanoic) (gm)
DR1TS100	DR2TS100	SFA 10:0 (Decanoic) (gm)
DR1TS120	DR2TS120	SFA 12:0 (Dodecanoic) (gm)
DR1TS140	DR2TS140	SFA 14:0 (Tetradecanoic) (gm)
DR1TS160	DR2TS160	SFA 16:0 (Hexadecanoic) (gm)
DR1TS180	DR2TS180	SFA 18:0 (Octadecanoic) (gm)
DR1TM161	DR2TM161	MFA 16:1 (Hexadecenoic) (gm)
DR1TM181	DR2TM181	MFA 18:1 (Octadecenoic) (gm)
DR1TM201	DR2TM201	MFA 20:1 (Eicosenoic) (gm)
DR1TM221	DR2TM221	MFA 22:1 (Docosenoic) (gm)
DR1TP182	DR2TP182	PFA 18:2 (Octadecadienoic) (gm)
DR1TP183	DR2TP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1TP184	DR2TP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1TP204	DR2TP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1TP205	DR2TP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1TP225	DR2TP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1TP226	DR2TP226	PFA 22:6 (Docosahexaenoic) (gm)
DR1_300	DR2_300	Compare food consumed yesterday to usual
DR1_320Z	DR2_320Z	Total plain water drank yesterday (gm)
DR1_330Z	DR2_330Z	Total tap water drank yesterday (gm)
DR1BWATZ	DR2BWATZ	Total bottled water drank yesterday (gm)
DR1TWSZ	DR2TWSZ	Tap water source
DRD340	N/A	Shellfish eaten during past 30 days
DRD350A	N/A	Clams eaten during past 30 days
DRD350AQ	N/A	# of times clams eaten in past 30 days
DRD350B	N/A	Crabs eaten during past 30 days
DRD350BQ	N/A	# of times crabs eaten in past 30 days
DRD350C	N/A	Crayfish eaten during past 30 days
DRD350CQ	N/A	# of times crayfish eaten past 30 days
DRD350D	N/A	Lobsters eaten during past 30 days
DRD350DQ	N/A	# of times lobsters eaten past 30 days
DRD350E	N/A	Mussels eaten during past 30 days
DRD350EQ	N/A	# of times mussels eaten in past 30 days
DRD350F	N/A	Oysters eaten during past 30 days
DRD350FQ	N/A	# of times oysters eaten in past 30 days
DRD350G	N/A	Scallops eaten during past 30 days
DRD350GQ	N/A	# of times scallops eaten past 30 days

Day1 Name	Day2 Name	Variable Label
DRD350H	N/A	Shrimp eaten during past 30 days
DRD350HQ	N/A	# of times shrimp eaten in past 30 days
DRD350I	N/A	Other shellfish eaten past 30 days
DRD350IQ	N/A	# of times other shellfish eaten
DRD350J	N/A	Other unknown shellfish eaten past 30 d
DRD350JQ	N/A	# of times other unknown shellfish eaten
DRD350K	N/A	Refused on shellfish eaten past 30 days
DRD360	N/A	Fish eaten during past 30 days
DRD370A	N/A	Breaded fish products eaten past 30 days
DRD370AQ	N/A	# of times breaded fish products eaten
DRD370B	N/A	Tuna eaten during past 30 days
DRD370BQ	N/A	# of times tuna eaten in past 30 days
DRD370C	N/A	Bass eaten during past 30 days
DRD370CQ	N/A	# of times bass eaten in past 30 days
DRD370D	N/A	Catfish eaten during past 30 days
DRD370DQ	N/A	# of times catfish eaten in past 30 days
DRD370E	N/A	Cod eaten during past 30 days
DRD370EQ	N/A	# of times cod eaten in past 30 days
DRD370F	N/A	Flatfish eaten during past 30 days
DRD370FQ	N/A	# of times flatfish eaten past 30 days
DRD370G	N/A	Haddock eaten during past 30 days
DRD370GQ	N/A	# of times haddock eaten in past 30 days
DRD370H	N/A	Mackerel eaten during past 30 days
DRD370HQ	N/A	# of times mackerel eaten past 30 days
DRD370I	N/A	Perch eaten during past 30 days
DRD370IQ	N/A	# of times perch eaten in past 30 days
DRD370J	N/A	Pike eaten during past 30 days
DRD370JQ	N/A	# of times pike eaten in past 30 days
DRD370K	N/A	Pollock eaten during past 30 days
DRD370KQ	N/A	# of times pollock eaten in past 30 days
DRD370L	N/A	Porgy eaten during past 30 days
DRD370LQ	N/A	# of times porgy eaten in past 30 days
DRD370M	N/A	Salmon eaten during past 30 days
DRD370MQ	N/A	# of times salmon eaten in past 30 days
DRD370N	N/A	Sardines eaten during past 30 days
DRD370NQ	N/A	# of times sardines eaten past 30 days
DRD370O	N/A	Sea bass eaten during past 30 days
DRD370OQ	N/A	# of times sea bass eaten past 30 days
DRD370P	N/A	Shark eaten during past 30 days

Day1 Name	Day2 Name	Variable Label
DRD370PQ	N/A	# of times shark eaten in past 30 days
DRD370Q	N/A	Swordfish eaten during past 30 days
DRD370QQ	N/A	# of times swordfish eaten past 30 days
DRD370R	N/A	Trout eaten during past 30 days
DRD370RQ	N/A	# of times trout eaten in past 30 days
DRD370S	N/A	Walleye eaten during past 30 days
DRD370SQ	N/A	# of times walleye eaten in past 30 days
DRD370T	N/A	Other fish eaten during past 30 days
DRD370TQ	N/A	# of times other fish eaten past 30 days
DRD370U	N/A	Other unknown fish eaten in past 30 days
DRD370UQ	N/A	# of times other unknown fish eaten
DRD370V	N/A	Refused on fish eaten past 30 days

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Interview - Individual Foods, Second Day (DR2IFF_L)

Data File: DR2IFF_L.xpt

First Published: October 2024

Last Revised: NA

Component Description

The objective of the dietary interview component is to obtain detailed dietary intake information from NHANES participants. The dietary intake data are used to estimate the types and amounts of foods and beverages (including all types of water) consumed during the 24-hour period prior to the interview (midnight to midnight), and to estimate intakes of energy, nutrients, and other food components from those foods and beverages. Following the dietary recall, participants are asked questions on salt use, whether the person's overall intake on the previous day was much more than usual, usual or much less than usual, and whether the participant is on any type of special diet. Questions on frequency of fish and shellfish consumed during the past 30 days are asked of participants 1 year or older, with the use of proxies for young children (see the Dietary Interview Procedure Manuals (cdc.gov) for more information on the proxy interview).

The dietary interview component, called What We Eat in America (WWEIA), is conducted as a partnership between the U.S. Department of Agriculture (USDA) and the U.S. Department of Health and Human Services (DHHS). Under this partnership, DHHS' National Center for Health Statistics (NCHS), Division of Health and Nutrition Examination Surveys is responsible for the survey sample design and all aspects of data collection and USDA's Food Surveys Research Group (FSRG) is responsible for the dietary data collection methodology, maintenance of the databases used to code and process the data, and data review and processing.

All NHANES participants are eligible for two 24-hour dietary recall interviews. Traditionally, the first dietary recall interview was collected in-person in the Mobile Examination Center (MEC) and the second interview was collected by telephone 3 to 10 days later. In August 2021-August 2023, the time participants spent in the MEC was limited to reduce risk of exposure to SARS-CoV-2. As a result, both dietary recall interviews were administered via telephone. Further information about changes in data collection are described in the Plan and Operations report of the NHANES August 2021-August 2023.

As in previous years, two types of dietary intake data are available for the August 2021-August 2023 survey cycle: Individual Foods files and Total Nutrient Intakes files.

What's New with the August 2021-August 2023 WWEIA Release:

In response to the COVID-19 pandemic, the mode of the first dietary recall changed from in-person during the MEC examination to telephone 3 to 7 days after the examination. Both dietary interviews were completed in English or Spanish. During the mobile examination, participants scheduled an appointment for the first telephone dietary interview and received a set of measuring guides to help estimate the amounts of foods and beverages consumed. **Appendix 1** provides a summary of changes among the 5 latest cycles of data collection. No new data variables were added in the August 2021-August 2023 WWEIA release.

Dietary Interview Data Files: Four data files were produced from the information collected in the dietary interviews: two Individual Foods files and two Total Nutrient Intakes files. Each

file includes one day of intake data. The number "1" or "2" in the file name identifies the day of the interview: 1 = first day, 2 = second day. File names are as follows:

File Names for Dietary Interview Data:

File	Day 1	Day 2
Individual Foods File	DR1IFF_L	DR2IFF_L
Total Nutrient Intakes File	DR1TOT_L	DR2TOT_L

The amounts in these files reflect only nutrients obtained from foods, beverages, and water, including tap and bottled water. They do not include nutrients obtained from dietary supplement intakes, antacids, or medications. Data on intake of dietary supplement use are available on the August 2021-August 2023 Dietary Data (cdc.gov) page.

Individual Foods Files (DR1IFF_L and DR2IFF_L): Detailed information about each food/beverage item (including the description, amount of, and nutrient content) reported by each participant is included in the Individual Foods files. The names for both Day 1 and Day 2 variables are listed in **Appendix 2**.

The Individual Foods files include, for each interview day, one record for each food/beverage consumed by a participant. Each record is uniquely numbered within a participant's set of records and contains the information listed below:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Whether the food/beverage was eaten in combination with other foods, such as in a sandwich;
- Time of eating occasion/when the food was eaten;
- Eating occasion name;
- Where the food/beverage was obtained;
- Whether the meal/snack was eaten at home or not;
- A USDA Food and Nutrient Database for Dietary Studies (FNDDS) code identifying the food/beverage;
- Amount of food/beverage consumed, in grams; and
- Food energy and 64 nutrients/food components (listed in **Appendix 3**) from each food/beverage as calculated using USDA's Food and Nutrient Database for Dietary Studies 2021-2023 (FNDDS 2021-2023).

Descriptions for the USDA FNDDS food codes are provided in the Food Code Description file (DRXFCD_L). The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code in the FNDDS 2021-2023. **Appendix 4** provides SAS code examples that may be used to link the food code description to the Individual Foods file.

Total Nutrient Intakes Files (DR1TOT_L and DR2TOT_L): For each participant, daily total energy and nutrient intakes from foods and beverages, and whether the amount of food consumed was usual, much more than usual, or much less than usual, are included in the Total Nutrient Intakes files. The Day 1 file also includes information on salt use in cooking and at the table; whether the participant is currently on any kind of diet to lose weight or for another health-related reason and, if so, the type of diet; and information on frequency of fish and shellfish consumption for participants aged 1 or older. The names for both Day 1 and Day 2 variables are listed in **Appendix 5**.

The Total Nutrient Intakes files provide a summary record of total nutrient intakes for each participant. Each total intake record contains the following information:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Type of salt used and how often added at the table and in food preparation (Day 1 file only);
- Use of salt at the table yesterday and the type of salt used;
- Whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet (Day 1 file only);
- Total number of foods and beverages including water reported for that participant for that day's intake;
- Daily aggregates of food energy and 64 nutrients/food components (listed in **Appendix 3**) from all foods/beverages as calculated using USDA's Food and Nutrient Database for Dietary Studies August 2021-August 2023 (FNDDS 2021- 2023)
- Whether the amount of food consumed was usual, much more than usual, or much less than usual;
- Total amount of tap and bottled water consumed (calculated as the sum of reports of water drunk by itself in the 24-hour recall) and the usual source of tap water; and
- Frequency of fish and shellfish consumption in the past 30 days (participants 1 year or older, Day 1 file only).

Eligible Sample

All participants in the August 2021-August 2023 sample who spoke English or Spanish were eligible. Participants aged 1 year or older were eligible for the frequency of fish and shellfish consumption questions following the 24-hour recall.

Protocol and Procedure

The examination protocol and data collection methods are fully documented in the NHANES dietary interviewer procedures manuals.

Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant’s intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child’s intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish.

During the MEC visit, participants were scheduled an appointment for the first dietary interview and were given a set of measuring guides (NHANES Measuring Guides for the Dietary Recall Interview) to use for reporting food amounts during the telephone interview. The measuring guides included measuring cups, spoons, a ruler, and a food model booklet, which contained two-dimensional drawings of glasses, mugs, bowls, mounds, circles, wedges, a shape chart and a chicken chart. The first telephone dietary interview was scheduled 3 to 7 days post MEC examination while the second interview was collected 3 to 10 days following the first dietary interview and were generally scheduled on a different day of the week as the first interview. In August 2021-August 2023, 204 participants were interviewed on the same day of the week for both day 1 and day 2 interviews due to their scheduling availability. Any participant who did not have a telephone was given a toll-free number to call so that the recall could be conducted.

What We Eat in America data were collected using USDA's dietary data collection instrument, the Automated Multiple Pass Method (AMPM), available at:

http://www.ars.usda.gov/nea/bhnrc/fsrg. The AMPM was designed to provide an efficient and accurate means of collecting intakes for large-scale national surveys. The AMPM is a fully computerized recall method that uses a 5-step interview outlined below:

1. **Quick List** - Participant recalls all foods and beverages consumed the day before the interview (midnight to midnight).
2. **Forgotten Foods** - Participant is asked about consumption of foods commonly forgotten during the Quick List step.
3. **Time and Occasion** - Time and eating occasion are collected for each food.
4. **Detail Cycle** - For each food, a detailed description, amount eaten, and additions to the food are collected. Eating occasions and times between eating occasions are reviewed to elicit forgotten foods.
5. **Final Probe** - Additional foods not remembered earlier are collected.

The AMPM includes an extensive compilation of standardized food-specific questions and possible response options. Routing of questions is based on previous responses. The AMPM is updated for each 2-year collection of WWEIA to reflect the changing food supply and to address research needs from the data user community. Additional information about the AMPM is provided in Raper et al. (Raper et al., 2004).

The AMPM was validated in a large study and shown to be an effective method for collecting accurate group energy intake of adults. Completed in 2004, this extensive research project included 524 healthy, weight-stable volunteers, aged 30-69 years. The accuracy of the AMPM was evaluated by comparing reported energy intake (EI) to total energy expenditure (TEE) using the doubly labeled water technique (Moshfegh et al., 2008). Among the findings were that EI compared to TEE was under-reported by 11% overall, by less than 3% for normal weight subjects with body mass index (BMI) < 25 and 16% for overweight subjects with BMI ≥ 25.

Additional studies provide evidence that the AMPM accurately measures group energy intake. Blanton (Blanton et al., 2006) reported that EI was not significantly different from TEE for a sample of 20 adult females. Rumpler (Rumpler et al., 2008) found that mean EIs were accurately reported for a sample of 12 adult males.

Additional evidence for the accuracy of AMPM has been provided by analysis of the 24-hour urinary sodium data collected in the AMPM Validation Study, which suggest the AMPM is a valid measure for estimating mean sodium intake in adults. Dietary sodium intake calculated from 24-hour recall data of 465 subjects collected via AMPM was compared with sodium values from 24-hour urine collections measured during the same 24-hour period. The AMPM-derived mean dietary sodium estimates reflected over 90% of the biomarker-based estimates (Rhodes et al., 2013).

For additional information about the dietary interview component and related survey protocols, please visit the August 2021-August 2023 Dietary Interviewer Procedures Manual page. A description of changes to the NHANES August 2021-August 2023 survey is provided in the Plan and Operations of the National Heath And Nutrition Examination Survey, August 2021-August 2023.

Quality Assurance & Quality Control

All dietary interviewers were required to complete an intensive one-week training course and to conduct supervised practice interviews before working independently in the field. Retraining sessions were conducted annually to reinforce the proper protocols and technique.

Interviewers were monitored throughout the data collection period. Monitoring consisted of the following:

- Reviews of audio recorded interviews were conducted for approximately 5% of each interviewer's work.

- Quality control of interviews, which were checked for completeness of the recalls, missing information, inconsistent reports, and unclear notes. Written notification and feedback were provided to the interviewers.

Data Processing and Editing

Interview data files were sent electronically from the field and were imported into Survey Net, a computer-assisted food coding and data management system developed by USDA (Raper et al., 2004).

USDA's Food and Nutrient Database for Dietary Studies (FNDDS) 2021-2023 was used to process the intakes reported by the August 2021-August 2023 sample. The FNDDS includes comprehensive information that can be used to code individual foods/beverages and portion sizes reported by participants and includes nutrient values for calculating nutrient intakes. FNDDS nutrient values as well as food codes and portion sizes are updated for every 2-year WWEIA, NHANES release cycle. The basis for the nutrient values as well as food codes and portion weights in FNDDS are detailed in the documentation for FNDDS 2021-2023 available at http://www.ars.usda.gov/nea/bhnrc/fsrg.

Coders were required to pass a certification test after the initial training. They were routinely monitored to ensure the quality and completeness of their work. Approximately 10 percent of the coder's work was randomly selected to be independently coded by another coder. Results from the two codings were compared and adjudicated, if necessary.

After intake data were coded, various types of reviews and quality assurance procedures were conducted to ensure the quality of the data. Examples of reviews include the following:

- Interviewers' and coders' questions and comments were reviewed to ensure that they have been addressed.
- Decisions made by coders about how to code new or unusual foods/beverages or quantities reported by participants were reviewed for accuracy, reasonableness, and consistency across intake data; items in question were resolved by coder supervisors.
- Specific data integrity checks for reasonableness, consistency, and logic were conducted.

Analytic Notes

Each Individual Foods file (Day 1 and Day 2) is comprised of food records. For most participants, there are multiple records in each file. For each Total Nutrient Intakes file (Day 1 and Day 2) there is one record for each participant. These files can be linked with other NHANES files by the respondent sequence number (SEQN).

Variable names: For data collected on both Day 1 and Day 2, variable names are differentiated by having the number "1" or "2" in the third position of the variable name to identify the collection day. For example, the USDA food code variable (in the Individual Foods File), which identifies the food reported by the participant, is named DR1IFDCD in the Day 1 file and DR2IFDCD in the Day 2 file. Appendices **2** and **5** list the Day 1 and Day 2 variable names for the Individual Foods file and the Total Nutrient Intakes file, respectively.

Names for the following variables are the same for both days in the Individual Foods file and the Total Nutrient Intakes file:

Variables with the Same Name for Both Days in the Dietary Interview Files

Day 1 and Day 2 variable name	Label
SEQN	Respondent sequence number
WTDRD1	Dietary day one sample weight
WTDR2D	Dietary two-day sample weight
DRABF	Breast-fed infant (either day)
DRDINT	Number of days of intake

Number of days of intake: A variable has been included to indicate the number of days of intake collected from each participant. The variable name is DRDINT. In the August 2021-August 2023 sample, 6,754 participants provided complete dietary intakes for Day 1. The 6,754 participants with a complete Day 1 intake include 21 participants who had an unreliable day 1 intake but reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall. Of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2.

Dietary recall status code: A status code (DR1DRSTZ or DR2DRSTZ) is used in both the Individual Foods and Total Nutrient Intakes files to indicate the quality and completeness of a survey participant's response to the dietary recall section. The codes are the following:

1 = Reliable and met the following minimum criteria:

- The first 4 steps of the 5-step AMPM completed.
- Food/beverages consumed for each reported eating occasion identified.

For individuals with a code 1, all relevant variables associated with the 24-hour dietary recall contain a value.

2 = Not reliable or did not meet the minimum criteria

Individuals with a code 2 have incomplete records. No data on total nutrient intakes and the total number of foods reported are provided for these cases. These individuals have no records in the Individual Foods files.

3 [Code 3 is not included in the current datasets. It was only used for data from the 1999-2000 survey cycle.]

4 = Reported consuming breast milk

For infants and children who consumed human milk, there is a record in the Individual Foods files for each report of human milk. However, because amounts of human milk intake are not quantified, these records contain missing values for the amount consumed and for the amounts of energy and nutrients from human milk. Also, records of human milk have a missing value for the food source variable (DR1FS, DR2FS) and the eaten at home variable (DR1_040Z, DR2_040Z) in the Individual Foods files. Records for any other foods and beverages consumed by breast-fed infants and children are included in the Individual Foods files along with their amounts and nutrient information. Because of the missing amount or quantity information for human milk, no total nutrient intakes (contained in the Total Nutrient Intakes files) were computed for participants with a code 4.

A variable that identifies breast-fed children, DRABF, is included. This variable has a code of 1 if a child consumed human milk in either intake day.

5 = Not done

This code is assigned when the dietary recall section of the interview did not take place due to various reasons (such as refusal, equipment failure, or unable to contact the participant). These individuals have no records in the Individual Foods files. These individuals have a record in the Total Nutrients file with values only for the following variables: the respondent sequence number (SEQN), the dietary recall status code (DR1DRSTZ or DR2DRSTZ) and for participants 1 year or older, the fish and shellfish questions in the DR1TOT_L file (DRD340, DRD350A-K, DRD350AQ-JQ, DRD360, DRD370A-V, and DRD370AQ-UQ).

Only codes 1 and 4 appear in the Individual Foods file.

Distinguishing Between Foods/Beverages and Dietary Supplements in NHANES: In August 2021-August 2023, there was no 24-hour dietary supplement use collection. The 30-day dietary supplement use collection changed from being collected during the home interview to collection after the first 24-hour dietary recall via telephone. All NHANES participants responding to the 24-hour dietary recall interview are eligible for the dietary supplement and non-prescription antacid use questions. Information is obtained on all vitamins, minerals, herbals, and other dietary supplements as well as non-prescription antacids that were consumed during a 30-day period, including the name and the amount of supplement or antacid taken.

Distinguishing between foods/beverages and supplements can be challenging. NCHS and FSRG review questionable items reported in the dietary supplement and dietary recall components to resolve disposition of these items into the appropriate component. Products that are labeled as a dietary supplement, that have a supplement facts panel on the label, and are in tablets, capsules, softgels, gelcaps, or other pill forms, are considered dietary supplements. Items that are powders or liquids can be hard to distinguish. General guidelines used state that if powders and liquid concentrates have product directions stating that they be added to a liquid, they are classified as beverages. Examples are teas and protein powders. An exception is made for fiber products, which are classified as dietary supplements. Along this same guideline, energy drinks are considered beverages, but “energy shot” type products are considered dietary supplements.

It is best to refer to the databases that detail every food/beverage and dietary supplement reported in NHANES to identify exact determination used. The databases are:

- 2021-2023 Food and Nutrient Database for Dietary Studies
- NHANES Dietary Supplement Database

Participants who reported consuming only water, no food or other beverages: Records are included in the Individual Foods file for participants who consumed only water. There are 5 such individuals in the August 2021-August 2023 datasets, 4 in the Day 1 data and 1 in the Day 2 data. Their dietary recall status variable for the day is coded as "1" (complete and reliable) in the Total Nutrients file and the total number of items is the number of times water was reported. Individuals with just water intake and no food intake will have zero energy intake for the day.

Participants who reported consuming no water, food or other beverages: There can be participants whose intakes are determined to be complete even though they reported no water, food, or other beverage records for the day. For such participants there are no records in the Individual Foods file, but their dietary recall status is coded as complete and reliable, and the Total Nutrients file will include records with zero values for all nutrients. In the August 2021-August 2023 datasets, there are 3 individuals in the day 1 data that reported no water, food, or other beverage records for the day.

Number of days between the intake day and the day of family interview: Each of the four intake files includes a variable (DR1DBIH for Day 1 files and DR2DBIH for Day 2 files) to indicate the number of days between the intake day (i.e., the period covered by the 24-hour recall) and the day that the family questionnaire was administered in the household. A positive value in DR1BHIH or DR2BHIH indicates the family interview occurred prior to the intake day. In the survey, most of the family interviews were done before the participant participated in the dietary interview. A value of "0" in DR1BHIH or DR2BHIH indicates the family interview

occurred on the same date as the intake day. A negative value (i.e., DR1BHIH<0 or DR2BHIH<0) means that the family interview occurred after the intake day.

Food source: The source from which each food/beverage was obtained (e.g., from a store, fast food restaurant, cafeteria) is identified by the variables DR1FS (day 1) and DR2FS (day 2) in the Individual Foods files.

The code descriptions for this variable are:

Code Descriptions for Source of Food Variable

Code	Description
1	Store grocery/supermarket
2	Restaurant with waiter/waitress
3	Restaurant fast food/Pizza
4	Bar/Tavern/Lounge
5	Restaurant, no additional information
6	Cafeteria NOT in a K-12 school
7	Cafeteria in a K-12 school
8	Child/Adult care center
9	Child/Adult home care
10	Soup kitchen/shelter/food pantry facility
11	Meals on Wheels Program
12	Community food program – other
13	Community program, no additional info
14	Vending machine
15	Common coffee pot or snack tray
16	From someone else/gift
17	Mail order purchase
18	Residential dining facility
19	Grown or caught by you or someone you know
20	Fish caught by you or someone you know
24	Sport, recreation, or entertainment
25	Street vendor, vending truck
26	Fundraiser sales
27	Store - convenience type
28	Store - no additional information
91	Other, specify

Eating occasion: The variables DR1_030Z and DR2_030Z are located in the Individual Foods file. The code descriptions for the eating occasion variables are shown in the table below.

Code Descriptions for Eating Occasion Variable

Code	Description
1	Breakfast
2	Lunch
3	Dinner
4	Supper
5	Brunch
6	Snack
7	Beverage/Drink
8	Feeding-infant only
9	Extended consumption
10	Desayuno
11	Almuerzo
12	Comida
13	Merienda
14	Cena
15	Entre comida
16	Botana
17	Bocadillo
18	Tentempie
19	Bebida
91	Other

Eating occasion was designated by the respondent. During the interview, a list of eating occasion names was available to the respondent for selection. However, eating occasion names were not defined for the respondent.

Foods and beverages coded as part of a combination: 41 percent of foods and beverages reported in the WWEIA, NHANES August 2021-August 2023 sample were identified as items consumed together as combinations. Items consumed as a combination were identified by one of sixteen unique "combination food types." Foods and beverages not coded in combination have the code "0" for the combination food type variable.

The combination types provide a linkage for:

- Foods or beverages with additions, such as cereal with milk, coffee with cream;
- Multi-component foods that have specific protocol for collection such as some salads and sandwiches; and
- Other combinations that do not have a unique code in the FNDDS.

Combination Type, Code, Examples, and Percent of Food and Beverages Reported by Type, August 2021-August 2023, Day 1

Combination Type	Code	Examples of Combination Type	% Items
Not in combination	0	NA	59
Beverage w/ additions	1	Coffee, tea with: milk, cream, sugar.	9
Cereal w/ additions	2	Cereals (ready-to-eat, cooked) with: milk, sugar, fruit, butter.	4
Bread/baked product w/additions	3	Breads, rolls, pancakes with: butter, jam, syrup, fruit. Cakes, pies with: ice cream, toppings. Crackers with: cheese, dip, peanut butter.	5
Salad	4	Components of salads that do not have a single code in FNDDS. It may also designate additional items to single code salads.	5
Sandwiches	5	Components of sandwiches that do not have a single code in FNDDS. It may also designate additional items added to single code sandwiches.	6
Soup	6	Soup with: crackers, croutons, cheese.	<1
Frozen meals	7	Components of a prepackaged frozen meal and additions to the meal.	<1
Ice cream/ frozen yogurt w/ additions	8	Ice cream with: syrup, nuts, toppings.	<1
Dried beans or Vegetable w/ additions	9	French fries, potatoes with: catsup, gravy, butter, toppings. Beans with: sauce, butter.	3
Fruit w/ additions	10	Fruit with: toppings, milk, honey. Components of fruit mixtures or salads that do not have a single code in FNDDS.	1
Tortilla products	11	Components of tacos and tortilla products that do not have a single code in FNDDS. It may also designate additional items to single code tacos or tortilla products.	2
Meat, Poultry, Fish	12	Meat, poultry, fish with: gravy, sauce, and condiments.	2
Lunchables®	13	Components of pre-packaged lunch kits.	<1
Chips w/ additions	14	Potato chips, corn chips with: dip, cheese, salsa.	1
Baby Toddler Food and Infant Formula	15	Baby Toddler cereal with: infant formula, fruit. Infant formula with: baby cereal.	<1
Other mixtures	90	Rice, pasta, spaghetti, eggs, other mixtures with: butter, gravy, sauce, condiments.	4

All items given a combination food type are given an additional variable to identify each of the items within the combination. This variable is the "combination food number" that is unique to the combination food type within the individual intake.

Variable Labels and Names for Combination Coding

Combination Coding	Variable Name, Day 1	Variable Name, Day 2
Combination food type	DR1CCMTX	DR2CCMTX
Combination food number	DR1CCMNM	DR2CCMNM

The What We Eat in America Food Categories, available on the FSRG website (http://www.ars.usda.gov/nea/bhnrc/fsrg), is a grouping scheme that combines foods and beverages together that have similar usage and nutrient content with the emphasis on

how they are commonly consumed in the American diet. There are more than 170 unique categories, and each is assigned a 4-digit number and description. The WWEIA Food Categories contain discrete food items and are not disaggregated (e.g., pizza vs. grain, cheese, tomatoes, etc.). Designed to be flexible, the categories can be combined as needed to address specific research questions. A new version of the WWEIA Food Categories is produced for each 2-year release cycle of WWEIA, NHANES.

Special diet: Information on whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet, was provided. The variable DRQSDIET identifies whether a participant was on a special diet. The variables DRQSDT1 through DRQSDT12 and DRQSDT91 identify the type of diet or diets that the participant was following. These variables can be found in the Total Nutrient Intakes file.

Sample weights for dietary intake data: The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, **the appropriate sample weights must be used.**

For the NHANES August 2021-August 2023 sample, there were 22,660 persons selected; of these 8,860 were considered participants to the MEC examination and data collection. A total of 6,754 MEC participants provided complete dietary intakes for Day 1, and of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2. The 6,754 participants who provided complete Day 1 intakes includes 21 participants who had an unreliable day 1 intake and reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall.

Most analyses of NHANES data use data collected in the MEC and the variable WTMEC2YR should be used for the sample weights. However, for the WWEIA dietary data, different sample weights are recommended for analysis. Because food intake can vary by weekdays and weekends, use of the MEC weights disproportionately represents intakes on weekends.

A set of weights (WTDRD1) is provided that should be used when an analysis uses the Day 1 dietary recall data (either alone or when Day 1 nutrient data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with Day 1 data. Day 1 weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These Day 1 weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using Day 1 data and Day 1 weights might be larger than standard errors for similar estimates based on MEC weights.

When analysis is based on both days of dietary intake, only 5,879 sample participants have complete data. The NHANES protocol requires an attempt to collect the second day of dietary data at least 3 days after the first day, but the actual number of days between the two interviews is variable. A set of adjusted weights, WTDR2D, is to be used when an analysis uses the smaller sample with completed Day 1 and Day 2 dietary data. This two-day weight was constructed for the 5,879 participants by taking the Day 1 weights (WTDRD1) and further adjusting for: (a) the additional non-response for the second recall; and (b) for the proportion of weekend (Friday through Sunday) and weekday (Monday through Thursday) combinations of Day 1 and Day 2 recalls.

NOTE: All sample weights are person-level weights, and each set of dietary weights should sum to the same overall population control total as the MEC weights (WTMEC2YR). For the August 2021-August 2023 cycle, the Day 1 dietary recall weights (WTDRD1) are appropriate to use in the analysis of the fish and shellfish consumption data (i.e., variables DRD340, DRD350A-K, DRD350AQ-JQ DRD360, DRD370A-V, and DRD370AQ-UQ) located in the Day 1 Total Nutrient Intake File (DR1TOT_L), irrespective of whether other dietary data are included in the analysis. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

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Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
English Instructions:	One of the key variables for the file. The primary key variables are SEQN and DR2ILINE.
Target:	Both males and females 0 YEARS - 150 YEARS

WTDRD1 - Dietary day one sample weight

Variable Name: WTDRD1

SAS Label: Dietary day one sample weight

English Text: Dietary day one sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5708.632624 to 394679.4226	Range of Values	88032	88032	
0	Day 1 dietary recall not done/incomplete	0	88032	
.	Missing	0	88032	

WTDR2D - Dietary two-day sample weight

Variable Name: WTDR2D

SAS Label: Dietary two-day sample weight

English Text: Dietary two-day sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6018.694738 to 759868.32478	Range of Values	88032	88032	
0	Day 2 dietary recall not done/incomplete	0	88032	
.	Missing	0	88032	

DR2ILINE - Food/Individual component number

Variable Name: DR2ILINE

SAS Label: Food/Individual component number

English Text: Food/Individual component number

English Instructions: One of the key variables for the file. The primary key variables are SEQN and DR2ILINE.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 40	Range of Values	88032	88032	
.	Missing	0	88032	

DR2DRSTZ - Dietary recall status

Variable Name: DR2DRSTZ

SAS Label: Dietary recall status

English Text: Dietary recall status

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Reliable and met the minimum criteria	87424	87424	
2	Not reliable or not met the minimum criteria	0	87424	
4	Reported consuming breast-milk	608	88032	
5	Not done	0	88032	
.	Missing	0	88032	

DR2EXMER - Interviewer ID code

Variable Name: DR2EXMER

SAS Label: Interviewer ID code

English Text: Interviewer ID code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 96	Range of Values	88032	88032	
.	Missing	0	88032	

DRABF - Breast-fed infant (either day)

Variable Name: DRABF

SAS Label: Breast-fed infant (either day)

English Text: Indicates whether the sample person was an infant who was breast fed on either of the two recall days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	624	624	
2	No	87408	88032	
.	Missing	0	88032	

DRDINT - Number of days of intake

Variable Name: DRDINT

SAS Label: Number of days of intake

English Text: Indicates whether the sample person has intake data for one or two days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Day 1 only	0	0	
2	Day 1 and day 2	88032	88032	
.	Missing	0	88032	

DR2DBIH - # of days b/w intake and HH interview

Variable Name: DR2DBIH

SAS Label: # of days b/w intake and HH interview

English Text: Number of days between intake day and the day of family questionnaire administered in the household.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
-21 to 167	Range of Values	83166	83166	
.	Missing	4866	88032	

DR2DAY - Intake day of the week

Variable Name: DR2DAY

SAS Label: Intake day of the week

English Text: Intake day of the week

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Sunday	16799	16799	
2	Monday	13762	30561	
3	Tuesday	12495	43056	
4	Wednesday	16047	59103	
5	Thursday	14595	73698	
6	Friday	6792	80490	
7	Saturday	7542	88032	
.	Missing	0	88032	

DR2LANG - Language respondent used mostly

Variable Name: DR2LANG

SAS Label: Language respondent used mostly

English Text: The respondent spoke mostly:

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	English	81271	81271	
2	Spanish	6579	87850	
3	English and Spanish	182	88032	
4	Other	0	88032	
.	Missing	0	88032	

DR2CCMNM - Combination food number

Variable Name: DR2CCMNM

SAS Label: Combination food number

English Text: Combination food number (sequential number)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 12	Range of Values	88032	88032	
.	Missing	0	88032	

DR2CCMTX - Combination food type

Variable Name: DR2CCMTX

SAS Label: Combination food type

English Text: Combination food type

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	Non-combination food	51380	51380	
1	Beverage w/ additions	7709	59089	
2	Cereal w/ additions	2980	62069	
3	Bread/baked products w/ additions	4544	66613	
4	Salad	4855	71468	
5	Sandwiches	5230	76698	
6	Soup	389	77087	
7	Frozen meals	39	77126	
8	Ice cream/frozen yogurt w/ additions	260	77386	
9	Dried beans and vegetable w/ additions	2341	79727	
10	Fruit w/ additions	635	80362	
11	Tortilla products	1951	82313	
12	Meat, poultry, fish	1750	84063	
13	Lunchables®	137	84200	
14	Chips w/ additions	485	84685	
15	Baby Toddler Food and Infant Formula	86	84771	
90	Other mixtures	3261	88032	
.	Missing	0	88032	

DR2_020 - Time of eating occasion (HH:MM)

Variable Name: DR2_020

SAS Label: Time of eating occasion (HH:MM)

English Text: What time did you begin to eat/drink the meal/food?

English Instructions: Format HHMM6.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 86340	Range of Values	88032	88032	
.	Missing	0	88032	

DR2_030Z - Name of eating occasion

Variable Name: DR2_030Z

SAS Label: Name of eating occasion

English Text: Name of eating occasion

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Breakfast	17334	17334	
2	Lunch	17934	35268	
3	Dinner	18882	54150	
4	Supper	3537	57687	
5	Brunch	601	58288	
6	Snack	14649	72937	
7	Drink	5920	78857	
8	Infant feeding	730	79587	
9	Extended consumption	2939	82526	
10	Desayano	1239	83765	
11	Almuerzo	1031	84796	
12	Comida	773	85569	
13	Merienda	556	86125	
14	Cena	1164	87289	
15	Entre comida	130	87419	
16	Botana	131	87550	
17	Bocadillo	156	87706	
18	Tentempie	25	87731	
19	Bebida	301	88032	
91	Other	0	88032	
99	Don't know	0	88032	
.	Missing	0	88032	

DR2FS - Source of food

Variable Name: DR2FS

SAS Label: Source of food

English Text: Where did you get (this/most of the ingredients for this) {FOODNAME}?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Store - grocery/supermarket	59861	59861	
2	Restaurant with waiter/waitress	4807	64668	
3	Restaurant fast food/pizza	6260	70928	
4	Bar/tavern/lounge	237	71165	
5	Restaurant no additional information	16	71181	
6	Cafeteria NOT in a K-12 school	567	71748	
7	Cafeteria in a K-12 school	1050	72798	
8	Child/Adult care center	171	72969	
9	Child/Adult home care	14	72983	
10	Soup kitchen/shelter/food pantry	52	73035	
11	Meals on Wheels	74	73109	
12	Community food program - other	126	73235	
13	Community program no additional information	1	73236	
14	Vending machine	181	73417	
15	Common coffee pot or snack tray	387	73804	
16	From someone else/gift	3032	76836	
17	Mail order purchase	936	77772	
18	Residential dining facility	78	77850	
19	Grown or caught by you or someone you know	622	78472	
20	Fish caught by you or someone you know	8	78480	
24	Sport, recreation, or entertainment facility	224	78704	
25	Street vendor, vending truck	215	78919	
26	Fundraiser sales	43	78962	
27	Store - convenience type	1922	80884	
28	Store - no additional info	298	81182	
91	Other, specify	0	81182	
99	Don't know	113	81295	
.	Missing	6737	88032	

DR2_040Z - Did you eat this meal at home?

Variable Name: DR2_040Z

SAS Label: Did you eat this meal at home?

English Text: Did you eat this meal at home?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	67446	67446	
2	No	20327	87773	
7	Refused	0	87773	
9	Don't know	5	87778	
.	Missing	254	88032	

DR2IFDCD - USDA food code

Variable Name: DR2IFDCD

SAS Label: USDA food code

English Text: USDA food code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11000000 to 95342000	Range of Values	88032	88032	
.	Missing	0	88032	

DR2IGRMS - Grams

Variable Name: DR2IGRMS

SAS Label: Grams

English Text: Gram weight of the food/individual component

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 44160	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IKCAL - Energy (kcal)

Variable Name: DR2IKCAL

SAS Label: Energy (kcal)

English Text: Energy (kcal)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5290	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IPROT - Protein (gm)

Variable Name: DR2IPROT

SAS Label: Protein (gm)

English Text: Protein (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 249.58	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICARB - Carbohydrate (gm)

Variable Name: DR2ICARB

SAS Label: Carbohydrate (gm)

English Text: Carbohydrate (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 724.59	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ISUGR - Total sugars (gm)

Variable Name: DR2ISUGR

SAS Label: Total sugars (gm)

English Text: Total sugars (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 479.88	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IFIBE - Dietary fiber (gm)

Variable Name: DR2IFIBE

SAS Label: Dietary fiber (gm)

English Text: Dietary fiber (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 94.8	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ITFAT - Total fat (gm)

Variable Name: DR2ITFAT

SAS Label: Total fat (gm)

English Text: Total fat (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 280.12	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ISFAT - Total saturated fatty acids (gm)

Variable Name: DR2ISFAT

SAS Label: Total saturated fatty acids (gm)

English Text: Total saturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 175.845	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IMFAT - Total monounsaturated fatty acids (gm)

Variable Name: DR2IMFAT

SAS Label: Total monounsaturated fatty acids (gm)

English Text: Total monounsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 123.9	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IPFAT - Total polyunsaturated fatty acids (gm)

Variable Name: DR2IPFAT

SAS Label: Total polyunsaturated fatty acids (gm)

English Text: Total polyunsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 106.152	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICHOL - Cholesterol (mg)

Variable Name: DR2ICHOL

SAS Label: Cholesterol (mg)

English Text: Cholesterol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2755	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IATOC - Vitamin E as alpha-tocopherol (mg)

Variable Name: DR2IATOC

SAS Label: Vitamin E as alpha-tocopherol (mg)

English Text: Vitamin E as alpha-tocopherol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 102.06	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IATOA - Added alpha-tocopherol (Vitamin E) (mg)

Variable Name: DR2IATOA

SAS Label: Added alpha-tocopherol (Vitamin E) (mg)

English Text: Added alpha-tocopherol (Vitamin E) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 102.06	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IRET - Retinol (mcg)

Variable Name: DR2IRET

SAS Label: Retinol (mcg)

English Text: Retinol (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 28581	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVARA - Vitamin A, RAE (mcg)

Variable Name: DR2IVARA

SAS Label: Vitamin A, RAE (mcg)

English Text: Vitamin A as retinol activity equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 28643	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IACAR - Alpha-carotene (mcg)

Variable Name: DR2IACAR

SAS Label: Alpha-carotene (mcg)

English Text: Alpha-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 15631	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IBCAR - Beta-carotene (mcg)

Variable Name: DR2IBCAR

SAS Label: Beta-carotene (mcg)

English Text: Beta-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 41934	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICRYP - Beta-cryptoxanthin (mcg)

Variable Name: DR2ICRYP

SAS Label: Beta-cryptoxanthin (mcg)

English Text: Beta-cryptoxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2460	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ILYCO - Lycopene (mcg)

Variable Name: DR2ILYCO

SAS Label: Lycopene (mcg)

English Text: Lycopene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 137752	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ILZ - Lutein + zeaxanthin (mcg)

Variable Name: DR2ILZ

SAS Label: Lutein + zeaxanthin (mcg)

English Text: Lutein + zeaxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 67265	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVB1 - Thiamin (Vitamin B1) (mg)

Variable Name: DR2IVB1

SAS Label: Thiamin (Vitamin B1) (mg)

English Text: Thiamin (Vitamin B1) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5.481	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVB2 - Riboflavin (Vitamin B2) (mg)

Variable Name: DR2IVB2

SAS Label: Riboflavin (Vitamin B2) (mg)

English Text: Riboflavin (Vitamin B2) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 23.808	Range of Values	87778	87778	
.	Missing	254	88032	

DR2INIAC - Niacin (mg)

Variable Name: DR2INIAC

SAS Label: Niacin (mg)

English Text: Niacin (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 146.122	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVB6 - Vitamin B6 (mg)

Variable Name: DR2IVB6

SAS Label: Vitamin B6 (mg)

English Text: Vitamin B6 (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 21.497	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IFOLA - Total folate (mcg)

Variable Name: DR2IFOLA

SAS Label: Total folate (mcg)

English Text: Total folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1612	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IFA - Folic acid (mcg)

Variable Name: DR2IFA

SAS Label: Folic acid (mcg)

English Text: Folic acid (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1456	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IFF - Food folate (mcg)

Variable Name: DR2IFF

SAS Label: Food folate (mcg)

English Text: Food folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1612	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IFDFE - Folate, DFE (mcg)

Variable Name: DR2IFDFE

SAS Label: Folate, DFE (mcg)

English Text: Folate as dietary folate equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2476	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICHL - Total choline (mg)

Variable Name: DR2ICHL

SAS Label: Total choline (mg)

English Text: Total choline (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1802.2	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVB12 - Vitamin B12 (mcg)

Variable Name: DR2IVB12

SAS Label: Vitamin B12 (mcg)

English Text: Vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 307.04	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IB12A - Added vitamin B12 (mcg)

Variable Name: DR2IB12A

SAS Label: Added vitamin B12 (mcg)

English Text: Added vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 56.1	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVC - Vitamin C (mg)

Variable Name: DR2IVC

SAS Label: Vitamin C (mg)

English Text: Vitamin C (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 766.3	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVD - Vitamin D (D2 + D3) (mcg)

Variable Name: DR2IVD

SAS Label: Vitamin D (D2 + D3) (mcg)

English Text: Vitamin D (D2 + D3) (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 62	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IVK - Vitamin K (mcg)

Variable Name: DR2IVK

SAS Label: Vitamin K (mcg)

English Text: Vitamin K (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3391.3	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICALC - Calcium (mg)

Variable Name: DR2ICALC

SAS Label: Calcium (mg)

English Text: Calcium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 7415	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IPHOS - Phosphorus (mg)

Variable Name: DR2IPHOS

SAS Label: Phosphorus (mg)

English Text: Phosphorus (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 11545	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IMAGN - Magnesium (mg)

Variable Name: DR2IMAGN

SAS Label: Magnesium (mg)

English Text: Magnesium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1275	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IIRON - Iron (mg)

Variable Name: DR2IIRON

SAS Label: Iron (mg)

English Text: Iron (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 61.21	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IZINC - Zinc (mg)

Variable Name: DR2IZINC

SAS Label: Zinc (mg)

English Text: Zinc (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 54.04	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICOPP - Copper (mg)

Variable Name: DR2ICOPP

SAS Label: Copper (mg)

English Text: Copper (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 54.031	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ISODI - Sodium (mg)

Variable Name: DR2ISODI

SAS Label: Sodium (mg)

English Text: Sodium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 21441	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IPOTA - Potassium (mg)

Variable Name: DR2IPOTA

SAS Label: Potassium (mg)

English Text: Potassium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 8084	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ISELE - Selenium (mcg)

Variable Name: DR2ISELE

SAS Label: Selenium (mcg)

English Text: Selenium (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 958.5	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ICAFF - Caffeine (mg)

Variable Name: DR2ICAFF

SAS Label: Caffeine (mg)

English Text: Caffeine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1440	Range of Values	87778	87778	
.	Missing	254	88032	

DR2ITHEO - Theobromine (mg)

Variable Name: DR2ITHEO

SAS Label: Theobromine (mg)

English Text: Theobromine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 652	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IALCO - Alcohol (gm)

Variable Name: DR2IALCO

SAS Label: Alcohol (gm)

English Text: Alcohol (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 237.1	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IMOIS - Moisture (gm)

Variable Name: DR2IMOIS

SAS Label: Moisture (gm)

English Text: Moisture (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 44151.17	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS040 - SFA 4:0 (Butanoic) (gm)

Variable Name: DR2IS040

SAS Label: SFA 4:0 (Butanoic) (gm)

English Text: SFA 4:0 (Butanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 9.078	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS060 - SFA 6:0 (Hexanoic) (gm)

Variable Name: DR2IS060

SAS Label: SFA 6:0 (Hexanoic) (gm)

English Text: SFA 6:0 (Hexanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5.383	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS080 - SFA 8:0 (Octanoic) (gm)

Variable Name: DR2IS080

SAS Label: SFA 8:0 (Octanoic) (gm)

English Text: SFA 8:0 (Octanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4.919	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS100 - SFA 10:0 (Decanoic) (gm)

Variable Name: DR2IS100

SAS Label: SFA 10:0 (Decanoic) (gm)

English Text: SFA 10:0 (Decanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 6.769	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS120 - SFA 12:0 (Dodecanoic) (gm)

Variable Name: DR2IS120

SAS Label: SFA 12:0 (Dodecanoic) (gm)

English Text: SFA 12:0 (Dodecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 39.762	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS140 - SFA 14:0 (Tetradecanoic) (gm)

Variable Name: DR2IS140

SAS Label: SFA 14:0 (Tetradecanoic) (gm)

English Text: SFA 14:0 (Tetradecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 28.778	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS160 - SFA 16:0 (Hexadecanoic) (gm)

Variable Name: DR2IS160

SAS Label: SFA 16:0 (Hexadecanoic) (gm)

English Text: SFA 16:0 (Hexadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 77.954	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IS180 - SFA 18:0 (Octadecanoic) (gm)

Variable Name: DR2IS180

SAS Label: SFA 18:0 (Octadecanoic) (gm)

English Text: SFA 18:0 (Octadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 32.116	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IM161 - MFA 16:1 (Hexadecenoic) (gm)

Variable Name: DR2IM161

SAS Label: MFA 16:1 (Hexadecenoic) (gm)

English Text: MFA 16:1 (Hexadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 8.399	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IM181 - MFA 18:1 (Octadecenoic) (gm)

Variable Name: DR2IM181

SAS Label: MFA 18:1 (Octadecenoic) (gm)

English Text: MFA 18:1 (Octadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 123.319	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IM201 - MFA 20:1 (Eicosenoic) (gm)

Variable Name: DR2IM201

SAS Label: MFA 20:1 (Eicosenoic) (gm)

English Text: MFA 20:1 (Eicosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.537	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IM221 - MFA 22:1 (Docosenoic) (gm)

Variable Name: DR2IM221

SAS Label: MFA 22:1 (Docosenoic) (gm)

English Text: MFA 22:1 (Docosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.024	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP182 - PFA 18:2 (Octadecadienoic) (gm)

Variable Name: DR2IP182

SAS Label: PFA 18:2 (Octadecadienoic) (gm)

English Text: PFA 18:2 (Octadecadienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 92.174	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP183 - PFA 18:3 (Octadecatrienoic) (gm)

Variable Name: DR2IP183

SAS Label: PFA 18:3 (Octadecatrienoic) (gm)

English Text: PFA 18:3 (Octadecatrienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 16.344	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP184 - PFA 18:4 (Octadecatetraenoic) (gm)

Variable Name: DR2IP184

SAS Label: PFA 18:4 (Octadecatetraenoic) (gm)

English Text: PFA 18:4 (Octadecatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 0.612	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP204 - PFA 20:4 (Eicosatetraenoic) (gm)

Variable Name: DR2IP204

SAS Label: PFA 20:4 (Eicosatetraenoic) (gm)

English Text: PFA 20:4 (Eicosatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.139	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP205 - PFA 20:5 (Eicosapentaenoic) (gm)

Variable Name: DR2IP205

SAS Label: PFA 20:5 (Eicosapentaenoic) (gm)

English Text: PFA 20:5 (Eicosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.901	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP225 - PFA 22:5 (Docosapentaenoic) (gm)

Variable Name: DR2IP225

SAS Label: PFA 22:5 (Docosapentaenoic) (gm)

English Text: PFA 22:5 (Docosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.728	Range of Values	87778	87778	
.	Missing	254	88032	

DR2IP226 - PFA 22:6 (Docosahexaenoic) (gm)

Variable Name: DR2IP226

SAS Label: PFA 22:6 (Docosahexaenoic) (gm)

English Text: PFA 22:6 (Docosahexaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5.135	Range of Values	87778	87778	
.	Missing	254	88032	

Appendix 1. Changes between WWEIA survey cycles 2013-2014 thru the August 2021-August 2023 sample

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Administration of 24-hour dietary recall	Same as 2003-2012: Day 1 in-person in MEC, Day 2 by telephone	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Day 1 by telephone, Day 2 by telephone
Language, Day 1	Same as 1999-2012: Interpreters assisted as needed	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Language, Day 2	Same as 2011-2012: English, Spanish, or Asian languages	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Person who did not provide reliable intake for Day 1 but did for Day 2	Same as 2003-2012: Neither intake was included in the release datasets	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	The single reliable intake was included in the release datasets as Day 1 intake (n=21)
Food source (where food was obtained)	Codes 8 and 9 revised descriptions.	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014
Combination food types	Same as 2003-2012: Values for 15 combination types	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	Values for 16 combination types; added “Baby Toddler Food and Infant Formula”
Number of intakes that include only water consumption for the day	6 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (all in Day 2 data), records are included in Individual Foods file.	2 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (1 in Day 1, 4 in Day 2 data), records are included in Individual Foods file.	5 intakes (4 in Day 1, 1 in Day 2), records are included in Individual Foods file.
Number of intakes that include no water or food consumption for the day	1 intake in Day 2 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	2 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.	3 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.
Tap water source	Same as 2003-2012: Reported in 6 categories.	Same as 2003-2004	Responses combined into 4 categories	Same as 2017-2018	Same as 2017-2018
Modification codes: DR1MC	All remaining modification	No modification	No modification	No modification	No modification

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Day 2 Modification codes: DR2MC Modification Code Description file: DRXMCD	codes deleted; new food codes addressing modifications added in FNDDS 2013-2014	codes	codes	codes	codes
Salt use at the table and in cooking or preparing foods and type	Same as 2003-2004	Same as 2003-2004	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Salt used at the table yesterday and type	Question asked about salt use at the table yesterday and kind of salt to coincide with 24-hour recall.	Same as 2013-2014	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Main respondent and person whom helped in responding for the interview	Same as 2003-2012	Response category changed	Same as 2015-2016	Same as 2015-2016	Same as 2015-2016

Appendix 2. Variables in the Individual Foods Files (DR1IFF_L and DR2IFF_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1ILINE	DR2ILINE	Food/Individual component number
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1CCMNM	DR2CCMNM	Combination food number
DR1CCMTX	DR2CCMTX	Combination food type
DR1_020	DR2_020	Time of eating occasion (HH:MM)
DR1_030Z	DR2_030Z	Name of eating occasion
DR1FS	DR2FS	Source of food
DR1_040Z	DR2_040Z	Did you eat this meal at home?
DR1IFDCD	DR2IFDCD	USDA food code
DR1IGRMS	DR2IGRMS	Grams
DR1IKCAL	DR2IKCAL	Energy (kcal)
DR1IPROT	DR2IPROT	Protein (gm)
DR1ICARB	DR2ICARB	Carbohydrate (gm)
DR1ISUGR	DR2ISUGR	Total sugars (gm)
DR1IFIBE	DR2IFIBE	Dietary fiber (gm)
DR1ITFAT	DR2ITFAT	Total fat (gm)
DR1ISFAT	DR2ISFAT	Total saturated fatty acids (gm)
DR1IMFAT	DR2IMFAT	Total monounsaturated fatty acids (gm)
DR1IPFAT	DR2IPFAT	Total polyunsaturated fatty acids (gm)
DR1ICHOL	DR2ICHOL	Cholesterol (mg)
DR1IATOC	DR2IATOC	Vitamin E as alpha-tocopherol (mg)
DR1IATOA	DR2IATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1IRET	DR2IRET	Retinol (mcg)
DR1IVARA	DR2IVARA	Vitamin A, RAE (mcg)
DR1IACAR	DR2IACAR	Alpha-carotene (mcg)
DR1IBCAR	DR2IBCAR	Beta-carotene (mcg)
DR1ICRYP	DR2ICRYP	Beta-cryptoxanthin (mcg)
DR1ILYCO	DR2ILYCO	Lycopene (mcg)
DR1ILZ	DR2ILZ	Lutein + zeaxanthin (mcg)
DR1IVB1	DR2IVB1	Thiamin (Vitamin B1) (mg)

Day1 Name	Day2 Name	Variable Label
DR1IVB2	DR2IVB2	Riboflavin (Vitamin B2) (mg)
DR1INIAC	DR2INIAC	Niacin (mg)
DR1IVB6	DR2IVB6	Vitamin B6 (mg)
DR1IFOLA	DR2IFOLA	Total folate (mcg)
DR1IFA	DR2IFA	Folic acid (mcg)
DR1IFF	DR2IFF	Food folate (mcg)
DR1IFDFE	DR2IFDFE	Folate, DFE (mcg)
DR1ICHL	DR2ICHL	Total choline (mg)
DR1IVB12	DR2IVB12	Vitamin B12 (mcg)
DR1IB12A	DR2IB12A	Added vitamin B12 (mcg)
DR1IVC	DR2IVC	Vitamin C (mg)
DR1IVD	DR2IVD	Vitamin D (D2 + D3) (mcg)
DR1IVK	DR2IVK	Vitamin K (mcg)
DR1ICALC	DR2ICALC	Calcium (mg)
DR1IPHOS	DR2IPHOS	Phosphorus (mg)
DR1IMAGN	DR2IMAGN	Magnesium (mg)
DR1IIRON	DR2IIRON	Iron (mg)
DR1IZINC	DR2IZINC	Zinc (mg)
DR1ICOPP	DR2ICOPP	Copper (mg)
DR1ISODI	DR2ISODI	Sodium (mg)
DR1IPOTA	DR2IPOTA	Potassium (mg)
DR1ISELE	DR2ISELE	Selenium (mcg)
DR1ICAFF	DR2ICAFF	Caffeine (mg)
DR1ITHEO	DR2ITHEO	Theobromine (mg)
DR1IALCO	DR2IALCO	Alcohol (gm)
DR1IMOIS	DR2IMOIS	Moisture (gm)
DR1IS040	DR2IS040	SFA 4:0 (Butanoic) (gm)
DR1IS060	DR2IS060	SFA 6:0 (Hexanoic) (gm)
DR1IS080	DR2IS080	SFA 8:0 (Octanoic) (gm)
DR1IS100	DR2IS100	SFA 10:0 (Decanoic) (gm)
DR1IS120	DR2IS120	SFA 12:0 (Dodecanoic) (gm)
DR1IS140	DR2IS140	SFA 14:0 (Tetradecanoic) (gm)
DR1IS160	DR2IS160	SFA 16:0 (Hexadecanoic) (gm)
DR1IS180	DR2IS180	SFA 18:0 (Octadecanoic) (gm)
DR1IM161	DR2IM161	MFA 16:1 (Hexadecenoic) (gm)
DR1IM181	DR2IM181	MFA 18:1 (Octadecenoic) (gm)
DR1IM201	DR2IM201	MFA 20:1 (Eicosenoic) (gm)
DR1IM221	DR2IM221	MFA 22:1 (Docosenoic) (gm)
DR1IP182	DR2IP182	PFA 18:2 (Octadecadienoic) (gm)

Day1 Name	Day2 Name	Variable Label
DR1IP183	DR2IP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1IP184	DR2IP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1IP204	DR2IP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1IP205	DR2IP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1IP225	DR2IP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1IP226	DR2IP226	PFA 22:6 (Docosahexaenoic) (gm)

Appendix 3. List of Nutrients/Food Components (Unit)

Energy and Macronutrients

Food energy (kcal)
Protein (gm)
Carbohydrate (gm)
Fat, total (gm)
Alcohol (gm)

Sugars, total (gm)
Dietary fiber, total (gm)
Water (moisture) (gm)*

Saturated fatty acids, total (gm)
Monounsaturated fatty acids, total (gm)
Polyunsaturated fatty acids, total (gm)
Cholesterol (mg)

Individual fatty acids:
  4:0 (gm)
  6:0 (gm)
  8:0 (gm)
  10:0 (gm)
  12:0 (gm)
  14:0 (gm)
  16:0 (gm)
  18:0 (gm)
  16:1 (gm)
  18:1 (gm)
  20:1 (gm)
  22:1 (gm)
  18:2 (gm)
  18:3 (gm)
  18:4 (gm)
  20:4 (gm)
  20:5 n-3 (gm)
  22:5 n-3 (gm)
  22:6 n-3 (gm)

Vitamins, Minerals, and Other Components

Vitamin A as retinol activity equivalents (mcg)
Retinol (mcg)
Carotenoids:
  Carotene, alpha (mcg)
  Carotene, beta (mcg)
  Cryptoxanthin, beta (mcg)
  Lycopene (mcg)
  Lutein + zeaxanthin (mcg)
Vitamin E as alpha-tocopherol (mg)
  **Added vitamin E as alpha-tocopherol (mg)
Vitamin D (D2 + D3) (mcg)
Vitamin K as phylloquinone (mcg)
Vitamin C (mg)
Thiamin (mg)
Riboflavin (mg)
Niacin (mg)
Vitamin B-6 (mg)
Folate, total (mcg)
  Folate as dietary folate equivalents (mcg)

Folic acid (mcg)
Food folate (mcg)
Choline, total (mg)
Vitamin B-12 (mcg)
***Added vitamin B-12 (mcg)

Calcium (mg)
Iron (mg)
Magnesium (mg)
Phosphorus (mg)
Potassium (mg)
Sodium (mg)
Zinc (mg)
Copper (mg)
Selenium (mcg)
Caffeine (mg)
Theobromine (mg)

** Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)*

*** Represents a synthetic subcomponent of vitamin E and is included in the vitamin E value.*

**** Represents a fortified subcomponent of vitamin B12 and is included in the vitamin B12 value.*

Appendix 4. Adding Food Code Descriptions to Your Files

One supporting file is included with the Individual Foods files: the Food Code Description file (DRXFCD_L).

The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code included in the Individual Foods files.

The Food Code Description file (DRXFCD_L) contains three variables:

DRXFDCD a numeric value corresponding to DR1IFDCD in the file DR1IFF_L or DR2IFDCD in the file DR2IFF_L;

DRXFCSD a short description (up to 60 characters) of the food code;

DRXFDLD a long description (up to 200 characters) of the food code.

The following SQL code is an example of appending the shorter food code description (here renamed DR1IFCSD) to one of the Individual Foods files using PROC SQL from SAS®. Other SQL implementations may be different.

```
proc sql;
  create table DR1IFF_L_PLUS as
  select iff.*, desc.DRXFCSD as DR1FCSD, desc.DRXFCLD as DR1FCLD
  from NHANES.DR1IFF_L iff
  left join NHANES.DRXFCD_L desc
  on iff.DR1IFDCD = desc.DRXFDCD
  order by SEQN, DR1ILINE;
quit;
```

SAS® users may wish to use Proc Format to assign labels to the food codes. The following example generates and saves a picture format for food codes and a separate format for each food code that includes both the food code itself and the short food code description. It is assumed that the user has stored the Individual Foods files and the Food Code Description file in a library called NHANES and wishes to store the formats there as well.

```
options fmtsearch = (NHANES);

proc format library = library;
  picture foodcode
  low - high = '000-00000';
quit;

data tmp;
  set NHANES.DRXFCD_L;
  length cfoodcode $9 label $72;
  cfoodcode = put(DRXFDCD, foodcode.);
  label = cfoodcode || '  ' || DRXFCSD;
run;

data fmt (keep = fmtname start label);
  set tmp;
  retain fmtname 'DRXFDCD';
  rename DRXFDCD = start;
run;
```

```
proc format cntlin = fmt library = library;
run;
```

Appendix 5. Variables in the Total Nutrients Files (DR1TOT_L and DR2TOT_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1MRESP	DR2MRESP	Main respondent for this interview
DR1HELP	DR2HELP	Helped in responding for this interview
DBQ095Z	N/A	Type of table salt used
DBD100	N/A	How often add salt to food at table
DRQSPREP	N/A	Salt used in preparation?
DR1STY	DR2STY	Salt used at table yesterday?
DR1SKY	DR2SKY	Type of salt used yesterday
DRQSDIET	N/A	On special diet?
DRQSDT1	N/A	Weight loss/Low calorie diet
DRQSDT2	N/A	Low fat/Low cholesterol diet
DRQSDT3	N/A	Low salt/Low sodium diet
DRQSDT4	N/A	Sugar free/Low sugar diet
DRQSDT5	N/A	Low fiber diet
DRQSDT6	N/A	High fiber diet
DRQSDT7	N/A	Diabetic diet
DRQSDT8	N/A	Weight gain/Muscle building diet
DRQSDT9	N/A	Low carbohydrate diet
DRQSDT10	N/A	High protein diet
DRQSDT11	N/A	Gluten-free/Celiac diet
DRQSDT12	N/A	Renal/Kidney diet
DRQSDT91	N/A	Other special diet
DR1TNUMF	DR2TNUMF	Number of foods/beverages reported
DR1TKCAL	DR2TKCAL	Energy (kcal)
DR1TPROT	DR2TPROT	Protein (gm)
DR1TCARB	DR2TCARB	Carbohydrate (gm)
DR1TSUGR	DR2TSUGR	Total sugars (gm)
DR1TFIBE	DR2TFIBE	Dietary fiber (gm)
DR1TTFAT	DR2TTFAT	Total fat (gm)
DR1TSFAT	DR2TSFAT	Total saturated fatty acids (gm)

Day1 Name	Day2 Name	Variable Label
DR1TMFAT	DR2TMFAT	Total monounsaturated fatty acids (gm)
DR1TPFAT	DR2TPFAT	Total polyunsaturated fatty acids (gm)
DR1TCHOL	DR2TCHOL	Cholesterol (mg)
DR1TATOC	DR2TATOC	Vitamin E as alpha-tocopherol (mg)
DR1TATOA	DR2TATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1TRET	DR2TRET	Retinol (mcg)
DR1TVARA	DR2TVARA	Vitamin A, RAE (mcg)
DR1TACAR	DR2TACAR	Alpha-carotene (mcg)
DR1TBCAR	DR2TBCAR	Beta-carotene (mcg)
DR1TCRYP	DR2TCRYP	Beta-cryptoxanthin (mcg)
DR1TLYCO	DR2TLYCO	Lycopene (mcg)
DR1TLZ	DR2TLZ	Lutein + zeaxanthin (mcg)
DR1TVB1	DR2TVB1	Thiamin (Vitamin B1) (mg)
DR1TVB2	DR2TVB2	Riboflavin (Vitamin B2) (mg)
DR1TNIAC	DR2TNIAC	Niacin (mg)
DR1TVB6	DR2TVB6	Vitamin B6 (mg)
DR1TFOLA	DR2TFOLA	Total folate (mcg)
DR1TFA	DR2TFA	Folic acid (mcg)
DR1TFF	DR2TFF	Food folate (mcg)
DR1TFDFE	DR2TFDFE	Folate, DFE (mcg)
DR1TCHL	DR2TCHL	Total choline (mg)
DR1TVB12	DR2TVB12	Vitamin B12 (mcg)
DR1TB12A	DR2TB12A	Added vitamin B12 (mcg)
DR1TVC	DR2TVC	Vitamin C (mg)
DR1TVD	DR2TVD	Vitamin D (D2 + D3) (mcg)
DR1TVK	DR2TVK	Vitamin K (mcg)
DR1TCALC	DR2TCALC	Calcium (mg)
DR1TPHOS	DR2TPHOS	Phosphorus (mg)
DR1TMAGN	DR2TMAGN	Magnesium (mg)
DR1TIRON	DR2TIRON	Iron (mg)
DR1TZINC	DR2TZINC	Zinc (mg)
DR1TCOPP	DR2TCOPP	Copper (mg)
DR1TSODI	DR2TSODI	Sodium (mg)
DR1TPOTA	DR2TPOTA	Potassium (mg)
DR1TSELE	DR2TSELE	Selenium (mcg)
DR1TCAFF	DR2TCAFF	Caffeine (mg)
DR1TTHEO	DR2TTHEO	Theobromine (mg)
DR1TALCO	DR2TALCO	Alcohol (gm)
DR1TMOIS	DR2TMOIS	Moisture (gm)

Day1 Name	Day2 Name	Variable Label
DR1TS040	DR2TS040	SFA 4:0 (Butanoic) (gm)
DR1TS060	DR2TS060	SFA 6:0 (Hexanoic) (gm)
DR1TS080	DR2TS080	SFA 8:0 (Octanoic) (gm)
DR1TS100	DR2TS100	SFA 10:0 (Decanoic) (gm)
DR1TS120	DR2TS120	SFA 12:0 (Dodecanoic) (gm)
DR1TS140	DR2TS140	SFA 14:0 (Tetradecanoic) (gm)
DR1TS160	DR2TS160	SFA 16:0 (Hexadecanoic) (gm)
DR1TS180	DR2TS180	SFA 18:0 (Octadecanoic) (gm)
DR1TM161	DR2TM161	MFA 16:1 (Hexadecenoic) (gm)
DR1TM181	DR2TM181	MFA 18:1 (Octadecenoic) (gm)
DR1TM201	DR2TM201	MFA 20:1 (Eicosenoic) (gm)
DR1TM221	DR2TM221	MFA 22:1 (Docosenoic) (gm)
DR1TP182	DR2TP182	PFA 18:2 (Octadecadienoic) (gm)
DR1TP183	DR2TP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1TP184	DR2TP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1TP204	DR2TP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1TP205	DR2TP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1TP225	DR2TP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1TP226	DR2TP226	PFA 22:6 (Docosahexaenoic) (gm)
DR1_300	DR2_300	Compare food consumed yesterday to usual
DR1_320Z	DR2_320Z	Total plain water drank yesterday (gm)
DR1_330Z	DR2_330Z	Total tap water drank yesterday (gm)
DR1BWATZ	DR2BWATZ	Total bottled water drank yesterday (gm)
DR1TWSZ	DR2TWSZ	Tap water source
DRD340	N/A	Shellfish eaten during past 30 days
DRD350A	N/A	Clams eaten during past 30 days
DRD350AQ	N/A	# of times clams eaten in past 30 days
DRD350B	N/A	Crabs eaten during past 30 days
DRD350BQ	N/A	# of times crabs eaten in past 30 days
DRD350C	N/A	Crayfish eaten during past 30 days
DRD350CQ	N/A	# of times crayfish eaten past 30 days
DRD350D	N/A	Lobsters eaten during past 30 days
DRD350DQ	N/A	# of times lobsters eaten past 30 days
DRD350E	N/A	Mussels eaten during past 30 days
DRD350EQ	N/A	# of times mussels eaten in past 30 days
DRD350F	N/A	Oysters eaten during past 30 days
DRD350FQ	N/A	# of times oysters eaten in past 30 days
DRD350G	N/A	Scallops eaten during past 30 days
DRD350GQ	N/A	# of times scallops eaten past 30 days

Day1 Name	Day2 Name	Variable Label
DRD350H	N/A	Shrimp eaten during past 30 days
DRD350HQ	N/A	# of times shrimp eaten in past 30 days
DRD350I	N/A	Other shellfish eaten past 30 days
DRD350IQ	N/A	# of times other shellfish eaten
DRD350J	N/A	Other unknown shellfish eaten past 30 d
DRD350JQ	N/A	# of times other unknown shellfish eaten
DRD350K	N/A	Refused on shellfish eaten past 30 days
DRD360	N/A	Fish eaten during past 30 days
DRD370A	N/A	Breaded fish products eaten past 30 days
DRD370AQ	N/A	# of times breaded fish products eaten
DRD370B	N/A	Tuna eaten during past 30 days
DRD370BQ	N/A	# of times tuna eaten in past 30 days
DRD370C	N/A	Bass eaten during past 30 days
DRD370CQ	N/A	# of times bass eaten in past 30 days
DRD370D	N/A	Catfish eaten during past 30 days
DRD370DQ	N/A	# of times catfish eaten in past 30 days
DRD370E	N/A	Cod eaten during past 30 days
DRD370EQ	N/A	# of times cod eaten in past 30 days
DRD370F	N/A	Flatfish eaten during past 30 days
DRD370FQ	N/A	# of times flatfish eaten past 30 days
DRD370G	N/A	Haddock eaten during past 30 days
DRD370GQ	N/A	# of times haddock eaten in past 30 days
DRD370H	N/A	Mackerel eaten during past 30 days
DRD370HQ	N/A	# of times mackerel eaten past 30 days
DRD370I	N/A	Perch eaten during past 30 days
DRD370IQ	N/A	# of times perch eaten in past 30 days
DRD370J	N/A	Pike eaten during past 30 days
DRD370JQ	N/A	# of times pike eaten in past 30 days
DRD370K	N/A	Pollock eaten during past 30 days
DRD370KQ	N/A	# of times pollock eaten in past 30 days
DRD370L	N/A	Porgy eaten during past 30 days
DRD370LQ	N/A	# of times porgy eaten in past 30 days
DRD370M	N/A	Salmon eaten during past 30 days
DRD370MQ	N/A	# of times salmon eaten in past 30 days
DRD370N	N/A	Sardines eaten during past 30 days
DRD370NQ	N/A	# of times sardines eaten past 30 days
DRD370O	N/A	Sea bass eaten during past 30 days
DRD370OQ	N/A	# of times sea bass eaten past 30 days
DRD370P	N/A	Shark eaten during past 30 days

Day1 Name	Day2 Name	Variable Label
DRD370PQ	N/A	# of times shark eaten in past 30 days
DRD370Q	N/A	Swordfish eaten during past 30 days
DRD370QQ	N/A	# of times swordfish eaten past 30 days
DRD370R	N/A	Trout eaten during past 30 days
DRD370RQ	N/A	# of times trout eaten in past 30 days
DRD370S	N/A	Walleye eaten during past 30 days
DRD370SQ	N/A	# of times walleye eaten in past 30 days
DRD370T	N/A	Other fish eaten during past 30 days
DRD370TQ	N/A	# of times other fish eaten past 30 days
DRD370U	N/A	Other unknown fish eaten in past 30 days
DRD370UQ	N/A	# of times other unknown fish eaten
DRD370V	N/A	Refused on fish eaten past 30 days

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Interview - Total Nutrient Intakes, Second Day (DR2TOT_L)

Data File: DR2TOT_L.xpt

First Published: October 2024

Last Revised: NA

Component Description

The objective of the dietary interview component is to obtain detailed dietary intake information from NHANES participants. The dietary intake data are used to estimate the types and amounts of foods and beverages (including all types of water) consumed during the 24-hour period prior to the interview (midnight to midnight), and to estimate intakes of energy, nutrients, and other food components from those foods and beverages. Following the dietary recall, participants are asked questions on salt use, whether the person's overall intake on the previous day was much more than usual, usual or much less than usual, and whether the participant is on any type of special diet. Questions on frequency of fish and shellfish consumed during the past 30 days are asked of participants 1 year or older, with the use of proxies for young children (see the Dietary Interview Procedure Manuals (cdc.gov) for more information on the proxy interview).

The dietary interview component, called What We Eat in America (WWEIA), is conducted as a partnership between the U.S. Department of Agriculture (USDA) and the U.S. Department of Health and Human Services (DHHS). Under this partnership, DHHS' National Center for Health Statistics (NCHS), Division of Health and Nutrition Examination Surveys is responsible for the survey sample design and all aspects of data collection and USDA's Food Surveys Research Group (FSRG) is responsible for the dietary data collection methodology, maintenance of the databases used to code and process the data, and data review and processing.

All NHANES participants are eligible for two 24-hour dietary recall interviews. Traditionally, the first dietary recall interview was collected in-person in the Mobile Examination Center (MEC) and the second interview was collected by telephone 3 to 10 days later. In August 2021-August 2023, the time participants spent in the MEC was limited to reduce risk of exposure to SARS-CoV-2. As a result, both dietary recall interviews were administered via telephone. Further information about changes in data collection are described in the Plan and Operations report of the NHANES August 2021-August 2023.

As in previous years, two types of dietary intake data are available for the August 2021-August 2023 survey cycle: Individual Foods files and Total Nutrient Intakes files.

What's New with the August 2021-August 2023 WWEIA Release:

In response to the COVID-19 pandemic, the mode of the first dietary recall changed from in-person during the MEC examination to telephone 3 to 7 days after the examination. Both dietary interviews were completed in English or Spanish. During the mobile examination, participants scheduled an appointment for the first telephone dietary interview and received a set of measuring guides to help estimate the amounts of foods and beverages consumed. **Appendix 1** provides a summary of changes among the 5 latest cycles of data collection. No new data variables were added in the August 2021-August 2023 WWEIA release.

Dietary Interview Data Files: Four data files were produced from the information collected in the dietary interviews: two Individual Foods files and two Total Nutrient Intakes files. Each

file includes one day of intake data. The number "1" or "2" in the file name identifies the day of the interview: 1 = first day, 2 = second day. File names are as follows:

File Names for Dietary Interview Data:

File	Day 1	Day 2
Individual Foods File	DR1IFF_L	DR2IFF_L
Total Nutrient Intakes File	DR1TOT_L	DR2TOT_L

The amounts in these files reflect only nutrients obtained from foods, beverages, and water, including tap and bottled water. They do not include nutrients obtained from dietary supplement intakes, antacids, or medications. Data on intake of dietary supplement use are available on the August 2021-August 2023 Dietary Data (cdc.gov) page.

Individual Foods Files (DR1IFF_L and DR2IFF_L): Detailed information about each food/beverage item (including the description, amount of, and nutrient content) reported by each participant is included in the Individual Foods files. The names for both Day 1 and Day 2 variables are listed in **Appendix 2**.

The Individual Foods files include, for each interview day, one record for each food/beverage consumed by a participant. Each record is uniquely numbered within a participant's set of records and contains the information listed below:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Whether the food/beverage was eaten in combination with other foods, such as in a sandwich;
- Time of eating occasion/when the food was eaten;
- Eating occasion name;
- Where the food/beverage was obtained;
- Whether the meal/snack was eaten at home or not;
- A USDA Food and Nutrient Database for Dietary Studies (FNDDS) code identifying the food/beverage;
- Amount of food/beverage consumed, in grams; and
- Food energy and 64 nutrients/food components (listed in **Appendix 3**) from each food/beverage as calculated using USDA's Food and Nutrient Database for Dietary Studies 2021-2023 (FNDDS 2021-2023).

Descriptions for the USDA FNDDS food codes are provided in the Food Code Description file (DRXFCD_L). The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code in the FNDDS 2021-2023. **Appendix 4** provides SAS code examples that may be used to link the food code description to the Individual Foods file.

Total Nutrient Intakes Files (DR1TOT_L and DR2TOT_L): For each participant, daily total energy and nutrient intakes from foods and beverages, and whether the amount of food consumed was usual, much more than usual, or much less than usual, are included in the Total Nutrient Intakes files. The Day 1 file also includes information on salt use in cooking and at the table; whether the participant is currently on any kind of diet to lose weight or for another health-related reason and, if so, the type of diet; and information on frequency of fish and shellfish consumption for participants aged 1 or older. The names for both Day 1 and Day 2 variables are listed in **Appendix 5**.

The Total Nutrient Intakes files provide a summary record of total nutrient intakes for each participant. Each total intake record contains the following information:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Type of salt used and how often added at the table and in food preparation (Day 1 file only);
- Use of salt at the table yesterday and the type of salt used;
- Whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet (Day 1 file only);
- Total number of foods and beverages including water reported for that participant for that day's intake;
- Daily aggregates of food energy and 64 nutrients/food components (listed in **Appendix 3**) from all foods/beverages as calculated using USDA's Food and Nutrient Database for Dietary Studies August 2021-August 2023 (FNDDS 2021- 2023)
- Whether the amount of food consumed was usual, much more than usual, or much less than usual;
- Total amount of tap and bottled water consumed (calculated as the sum of reports of water drunk by itself in the 24-hour recall) and the usual source of tap water; and
- Frequency of fish and shellfish consumption in the past 30 days (participants 1 year or older, Day 1 file only).

Eligible Sample

All participants in the August 2021-August 2023 sample who spoke English or Spanish were eligible. Participants aged 1 year or older were eligible for the frequency of fish and shellfish consumption questions following the 24-hour recall.

Protocol and Procedure

The examination protocol and data collection methods are fully documented in the NHANES dietary interviewer procedures manuals.

Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant's intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child's intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish.

During the MEC visit, participants were scheduled an appointment for the first dietary interview and were given a set of measuring guides (NHANES Measuring Guides for the Dietary Recall Interview) to use for reporting food amounts during the telephone interview. The measuring guides included measuring cups, spoons, a ruler, and a food model booklet, which contained two-dimensional drawings of glasses, mugs, bowls, mounds, circles, wedges, a shape chart and a chicken chart. The first telephone dietary interview was scheduled 3 to 7 days post MEC examination while the second interview was collected 3 to 10 days following the first dietary interview and were generally scheduled on a different day of the week as the first interview. In August 2021-August 2023, 204 participants were interviewed on the same day of the week for both day 1 and day 2 interviews due to their scheduling availability. Any participant who did not have a telephone was given a toll-free number to call so that the recall could be conducted.

What We Eat in America data were collected using USDA's dietary data collection instrument, the Automated Multiple Pass Method (AMPM), available at:

http://www.ars.usda.gov/nea/bhnrc/fsrg. The AMPM was designed to provide an efficient and accurate means of collecting intakes for large-scale national surveys. The AMPM is a fully computerized recall method that uses a 5-step interview outlined below:

1. **Quick List** - Participant recalls all foods and beverages consumed the day before the interview (midnight to midnight).
2. **Forgotten Foods** - Participant is asked about consumption of foods commonly forgotten during the Quick List step.
3. **Time and Occasion** - Time and eating occasion are collected for each food.
4. **Detail Cycle** - For each food, a detailed description, amount eaten, and additions to the food are collected. Eating occasions and times between eating occasions are reviewed to elicit forgotten foods.
5. **Final Probe** - Additional foods not remembered earlier are collected.

The AMPM includes an extensive compilation of standardized food-specific questions and possible response options. Routing of questions is based on previous responses. The AMPM is updated for each 2-year collection of WWEIA to reflect the changing food supply and to address research needs from the data user community. Additional information about the AMPM is provided in Raper et al. (Raper et al., 2004).

The AMPM was validated in a large study and shown to be an effective method for collecting accurate group energy intake of adults. Completed in 2004, this extensive research project included 524 healthy, weight-stable volunteers, aged 30-69 years. The accuracy of the AMPM was evaluated by comparing reported energy intake (EI) to total energy expenditure (TEE) using the doubly labeled water technique (Moshfegh et al., 2008). Among the findings were that EI compared to TEE was under-reported by 11% overall, by less than 3% for normal weight subjects with body mass index (BMI) < 25 and 16% for overweight subjects with BMI ≥ 25.

Additional studies provide evidence that the AMPM accurately measures group energy intake. Blanton (Blanton et al., 2006) reported that EI was not significantly different from TEE for a sample of 20 adult females. Rumpler (Rumpler et al., 2008) found that mean EIs were accurately reported for a sample of 12 adult males.

Additional evidence for the accuracy of AMPM has been provided by analysis of the 24-hour urinary sodium data collected in the AMPM Validation Study, which suggest the AMPM is a valid measure for estimating mean sodium intake in adults. Dietary sodium intake calculated from 24-hour recall data of 465 subjects collected via AMPM was compared with sodium values from 24-hour urine collections measured during the same 24-hour period. The AMPM-derived mean dietary sodium estimates reflected over 90% of the biomarker-based estimates (Rhodes et al., 2013).

For additional information about the dietary interview component and related survey protocols, please visit the August 2021-August 2023 Dietary Interviewer Procedures Manual page. A description of changes to the NHANES August 2021-August 2023 survey is provided in the Plan and Operations of the National Heath And Nutrition Examination Survey, August 2021-August 2023.

Quality Assurance & Quality Control

All dietary interviewers were required to complete an intensive one-week training course and to conduct supervised practice interviews before working independently in the field. Retraining sessions were conducted annually to reinforce the proper protocols and technique.

Interviewers were monitored throughout the data collection period. Monitoring consisted of the following:

- Reviews of audio recorded interviews were conducted for approximately 5% of each interviewer's work.

- Quality control of interviews, which were checked for completeness of the recalls, missing information, inconsistent reports, and unclear notes. Written notification and feedback were provided to the interviewers.

Data Processing and Editing

Interview data files were sent electronically from the field and were imported into Survey Net, a computer-assisted food coding and data management system developed by USDA (Raper et al., 2004).

USDA's Food and Nutrient Database for Dietary Studies (FNDDS) 2021-2023 was used to process the intakes reported by the August 2021-August 2023 sample. The FNDDS includes comprehensive information that can be used to code individual foods/beverages and portion sizes reported by participants and includes nutrient values for calculating nutrient intakes. FNDDS nutrient values as well as food codes and portion sizes are updated for every 2-year WWEIA, NHANES release cycle. The basis for the nutrient values as well as food codes and portion weights in FNDDS are detailed in the documentation for FNDDS 2021-2023 available at http://www.ars.usda.gov/nea/bhnrc/fsrg.

Coders were required to pass a certification test after the initial training. They were routinely monitored to ensure the quality and completeness of their work. Approximately 10 percent of the coder's work was randomly selected to be independently coded by another coder. Results from the two codings were compared and adjudicated, if necessary.

After intake data were coded, various types of reviews and quality assurance procedures were conducted to ensure the quality of the data. Examples of reviews include the following:

- Interviewers' and coders' questions and comments were reviewed to ensure that they have been addressed.
- Decisions made by coders about how to code new or unusual foods/beverages or quantities reported by participants were reviewed for accuracy, reasonableness, and consistency across intake data; items in question were resolved by coder supervisors.
- Specific data integrity checks for reasonableness, consistency, and logic were conducted.

Analytic Notes

Each Individual Foods file (Day 1 and Day 2) is comprised of food records. For most participants, there are multiple records in each file. For each Total Nutrient Intakes file (Day 1 and Day 2) there is one record for each participant. These files can be linked with other NHANES files by the respondent sequence number (SEQN).

Variable names: For data collected on both Day 1 and Day 2, variable names are differentiated by having the number "1" or "2" in the third position of the variable name to identify the collection day. For example, the USDA food code variable (in the Individual Foods File), which identifies the food reported by the participant, is named DR1IFDCD in the Day 1 file and DR2IFDCD in the Day 2 file. Appendices **2** and **5** list the Day 1 and Day 2 variable names for the Individual Foods file and the Total Nutrient Intakes file, respectively.

Names for the following variables are the same for both days in the Individual Foods file and the Total Nutrient Intakes file:

Variables with the Same Name for Both Days in the Dietary Interview Files

Day 1 and Day 2 variable name	Label
SEQN	Respondent sequence number
WTDRD1	Dietary day one sample weight
WTDR2D	Dietary two-day sample weight
DRABF	Breast-fed infant (either day)
DRDINT	Number of days of intake

Number of days of intake: A variable has been included to indicate the number of days of intake collected from each participant. The variable name is DRDINT. In the August 2021-August 2023 sample, 6,754 participants provided complete dietary intakes for Day 1. The 6,754 participants with a complete Day 1 intake include 21 participants who had an unreliable day 1 intake but reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall. Of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2.

Dietary recall status code: A status code (DR1DRSTZ or DR2DRSTZ) is used in both the Individual Foods and Total Nutrient Intakes files to indicate the quality and completeness of a survey participant's response to the dietary recall section. The codes are the following:

1 = Reliable and met the following minimum criteria:

- The first 4 steps of the 5-step AMPM completed.
- Food/beverages consumed for each reported eating occasion identified.

For individuals with a code 1, all relevant variables associated with the 24-hour dietary recall contain a value.

2 = Not reliable or did not meet the minimum criteria

Individuals with a code 2 have incomplete records. No data on total nutrient intakes and the total number of foods reported are provided for these cases. These individuals have no records in the Individual Foods files.

3   [Code 3 is not included in the current datasets. It was only used for data from the 1999-2000 survey cycle.]

4 = Reported consuming breast milk

For infants and children who consumed human milk, there is a record in the Individual Foods files for each report of human milk. However, because amounts of human milk intake are not quantified, these records contain missing values for the amount consumed and for the amounts of energy and nutrients from human milk. Also, records of human milk have a missing value for the food source variable (DR1FS, DR2FS) and the eaten at home variable (DR1_040Z, DR2_040Z) in the Individual Foods files. Records for any other foods and beverages consumed by breast-fed infants and children are included in the Individual Foods files along with their amounts and nutrient information. Because of the missing amount or quantity information for human milk, no total nutrient intakes (contained in the Total Nutrient Intakes files) were computed for participants with a code 4.

A variable that identifies breast-fed children, DRABF, is included. This variable has a code of 1 if a child consumed human milk in either intake day.

5 = Not done

This code is assigned when the dietary recall section of the interview did not take place due to various reasons (such as refusal, equipment failure, or unable to contact the participant). These individuals have no records in the Individual Foods files. These individuals have a record in the Total Nutrients file with values only for the following variables: the respondent sequence number (SEQN), the dietary recall status code (DR1DRSTZ or DR2DRSTZ) and for participants 1 year or older, the fish and shellfish questions in the DR1TOT_L file (DRD340, DRD350A-K, DRD350AQ-JQ, DRD360, DRD370A-V, and DRD370AQ-UQ).

Only codes 1 and 4 appear in the Individual Foods file.

Distinguishing Between Foods/Beverages and Dietary Supplements in NHANES: In August 2021-August 2023, there was no 24-hour dietary supplement use collection. The 30-day dietary supplement use collection changed from being collected during the home interview to collection after the first 24-hour dietary recall via telephone. All NHANES participants responding to the 24-hour dietary recall interview are eligible for the dietary supplement and non-prescription antacid use questions. Information is obtained on all vitamins, minerals, herbals, and other dietary supplements as well as non-prescription antacids that were consumed during a 30-day period, including the name and the amount of supplement or antacid taken.

Distinguishing between foods/beverages and supplements can be challenging. NCHS and FSRG review questionable items reported in the dietary supplement and dietary recall components to resolve disposition of these items into the appropriate component. Products that are labeled as a dietary supplement, that have a supplement facts panel on the label, and are in tablets, capsules, softgels, gelcaps, or other pill forms, are considered dietary supplements. Items that are powders or liquids can be hard to distinguish. General guidelines used state that if powders and liquid concentrates have product directions stating that they be added to a liquid, they are classified as beverages. Examples are teas and protein powders. An exception is made for fiber products, which are classified as dietary supplements. Along this same guideline, energy drinks are considered beverages, but “energy shot” type products are considered dietary supplements.

It is best to refer to the databases that detail every food/beverage and dietary supplement reported in NHANES to identify exact determination used. The databases are:

- 2021-2023 Food and Nutrient Database for Dietary Studies
- NHANES Dietary Supplement Database

Participants who reported consuming only water, no food or other beverages: Records are included in the Individual Foods file for participants who consumed only water. There are 5 such individuals in the August 2021-August 2023 datasets, 4 in the Day 1 data and 1 in the Day 2 data. Their dietary recall status variable for the day is coded as "1" (complete and reliable) in the Total Nutrients file and the total number of items is the number of times water was reported. Individuals with just water intake and no food intake will have zero energy intake for the day.

Participants who reported consuming no water, food or other beverages: There can be participants whose intakes are determined to be complete even though they reported no water, food, or other beverage records for the day. For such participants there are no records in the Individual Foods file, but their dietary recall status is coded as complete and reliable, and the Total Nutrients file will include records with zero values for all nutrients. In the August 2021-August 2023 datasets, there are 3 individuals in the day 1 data that reported no water, food, or other beverage records for the day.

Number of days between the intake day and the day of family interview: Each of the four intake files includes a variable (DR1DBIH for Day 1 files and DR2DBIH for Day 2 files) to indicate the number of days between the intake day (i.e., the period covered by the 24-hour recall) and the day that the family questionnaire was administered in the household. A positive value in DR1BHIH or DR2BHIH indicates the family interview occurred prior to the intake day. In the survey, most of the family interviews were done before the participant participated in the dietary interview. A value of "0" in DR1BHIH or DR2BHIH indicates the family interview

occurred on the same date as the intake day. A negative value (i.e., DR1BHIH<0 or DR2BHIH<0) means that the family interview occurred after the intake day.

Food source: The source from which each food/beverage was obtained (e.g., from a store, fast food restaurant, cafeteria) is identified by the variables DR1FS (day 1) and DR2FS (day 2) in the Individual Foods files.

The code descriptions for this variable are:

Code Descriptions for Source of Food Variable

Code	Description
1	Store grocery/supermarket
2	Restaurant with waiter/waitress
3	Restaurant fast food/Pizza
4	Bar/Tavern/Lounge
5	Restaurant, no additional information
6	Cafeteria NOT in a K-12 school
7	Cafeteria in a K-12 school
8	Child/Adult care center
9	Child/Adult home care
10	Soup kitchen/shelter/food pantry facility
11	Meals on Wheels Program
12	Community food program – other
13	Community program, no additional info
14	Vending machine
15	Common coffee pot or snack tray
16	From someone else/gift
17	Mail order purchase
18	Residential dining facility
19	Grown or caught by you or someone you know
20	Fish caught by you or someone you know
24	Sport, recreation, or entertainment
25	Street vendor, vending truck
26	Fundraiser sales
27	Store - convenience type
28	Store - no additional information
91	Other, specify

Eating occasion: The variables DR1_030Z and DR2_030Z are located in the Individual Foods file. The code descriptions for the eating occasion variables are shown in the table below.

Code Descriptions for Eating Occasion Variable

Code	Description
1	Breakfast
2	Lunch
3	Dinner
4	Supper
5	Brunch
6	Snack
7	Beverage/Drink
8	Feeding-infant only
9	Extended consumption
10	Desayuno
11	Almuerzo
12	Comida
13	Merienda
14	Cena
15	Entre comida
16	Botana
17	Bocadillo
18	Tentempie
19	Bebida
91	Other

Eating occasion was designated by the respondent. During the interview, a list of eating occasion names was available to the respondent for selection. However, eating occasion names were not defined for the respondent.

Foods and beverages coded as part of a combination: 41 percent of foods and beverages reported in the WWEIA, NHANES August 2021-August 2023 sample were identified as items consumed together as combinations. Items consumed as a combination were identified by one of sixteen unique "combination food types." Foods and beverages not coded in combination have the code "0" for the combination food type variable.

The combination types provide a linkage for:

- Foods or beverages with additions, such as cereal with milk, coffee with cream;
- Multi-component foods that have specific protocol for collection such as some salads and sandwiches; and
- Other combinations that do not have a unique code in the FNDDS.

Combination Type, Code, Examples, and Percent of Food and Beverages Reported by Type, August 2021-August 2023, Day 1

Combination Type	Code	Examples of Combination Type	% Items
Not in combination	0	NA	59
Beverage w/ additions	1	Coffee, tea with: milk, cream, sugar.	9
Cereal w/ additions	2	Cereals (ready-to-eat, cooked) with: milk, sugar, fruit, butter.	4
Bread/baked product w/additions	3	Breads, rolls, pancakes with: butter, jam, syrup, fruit. Cakes, pies with: ice cream, toppings. Crackers with: cheese, dip, peanut butter.	5
Salad	4	Components of salads that do not have a single code in FNDDS. It may also designate additional items to single code salads.	5
Sandwiches	5	Components of sandwiches that do not have a single code in FNDDS. It may also designate additional items added to single code sandwiches.	6
Soup	6	Soup with: crackers, croutons, cheese.	<1
Frozen meals	7	Components of a prepackaged frozen meal and additions to the meal.	<1
Ice cream/ frozen yogurt w/ additions	8	Ice cream with: syrup, nuts, toppings.	<1
Dried beans or Vegetable w/ additions	9	French fries, potatoes with: catsup, gravy, butter, toppings. Beans with: sauce, butter.	3
Fruit w/ additions	10	Fruit with: toppings, milk, honey. Components of fruit mixtures or salads that do not have a single code in FNDDS.	1
Tortilla products	11	Components of tacos and tortilla products that do not have a single code in FNDDS. It may also designate additional items to single code tacos or tortilla products.	2
Meat, Poultry, Fish	12	Meat, poultry, fish with: gravy, sauce, and condiments.	2
Lunchables®	13	Components of pre-packaged lunch kits.	<1
Chips w/ additions	14	Potato chips, corn chips with: dip, cheese, salsa.	1
Baby Toddler Food and Infant Formula	15	Baby Toddler cereal with: infant formula, fruit. Infant formula with: baby cereal.	<1
Other mixtures	90	Rice, pasta, spaghetti, eggs, other mixtures with: butter, gravy, sauce, condiments.	4

All items given a combination food type are given an additional variable to identify each of the items within the combination. This variable is the "combination food number" that is unique to the combination food type within the individual intake.

Variable Labels and Names for Combination Coding

Combination Coding	Variable Name, Day 1	Variable Name, Day 2
Combination food type	DR1CCMTX	DR2CCMTX
Combination food number	DR1CCMNM	DR2CCMNM

The What We Eat in America Food Categories, available on the FSRG website (http://www.ars.usda.gov/nea/bhnrc/fsrg), is a grouping scheme that combines foods and beverages together that have similar usage and nutrient content with the emphasis on

how they are commonly consumed in the American diet. There are more than 170 unique categories, and each is assigned a 4-digit number and description. The WWEIA Food Categories contain discrete food items and are not disaggregated (e.g., pizza vs. grain, cheese, tomatoes, etc.). Designed to be flexible, the categories can be combined as needed to address specific research questions. A new version of the WWEIA Food Categories is produced for each 2-year release cycle of WWEIA, NHANES.

Special diet: Information on whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet, was provided. The variable DRQSDIET identifies whether a participant was on a special diet. The variables DRQSDT1 through DRQSDT12 and DRQSDT91 identify the type of diet or diets that the participant was following. These variables can be found in the Total Nutrient Intakes file.

Sample weights for dietary intake data: The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, **the appropriate sample weights must be used.**

For the NHANES August 2021-August 2023 sample, there were 22,660 persons selected; of these 8,860 were considered participants to the MEC examination and data collection. A total of 6,754 MEC participants provided complete dietary intakes for Day 1, and of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2. The 6,754 participants who provided complete Day 1 intakes includes 21 participants who had an unreliable day 1 intake and reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall.

Most analyses of NHANES data use data collected in the MEC and the variable WTMEC2YR should be used for the sample weights. However, for the WWEIA dietary data, different sample weights are recommended for analysis. Because food intake can vary by weekdays and weekends, use of the MEC weights disproportionately represents intakes on weekends.

A set of weights (WTDRD1) is provided that should be used when an analysis uses the Day 1 dietary recall data (either alone or when Day 1 nutrient data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with Day 1 data. Day 1 weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These Day 1 weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using Day 1 data and Day 1 weights might be larger than standard errors for similar estimates based on MEC weights.

When analysis is based on both days of dietary intake, only 5,879 sample participants have complete data. The NHANES protocol requires an attempt to collect the second day of dietary data at least 3 days after the first day, but the actual number of days between the two interviews is variable. A set of adjusted weights, WTDR2D, is to be used when an analysis uses the smaller sample with completed Day 1 and Day 2 dietary data. This two-day weight was constructed for the 5,879 participants by taking the Day 1 weights (WTDRD1) and further adjusting for: (a) the additional non-response for the second recall; and (b) for the proportion of weekend (Friday through Sunday) and weekday (Monday through Thursday) combinations of Day 1 and Day 2 recalls.

NOTE: All sample weights are person-level weights, and each set of dietary weights should sum to the same overall population control total as the MEC weights (WTMEC2YR). For the August 2021-August 2023 cycle, the Day 1 dietary recall weights (WTDRD1) are appropriate to use in the analysis of the fish and shellfish consumption data (i.e., variables DRD340, DRD350A-K, DRD350AQ-JQ DRD360, DRD370A-V, and DRD370AQ-UQ) located in the Day 1 Total Nutrient Intake File (DR1TOT_L), irrespective of whether other dietary data are included in the analysis. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

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Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

WTDRD1 - Dietary day one sample weight

Variable Name: WTDRD1

SAS Label: Dietary day one sample weight

English Text: Dietary day one sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5708.632624 to 408505.81364	Range of Values	6754	6754	
0	Day 1 dietary recall not done/incomplete	2106	8860	
.	Missing	0	8860	

WTDR2D - Dietary two-day sample weight

Variable Name: WTDR2D

SAS Label: Dietary two-day sample weight

English Text: Dietary two-day sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
6018.694738 to 759868.32478	Range of Values	5879	5879	
0	Day 2 dietary recall not done/incomplete	875	6754	
.	Missing	2106	8860	

DR2DRSTZ - Dietary recall status

Variable Name: DR2DRSTZ

SAS Label: Dietary recall status

English Text: Dietary recall status

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Reliable and met the minimum criteria	5830	5830	
2	Not reliable or not met the minimum criteria	23	5853	
4	Reported consuming breast-milk	49	5902	
5	Not done	2958	8860	
.	Missing	0	8860	

DR2EXMER - Interviewer ID code

Variable Name: DR2EXMER

SAS Label: Interviewer ID code

English Text: Interviewer ID code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 96	Range of Values	5902	5902	
.	Missing	2958	8860	

DRABF - Breast-fed infant (either day)

Variable Name: DRABF

SAS Label: Breast-fed infant (either day)

English Text: Indicates whether the sample person was an infant who was breast-fed on either of the two recall days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	60	60	
2	No	6674	6734	
.	Missing	2126	8860	

DRDINT - Number of days of intake

Variable Name: DRDINT

SAS Label: Number of days of intake

English Text: Indicates whether the sample person has intake data for one or two days.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Day 1 only	875	875	
2	Day 1 and day 2	5879	6754	
.	Missing	2106	8860	

DR2DBIH - # of days b/w intake and HH interview

Variable Name: DR2DBIH

SAS Label: # of days b/w intake and HH interview

English Text: Number of days between intake day and the day of family questionnaire administered in the household.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
-21 to 167	Range of Values	5550	5550	
.	Missing	3310	8860	

DR2DAY - Intake day of the week

Variable Name: DR2DAY

SAS Label: Intake day of the week

English Text: Intake day of the week

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Sunday	1117	1117	
2	Monday	910	2027	
3	Tuesday	833	2860	
4	Wednesday	1074	3934	
5	Thursday	985	4919	
6	Friday	459	5378	
7	Saturday	524	5902	
.	Missing	2958	8860	

DR2LANG - Language respondent used mostly

Variable Name: DR2LANG

SAS Label: Language respondent used mostly

English Text: The respondent spoke mostly:

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	English	5455	5455	
2	Spanish	460	5915	
3	English and Spanish	14	5929	
4	Other	0	5929	
.	Missing	2931	8860	

DR2MRESP - Main respondent for this interview

Variable Name: DR2MRESP

SAS Label: Main respondent for this interview

English Text: Who was the main respondent for this interview?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	SP	5015	5015	
2	Mother of SP	698	5713	
3	Father of SP	123	5836	
5	Spouse of SP	11	5847	
6	Child of SP	12	5859	
7	Grandparent of SP	13	5872	
8	Friend, partner, non-relative	0	5872	
9	Translator, not a HH member	0	5872	
10	Child care provider, caretaker	3	5875	
11	Other relative	4	5879	
77	Refused	0	5879	
99	Don't know	0	5879	
.	Missing	2981	8860	

DR2HELP - Helped in responding for this interview

Variable Name: DR2HELP

SAS Label: Helped in responding for this interview

English Text: Who helped in responding for this interview

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	SP	217	217	
4	Parent of SP	283	500	
5	Spouse of SP	86	586	
6	Child of SP	13	599	
7	Grandparent of SP	6	605	
8	Friend, partner, non-relative	5	610	
9	Translator, not a HH member	0	610	
10	Child care provider, caretaker	1	611	
11	Other relative	11	622	
12	No One	5237	5859	
77	Refused	0	5859	
99	Don't know	17	5876	
.	Missing	2984	8860	

DR2TNUMF - Number of foods/beverages reported

Variable Name: DR2TNUMF

SAS Label: Number of foods/beverages reported

English Text: Total number of foods/beverages reported in the individual foods file

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 40	Range of Values	5879	5879	
.	Missing	2981	8860	

DR2STY - Salt used at table yesterday?

Variable Name: DR2STY

SAS Label: Salt used at table yesterday?

English Text: Did {you/SP} add any salt to {your/her/his} food at the table yesterday? Salt includes ordinary or sea salt, lite salt, or a salt substitute.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	1074	1074	
2	No	4818	5892	DR2TKCAL
9	Don't know	10	5902	DR2TKCAL
.	Missing	2958	8860	

DR2SKY - Type of salt used yesterday

Variable Name: DR2SKY

SAS Label: Type of salt used yesterday

English Text: What type of salt was it? (Was it ordinary or sea salt, lite salt, or a salt substitute?)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Ordinary, sea, seasoned, or other flavored salt	1018	1018	
2	Lite salt	39	1057	
3	Salt substitute	15	1072	
91	Other	0	1072	
99	Don't know	2	1074	
.	Missing	7786	8860	

DR2TKCAL - Energy (kcal)

Variable Name: DR2TKCAL

SAS Label: Energy (kcal)

English Text: Energy (kcal)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 10233	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TPROT - Protein (gm)

Variable Name: DR2TPROT

SAS Label: Protein (gm)

English Text: Protein (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 337.77	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCARB - Carbohydrate (gm)

Variable Name: DR2TCARB

SAS Label: Carbohydrate (gm)

English Text: Carbohydrate (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1300.59	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TSUGR - Total sugars (gm)

Variable Name: DR2TSUGR

SAS Label: Total sugars (gm)

English Text: Total sugars (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 716.29	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TFIBE - Dietary fiber (gm)

Variable Name: DR2TFIBE

SAS Label: Dietary fiber (gm)

English Text: Dietary fiber (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 102.6	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TTFAT - Total fat (gm)

Variable Name: DR2TTFAT

SAS Label: Total fat (gm)

English Text: Total fat (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 545.82	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TSFAT - Total saturated fatty acids (gm)

Variable Name: DR2TSFAT

SAS Label: Total saturated fatty acids (gm)

English Text: Total saturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 227.727	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TMFAT - Total monounsaturated fatty acids (gm)

Variable Name: DR2TMFAT

SAS Label: Total monounsaturated fatty acids (gm)

English Text: Total monounsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 204.475	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TPFAT - Total polyunsaturated fatty acids (gm)

Variable Name: DR2TPFAT

SAS Label: Total polyunsaturated fatty acids (gm)

English Text: Total polyunsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 137.68	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCHOL - Cholesterol (mg)

Variable Name: DR2TCHOL

SAS Label: Cholesterol (mg)

English Text: Cholesterol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2843	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TATOC - Vitamin E as alpha-tocopherol (mg)

Variable Name: DR2TATOC

SAS Label: Vitamin E as alpha-tocopherol (mg)

English Text: Vitamin E as alpha-tocopherol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 127.68	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TATOA - Added alpha-tocopherol (Vitamin E) (mg)

Variable Name: DR2TATOA

SAS Label: Added alpha-tocopherol (Vitamin E) (mg)

English Text: Added alpha-tocopherol (Vitamin E) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 109.23	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TRET - Retinol (mcg)

Variable Name: DR2TRET

SAS Label: Retinol (mcg)

English Text: Retinol (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 28666	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVARA - Vitamin A, RAE (mcg)

Variable Name: DR2TVARA

SAS Label: Vitamin A, RAE (mcg)

English Text: Vitamin A as retinol activity equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 29436	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TACAR - Alpha-carotene (mcg)

Variable Name: DR2TACAR

SAS Label: Alpha-carotene (mcg)

English Text: Alpha-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 21552	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TBCAR - Beta-carotene (mcg)

Variable Name: DR2TBCAR

SAS Label: Beta-carotene (mcg)

English Text: Beta-carotene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 54134	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCRYP - Beta-cryptoxanthin (mcg)

Variable Name: DR2TCRYP

SAS Label: Beta-cryptoxanthin (mcg)

English Text: Beta-cryptoxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5374	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TLYCO - Lycopene (mcg)

Variable Name: DR2TLYCO

SAS Label: Lycopene (mcg)

English Text: Lycopene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 137752	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TLZ - Lutein + zeaxanthin (mcg)

Variable Name: DR2TLZ

SAS Label: Lutein + zeaxanthin (mcg)

English Text: Lutein + zeaxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 68086	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVB1 - Thiamin (Vitamin B1) (mg)

Variable Name: DR2TVB1

SAS Label: Thiamin (Vitamin B1) (mg)

English Text: Thiamin (Vitamin B1) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 12.846	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVB2 - Riboflavin (Vitamin B2) (mg)

Variable Name: DR2TVB2

SAS Label: Riboflavin (Vitamin B2) (mg)

English Text: Riboflavin (Vitamin B2) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 47.712	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TNIAC - Niacin (mg)

Variable Name: DR2TNIAC

SAS Label: Niacin (mg)

English Text: Niacin (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 299.163	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVB6 - Vitamin B6 (mg)

Variable Name: DR2TVB6

SAS Label: Vitamin B6 (mg)

English Text: Vitamin B6 (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 40.491	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TFOLA - Total folate (mcg)

Variable Name: DR2TFOLA

SAS Label: Total folate (mcg)

English Text: Total folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2294	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TFA - Folic acid (mcg)

Variable Name: DR2TFA

SAS Label: Folic acid (mcg)

English Text: Folic acid (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1697	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TFF - Food folate (mcg)

Variable Name: DR2TFF

SAS Label: Food folate (mcg)

English Text: Food folate (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1750	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TFDFE - Folate, DFE (mcg)

Variable Name: DR2TFDFE

SAS Label: Folate, DFE (mcg)

English Text: Folate as dietary folate equivalents (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3217	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCHL - Total choline (mg)

Variable Name: DR2TCHL

SAS Label: Total choline (mg)

English Text: Total choline (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2219.3	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVB12 - Vitamin B12 (mcg)

Variable Name: DR2TVB12

SAS Label: Vitamin B12 (mcg)

English Text: Vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 307.41	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TB12A - Added vitamin B12 (mcg)

Variable Name: DR2TB12A

SAS Label: Added vitamin B12 (mcg)

English Text: Added vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 112.2	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVC - Vitamin C (mg)

Variable Name: DR2TVC

SAS Label: Vitamin C (mg)

English Text: Vitamin C (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1055.5	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVD - Vitamin D (D2 + D3) (mcg)

Variable Name: DR2TVD

SAS Label: Vitamin D (D2 + D3) (mcg)

English Text: Vitamin D (D2 + D3) (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 63.4	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TVK - Vitamin K (mcg)

Variable Name: DR2TVK

SAS Label: Vitamin K (mcg)

English Text: Vitamin K (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3398.9	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCALC - Calcium (mg)

Variable Name: DR2TCALC

SAS Label: Calcium (mg)

English Text: Calcium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 8419	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TPHOS - Phosphorus (mg)

Variable Name: DR2TPHOS

SAS Label: Phosphorus (mg)

English Text: Phosphorus (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 13674	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TMAGN - Magnesium (mg)

Variable Name: DR2TMAGN

SAS Label: Magnesium (mg)

English Text: Magnesium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 1667	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TIRON - Iron (mg)

Variable Name: DR2TIRON

SAS Label: Iron (mg)

English Text: Iron (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 77.17	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TZINC - Zinc (mg)

Variable Name: DR2TZINC

SAS Label: Zinc (mg)

English Text: Zinc (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.04 to 67.76	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCOPP - Copper (mg)

Variable Name: DR2TCOPP

SAS Label: Copper (mg)

English Text: Copper (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.03 to 54.725	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TSODI - Sodium (mg)

Variable Name: DR2TSODI

SAS Label: Sodium (mg)

English Text: Sodium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 26274	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TPOTA - Potassium (mg)

Variable Name: DR2TPOTA

SAS Label: Potassium (mg)

English Text: Potassium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 16193	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TSELE - Selenium (mcg)

Variable Name: DR2TSELE

SAS Label: Selenium (mcg)

English Text: Selenium (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1035.1	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TCAFF - Caffeine (mg)

Variable Name: DR2TCAFF

SAS Label: Caffeine (mg)

English Text: Caffeine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1466	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TTHEO - Theobromine (mg)

Variable Name: DR2TTHEO

SAS Label: Theobromine (mg)

English Text: Theobromine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 832	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TALCO - Alcohol (gm)

Variable Name: DR2TALCO

SAS Label: Alcohol (gm)

English Text: Alcohol (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 329.5	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TMOIS - Moisture (gm)

Variable Name: DR2TMOIS

SAS Label: Moisture (gm)

English Text: Moisture (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
85.36 to 46341.78	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS040 - SFA 4:0 (Butanoic) (gm)

Variable Name: DR2TS040

SAS Label: SFA 4:0 (Butanoic) (gm)

English Text: SFA 4:0 (Butanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 9.468	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS060 - SFA 6:0 (Hexanoic) (gm)

Variable Name: DR2TS060

SAS Label: SFA 6:0 (Hexanoic) (gm)

English Text: SFA 6:0 (Hexanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5.694	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS080 - SFA 8:0 (Octanoic) (gm)

Variable Name: DR2TS080

SAS Label: SFA 8:0 (Octanoic) (gm)

English Text: SFA 8:0 (Octanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 13.867	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS100 - SFA 10:0 (Decanoic) (gm)

Variable Name: DR2TS100

SAS Label: SFA 10:0 (Decanoic) (gm)

English Text: SFA 10:0 (Decanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 8.408	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS120 - SFA 12:0 (Dodecanoic) (gm)

Variable Name: DR2TS120

SAS Label: SFA 12:0 (Dodecanoic) (gm)

English Text: SFA 12:0 (Dodecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 44.439	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS140 - SFA 14:0 (Tetradecanoic) (gm)

Variable Name: DR2TS140

SAS Label: SFA 14:0 (Tetradecanoic) (gm)

English Text: SFA 14:0 (Tetradecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 31.043	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS160 - SFA 16:0 (Hexadecanoic) (gm)

Variable Name: DR2TS160

SAS Label: SFA 16:0 (Hexadecanoic) (gm)

English Text: SFA 16:0 (Hexadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 107.752	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TS180 - SFA 18:0 (Octadecanoic) (gm)

Variable Name: DR2TS180

SAS Label: SFA 18:0 (Octadecanoic) (gm)

English Text: SFA 18:0 (Octadecanoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 45.847	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TM161 - MFA 16:1 (Hexadecenoic) (gm)

Variable Name: DR2TM161

SAS Label: MFA 16:1 (Hexadecenoic) (gm)

English Text: MFA 16:1 (Hexadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 9.44	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TM181 - MFA 18:1 (Octadecenoic) (gm)

Variable Name: DR2TM181

SAS Label: MFA 18:1 (Octadecenoic) (gm)

English Text: MFA 18:1 (Octadecenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 198.206	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TM201 - MFA 20:1 (Eicosenoic) (gm)

Variable Name: DR2TM201

SAS Label: MFA 20:1 (Eicosenoic) (gm)

English Text: MFA 20:1 (Eicosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4.701	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TM221 - MFA 22:1 (Docosenoic) (gm)

Variable Name: DR2TM221

SAS Label: MFA 22:1 (Docosenoic) (gm)

English Text: MFA 22:1 (Docosenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.081	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP182 - PFA 18:2 (Octadecadienoic) (gm)

Variable Name: DR2TP182

SAS Label: PFA 18:2 (Octadecadienoic) (gm)

English Text: PFA 18:2 (Octadecadienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 125.161	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP183 - PFA 18:3 (Octadecatrienoic) (gm)

Variable Name: DR2TP183

SAS Label: PFA 18:3 (Octadecatrienoic) (gm)

English Text: PFA 18:3 (Octadecatrienoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 18.276	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP184 - PFA 18:4 (Octadecatetraenoic) (gm)

Variable Name: DR2TP184

SAS Label: PFA 18:4 (Octadecatetraenoic) (gm)

English Text: PFA 18:4 (Octadecatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 0.612	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP204 - PFA 20:4 (Eicosatetraenoic) (gm)

Variable Name: DR2TP204

SAS Label: PFA 20:4 (Eicosatetraenoic) (gm)

English Text: PFA 20:4 (Eicosatetraenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.184	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP205 - PFA 20:5 (Eicosapentaenoic) (gm)

Variable Name: DR2TP205

SAS Label: PFA 20:5 (Eicosapentaenoic) (gm)

English Text: PFA 20:5 (Eicosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3.901	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP225 - PFA 22:5 (Docosapentaenoic) (gm)

Variable Name: DR2TP225

SAS Label: PFA 22:5 (Docosapentaenoic) (gm)

English Text: PFA 22:5 (Docosapentaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1.728	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2TP226 - PFA 22:6 (Docosahexaenoic) (gm)

Variable Name: DR2TP226

SAS Label: PFA 22:6 (Docosahexaenoic) (gm)

English Text: PFA 22:6 (Docosahexaenoic) (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 5.135	Range of Values	5830	5830	
.	Missing	3030	8860	

DR2_300 - Compare food consumed yesterday to usual

Variable Name: DR2_300

SAS Label: Compare food consumed yesterday to usual

English Text: Was the amount of food that {you/NAME} ate yesterday much more than usual, usual, or much less than usual?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Much more than usual	343	343	
2	Usual	4924	5267	
3	Much less than usual	620	5887	
7	Refused	0	5887	
9	Don't know	15	5902	
.	Missing	2958	8860	

DR2_320Z - Total plain water drank yesterday (gm)

Variable Name: DR2_320Z

SAS Label: Total plain water drank yesterday (gm)

English Text: Total plain water drank yesterday - including plain tap water, water from a drinking fountain, water from a water cooler, bottled water, and spring water.

English Instructions: Calculated from water consumption records reported as part of the 24-hour dietary recall interview.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 44160	Range of Values	5879	5879	
.	Missing	2981	8860	

DR2_330Z - Total tap water drank yesterday (gm)

Variable Name: DR2_330Z

SAS Label: Total tap water drank yesterday (gm)

English Text: Total tap water drank yesterday - including filtered tap water and water from a drinking fountain.

English Instructions: Calculated from tap water consumption records reported as part of the 24-hour dietary recall interview.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 17160	Range of Values	5879	5879	
.	Missing	2981	8860	

DR2BWATZ - Total bottled water drank yesterday (gm)

Variable Name: DR2BWATZ

SAS Label: Total bottled water drank yesterday (gm)

English Text: Total bottled water drank yesterday. (gm)

English Instructions: Calculated from bottle water consumption records reported as part of the 24-hour dietary recall interview.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 44160	Range of Values	5879	5879	
.	Missing	2981	8860	

DR2TWSZ - Tap water source

Variable Name: DR2TWSZ

SAS Label: Tap water source

English Text: When you drink tap water, what is the main source of the tap water?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Community supply	3932	3932	
4	Don't drink tap water	1151	5083	
91	Other	676	5759	
99	Don't know	143	5902	
.	Missing	2958	8860	

Appendix 1. Changes between WWEIA survey cycles 2013-2014 thru the August 2021-August 2023 sample

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Administration of 24-hour dietary recall	Same as 2003-2012: Day 1 in-person in MEC, Day 2 by telephone	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Day 1 by telephone, Day 2 by telephone
Language, Day 1	Same as 1999-2012: Interpreters assisted as needed	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Language, Day 2	Same as 2011-2012: English, Spanish, or Asian languages	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Person who did not provide reliable intake for Day 1 but did for Day 2	Same as 2003-2012: Neither intake was included in the release datasets	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	The single reliable intake was included in the release datasets as Day 1 intake (n=21)
Food source (where food was obtained)	Codes 8 and 9 revised descriptions.	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014
Combination food types	Same as 2003-2012: Values for 15 combination types	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	Values for 16 combination types; added "Baby Toddler Food and Infant Formula"
Number of intakes that include only water consumption for the day	6 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (all in Day 2 data), records are included in Individual Foods file.	2 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (1 in Day 1, 4 in Day 2 data), records are included in Individual Foods file.	5 intakes (4 in Day 1, 1 in Day 2), records are included in Individual Foods file.
Number of intakes that include no water or food consumption for the day	1 intake in Day 2 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	2 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.	3 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.
Tap water source	Same as 2003-2012: Reported in 6 categories.	Same as 2003-2004	Responses combined into 4 categories	Same as 2017-2018	Same as 2017-2018
Modification codes: DR1MC	All remaining modification	No modification	No modification	No modification	No modification

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Day 2 Modification codes: DR2MC Modification Code Description file: DRXMCD	codes deleted; new food codes addressing modifications added in FNDDS 2013-2014	codes	codes	codes	codes
Salt use at the table and in cooking or preparing foods and type	Same as 2003-2004	Same as 2003-2004	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Salt used at the table yesterday and type	Question asked about salt use at the table yesterday and kind of salt to coincide with 24-hour recall.	Same as 2013-2014	Question wording changed slightly by removing, "seasoned salt", and adding, "sea salt".	Same as 2017-2018	Same as 2017-2018
Main respondent and person whom helped in responding for the interview	Same as 2003-2012	Response category changed	Same as 2015-2016	Same as 2015-2016	Same as 2015-2016

Appendix 2. Variables in the Individual Foods Files (DR1IFF_L and DR2IFF_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1ILINE	DR2ILINE	Food/Individual component number
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1CCMNM	DR2CCMNM	Combination food number
DR1CCMTX	DR2CCMTX	Combination food type
DR1_020	DR2_020	Time of eating occasion (HH:MM)
DR1_030Z	DR2_030Z	Name of eating occasion
DR1FS	DR2FS	Source of food
DR1_040Z	DR2_040Z	Did you eat this meal at home?
DR1IFDCD	DR2IFDCD	USDA food code
DR1IGRMS	DR2IGRMS	Grams
DR1IKCAL	DR2IKCAL	Energy (kcal)
DR1IPROT	DR2IPROT	Protein (gm)
DR1ICARB	DR2ICARB	Carbohydrate (gm)
DR1ISUGR	DR2ISUGR	Total sugars (gm)
DR1IFIBE	DR2IFIBE	Dietary fiber (gm)
DR1ITFAT	DR2ITFAT	Total fat (gm)
DR1ISFAT	DR2ISFAT	Total saturated fatty acids (gm)
DR1IMFAT	DR2IMFAT	Total monounsaturated fatty acids (gm)
DR1IPFAT	DR2IPFAT	Total polyunsaturated fatty acids (gm)
DR1ICHOL	DR2ICHOL	Cholesterol (mg)
DR1IATOC	DR2IATOC	Vitamin E as alpha-tocopherol (mg)
DR1IATOA	DR2IATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1IRET	DR2IRET	Retinol (mcg)
DR1IVARA	DR2IVARA	Vitamin A, RAE (mcg)
DR1IACAR	DR2IACAR	Alpha-carotene (mcg)
DR1IBCAR	DR2IBCAR	Beta-carotene (mcg)
DR1ICRYP	DR2ICRYP	Beta-cryptoxanthin (mcg)
DR1ILYCO	DR2ILYCO	Lycopene (mcg)
DR1ILZ	DR2ILZ	Lutein + zeaxanthin (mcg)
DR1IVB1	DR2IVB1	Thiamin (Vitamin B1) (mg)

Day1 Name	Day2 Name	Variable Label
DR1IVB2	DR2IVB2	Riboflavin (Vitamin B2) (mg)
DR1INIAC	DR2INIAC	Niacin (mg)
DR1IVB6	DR2IVB6	Vitamin B6 (mg)
DR1IFOLA	DR2IFOLA	Total folate (mcg)
DR1IFA	DR2IFA	Folic acid (mcg)
DR1IFF	DR2IFF	Food folate (mcg)
DR1IFDFE	DR2IFDFE	Folate, DFE (mcg)
DR1ICHL	DR2ICHL	Total choline (mg)
DR1IVB12	DR2IVB12	Vitamin B12 (mcg)
DR1IB12A	DR2IB12A	Added vitamin B12 (mcg)
DR1IVC	DR2IVC	Vitamin C (mg)
DR1IVD	DR2IVD	Vitamin D (D2 + D3) (mcg)
DR1IVK	DR2IVK	Vitamin K (mcg)
DR1ICALC	DR2ICALC	Calcium (mg)
DR1IPHOS	DR2IPHOS	Phosphorus (mg)
DR1IMAGN	DR2IMAGN	Magnesium (mg)
DR1IIRON	DR2IIRON	Iron (mg)
DR1IZINC	DR2IZINC	Zinc (mg)
DR1ICOPP	DR2ICOPP	Copper (mg)
DR1ISODI	DR2ISODI	Sodium (mg)
DR1IPOTA	DR2IPOTA	Potassium (mg)
DR1ISELE	DR2ISELE	Selenium (mcg)
DR1ICAFF	DR2ICAFF	Caffeine (mg)
DR1ITHEO	DR2ITHEO	Theobromine (mg)
DR1IALCO	DR2IALCO	Alcohol (gm)
DR1IMOIS	DR2IMOIS	Moisture (gm)
DR1IS040	DR2IS040	SFA 4:0 (Butanoic) (gm)
DR1IS060	DR2IS060	SFA 6:0 (Hexanoic) (gm)
DR1IS080	DR2IS080	SFA 8:0 (Octanoic) (gm)
DR1IS100	DR2IS100	SFA 10:0 (Decanoic) (gm)
DR1IS120	DR2IS120	SFA 12:0 (Dodecanoic) (gm)
DR1IS140	DR2IS140	SFA 14:0 (Tetradecanoic) (gm)
DR1IS160	DR2IS160	SFA 16:0 (Hexadecanoic) (gm)
DR1IS180	DR2IS180	SFA 18:0 (Octadecanoic) (gm)
DR1IM161	DR2IM161	MFA 16:1 (Hexadecenoic) (gm)
DR1IM181	DR2IM181	MFA 18:1 (Octadecenoic) (gm)
DR1IM201	DR2IM201	MFA 20:1 (Eicosenoic) (gm)
DR1IM221	DR2IM221	MFA 22:1 (Docosenoic) (gm)
DR1IP182	DR2IP182	PFA 18:2 (Octadecadienoic) (gm)

Day1 Name	Day2 Name	Variable Label
DR1IP183	DR2IP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1IP184	DR2IP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1IP204	DR2IP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1IP205	DR2IP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1IP225	DR2IP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1IP226	DR2IP226	PFA 22:6 (Docosahexaenoic) (gm)

Appendix 3. List of Nutrients/Food Components (Unit)

Energy and Macronutrients

Food energy (kcal)
Protein (gm)
Carbohydrate (gm)
Fat, total (gm)
Alcohol (gm)

Sugars, total (gm)
Dietary fiber, total (gm)
Water (moisture) (gm)*

Saturated fatty acids, total (gm)
Monounsaturated fatty acids, total (gm)
Polyunsaturated fatty acids, total (gm)
Cholesterol (mg)

Individual fatty acids:
- 4:0 (gm)
- 6:0 (gm)
- 8:0 (gm)
- 10:0 (gm)
- 12:0 (gm)
- 14:0 (gm)
- 16:0 (gm)
- 18:0 (gm)
- 16:1 (gm)
- 18:1 (gm)
- 20:1 (gm)
- 22:1 (gm)
- 18:2 (gm)
- 18:3 (gm)
- 18:4 (gm)
- 20:4 (gm)
- 20:5 n-3 (gm)
- 22:5 n-3 (gm)
- 22:6 n-3 (gm)

Vitamins, Minerals, and Other Components

Vitamin A as retinol activity equivalents (mcg)
Retinol (mcg)
Carotenoids:
- Carotene, alpha (mcg)
- Carotene, beta (mcg)
- Cryptoxanthin, beta (mcg)
- Lycopene (mcg)
- Lutein + zeaxanthin (mcg)

Vitamin E as alpha-tocopherol (mg)
- **Added vitamin E as alpha-tocopherol (mg)

Vitamin D (D2 + D3) (mcg)
Vitamin K as phylloquinone (mcg)
Vitamin C (mg)
Thiamin (mg)
Riboflavin (mg)
Niacin (mg)
Vitamin B-6 (mg)
Folate, total (mcg)
- Folate as dietary folate equivalents (mcg)

Folic acid (mcg)
Food folate (mcg)
Choline, total (mg)
Vitamin B-12 (mcg)
***Added vitamin B-12 (mcg)

Calcium (mg)
Iron (mg)
Magnesium (mg)
Phosphorus (mg)
Potassium (mg)
Sodium (mg)
Zinc (mg)
Copper (mg)
Selenium (mcg)
Caffeine (mg)
Theobromine (mg)

** Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)*

*** Represents a synthetic subcomponent of vitamin E and is included in the vitamin E value.*

**** Represents a fortified subcomponent of vitamin B12 and is included in the vitamin B12 value.*

Appendix 4. Adding Food Code Descriptions to Your Files

One supporting file is included with the Individual Foods files: the Food Code Description file (DRXFCD_L).

The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code included in the Individual Foods files.

The Food Code Description file (DRXFCD_L) contains three variables:

DRXFDCD a numeric value corresponding to DR1IFDCD in the file DR1IFF_L or DR2IFDCD in the file DR2IFF_L;

DRXFCSD a short description (up to 60 characters) of the food code;

DRXFDLD a long description (up to 200 characters) of the food code.

The following SQL code is an example of appending the shorter food code description (here renamed DR1IFCSD) to one of the Individual Foods files using PROC SQL from SAS®. Other SQL implementations may be different.

```
proc sql;
  create table DR1IFF_L_PLUS as
  select iff.*, desc.DRXFCSD as DR1FCSD, desc.DRXFCLD as DR1FCLD
  from NHANES.DR1IFF_L iff
  left join NHANES.DRXFCD_L desc
  on iff.DR1IFDCD = desc.DRXFDCD
  order by SEQN, DR1ILINE;
quit;
```

SAS® users may wish to use Proc Format to assign labels to the food codes. The following example generates and saves a picture format for food codes and a separate format for each food code that includes both the food code itself and the short food code description. It is assumed that the user has stored the Individual Foods files and the Food Code Description file in a library called NHANES and wishes to store the formats there as well.

```
options fmtsearch = (NHANES);

proc format library = library;
  picture foodcode
  low - high = '000-00000';
quit;

data tmp;
  set NHANES.DRXFCD_L;
  length cfoodcode $9 label $72;
  cfoodcode = put(DRXFDCD, foodcode.);
  label = cfoodcode || '  ' || DRXFCSD;
run;

data fmt (keep = fmtname start label);
  set tmp;
  retain fmtname 'DRXFDCD';
  rename DRXFDCD = start;
run;
```

```
proc format cntlin = fmt library = library;
run;
```

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Appendix 5. Variables in the Total Nutrients Files (DR1TOT_L and DR2TOT_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1MRESP	DR2MRESP	Main respondent for this interview
DR1HELP	DR2HELP	Helped in responding for this interview
DBQ095Z	N/A	Type of table salt used
DBD100	N/A	How often add salt to food at table
DRQSPREP	N/A	Salt used in preparation?
DR1STY	DR2STY	Salt used at table yesterday?
DR1SKY	DR2SKY	Type of salt used yesterday
DRQSDIET	N/A	On special diet?
DRQSDT1	N/A	Weight loss/Low calorie diet
DRQSDT2	N/A	Low fat/Low cholesterol diet
DRQSDT3	N/A	Low salt/Low sodium diet
DRQSDT4	N/A	Sugar free/Low sugar diet
DRQSDT5	N/A	Low fiber diet
DRQSDT6	N/A	High fiber diet
DRQSDT7	N/A	Diabetic diet
DRQSDT8	N/A	Weight gain/Muscle building diet
DRQSDT9	N/A	Low carbohydrate diet
DRQSDT10	N/A	High protein diet
DRQSDT11	N/A	Gluten-free/Celiac diet
DRQSDT12	N/A	Renal/Kidney diet
DRQSDT91	N/A	Other special diet
DR1TNUMF	DR2TNUMF	Number of foods/beverages reported
DR1TKCAL	DR2TKCAL	Energy (kcal)
DR1TPROT	DR2TPROT	Protein (gm)
DR1TCARB	DR2TCARB	Carbohydrate (gm)
DR1TSUGR	DR2TSUGR	Total sugars (gm)
DR1TFIBE	DR2TFIBE	Dietary fiber (gm)
DR1TTFAT	DR2TTFAT	Total fat (gm)
DR1TSFAT	DR2TSFAT	Total saturated fatty acids (gm)

Day1 Name	Day2 Name	Variable Label
DR1TMFAT	DR2TMFAT	Total monounsaturated fatty acids (gm)
DR1TPFAT	DR2TPFAT	Total polyunsaturated fatty acids (gm)
DR1TCHOL	DR2TCHOL	Cholesterol (mg)
DR1TATOC	DR2TATOC	Vitamin E as alpha-tocopherol (mg)
DR1TATOA	DR2TATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1TRET	DR2TRET	Retinol (mcg)
DR1TVARA	DR2TVARA	Vitamin A, RAE (mcg)
DR1TACAR	DR2TACAR	Alpha-carotene (mcg)
DR1TBCAR	DR2TBCAR	Beta-carotene (mcg)
DR1TCRYP	DR2TCRYP	Beta-cryptoxanthin (mcg)
DR1TLYCO	DR2TLYCO	Lycopene (mcg)
DR1TLZ	DR2TLZ	Lutein + zeaxanthin (mcg)
DR1TVB1	DR2TVB1	Thiamin (Vitamin B1) (mg)
DR1TVB2	DR2TVB2	Riboflavin (Vitamin B2) (mg)
DR1TNIAC	DR2TNIAC	Niacin (mg)
DR1TVB6	DR2TVB6	Vitamin B6 (mg)
DR1TFOLA	DR2TFOLA	Total folate (mcg)
DR1TFA	DR2TFA	Folic acid (mcg)
DR1TFF	DR2TFF	Food folate (mcg)
DR1TFDFE	DR2TFDFE	Folate, DFE (mcg)
DR1TCHL	DR2TCHL	Total choline (mg)
DR1TVB12	DR2TVB12	Vitamin B12 (mcg)
DR1TB12A	DR2TB12A	Added vitamin B12 (mcg)
DR1TVC	DR2TVC	Vitamin C (mg)
DR1TVD	DR2TVD	Vitamin D (D2 + D3) (mcg)
DR1TVK	DR2TVK	Vitamin K (mcg)
DR1TCALC	DR2TCALC	Calcium (mg)
DR1TPHOS	DR2TPHOS	Phosphorus (mg)
DR1TMAGN	DR2TMAGN	Magnesium (mg)
DR1TIRON	DR2TIRON	Iron (mg)
DR1TZINC	DR2TZINC	Zinc (mg)
DR1TCOPP	DR2TCOPP	Copper (mg)
DR1TSODI	DR2TSODI	Sodium (mg)
DR1TPOTA	DR2TPOTA	Potassium (mg)
DR1TSELE	DR2TSELE	Selenium (mcg)
DR1TCAFF	DR2TCAFF	Caffeine (mg)
DR1TTHEO	DR2TTHEO	Theobromine (mg)
DR1TALCO	DR2TALCO	Alcohol (gm)
DR1TMOIS	DR2TMOIS	Moisture (gm)

Day1 Name	Day2 Name	Variable Label
DR1TS040	DR2TS040	SFA 4:0 (Butanoic) (gm)
DR1TS060	DR2TS060	SFA 6:0 (Hexanoic) (gm)
DR1TS080	DR2TS080	SFA 8:0 (Octanoic) (gm)
DR1TS100	DR2TS100	SFA 10:0 (Decanoic) (gm)
DR1TS120	DR2TS120	SFA 12:0 (Dodecanoic) (gm)
DR1TS140	DR2TS140	SFA 14:0 (Tetradecanoic) (gm)
DR1TS160	DR2TS160	SFA 16:0 (Hexadecanoic) (gm)
DR1TS180	DR2TS180	SFA 18:0 (Octadecanoic) (gm)
DR1TM161	DR2TM161	MFA 16:1 (Hexadecenoic) (gm)
DR1TM181	DR2TM181	MFA 18:1 (Octadecenoic) (gm)
DR1TM201	DR2TM201	MFA 20:1 (Eicosenoic) (gm)
DR1TM221	DR2TM221	MFA 22:1 (Docosenoic) (gm)
DR1TP182	DR2TP182	PFA 18:2 (Octadecadienoic) (gm)
DR1TP183	DR2TP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1TP184	DR2TP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1TP204	DR2TP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1TP205	DR2TP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1TP225	DR2TP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1TP226	DR2TP226	PFA 22:6 (Docosahexaenoic) (gm)
DR1_300	DR2_300	Compare food consumed yesterday to usual
DR1_320Z	DR2_320Z	Total plain water drank yesterday (gm)
DR1_330Z	DR2_330Z	Total tap water drank yesterday (gm)
DR1BWATZ	DR2BWATZ	Total bottled water drank yesterday (gm)
DR1TWSZ	DR2TWSZ	Tap water source
DRD340	N/A	Shellfish eaten during past 30 days
DRD350A	N/A	Clams eaten during past 30 days
DRD350AQ	N/A	# of times clams eaten in past 30 days
DRD350B	N/A	Crabs eaten during past 30 days
DRD350BQ	N/A	# of times crabs eaten in past 30 days
DRD350C	N/A	Crayfish eaten during past 30 days
DRD350CQ	N/A	# of times crayfish eaten past 30 days
DRD350D	N/A	Lobsters eaten during past 30 days
DRD350DQ	N/A	# of times lobsters eaten past 30 days
DRD350E	N/A	Mussels eaten during past 30 days
DRD350EQ	N/A	# of times mussels eaten in past 30 days
DRD350F	N/A	Oysters eaten during past 30 days
DRD350FQ	N/A	# of times oysters eaten in past 30 days
DRD350G	N/A	Scallops eaten during past 30 days
DRD350GQ	N/A	# of times scallops eaten past 30 days

Day1 Name	Day2 Name	Variable Label
DRD350H	N/A	Shrimp eaten during past 30 days
DRD350HQ	N/A	# of times shrimp eaten in past 30 days
DRD350I	N/A	Other shellfish eaten past 30 days
DRD350IQ	N/A	# of times other shellfish eaten
DRD350J	N/A	Other unknown shellfish eaten past 30 d
DRD350JQ	N/A	# of times other unknown shellfish eaten
DRD350K	N/A	Refused on shellfish eaten past 30 days
DRD360	N/A	Fish eaten during past 30 days
DRD370A	N/A	Breaded fish products eaten past 30 days
DRD370AQ	N/A	# of times breaded fish products eaten
DRD370B	N/A	Tuna eaten during past 30 days
DRD370BQ	N/A	# of times tuna eaten in past 30 days
DRD370C	N/A	Bass eaten during past 30 days
DRD370CQ	N/A	# of times bass eaten in past 30 days
DRD370D	N/A	Catfish eaten during past 30 days
DRD370DQ	N/A	# of times catfish eaten in past 30 days
DRD370E	N/A	Cod eaten during past 30 days
DRD370EQ	N/A	# of times cod eaten in past 30 days
DRD370F	N/A	Flatfish eaten during past 30 days
DRD370FQ	N/A	# of times flatfish eaten past 30 days
DRD370G	N/A	Haddock eaten during past 30 days
DRD370GQ	N/A	# of times haddock eaten in past 30 days
DRD370H	N/A	Mackerel eaten during past 30 days
DRD370HQ	N/A	# of times mackerel eaten past 30 days
DRD370I	N/A	Perch eaten during past 30 days
DRD370IQ	N/A	# of times perch eaten in past 30 days
DRD370J	N/A	Pike eaten during past 30 days
DRD370JQ	N/A	# of times pike eaten in past 30 days
DRD370K	N/A	Pollock eaten during past 30 days
DRD370KQ	N/A	# of times pollock eaten in past 30 days
DRD370L	N/A	Porgy eaten during past 30 days
DRD370LQ	N/A	# of times porgy eaten in past 30 days
DRD370M	N/A	Salmon eaten during past 30 days
DRD370MQ	N/A	# of times salmon eaten in past 30 days
DRD370N	N/A	Sardines eaten during past 30 days
DRD370NQ	N/A	# of times sardines eaten past 30 days
DRD370O	N/A	Sea bass eaten during past 30 days
DRD370OQ	N/A	# of times sea bass eaten past 30 days
DRD370P	N/A	Shark eaten during past 30 days

Day1 Name	Day2 Name	Variable Label
DRD370PQ	N/A	# of times shark eaten in past 30 days
DRD370Q	N/A	Swordfish eaten during past 30 days
DRD370QQ	N/A	# of times swordfish eaten past 30 days
DRD370R	N/A	Trout eaten during past 30 days
DRD370RQ	N/A	# of times trout eaten in past 30 days
DRD370S	N/A	Walleye eaten during past 30 days
DRD370SQ	N/A	# of times walleye eaten in past 30 days
DRD370T	N/A	Other fish eaten during past 30 days
DRD370TQ	N/A	# of times other fish eaten past 30 days
DRD370U	N/A	Other unknown fish eaten in past 30 days
DRD370UQ	N/A	# of times other unknown fish eaten
DRD370V	N/A	Refused on fish eaten past 30 days

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Interview Technical Support File - Food Codes (DRXFCD_L)

Data File: DRXFCD_L.xpt

First Published: October 2024

Last Revised: NA

Component Description

The objective of the dietary interview component is to obtain detailed dietary intake information from NHANES participants. The dietary intake data are used to estimate the types and amounts of foods and beverages (including all types of water) consumed during the 24-hour period prior to the interview (midnight to midnight), and to estimate intakes of energy, nutrients, and other food components from those foods and beverages. Following the dietary recall, participants are asked questions on salt use, whether the person's overall intake on the previous day was much more than usual, usual or much less than usual, and whether the participant is on any type of special diet. Questions on frequency of fish and shellfish consumed during the past 30 days are asked of participants 1 year or older, with the use of proxies for young children (see the Dietary Interview Procedure Manuals (cdc.gov) for more information on the proxy interview).

The dietary interview component, called What We Eat in America (WWEIA), is conducted as a partnership between the U.S. Department of Agriculture (USDA) and the U.S. Department of Health and Human Services (DHHS). Under this partnership, DHHS' National Center for Health Statistics (NCHS), Division of Health and Nutrition Examination Surveys is responsible for the survey sample design and all aspects of data collection and USDA's Food Surveys Research Group (FSRG) is responsible for the dietary data collection methodology, maintenance of the databases used to code and process the data, and data review and processing.

All NHANES participants are eligible for two 24-hour dietary recall interviews. Traditionally, the first dietary recall interview was collected in-person in the Mobile Examination Center (MEC) and the second interview was collected by telephone 3 to 10 days later. In August 2021-August 2023, the time participants spent in the MEC was limited to reduce risk of exposure to SARS-CoV-2. As a result, both dietary recall interviews were administered via telephone. Further information about changes in data collection are described in the Plan and Operations report of the NHANES August 2021-August 2023.

As in previous years, two types of dietary intake data are available for the August 2021-August 2023 survey cycle: Individual Foods files and Total Nutrient Intakes files.

What's New with the August 2021-August 2023 WWEIA Release:

In response to the COVID-19 pandemic, the mode of the first dietary recall changed from in-person during the MEC examination to telephone 3 to 7 days after the examination. Both dietary interviews were completed in English or Spanish. During the mobile examination, participants scheduled an appointment for the first telephone dietary interview and received a set of measuring guides to help estimate the amounts of foods and beverages consumed. **Appendix 1** provides a summary of changes among the 5 latest cycles of data collection. No new data variables were added in the August 2021-August 2023 WWEIA release.

Dietary Interview Data Files: Four data files were produced from the information collected in the dietary interviews: two Individual Foods files and two Total Nutrient Intakes files. Each

file includes one day of intake data. The number "1" or "2" in the file name identifies the day of the interview: 1 = first day, 2 = second day. File names are as follows:

File Names for Dietary Interview Data:

File	Day 1	Day 2
Individual Foods File	DR1IFF_L	DR2IFF_L
Total Nutrient Intakes File	DR1TOT_L	DR2TOT_L

The amounts in these files reflect only nutrients obtained from foods, beverages, and water, including tap and bottled water. They do not include nutrients obtained from dietary supplement intakes, antacids, or medications. Data on intake of dietary supplement use are available on the August 2021-August 2023 Dietary Data (cdc.gov) page.

Individual Foods Files (DR1IFF_L and DR2IFF_L): Detailed information about each food/beverage item (including the description, amount of, and nutrient content) reported by each participant is included in the Individual Foods files. The names for both Day 1 and Day 2 variables are listed in **Appendix 2**.

The Individual Foods files include, for each interview day, one record for each food/beverage consumed by a participant. Each record is uniquely numbered within a participant's set of records and contains the information listed below:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Whether the food/beverage was eaten in combination with other foods, such as in a sandwich;
- Time of eating occasion/when the food was eaten;
- Eating occasion name;
- Where the food/beverage was obtained;
- Whether the meal/snack was eaten at home or not;
- A USDA Food and Nutrient Database for Dietary Studies (FNDDS) code identifying the food/beverage;
- Amount of food/beverage consumed, in grams; and
- Food energy and 64 nutrients/food components (listed in **Appendix 3**) from each food/beverage as calculated using USDA's Food and Nutrient Database for Dietary Studies 2021-2023 (FNDDS 2021-2023).

Descriptions for the USDA FNDDS food codes are provided in the Food Code Description file (DRXFCD_L). The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code in the FNDDS 2021-2023. **Appendix 4** provides SAS code examples that may be used to link the food code description to the Individual Foods file.

Total Nutrient Intakes Files (DR1TOT_L and DR2TOT_L): For each participant, daily total energy and nutrient intakes from foods and beverages, and whether the amount of food consumed was usual, much more than usual, or much less than usual, are included in the Total Nutrient Intakes files. The Day 1 file also includes information on salt use in cooking and at the table; whether the participant is currently on any kind of diet to lose weight or for another health-related reason and, if so, the type of diet; and information on frequency of fish and shellfish consumption for participants aged 1 or older. The names for both Day 1 and Day 2 variables are listed in **Appendix 5**.

The Total Nutrient Intakes files provide a summary record of total nutrient intakes for each participant. Each total intake record contains the following information:

- Number of days of complete intake obtained from participant;
- Day of the week of the intake;
- Type of salt used and how often added at the table and in food preparation (Day 1 file only);
- Use of salt at the table yesterday and the type of salt used;
- Whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet (Day 1 file only);
- Total number of foods and beverages including water reported for that participant for that day's intake;
- Daily aggregates of food energy and 64 nutrients/food components (listed in **Appendix 3**) from all foods/beverages as calculated using USDA's Food and Nutrient Database for Dietary Studies August 2021-August 2023 (FNDDS 2021- 2023)
- Whether the amount of food consumed was usual, much more than usual, or much less than usual;
- Total amount of tap and bottled water consumed (calculated as the sum of reports of water drunk by itself in the 24-hour recall) and the usual source of tap water; and
- Frequency of fish and shellfish consumption in the past 30 days (participants 1 year or older, Day 1 file only).

Eligible Sample

All participants in the August 2021-August 2023 sample who spoke English or Spanish were eligible. Participants aged 1 year or older were eligible for the frequency of fish and shellfish consumption questions following the 24-hour recall.

Protocol and Procedure

The examination protocol and data collection methods are fully documented in the NHANES dietary interviewer procedures manuals.

Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant’s intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child’s intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish.

During the MEC visit, participants were scheduled an appointment for the first dietary interview and were given a set of measuring guides (NHANES Measuring Guides for the Dietary Recall Interview) to use for reporting food amounts during the telephone interview. The measuring guides included measuring cups, spoons, a ruler, and a food model booklet, which contained two-dimensional drawings of glasses, mugs, bowls, mounds, circles, wedges, a shape chart and a chicken chart. The first telephone dietary interview was scheduled 3 to 7 days post MEC examination while the second interview was collected 3 to 10 days following the first dietary interview and were generally scheduled on a different day of the week as the first interview. In August 2021-August 2023, 204 participants were interviewed on the same day of the week for both day 1 and day 2 interviews due to their scheduling availability. Any participant who did not have a telephone was given a toll-free number to call so that the recall could be conducted.

What We Eat in America data were collected using USDA's dietary data collection instrument, the Automated Multiple Pass Method (AMPM), available at:

http://www.ars.usda.gov/nea/bhnrc/fsrg. The AMPM was designed to provide an efficient and accurate means of collecting intakes for large-scale national surveys. The AMPM is a fully computerized recall method that uses a 5-step interview outlined below:

1. **Quick List** - Participant recalls all foods and beverages consumed the day before the interview (midnight to midnight).
2. **Forgotten Foods** - Participant is asked about consumption of foods commonly forgotten during the Quick List step.
3. **Time and Occasion** - Time and eating occasion are collected for each food.
4. **Detail Cycle** - For each food, a detailed description, amount eaten, and additions to the food are collected. Eating occasions and times between eating occasions are reviewed to elicit forgotten foods.
5. **Final Probe** - Additional foods not remembered earlier are collected.

The AMPM includes an extensive compilation of standardized food-specific questions and possible response options. Routing of questions is based on previous responses. The AMPM is updated for each 2-year collection of WWEIA to reflect the changing food supply and to address research needs from the data user community. Additional information about the AMPM is provided in Raper et al. (Raper et al., 2004).

The AMPM was validated in a large study and shown to be an effective method for collecting accurate group energy intake of adults. Completed in 2004, this extensive research project included 524 healthy, weight-stable volunteers, aged 30-69 years. The accuracy of the AMPM was evaluated by comparing reported energy intake (EI) to total energy expenditure (TEE) using the doubly labeled water technique (Moshfegh et al., 2008). Among the findings were that EI compared to TEE was under-reported by 11% overall, by less than 3% for normal weight subjects with body mass index (BMI) < 25 and 16% for overweight subjects with BMI ≥ 25.

Additional studies provide evidence that the AMPM accurately measures group energy intake. Blanton (Blanton et al., 2006) reported that EI was not significantly different from TEE for a sample of 20 adult females. Rumpler (Rumpler et al., 2008) found that mean EIs were accurately reported for a sample of 12 adult males.

Additional evidence for the accuracy of AMPM has been provided by analysis of the 24-hour urinary sodium data collected in the AMPM Validation Study, which suggest the AMPM is a valid measure for estimating mean sodium intake in adults. Dietary sodium intake calculated from 24-hour recall data of 465 subjects collected via AMPM was compared with sodium values from 24-hour urine collections measured during the same 24-hour period. The AMPM-derived mean dietary sodium estimates reflected over 90% of the biomarker-based estimates (Rhodes et al., 2013).

For additional information about the dietary interview component and related survey protocols, please visit the August 2021-August 2023 Dietary Interviewer Procedures Manual page. A description of changes to the NHANES August 2021-August 2023 survey is provided in the Plan and Operations of the National Heath And Nutrition Examination Survey, August 2021-August 2023.

Quality Assurance & Quality Control

All dietary interviewers were required to complete an intensive one-week training course and to conduct supervised practice interviews before working independently in the field. Retraining sessions were conducted annually to reinforce the proper protocols and technique.

Interviewers were monitored throughout the data collection period. Monitoring consisted of the following:

- Reviews of audio recorded interviews were conducted for approximately 5% of each interviewer's work.

- Quality control of interviews, which were checked for completeness of the recalls, missing information, inconsistent reports, and unclear notes. Written notification and feedback were provided to the interviewers.

Data Processing and Editing

Interview data files were sent electronically from the field and were imported into Survey Net, a computer-assisted food coding and data management system developed by USDA (Raper et al., 2004).

USDA's Food and Nutrient Database for Dietary Studies (FNDDS) 2021-2023 was used to process the intakes reported by the August 2021-August 2023 sample. The FNDDS includes comprehensive information that can be used to code individual foods/beverages and portion sizes reported by participants and includes nutrient values for calculating nutrient intakes. FNDDS nutrient values as well as food codes and portion sizes are updated for every 2-year WWEIA, NHANES release cycle. The basis for the nutrient values as well as food codes and portion weights in FNDDS are detailed in the documentation for FNDDS 2021-2023 available at http://www.ars.usda.gov/nea/bhnrc/fsrg.

Coders were required to pass a certification test after the initial training. They were routinely monitored to ensure the quality and completeness of their work. Approximately 10 percent of the coder's work was randomly selected to be independently coded by another coder. Results from the two codings were compared and adjudicated, if necessary.

After intake data were coded, various types of reviews and quality assurance procedures were conducted to ensure the quality of the data. Examples of reviews include the following:

- Interviewers' and coders' questions and comments were reviewed to ensure that they have been addressed.
- Decisions made by coders about how to code new or unusual foods/beverages or quantities reported by participants were reviewed for accuracy, reasonableness, and consistency across intake data; items in question were resolved by coder supervisors.
- Specific data integrity checks for reasonableness, consistency, and logic were conducted.

Analytic Notes

Each Individual Foods file (Day 1 and Day 2) is comprised of food records. For most participants, there are multiple records in each file. For each Total Nutrient Intakes file (Day 1 and Day 2) there is one record for each participant. These files can be linked with other NHANES files by the respondent sequence number (SEQN).

Variable names: For data collected on both Day 1 and Day 2, variable names are differentiated by having the number "1" or "2" in the third position of the variable name to identify the collection day. For example, the USDA food code variable (in the Individual Foods File), which identifies the food reported by the participant, is named DR1IFDCD in the Day 1 file and DR2IFDCD in the Day 2 file. Appendices **2** and **5** list the Day 1 and Day 2 variable names for the Individual Foods file and the Total Nutrient Intakes file, respectively.

Names for the following variables are the same for both days in the Individual Foods file and the Total Nutrient Intakes file:

Variables with the Same Name for Both Days in the Dietary Interview Files

Day 1 and Day 2 variable name	Label
SEQN	Respondent sequence number
WTDRD1	Dietary day one sample weight
WTDR2D	Dietary two-day sample weight
DRABF	Breast-fed infant (either day)
DRDINT	Number of days of intake

Number of days of intake: A variable has been included to indicate the number of days of intake collected from each participant. The variable name is DRDINT. In the August 2021-August 2023 sample, 6,754 participants provided complete dietary intakes for Day 1. The 6,754 participants with a complete Day 1 intake include 21 participants who had an unreliable day 1 intake but reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall. Of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2.

Dietary recall status code: A status code (DR1DRSTZ or DR2DRSTZ) is used in both the Individual Foods and Total Nutrient Intakes files to indicate the quality and completeness of a survey participant's response to the dietary recall section. The codes are the following:

1 = Reliable and met the following minimum criteria:

- The first 4 steps of the 5-step AMPM completed.
- Food/beverages consumed for each reported eating occasion identified.

For individuals with a code 1, all relevant variables associated with the 24-hour dietary recall contain a value.

2 = Not reliable or did not meet the minimum criteria

Individuals with a code 2 have incomplete records. No data on total nutrient intakes and the total number of foods reported are provided for these cases. These individuals have no records in the Individual Foods files.

3 [Code 3 is not included in the current datasets. It was only used for data from the 1999-2000 survey cycle.]

4 = Reported consuming breast milk

For infants and children who consumed human milk, there is a record in the Individual Foods files for each report of human milk. However, because amounts of human milk intake are not quantified, these records contain missing values for the amount consumed and for the amounts of energy and nutrients from human milk. Also, records of human milk have a missing value for the food source variable (DR1FS, DR2FS) and the eaten at home variable (DR1_040Z, DR2_040Z) in the Individual Foods files. Records for any other foods and beverages consumed by breast-fed infants and children are included in the Individual Foods files along with their amounts and nutrient information. Because of the missing amount or quantity information for human milk, no total nutrient intakes (contained in the Total Nutrient Intakes files) were computed for participants with a code 4.

A variable that identifies breast-fed children, DRABF, is included. This variable has a code of 1 if a child consumed human milk in either intake day.

5 = Not done

This code is assigned when the dietary recall section of the interview did not take place due to various reasons (such as refusal, equipment failure, or unable to contact the participant). These individuals have no records in the Individual Foods files. These individuals have a record in the Total Nutrients file with values only for the following variables: the respondent sequence number (SEQN), the dietary recall status code (DR1DRSTZ or DR2DRSTZ) and for participants 1 year or older, the fish and shellfish questions in the DR1TOT_L file (DRD340, DRD350A-K, DRD350AQ-JQ, DRD360, DRD370A-V, and DRD370AQ-UQ).

Only codes 1 and 4 appear in the Individual Foods file.

Distinguishing Between Foods/Beverages and Dietary Supplements in NHANES: In August 2021-August 2023, there was no 24-hour dietary supplement use collection. The 30-day dietary supplement use collection changed from being collected during the home interview to collection after the first 24-hour dietary recall via telephone. All NHANES participants responding to the 24-hour dietary recall interview are eligible for the dietary supplement and non-prescription antacid use questions. Information is obtained on all vitamins, minerals, herbals, and other dietary supplements as well as non-prescription antacids that were consumed during a 30-day period, including the name and the amount of supplement or antacid taken.

Distinguishing between foods/beverages and supplements can be challenging. NCHS and FSRG review questionable items reported in the dietary supplement and dietary recall components to resolve disposition of these items into the appropriate component. Products that are labeled as a dietary supplement, that have a supplement facts panel on the label, and are in tablets, capsules, softgels, gelcaps, or other pill forms, are considered dietary supplements. Items that are powders or liquids can be hard to distinguish. General guidelines used state that if powders and liquid concentrates have product directions stating that they be added to a liquid, they are classified as beverages. Examples are teas and protein powders. An exception is made for fiber products, which are classified as dietary supplements. Along this same guideline, energy drinks are considered beverages, but “energy shot” type products are considered dietary supplements.

It is best to refer to the databases that detail every food/beverage and dietary supplement reported in NHANES to identify exact determination used. The databases are:

- 2021-2023 Food and Nutrient Database for Dietary Studies
- NHANES Dietary Supplement Database

Participants who reported consuming only water, no food or other beverages: Records are included in the Individual Foods file for participants who consumed only water. There are 5 such individuals in the August 2021-August 2023 datasets, 4 in the Day 1 data and 1 in the Day 2 data. Their dietary recall status variable for the day is coded as "1" (complete and reliable) in the Total Nutrients file and the total number of items is the number of times water was reported. Individuals with just water intake and no food intake will have zero energy intake for the day.

Participants who reported consuming no water, food or other beverages: There can be participants whose intakes are determined to be complete even though they reported no water, food, or other beverage records for the day. For such participants there are no records in the Individual Foods file, but their dietary recall status is coded as complete and reliable, and the Total Nutrients file will include records with zero values for all nutrients. In the August 2021-August 2023 datasets, there are 3 individuals in the day 1 data that reported no water, food, or other beverage records for the day.

Number of days between the intake day and the day of family interview: Each of the four intake files includes a variable (DR1DBIH for Day 1 files and DR2DBIH for Day 2 files) to indicate the number of days between the intake day (i.e., the period covered by the 24-hour recall) and the day that the family questionnaire was administered in the household. A positive value in DR1BHIH or DR2BHIH indicates the family interview occurred prior to the intake day. In the survey, most of the family interviews were done before the participant participated in the dietary interview. A value of "0" in DR1BHIH or DR2BHIH indicates the family interview

occurred on the same date as the intake day. A negative value (i.e., DR1BHIH<0 or DR2BHIH<0) means that the family interview occurred after the intake day.

Food source: The source from which each food/beverage was obtained (e.g., from a store, fast food restaurant, cafeteria) is identified by the variables DR1FS (day 1) and DR2FS (day 2) in the Individual Foods files.

The code descriptions for this variable are:

Code Descriptions for Source of Food Variable

Code	Description
1	Store grocery/supermarket
2	Restaurant with waiter/waitress
3	Restaurant fast food/Pizza
4	Bar/Tavern/Lounge
5	Restaurant, no additional information
6	Cafeteria NOT in a K-12 school
7	Cafeteria in a K-12 school
8	Child/Adult care center
9	Child/Adult home care
10	Soup kitchen/shelter/food pantry facility
11	Meals on Wheels Program
12	Community food program – other
13	Community program, no additional info
14	Vending machine
15	Common coffee pot or snack tray
16	From someone else/gift
17	Mail order purchase
18	Residential dining facility
19	Grown or caught by you or someone you know
20	Fish caught by you or someone you know
24	Sport, recreation, or entertainment
25	Street vendor, vending truck
26	Fundraiser sales
27	Store - convenience type
28	Store - no additional information
91	Other, specify

Eating occasion: The variables DR1_030Z and DR2_030Z are located in the Individual Foods file. The code descriptions for the eating occasion variables are shown in the table below.

Code Descriptions for Eating Occasion Variable

Code	Description
1	Breakfast
2	Lunch
3	Dinner
4	Supper
5	Brunch
6	Snack
7	Beverage/Drink
8	Feeding-infant only
9	Extended consumption
10	Desayuno
11	Almuerzo
12	Comida
13	Merienda
14	Cena
15	Entre comida
16	Botana
17	Bocadillo
18	Tentempie
19	Bebida
91	Other

Eating occasion was designated by the respondent. During the interview, a list of eating occasion names was available to the respondent for selection. However, eating occasion names were not defined for the respondent.

Foods and beverages coded as part of a combination: 41 percent of foods and beverages reported in the WWEIA, NHANES August 2021-August 2023 sample were identified as items consumed together as combinations. Items consumed as a combination were identified by one of sixteen unique "combination food types." Foods and beverages not coded in combination have the code "0" for the combination food type variable.

The combination types provide a linkage for:

- Foods or beverages with additions, such as cereal with milk, coffee with cream;
- Multi-component foods that have specific protocol for collection such as some salads and sandwiches; and
- Other combinations that do not have a unique code in the FNDDS.

Combination Type, Code, Examples, and Percent of Food and Beverages Reported by Type, August 2021-August 2023, Day 1

Combination Type	Code	Examples of Combination Type	% Items
Not in combination	0	NA	59
Beverage w/ additions	1	Coffee, tea with: milk, cream, sugar.	9
Cereal w/ additions	2	Cereals (ready-to-eat, cooked) with: milk, sugar, fruit, butter.	4
Bread/baked product w/additions	3	Breads, rolls, pancakes with: butter, jam, syrup, fruit. Cakes, pies with: ice cream, toppings. Crackers with: cheese, dip, peanut butter.	5
Salad	4	Components of salads that do not have a single code in FNDDS. It may also designate additional items to single code salads.	5
Sandwiches	5	Components of sandwiches that do not have a single code in FNDDS. It may also designate additional items added to single code sandwiches.	6
Soup	6	Soup with: crackers, croutons, cheese.	<1
Frozen meals	7	Components of a prepackaged frozen meal and additions to the meal.	<1
Ice cream/ frozen yogurt w/ additions	8	Ice cream with: syrup, nuts, toppings.	<1
Dried beans or Vegetable w/ additions	9	French fries, potatoes with: catsup, gravy, butter, toppings. Beans with: sauce, butter.	3
Fruit w/ additions	10	Fruit with: toppings, milk, honey. Components of fruit mixtures or salads that do not have a single code in FNDDS.	1
Tortilla products	11	Components of tacos and tortilla products that do not have a single code in FNDDS. It may also designate additional items to single code tacos or tortilla products.	2
Meat, Poultry, Fish	12	Meat, poultry, fish with: gravy, sauce, and condiments.	2
Lunchables®	13	Components of pre-packaged lunch kits.	<1
Chips w/ additions	14	Potato chips, corn chips with: dip, cheese, salsa.	1
Baby Toddler Food and Infant Formula	15	Baby Toddler cereal with: infant formula, fruit. Infant formula with: baby cereal.	<1
Other mixtures	90	Rice, pasta, spaghetti, eggs, other mixtures with: butter, gravy, sauce, condiments.	4

All items given a combination food type are given an additional variable to identify each of the items within the combination. This variable is the "combination food number" that is unique to the combination food type within the individual intake.

Variable Labels and Names for Combination Coding

Combination Coding	Variable Name, Day 1	Variable Name, Day 2
Combination food type	DR1CCMTX	DR2CCMTX
Combination food number	DR1CCMNM	DR2CCMNM

The What We Eat in America Food Categories, available on the FSRG website (http://www.ars.usda.gov/nea/bhnrc/fsrg), is a grouping scheme that combines foods and beverages together that have similar usage and nutrient content with the emphasis on

how they are commonly consumed in the American diet. There are more than 170 unique categories, and each is assigned a 4-digit number and description. The WWEIA Food Categories contain discrete food items and are not disaggregated (e.g., pizza vs. grain, cheese, tomatoes, etc.). Designed to be flexible, the categories can be combined as needed to address specific research questions. A new version of the WWEIA Food Categories is produced for each 2-year release cycle of WWEIA, NHANES.

Special diet: Information on whether the participant is currently on any kind of diet to lose weight or for other health-related reason and, if so, the type of diet, was provided. The variable DRQSDIET identifies whether a participant was on a special diet. The variables DRQSDT1 through DRQSDT12 and DRQSDT91 identify the type of diet or diets that the participant was following. These variables can be found in the Total Nutrient Intakes file.

Sample weights for dietary intake data: The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, **the appropriate sample weights must be used.**

For the NHANES August 2021-August 2023 sample, there were 22,660 persons selected; of these 8,860 were considered participants to the MEC examination and data collection. A total of 6,754 MEC participants provided complete dietary intakes for Day 1, and of those providing the Day 1 data, 5,879 provided complete dietary intakes for Day 2. The 6,754 participants who provided complete Day 1 intakes includes 21 participants who had an unreliable day 1 intake and reliable day 2 intake. For these 21 individuals, the single reliable recall is included in the data set as Day 1 recall.

Most analyses of NHANES data use data collected in the MEC and the variable WTMEC2YR should be used for the sample weights. However, for the WWEIA dietary data, different sample weights are recommended for analysis. Because food intake can vary by weekdays and weekends, use of the MEC weights disproportionately represents intakes on weekends.

A set of weights (WTDRD1) is provided that should be used when an analysis uses the Day 1 dietary recall data (either alone or when Day 1 nutrient data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with Day 1 data. Day 1 weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These Day 1 weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using Day 1 data and Day 1 weights might be larger than standard errors for similar estimates based on MEC weights.

When analysis is based on both days of dietary intake, only 5,879 sample participants have complete data. The NHANES protocol requires an attempt to collect the second day of dietary data at least 3 days after the first day, but the actual number of days between the two interviews is variable. A set of adjusted weights, WTDR2D, is to be used when an analysis uses the smaller sample with completed Day 1 and Day 2 dietary data. This two-day weight was constructed for the 5,879 participants by taking the Day 1 weights (WTDRD1) and further adjusting for: (a) the additional non-response for the second recall; and (b) for the proportion of weekend (Friday through Sunday) and weekday (Monday through Thursday) combinations of Day 1 and Day 2 recalls.

NOTE: All sample weights are person-level weights, and each set of dietary weights should sum to the same overall population control total as the MEC weights (WTMEC2YR). For the August 2021-August 2023 cycle, the Day 1 dietary recall weights (WTDRD1) are appropriate to use in the analysis of the fish and shellfish consumption data (i.e., variables DRD340, DRD350A-K, DRD350AQ-JQ DRD360, DRD370A-V, and DRD370AQ-UQ) located in the Day 1 Total Nutrient Intake File (DR1TOT_L), irrespective of whether other dietary data are included in the analysis. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

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Codebook and Frequencies

DRXFDCD - Food Code

Variable Name: DRXFDCD

SAS Label: Food Code

English Text: Food Code

Code or Value	Value Description	Count	Cumulative	Skip to Item
11000000 to 99998210	Range of Values	5432	5432	
.	Missing	0	5432	

DRXFCSD - Short Food Code Description

Variable Name: DRXFCSD

SAS Label: Short Food Code Description

English Text: Short Food Code Description

Code or Value	Value Description	Count	Cumulative	Skip to Item
Short Food Code Description	Value was recorded	5432	5432	
< blank >	Missing	0	5432	

DRXFCLD - Long Food Code Description

Variable Name: DRXFCLD

SAS Label: Long Food Code Description

English Text: Long Food Code Description

Code or Value	Value Description	Count	Cumulative	Skip to Item
Long Food Code Description	Value was recorded	5432	5432	
< blank >	Missing	0	5432	

Appendix 1. Changes between WWEIA survey cycles 2013-2014 thru the August 2021-August 2023 sample

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Administration of 24-hour dietary recall	Same as 2003-2012: Day 1 in-person in MEC, Day 2 by telephone	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Day 1 by telephone, Day 2 by telephone
Language, Day 1	Same as 1999-2012: Interpreters assisted as needed	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Language, Day 2	Same as 2011-2012: English, Spanish, or Asian languages	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	English or Spanish language
Person who did not provide reliable intake for Day 1 but did for Day 2	Same as 2003-2012: Neither intake was included in the release datasets	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	The single reliable intake was included in the release datasets as Day 1 intake (n=21)
Food source (where food was obtained)	Codes 8 and 9 revised descriptions.	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014	Same as 2013-2014
Combination food types	Same as 2003-2012: Values for 15 combination types	Same as 2003-2014	Same as 2003-2014	Same as 2003-2014	Values for 16 combination types; added “Baby Toddler Food and Infant Formula”
Number of intakes that include only water consumption for the day	6 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (all in Day 2 data), records are included in Individual Foods file.	2 intakes (all in Day 2 data), records are included in Individual Foods file.	5 intakes (1 in Day 1, 4 in Day 2 data), records are included in Individual Foods file.	5 intakes (4 in Day 1, 1 in Day 2), records are included in Individual Foods file.
Number of intakes that include no water or food consumption for the day	1 intake in Day 2 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	1 intake in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for this intake.	2 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.	3 intakes in Day 1 with no food or water records for the day. Record is not included in the Individual Foods File for these intakes.
Tap water source	Same as 2003-2012: Reported in 6 categories.	Same as 2003-2004	Responses combined into 4 categories	Same as 2017-2018	Same as 2017-2018
Modification codes: DR1MC	All remaining modification	No modification	No modification	No modification	No modification

Variable or feature	WWEIA 2013-2014	WWEIA 2015-2016	WWEIA 2017-2018	WWEIA 2017-March 2020 Pre-Pandemic	WWEIA Aug 2021-Aug 2023
Day 2 Modification codes: DR2MC Modification Code Description file: DRXMCD	codes deleted; new food codes addressing modifications added in FNDDS 2013-2014	codes	codes	codes	codes
Salt use at the table and in cooking or preparing foods and type	Same as 2003-2004	Same as 2003-2004	Question wording changed slightly by removing, “seasoned salt”, and adding, “sea salt”.	Same as 2017-2018	Same as 2017-2018
Salt used at the table yesterday and type	Question asked about salt use at the table yesterday and kind of salt to coincide with 24-hour recall.	Same as 2013-2014	Question wording changed slightly by removing, “seasoned salt”, and adding, “sea salt”.	Same as 2017-2018	Same as 2017-2018
Main respondent and person whom helped in responding for the interview	Same as 2003-2012	Response category changed	Same as 2015-2016	Same as 2015-2016	Same as 2015-2016

Appendix 2. Variables in the Individual Foods Files (DR1IFF_L and DR2IFF_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1ILINE	DR2ILINE	Food/Individual component number
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1CCMNM	DR2CCMNM	Combination food number
DR1CCMTX	DR2CCMTX	Combination food type
DR1_020	DR2_020	Time of eating occasion (HH:MM)
DR1_030Z	DR2_030Z	Name of eating occasion
DR1FS	DR2FS	Source of food
DR1_040Z	DR2_040Z	Did you eat this meal at home?
DR1IFDCD	DR2IFDCD	USDA food code
DR1IGRMS	DR2IGRMS	Grams
DR1IKCAL	DR2IKCAL	Energy (kcal)
DR1IPROT	DR2IPROT	Protein (gm)
DR1ICARB	DR2ICARB	Carbohydrate (gm)
DR1ISUGR	DR2ISUGR	Total sugars (gm)
DR1IFIBE	DR2IFIBE	Dietary fiber (gm)
DR1ITFAT	DR2ITFAT	Total fat (gm)
DR1ISFAT	DR2ISFAT	Total saturated fatty acids (gm)
DR1IMFAT	DR2IMFAT	Total monounsaturated fatty acids (gm)
DR1IPFAT	DR2IPFAT	Total polyunsaturated fatty acids (gm)
DR1ICHOL	DR2ICHOL	Cholesterol (mg)
DR1IATOC	DR2IATOC	Vitamin E as alpha-tocopherol (mg)
DR1IATOA	DR2IATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1IRET	DR2IRET	Retinol (mcg)
DR1IVARA	DR2IVARA	Vitamin A, RAE (mcg)
DR1IACAR	DR2IACAR	Alpha-carotene (mcg)
DR1IBCAR	DR2IBCAR	Beta-carotene (mcg)
DR1ICRYP	DR2ICRYP	Beta-cryptoxanthin (mcg)
DR1ILYCO	DR2ILYCO	Lycopene (mcg)
DR1ILZ	DR2ILZ	Lutein + zeaxanthin (mcg)
DR1IVB1	DR2IVB1	Thiamin (Vitamin B1) (mg)

Day1 Name	Day2 Name	Variable Label
DR1IVB2	DR2IVB2	Riboflavin (Vitamin B2) (mg)
DR1INIAC	DR2INIAC	Niacin (mg)
DR1IVB6	DR2IVB6	Vitamin B6 (mg)
DR1IFOLA	DR2IFOLA	Total folate (mcg)
DR1IFA	DR2IFA	Folic acid (mcg)
DR1IFF	DR2IFF	Food folate (mcg)
DR1IFDFE	DR2IFDFE	Folate, DFE (mcg)
DR1ICHL	DR2ICHL	Total choline (mg)
DR1IVB12	DR2IVB12	Vitamin B12 (mcg)
DR1IB12A	DR2IB12A	Added vitamin B12 (mcg)
DR1IVC	DR2IVC	Vitamin C (mg)
DR1IVD	DR2IVD	Vitamin D (D2 + D3) (mcg)
DR1IVK	DR2IVK	Vitamin K (mcg)
DR1ICALC	DR2ICALC	Calcium (mg)
DR1IPHOS	DR2IPHOS	Phosphorus (mg)
DR1IMAGN	DR2IMAGN	Magnesium (mg)
DR1IIRON	DR2IIRON	Iron (mg)
DR1IZINC	DR2IZINC	Zinc (mg)
DR1ICOPP	DR2ICOPP	Copper (mg)
DR1ISODI	DR2ISODI	Sodium (mg)
DR1IPOTA	DR2IPOTA	Potassium (mg)
DR1ISELE	DR2ISELE	Selenium (mcg)
DR1ICAFF	DR2ICAFF	Caffeine (mg)
DR1ITHEO	DR2ITHEO	Theobromine (mg)
DR1IALCO	DR2IALCO	Alcohol (gm)
DR1IMOIS	DR2IMOIS	Moisture (gm)
DR1IS040	DR2IS040	SFA 4:0 (Butanoic) (gm)
DR1IS060	DR2IS060	SFA 6:0 (Hexanoic) (gm)
DR1IS080	DR2IS080	SFA 8:0 (Octanoic) (gm)
DR1IS100	DR2IS100	SFA 10:0 (Decanoic) (gm)
DR1IS120	DR2IS120	SFA 12:0 (Dodecanoic) (gm)
DR1IS140	DR2IS140	SFA 14:0 (Tetradecanoic) (gm)
DR1IS160	DR2IS160	SFA 16:0 (Hexadecanoic) (gm)
DR1IS180	DR2IS180	SFA 18:0 (Octadecanoic) (gm)
DR1IM161	DR2IM161	MFA 16:1 (Hexadecenoic) (gm)
DR1IM181	DR2IM181	MFA 18:1 (Octadecenoic) (gm)
DR1IM201	DR2IM201	MFA 20:1 (Eicosenoic) (gm)
DR1IM221	DR2IM221	MFA 22:1 (Docosenoic) (gm)
DR1IP182	DR2IP182	PFA 18:2 (Octadecadienoic) (gm)

Day1 Name	Day2 Name	Variable Label
DR1IP183	DR2IP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1IP184	DR2IP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1IP204	DR2IP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1IP205	DR2IP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1IP225	DR2IP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1IP226	DR2IP226	PFA 22:6 (Docosahexaenoic) (gm)

Appendix 3. List of Nutrients/Food Components (Unit)

Energy and Macronutrients

Food energy (kcal)
Protein (gm)
Carbohydrate (gm)
Fat, total (gm)
Alcohol (gm)

Sugars, total (gm)
Dietary fiber, total (gm)
Water (moisture) (gm)*

Saturated fatty acids, total (gm)
Monounsaturated fatty acids, total (gm)
Polyunsaturated fatty acids, total (gm)
Cholesterol (mg)

Individual fatty acids:
  4:0 (gm)
  6:0 (gm)
  8:0 (gm)
  10:0 (gm)
  12:0 (gm)
  14:0 (gm)
  16:0 (gm)
  18:0 (gm)
  16:1 (gm)
  18:1 (gm)
  20:1 (gm)
  22:1 (gm)
  18:2 (gm)
  18:3 (gm)
  18:4 (gm)
  20:4 (gm)
  20:5 n-3 (gm)
  22:5 n-3 (gm)
  22:6 n-3 (gm)

Vitamins, Minerals, and Other Components

Vitamin A as retinol activity equivalents (mcg)
Retinol (mcg)
Carotenoids:
  Carotene, alpha (mcg)
  Carotene, beta (mcg)
  Cryptoxanthin, beta (mcg)
  Lycopene (mcg)
  Lutein + zeaxanthin (mcg)
Vitamin E as alpha-tocopherol (mg)
  **Added vitamin E as alpha-tocopherol (mg)
Vitamin D (D2 + D3) (mcg)
Vitamin K as phylloquinone (mcg)
Vitamin C (mg)
Thiamin (mg)
Riboflavin (mg)
Niacin (mg)
Vitamin B-6 (mg)
Folate, total (mcg)
  Folate as dietary folate equivalents (mcg)

Folic acid (mcg)
Food folate (mcg)
Choline, total (mg)
Vitamin B-12 (mcg)
***Added vitamin B-12 (mcg)

Calcium (mg)
Iron (mg)
Magnesium (mg)
Phosphorus (mg)
Potassium (mg)
Sodium (mg)
Zinc (mg)
Copper (mg)
Selenium (mcg)
Caffeine (mg)
Theobromine (mg)

** Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)*

*** Represents a synthetic subcomponent of vitamin E and is included in the vitamin E value.*

**** Represents a fortified subcomponent of vitamin B12 and is included in the vitamin B12 value.*

Appendix 4. Adding Food Code Descriptions to Your Files

One supporting file is included with the Individual Foods files: the Food Code Description file (DRXFCD_L).

The DRXFCD_L file includes abbreviated descriptions (up to 60 characters) and complete descriptions (up to 200 characters) associated with each USDA food code included in the Individual Foods files.

The Food Code Description file (DRXFCD_L) contains three variables:

DRXFDCD a numeric value corresponding to DR1IFDCD in the file DR1IFF_L or DR2IFDCD in the file DR2IFF_L;

DRXFCSD a short description (up to 60 characters) of the food code;

DRXFDLD a long description (up to 200 characters) of the food code.

The following SQL code is an example of appending the shorter food code description (here renamed DR1IFCSD) to one of the Individual Foods files using PROC SQL from SAS®. Other SQL implementations may be different.

```
proc sql;
  create table DR1IFF_L_PLUS as
  select iff.*, desc.DRXFCSD as DR1FCSD, desc.DRXFCLD as DR1FCLD
  from NHANES.DR1IFF_L iff
  left join NHANES.DRXFCD_L desc
  on iff.DR1IFDCD = desc.DRXFDCD
  order by SEQN, DR1ILINE;
quit;
```

SAS® users may wish to use Proc Format to assign labels to the food codes. The following example generates and saves a picture format for food codes and a separate format for each food code that includes both the food code itself and the short food code description. It is assumed that the user has stored the Individual Foods files and the Food Code Description file in a library called NHANES and wishes to store the formats there as well.

```
options fmtsearch = (NHANES);

proc format library = library;
  picture foodcode
  low - high = '000-00000';
quit;

data tmp;
  set NHANES.DRXFCD_L;
  length cfoodcode $9 label $72;
  cfoodcode = put(DRXFDCD, foodcode.);
  label = cfoodcode || '  ' || DRXFCSD;
run;

data fmt (keep = fmtname start label);
  set tmp;
  retain fmtname 'DRXFDCD';
  rename DRXFDCD = start;
run;
```

```
proc format cntlin = fmt library = library;
run;
```

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Appendix 5. Variables in the Total Nutrients Files (DR1TOT_L and DR2TOT_L) by Position

Day1 Name	Day2 Name	Variable Label
SEQN	SEQN	Respondent sequence number
WTDRD1	WTDRD1	Dietary day one sample weight
WTDR2D	WTDR2D	Dietary two-day sample weight
DR1DRSTZ	DR2DRSTZ	Dietary recall status
DR1EXMER	DR2EXMER	Interviewer ID code
DRABF	DRABF	Breast-fed infant (either day)
DRDINT	DRDINT	Number of days of intake
DR1DBIH	DR2DBIH	# of days b/w intake and HH interview
DR1DAY	DR2DAY	Intake day of the week
DR1LANG	DR2LANG	Language respondent used mostly
DR1MRESP	DR2MRESP	Main respondent for this interview
DR1HELP	DR2HELP	Helped in responding for this interview
DBQ095Z	N/A	Type of table salt used
DBD100	N/A	How often add salt to food at table
DRQSPREP	N/A	Salt used in preparation?
DR1STY	DR2STY	Salt used at table yesterday?
DR1SKY	DR2SKY	Type of salt used yesterday
DRQSDIET	N/A	On special diet?
DRQSDT1	N/A	Weight loss/Low calorie diet
DRQSDT2	N/A	Low fat/Low cholesterol diet
DRQSDT3	N/A	Low salt/Low sodium diet
DRQSDT4	N/A	Sugar free/Low sugar diet
DRQSDT5	N/A	Low fiber diet
DRQSDT6	N/A	High fiber diet
DRQSDT7	N/A	Diabetic diet
DRQSDT8	N/A	Weight gain/Muscle building diet
DRQSDT9	N/A	Low carbohydrate diet
DRQSDT10	N/A	High protein diet
DRQSDT11	N/A	Gluten-free/Celiac diet
DRQSDT12	N/A	Renal/Kidney diet
DRQSDT91	N/A	Other special diet
DR1TNUMF	DR2TNUMF	Number of foods/beverages reported
DR1TKCAL	DR2TKCAL	Energy (kcal)
DR1TPROT	DR2TPROT	Protein (gm)
DR1TCARB	DR2TCARB	Carbohydrate (gm)
DR1TSUGR	DR2TSUGR	Total sugars (gm)
DR1TFIBE	DR2TFIBE	Dietary fiber (gm)
DR1TTFAT	DR2TTFAT	Total fat (gm)
DR1TSFAT	DR2TSFAT	Total saturated fatty acids (gm)

Day1 Name	Day2 Name	Variable Label
DR1TMFAT	DR2TMFAT	Total monounsaturated fatty acids (gm)
DR1TPFAT	DR2TPFAT	Total polyunsaturated fatty acids (gm)
DR1TCHOL	DR2TCHOL	Cholesterol (mg)
DR1TATOC	DR2TATOC	Vitamin E as alpha-tocopherol (mg)
DR1TATOA	DR2TATOA	Added alpha-tocopherol (Vitamin E) (mg)
DR1TRET	DR2TRET	Retinol (mcg)
DR1TVARA	DR2TVARA	Vitamin A, RAE (mcg)
DR1TACAR	DR2TACAR	Alpha-carotene (mcg)
DR1TBCAR	DR2TBCAR	Beta-carotene (mcg)
DR1TCRYP	DR2TCRYP	Beta-cryptoxanthin (mcg)
DR1TLYCO	DR2TLYCO	Lycopene (mcg)
DR1TLZ	DR2TLZ	Lutein + zeaxanthin (mcg)
DR1TVB1	DR2TVB1	Thiamin (Vitamin B1) (mg)
DR1TVB2	DR2TVB2	Riboflavin (Vitamin B2) (mg)
DR1TNIAC	DR2TNIAC	Niacin (mg)
DR1TVB6	DR2TVB6	Vitamin B6 (mg)
DR1TFOLA	DR2TFOLA	Total folate (mcg)
DR1TFA	DR2TFA	Folic acid (mcg)
DR1TFF	DR2TFF	Food folate (mcg)
DR1TFDFE	DR2TFDFE	Folate, DFE (mcg)
DR1TCHL	DR2TCHL	Total choline (mg)
DR1TVB12	DR2TVB12	Vitamin B12 (mcg)
DR1TB12A	DR2TB12A	Added vitamin B12 (mcg)
DR1TVC	DR2TVC	Vitamin C (mg)
DR1TVD	DR2TVD	Vitamin D (D2 + D3) (mcg)
DR1TVK	DR2TVK	Vitamin K (mcg)
DR1TCALC	DR2TCALC	Calcium (mg)
DR1TPHOS	DR2TPHOS	Phosphorus (mg)
DR1TMAGN	DR2TMAGN	Magnesium (mg)
DR1TIRON	DR2TIRON	Iron (mg)
DR1TZINC	DR2TZINC	Zinc (mg)
DR1TCOPP	DR2TCOPP	Copper (mg)
DR1TSODI	DR2TSODI	Sodium (mg)
DR1TPOTA	DR2TPOTA	Potassium (mg)
DR1TSELE	DR2TSELE	Selenium (mcg)
DR1TCAFF	DR2TCAFF	Caffeine (mg)
DR1TTHEO	DR2TTHEO	Theobromine (mg)
DR1TALCO	DR2TALCO	Alcohol (gm)
DR1TMOIS	DR2TMOIS	Moisture (gm)

Day1 Name	Day2 Name	Variable Label
DR1TS040	DR2TS040	SFA 4:0 (Butanoic) (gm)
DR1TS060	DR2TS060	SFA 6:0 (Hexanoic) (gm)
DR1TS080	DR2TS080	SFA 8:0 (Octanoic) (gm)
DR1TS100	DR2TS100	SFA 10:0 (Decanoic) (gm)
DR1TS120	DR2TS120	SFA 12:0 (Dodecanoic) (gm)
DR1TS140	DR2TS140	SFA 14:0 (Tetradecanoic) (gm)
DR1TS160	DR2TS160	SFA 16:0 (Hexadecanoic) (gm)
DR1TS180	DR2TS180	SFA 18:0 (Octadecanoic) (gm)
DR1TM161	DR2TM161	MFA 16:1 (Hexadecenoic) (gm)
DR1TM181	DR2TM181	MFA 18:1 (Octadecenoic) (gm)
DR1TM201	DR2TM201	MFA 20:1 (Eicosenoic) (gm)
DR1TM221	DR2TM221	MFA 22:1 (Docosenoic) (gm)
DR1TP182	DR2TP182	PFA 18:2 (Octadecadienoic) (gm)
DR1TP183	DR2TP183	PFA 18:3 (Octadecatrienoic) (gm)
DR1TP184	DR2TP184	PFA 18:4 (Octadecatetraenoic) (gm)
DR1TP204	DR2TP204	PFA 20:4 (Eicosatetraenoic) (gm)
DR1TP205	DR2TP205	PFA 20:5 (Eicosapentaenoic) (gm)
DR1TP225	DR2TP225	PFA 22:5 (Docosapentaenoic) (gm)
DR1TP226	DR2TP226	PFA 22:6 (Docosahexaenoic) (gm)
DR1_300	DR2_300	Compare food consumed yesterday to usual
DR1_320Z	DR2_320Z	Total plain water drank yesterday (gm)
DR1_330Z	DR2_330Z	Total tap water drank yesterday (gm)
DR1BWATZ	DR2BWATZ	Total bottled water drank yesterday (gm)
DR1TWSZ	DR2TWSZ	Tap water source
DRD340	N/A	Shellfish eaten during past 30 days
DRD350A	N/A	Clams eaten during past 30 days
DRD350AQ	N/A	# of times clams eaten in past 30 days
DRD350B	N/A	Crabs eaten during past 30 days
DRD350BQ	N/A	# of times crabs eaten in past 30 days
DRD350C	N/A	Crayfish eaten during past 30 days
DRD350CQ	N/A	# of times crayfish eaten past 30 days
DRD350D	N/A	Lobsters eaten during past 30 days
DRD350DQ	N/A	# of times lobsters eaten past 30 days
DRD350E	N/A	Mussels eaten during past 30 days
DRD350EQ	N/A	# of times mussels eaten in past 30 days
DRD350F	N/A	Oysters eaten during past 30 days
DRD350FQ	N/A	# of times oysters eaten in past 30 days
DRD350G	N/A	Scallops eaten during past 30 days
DRD350GQ	N/A	# of times scallops eaten past 30 days

Day1 Name	Day2 Name	Variable Label
DRD350H	N/A	Shrimp eaten during past 30 days
DRD350HQ	N/A	# of times shrimp eaten in past 30 days
DRD350I	N/A	Other shellfish eaten past 30 days
DRD350IQ	N/A	# of times other shellfish eaten
DRD350J	N/A	Other unknown shellfish eaten past 30 d
DRD350JQ	N/A	# of times other unknown shellfish eaten
DRD350K	N/A	Refused on shellfish eaten past 30 days
DRD360	N/A	Fish eaten during past 30 days
DRD370A	N/A	Breaded fish products eaten past 30 days
DRD370AQ	N/A	# of times breaded fish products eaten
DRD370B	N/A	Tuna eaten during past 30 days
DRD370BQ	N/A	# of times tuna eaten in past 30 days
DRD370C	N/A	Bass eaten during past 30 days
DRD370CQ	N/A	# of times bass eaten in past 30 days
DRD370D	N/A	Catfish eaten during past 30 days
DRD370DQ	N/A	# of times catfish eaten in past 30 days
DRD370E	N/A	Cod eaten during past 30 days
DRD370EQ	N/A	# of times cod eaten in past 30 days
DRD370F	N/A	Flatfish eaten during past 30 days
DRD370FQ	N/A	# of times flatfish eaten past 30 days
DRD370G	N/A	Haddock eaten during past 30 days
DRD370GQ	N/A	# of times haddock eaten in past 30 days
DRD370H	N/A	Mackerel eaten during past 30 days
DRD370HQ	N/A	# of times mackerel eaten past 30 days
DRD370I	N/A	Perch eaten during past 30 days
DRD370IQ	N/A	# of times perch eaten in past 30 days
DRD370J	N/A	Pike eaten during past 30 days
DRD370JQ	N/A	# of times pike eaten in past 30 days
DRD370K	N/A	Pollock eaten during past 30 days
DRD370KQ	N/A	# of times pollock eaten in past 30 days
DRD370L	N/A	Porgy eaten during past 30 days
DRD370LQ	N/A	# of times porgy eaten in past 30 days
DRD370M	N/A	Salmon eaten during past 30 days
DRD370MQ	N/A	# of times salmon eaten in past 30 days
DRD370N	N/A	Sardines eaten during past 30 days
DRD370NQ	N/A	# of times sardines eaten past 30 days
DRD370O	N/A	Sea bass eaten during past 30 days
DRD370OQ	N/A	# of times sea bass eaten past 30 days
DRD370P	N/A	Shark eaten during past 30 days

Day1 Name	Day2 Name	Variable Label
DRD370PQ	N/A	# of times shark eaten in past 30 days
DRD370Q	N/A	Swordfish eaten during past 30 days
DRD370QQ	N/A	# of times swordfish eaten past 30 days
DRD370R	N/A	Trout eaten during past 30 days
DRD370RQ	N/A	# of times trout eaten in past 30 days
DRD370S	N/A	Walleye eaten during past 30 days
DRD370SQ	N/A	# of times walleye eaten in past 30 days
DRD370T	N/A	Other fish eaten during past 30 days
DRD370TQ	N/A	# of times other fish eaten past 30 days
DRD370U	N/A	Other unknown fish eaten in past 30 days
DRD370UQ	N/A	# of times other unknown fish eaten
DRD370V	N/A	Refused on fish eaten past 30 days

National Health and Nutrition Examination Survey

1999-2023 Data Documentation, Codebook, and Frequencies

Dietary Supplement Database - Blend Information (DSBI)

Data File: DSBI.xpt

First Published: February 2025

Last Revised: NA

Component Description

The NHANES Dietary Supplement Database (NHANES-DSD) contains detailed information on the dietary supplements (DS) and non-prescription antacids containing calcium and/or magnesium (antacids) reported by survey participants since NHANES 1999. The NHANES-DSD release consists of three datasets containing information on the following: Dietary Supplement Product Information (DSPI), Dietary Supplement Ingredient Information (DSII), and Dietary Supplement Blend Information (DSBI).

Dietary supplement information from the in-house NCHS Product Label Database (PLD) is publicly released in three files that make up the NHANES-DSD. These files incorporate all products that have been reported by respondents since 1999 from the PLD. With subsequent releases, new products reported will be appended to the NHANES-DSD files.

The in-house PLD database is maintained by NCHS nutritionists. NCHS attempts to obtain a product label for all dietary supplements or antacids reported by participants from sources such as the manufacturer or retailer, the Internet, company catalogs, and the Physician's Desk Reference (PDR). Label information has also been obtained from the Dietary Supplement Label Database (DSLD), which is a joint project of the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) and National Library of Medicine (NLM). The DSLD contains the full label contents from a sample of dietary supplement products marketed in the U.S. Selected label information is then entered into the PLD including, but not limited to: supplement name; manufacturer and/or distributor; serving size; form of serving size; and ingredients and amounts. The ingredient information entered into the database is taken directly from the supplement facts box on the label or carton. The PLD is used to assist with data editing and then publicly released in the data files (DSPI, DSII, and DSBI).

In addition to entering labels into the PLD database, NCHS created generic and default dietary supplements and antacids, which are also maintained in the database. Generics were created in the database when we had information about a reported supplement, such as the strength of all ingredients, but no brand name. These were generally single ingredient supplements, which included a strength (e.g., vitamin C 500 mg) or multiple vitamins and/or mineral supplements made by a private label manufacturer that was known to us and for which we had a label with identical ingredients and strengths (e.g., brand X all-purpose multivitamin was reported, and we had a label for brand Y, made by the same manufacturer). When all ingredient strengths were unknown, a default supplement was created in the database. Defaults were created for single ingredient and multiple ingredient supplements based on our own data of most frequently reported supplements of that type, based on the survey cycle in which the data was collected. Created default products and the actual products or strengths that were assigned to these defaults are listed in the documentation for the files associated with participants' use of dietary supplements, located on the NHANES website under the dietary data links.

In 2019, the application used to manage and access NHANES dietary supplement product database was updated to adapt to the .NET environment. This resulted in the new structure of dietary supplement identifiers. Variables DSDPID, DSDIID, DSDBID were added to indicate the updated supplement ID, ingredient ID, and blend component ID, respectively. The variables

DSDSUPID, DSDINGID, and DSDBCID now indicate the old versions for supplement ID, ingredient ID, and blend component ID, respectively. In addition, six new variables were included to provide information on supplement reformulations: DSDPRDT, DSDPREID, DSDORGID, DSDSGPF, DSDSEQF, DSDLINRF to indicate the product type, previous product ID, original product ID, sequential group formulation, sequential formulation, and linear formulation, respectively.

Data Processing and Editing

The in-house PLD was used for processing and editing of the dietary supplement data since 1999.

Variables pulled directly from the in-house PLD

DSDSUPP: Name of Supplement
This is the name from the front of the product label. The brand name is generally entered first (i.e., Nature's Way) and then the actual product name (i.e., lutein). Information such as the strength (i.e., 60 mg extract) of the product and other qualifiers that help distinguish a product from a similar product (i.e., mega, super, high potency, time release, chewable, extract) are also listed if they are on the front of the product label. Words like "dietary supplement" and "health claims" are not entered as part of the name.

DSDPID: Supplement ID Number
New Supplement ID is a unique number assigned to each product entered in the inhouse PLD.

DSDPRDT: Product Type
If the supplement was entered by NCHS as a regular product, it is equal to 1. If the supplement was created by NCHS as a generic product, it is equal to 2. Otherwise, if the supplement was created by NCHS as a default product, it is equal to 3. Previous Variable DSDGENRC was replaced by code "2" in DSDPRDT.

DSDSUPID: Supplement ID Number-Old version
Old Supplement ID is a 10-digit identifier assigned to each product entered into the in-house PLD. All identifiable products have a supplement ID beginning with the number "1". The next 3 digits (positions 2-4) are "888" or "881" if the supplement was created by NCHS as a generic or default product; otherwise the digits in positions 2-4 are coded "000" or "001". The next 4 digits (positions 5-8) are sequentially assigned by the system for each product. The last 2 digits (positions 9-10) indicate formulations of the same supplement: the first formulation entered into the database = 00, the first reformulation = 01, the next = 02, etc. Note that these are reformulations of the same product. Different versions of products (e.g., liquid vs. tablet, with iron vs. without iron, regular vs. high potency) are considered as different products thus have different 4-digit numbers in positions 5 to 8. When a product name was entered as "no product information available," "refused," or "don't know," its ID number starts with a string of 6s.

DSDSRCE: Supplement Information Source
The source of each product label is recorded into the PLD. Generic and default products do not have a source code.

DSDTYPE: Supplement Type
The type of supplements is recorded into the database. The supplement types are: infant/pediatric, prenatal, mature, or standard . Products are coded as infant/pediatric when the product name states "infant, children, child, or kid/s" or has an indication in the title, label, or the form (e.g., animal shapes) of the supplement that it is intended for children. If this is not the case, but the suggested dose or directions indicate dosage for children only, then the code is infant/pediatric, but if dosages for adults are also included, then the product is coded as standard. Products are coded as "Prenatal" when the product name states prenatal or a derivative of this name or has an indication in the title or label that it is intended for pregnant women. If this is not the case, but the suggested dosage or directions indicate dosage for pregnant women only, the product is coded as a prenatal, but if dosages for non-pregnant adults are also included, the product is coded as standard. Products are coded as "mature"

when the product name or label includes words such as "mature, senior, geriatric, post-menopausal, or silver" or indicates with other words that it is intended for individuals 50 years and over. All other products are coded as "standard".

DSDSERVQ: Serving Size Quantity
This is the "serving size quantity," which is recorded from the product label's supplement facts panel for which the nutrient amounts are based on.

DSDSERVU: Serving Size Unit
This is the "serving size unit," which is recorded from the product label's supplement facts panel.

DSDPREID: Previous product ID
This represents the parent product from which this current product was derived.

DSDORGID: Original product ID
This represents the original or root product from which this formulation or product was derived.

DSDSGPF: Sequential group formulation
This represents the sequential order of creation from the original product within the group of identical product types.

DSDSEQF: Sequential formulation
This represents the sequential order of creation from the original product.

DSDLINRF: Linear formulation
This represents the direct line of descendancy order of creation from the original product. For example- original product being level 1, a product derived from original level 1 product being level 2 and a product derived from the level 2 product being level 3, and so on.

DSDIID: Ingredient ID
This is the new unique ingredient ID created by the PLD for each ingredient recorded from the product label's supplement facts panel.

DSDINGID: Ingredient ID-Old version
This is the old unique ingredient ID created by the PLD for each ingredient recorded from the product label's supplement facts panel.

DSDINGR: Ingredient Name
Ingredient names are recorded from the product label's supplement facts panel.

DSDOPER: Ingredient Operator
This is a symbol =, <, or > that comes from the product label's supplement facts panel.

DSDQTY: Ingredient Quantity
Ingredient quantity is recorded for each ingredient listed from the product label's supplement facts panel. Some nutrient amounts are for a nutrient compound (generally a foreign-made product or an antacid) and these must be converted to an elemental nutrient amount (See Appendix 2 for recommended conversions).

DSDUNIT: Ingredient Unit
Ingredient unit is recorded for each ingredient listed from the product label's supplement facts panel.

NHANES dietary supplement files have vitamin A and E expressed in different units based on what was reported on the product label's supplement facts panel. Vitamin E may be reported in International Units (IU) or micrograms (μg). Vitamin A may be reported as International Units

(IU) or as micrograms (μg) of retinol activity equivalents (RAE). Users should be aware of changes in dietary supplement labeling and are advised to use appropriate unit conversions, Unit Conversions.

DSDCAT: Ingredient Category
There are ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

DSDBCNAM: Blend Component Name
These are the ingredient names found within a blend. Blends in products will not give the actual breakdown of ingredient quantities in the blend. The ingredients will usually just be listed, and most of the time a total blend amount is given. The blend ingredients are released in file DSBI.

DSDBID: Blend Component ID
These are new unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCLID: Blend Component ID-Old version
These are old unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCCAT: Blend Component Category
There are blend ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

Variables created for the NHANES-DSD public release files

DSDBLFLG: Blend Flag
This indicator variable denotes whether an ingredient is a blend or not a blend. If the ingredient is a blend, then blend ingredients are contained in file DSBI.

DSDCNTV: Number of Vitamin(s) in Supplement
This variable indicates the number of vitamins in the supplement, including in blends. All ingredients categorized as "vitamins" in variables DSDCAT and DSDBCCAT are added up to get the number of vitamins in the product. If a product has the same vitamin listed as an ingredient as well as a blend, this vitamin was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTM: Number of Mineral(s) in Supplement
This variable indicates the number of minerals in the supplement, including in blends. All ingredients categorized as "minerals" in variables DSDCAT and DSDBCCAT are added up to get the number of minerals in the product. If a product has the same mineral listed as an ingredient as well as a blend, this mineral was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTA: Number of Amino Acid(s) in Supplement
This variable indicates the number of amino acids in the supplement, including in blends. All ingredients categorized as "amino acids" in variables DSDCAT and DSDBCCAT are added up to get the number of amino acids in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTB: Number of Botanical(s) in Supplement
This variable indicates the number of botanicals in the supplement, including in blends. All ingredients categorized as "botanicals" in variables DSDCAT and DSDBCCAT are added up to get the number of botanicals in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTO: Number of Other Ingredients in Supplement
This variable indicates the number of other ingredients in the supplement, including in blends. All ingredients categorized as “other” in variables DSDCAT and DSDBCCAT are added up to get the number of other ingredients in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

The NCHS-DSD file may be linked to both the Household dietary supplement data use files and the 24-hour dietary recall interview dietary supplement files by the supplement ID number (DSDPID).

NCHS-DSD contains information on all DS and antacids reported from 1999-2020. New products will be appended as they are reported in future data releases. There are three files. The “Supplement Information” file (DSPI) and the “Ingredient Information” file (DSII) can be linked by supplement ID number (DSDPID), which is a unique product identifier. The “Ingredient Information” file (DSII) and the “Supplement Blend” file (DSBI) can then be linked by the ingredient ID number (DSDIID).

Layout of DSQ_DSD

(DSPI): Supplement Product Information

Variable Name	Label
DSDPID	Supplement ID number
DSDPRDT	Product Type
DSDSUPP	Supplement Name
DSDSRCE	Supplement Information Source
DSDTYPE	Supplement Type
DSDSERVQ	Serving Size Quantity
DSDSERVU	Serving Size Unit
DSDPREID	Previous product ID
DSDORGID	Original product ID
DSDSGPF	Sequential group formulation
DSDSEQF	Sequential formulation
DSDLINRF	Linear formulation
DSDCNTV	Count of Vitamins in the Supplement
DSDCNTM	Count of Minerals in the Supplement
DSDCNTA	Count of Amino Acids in the Supplement
DSDCNTB	Count of Botanicals in the Supplement
DSDCNTO	Count of Other Ingredients in the Supplement
DSDSUPID	Supplement ID number-Old version

(DSII): Supplement Ingredient Information

Variable Name	Label
DSDPID	Supplement ID Number
DSDSUPP	Supplement Name
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDOPER	Ingredient Operator (<, =, >)
DSDQTY	Ingredient Quantity
DSDUNIT	Ingredient Unit
DSDCAT	Ingredient Category
DSDBLFLG	Blend Flag
DSDINGID	Ingredient ID Number- Old version

(DSBI): Supplement Blend Information

Variable Name	Label
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDBID	Blend Component ID Number
DSDBCNAM	Blend Component Name
DSDBCCAT	Blend Component Category
DSDBCID	Blend Component ID Number-Old version

Analytic Notes

Source of Product Information
The best source of product information is the label itself, but when this cannot be obtained, other sources are used. Information from other sources may not always be an accurate

reflection of what is actually on the supplement label. This is true for the supplement's apparent name as well as for the ingredients. The apparent name on the container is most important, since interviewers see the supplement container and record the name as it appears to them. Differences from what appears on the label are particularly noted for information from the Internet (name and ingredients), the PDR (name), and supplement carton (name). In addition, supplement companies may change the appearance of a label and thus the apparent name without changing the content or may change content with minimal change to the label, or may change both. NCHS attempts to obtain updated labels as they come onto the market, but cannot guarantee complete success. The source of the supplement information is included in the data release.

Using Self-Reported Data
NHANES data are self-reported and recorded by interviewers, and thus may contain inconsistencies or errors. Some inconsistencies have been edited; however, users may notice situations that still need editing. Users are advised to assess the data and edit it as deemed appropriate for the analyses being undertaken.

Codebook and Frequencies

DSDIID - INGREDIENT ID NUMBER

Variable Name: DSDIID

SAS Label: INGREDIENT ID NUMBER

English Text: Ingredient ID number

Code or Value	Value Description	Count	Cumulative	Skip to Item
88 to 19815	Range of Values	35942	35942	
.	Missing	0	35942	

DSDINGR - INGREDIENT NAME

Variable Name: DSDINGR

SAS Label: INGREDIENT NAME

English Text: Ingredient Name

Code or Value	Value Description	Count	Cumulative	Skip to Item
INGREDIENT NAME	Value was recorded	35942	35942	
< blank >	Missing	0	35942	

DSDBID - BLEND COMPONENT ID

Variable Name: DSDBID

SAS Label: BLEND COMPONENT ID

English Text: Blend Component ID

Code or Value	Value Description	Count	Cumulative	Skip to Item
3 to 19809	Range of Values	35942	35942	
.	Missing	0	35942	

DSDBCNAM - BLEND COMPONENT NAME

Variable Name: DSDBCNAM

SAS Label: BLEND COMPONENT NAME

English Text: Blend component name

Code or Value	Value Description	Count	Cumulative	Skip to Item
BLEND COMPONENT NAME	Value was recorded	35942	35942	
< blank >	Missing	0	35942	

DSDBCCAT - BLEND COMPONENT CATEGORY

Variable Name: DSDBCCAT

SAS Label: BLEND COMPONENT CATEGORY

English Text: Blend component category

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Vitamin	449	449	
2	Mineral	1921	2370	
3	Botanical	20870	23240	
4	Other	10667	33907	
5	Amino acid	2035	35942	
.	Missing	0	35942	

DSDBCID - BLEND COMPONENT ID - OLD VERSION

Variable Name: DSDBCID

SAS Label: BLEND COMPONENT ID - OLD VERSION

English Text: Blend Component ID-old version

Code or Value	Value Description	Count	Cumulative	Skip to Item
BLEND COMPONENT ID - OLD VERSION	Value was recorded	24976	24976	
< blank >	Missing	10966	35942	

Appendix 1: Rules for Classifying Ingredients

VITAMINS
An ingredient is classified as a vitamin if it is:

- A single vitamin listed by its name (e.g., vitamin A)
- A standard chemical form of the vitamin (retinol, retinal, retinoic acid) in synthetic or natural form

A vitamin will be classified as "other" when it exists as:

- A precursor or provitamin to the active form of the vitamin (e.g., 7-dehydrocholesterol, a precursor to Vitamin D)
- A complex, since the ingredient content is unclear (e.g., B-complex)

The following appear in supplements as a source of vitamins and are therefore classified as a vitamin:

- Vitamin A: palmitate, vitamin A acetate, vitamin A palmitate
- Vitamin B-1/Thiamin: thiamin monophosphate or TMP, thiamin mononitrate, thiamin hydrochloride
- Vitamin B-2/Riboflavin: riboflavin mononitrate, riboflavin-5-phosphate sodium
- Vitamin B-3/Niacin
- Vitamin B-5/Pantothenic Acid: pantothenate, calcium pantothenate
- Vitamin B-6: pyridoxine hydrochloride, vitamin B6 hydrochloride
- Vitamin B-12/Cobalamin: cyanocobalamin
- Vitamin C/Ascorbic Acid: ascorbyl palmitate, sodium ascorbate
- Vitamin D/Calciferol: cholecalciferol, ergocalciferol, calcitriol
- Vitamin E/Tocopherol: d-alpha tocopheryl acid succinate, dl-alpha tocopheryl acetate, d-alpha tocopheryl acetate, d-alpha tocopherol, d-alpha tocopheryl, tocopherols, mixed tocopherols, vitamin E acetate, tocotrienol
- Vitamin K/Menadione: phytonadione
- Biotin: Choline, choline bitartrate
- Folic Acid/Folate

MINERALS

An ingredient is classified as a mineral if it is a macro or micromineral (trace element):

- in its elemental form (e.g., iron)
- is the source of the mineral in a supplement (e.g., ferrous gluconate, potassium iodide, nickel chloride)

An ingredient containing a mineral is classified as "other" when it is:

- an enzyme (e.g., boron protease)
- a complex, since the ingredient content is unclear (e.g., Trace Mineral Complex) used as an electrolyte (chloride, potassium, sodium)

The following are classified as minerals:

- Arsenic
- Copper
- Phosphorus

- Barium
- Fluoride
- Selenium
- Boron
- Iodine
- Silicon
- Bromine
- Iron
- Strontium
- Cadmium
- Magnesium
- Sulfur
- Calcium
- Manganese
- Tin
- Chromium
- Molybdenum
- Vanadium
- Cobalt
- Nickel
- Zinc

BOTANICALS

An ingredient is classified as a botanical if it is:

- part of a plant, tree, shrub, herb, etc.

Botanicals may include the following words:

- Extract, Powder
- Leaf, Root, Flower, Stem, Peel, Fruit
- Component of a botanical that specifically mentions it is from the plant (e.g., soy isoflavones, citrus bioflavonoids)

An ingredient containing a botanical is classified as "other" if it is:

- listed only as an unspecified blend
- a chemical structure derived originally from a botanical (e.g., bromelain, the enzyme found in pineapple; Alliin, a phytochemical in garlic; apple cider vinegar)

AMINO ACIDS

An ingredient is classified as an amino acid if it is an essential or nonessential amino acid. It can exist in:

- it’s free form (e.g., lycine, glutamine)
- its post-translational form with one or two added groups (e.g., Cystine, Hydroxylysine, Hydroxyproline, Dimethylglycine, and 3-methylhistidine)
- one of its isomeric forms (e.g., l-tyrosine)
- the source of an amino acid in a supplement (e.g., l-lysine monohydrochloride, glutamic acid hydrochloride)

An amino acid would be classified as "other" if it is:

- in its post-translational form with three or more added groups (Trimethylglycine, Tetramethylglycine, etc.)
- an alpha-keto acid (an amino acid with its amino group, NH3, replaced by a keto group) (e.g., -ketoglutarate)
- a residue of an amino acid (-carboxyglutamic acid also known as GLA)
- as a complex of amino acids (e.g., natural amino acid complex), since the ingredient content is unclear

The following are classified as amino acids:

- Alanine
- Glycine
- Proline
- Arginine
- Histidine
- Serine
- Asparagine
- Isoleucine
- Taurine
- Aspartic
- Acid
- Leucine
- Threonine
- Cysteine
- Lysine
- Tryptophan
- Glutamic
- Acid
- Methionine
- Tyrosine
- Glutamine
- Phenylalanine
- Valine

OTHER

The following are examples of ingredients that would be classified as "other":

- an electrolyte (e.g., chloride, potassium, sodium)
- a hormone (e.g., DHEA, cholesterol), unless if it is the active form of a vitamin (calcitriol)
- an enzyme (e.g., cellulase, glucoamylase)
- Complexes and blends (unless all components are of the same type ex. amino acid blend)
- Bioflavonoids and Isoflavones (if not associated with a plant, in which case it would be classified as a Botanical)
- Vinegars
- Phytochemicals (e.g., lutein, allin)
- Vitamin precursors (e.g., some carotenoids)

Appendix 2: Conversion Factors for Supplement Nutrient Units to Food Units and for Nutrient Compounds to Elemental Nutrients

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
Vitamin A Conversion Factors		
ALPHA CAROTENE	587	1 IU alpha carotene = 7.2 mcg vitamin A
ALPHA CAROTENE	587	1 RAE = 24 mcg alpha carotene
BETA CAROTENE	392	1 IU beta carotene = 0.6 mcg vitamin A
BETA CAROTENE	392	1667 IU beta carotene = 1 mg beta carotene
BETA CAROTENE	392	1 RAE = 12 mcg beta carotene
VITAMIN A*	360	1 IU = 0.3 mcg vitamin A
VITAMIN A*	360	1 RAE = 1 mcg vitamin A
CRYPTOXANTHIN	614	1 RAE = 24 mcg cryptoxanthin
Vitamin D Conversion Factor		
VITAMIN D†	364	40 IU vitamin D = 1 mcg
Vitamin E Conversion Factor		
VITAMIN E‡	365	1 IU = 0.67 mg vitamin E
Calcium Conversion Factors		
CALCIUM CARBONATE	543	40% elemental calcium
CALCIUM L-THREONATE	4003	12.9 % elemental calcium
CALCIUM PANTOTHENATE	395	91.6% pantothenate
Iron Conversion Factors		
FERROUS FUMARATE	782	32.9% elemental iron
IRON FERROCHEL	13380	27.65% elemental iron
Glucosamine Conversion Factors		
GLUCOSAMINE HYDROCHLORIDE	410	83.0% glucosamine
GLUCOSAMINE SULFATE	143	65% glucosamine
GLUCOSAMINE SULFATE .2 KCL	850	29.6% glucosamine
D-GLUCOSAMINE SULFATE.2 NACL	1017	31.3% glucosamine
Magnesium Conversion Factors		
MAGNESIUM ASPARTATE	13643	8.42% elemental magnesium
MAGNESIUM CARBONATE	556	28.9% elemental magnesium
MAGNESIUM HYDROXIDE	544	41.7% elemental magnesium
MAGNESIUM PHOSPHATE TRIBASIC	616	27.7% elemental magnesium
MAGNESIUM TRISILICATE	2054	18.3 % elemental magnesium
Vitamin B-6 Conversion Factor		
PYRIDOXINE HYDROCHLORIDE	470	82% vitamin B-6
Other		
ALUMINUM HYDROXIDE GEL	14221	34.59% elemental aluminum

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
CHROMIUM NICOTINATE	14270	12.43% elemental chromium
CHROMIUM PICOLINATE	487	12.4% elemental chromium
CHOLINE BITARATE	82	41% choline
CHOLINE CITRATE	2248	41% choline
CREATINE MONOHYDRATE	480	88% creatine
CYSTEINE HCL	776	76.9% cysteine
DOCUSATE SODIUM	109	5.1% sodium
GLUTAMIC ACID HYDROCLORIDE	653	80.1% glutamic acid
L-CYSTEINE HCL	488	69.0% cysteine
L-GLUTAMIC ACID HCL	611	80.1% glutamic acid
L-LYSINE HCL	743	80.03% lysine
LYSINE HYDROCHLORIDE	2088	80.03% lysine
POTASSIUM CHOLRIDE	287	52.5% elemental potassium
POTASSIUM PHOSPHATE	575	28.7% elemental potassium
POTASSIUM PHOSPHATE MONOBASIC	615	28.7% elemental potassium
THIAMIN MONONITRATE	468	92% thiamin
ZINC PICOLINATE	2629	21.1% elemental zinc
Basic Unit Conversion		
1 gm = 1000 mg		
1 mg = 1000 mcg		

* Conversion factor used for Vitamin A is Retinol, most common form

† Conversion factor for Calciferol

‡ Conversion factor for Alpha Tocopherol, most common form

National Health and Nutrition Examination Survey

1999-2023 Data Documentation, Codebook, and Frequencies

Dietary Supplement Database - Ingredient Information (DSII)

Data File: DSII.xpt

First Published: February 2025

Last Revised: NA

Component Description

The NHANES Dietary Supplement Database (NHANES-DSD) contains detailed information on the dietary supplements (DS) and non-prescription antacids containing calcium and/or magnesium (antacids) reported by survey participants since NHANES 1999. The NHANES-DSD release consists of three datasets containing information on the following: Dietary Supplement Product Information (DSPI), Dietary Supplement Ingredient Information (DSII), and Dietary Supplement Blend Information (DSBI).

Dietary supplement information from the in-house NCHS Product Label Database (PLD) is publicly released in three files that make up the NHANES-DSD. These files incorporate all products that have been reported by respondents since 1999 from the PLD. With subsequent releases, new products reported will be appended to the NHANES-DSD files.

The in-house PLD database is maintained by NCHS nutritionists. NCHS attempts to obtain a product label for all dietary supplements or antacids reported by participants from sources such as the manufacturer or retailer, the Internet, company catalogs, and the Physician's Desk Reference (PDR). Label information has also been obtained from the Dietary Supplement Label Database (DSLD), which is a joint project of the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) and National Library of Medicine (NLM). The DSLD contains the full label contents from a sample of dietary supplement products marketed in the U.S. Selected label information is then entered into the PLD including, but not limited to: supplement name; manufacturer and/or distributor; serving size; form of serving size; and ingredients and amounts. The ingredient information entered into the database is taken directly from the supplement facts box on the label or carton. The PLD is used to assist with data editing and then publicly released in the data files (DSPI, DSII, and DSBI).

In addition to entering labels into the PLD database, NCHS created generic and default dietary supplements and antacids, which are also maintained in the database. Generics were created in the database when we had information about a reported supplement, such as the strength of all ingredients, but no brand name. These were generally single ingredient supplements, which included a strength (e.g., vitamin C 500 mg) or multiple vitamins and/or mineral supplements made by a private label manufacturer that was known to us and for which we had a label with identical ingredients and strengths (e.g., brand X all-purpose multivitamin was reported, and we had a label for brand Y, made by the same manufacturer). When all ingredient strengths were unknown, a default supplement was created in the database. Defaults were created for single ingredient and multiple ingredient supplements based on our own data of most frequently reported supplements of that type, based on the survey cycle in which the data was collected. Created default products and the actual products or strengths that were assigned to these defaults are listed in the documentation for the files associated with participants' use of dietary supplements, located on the NHANES website under the dietary data links.

In 2019, the application used to manage and access NHANES dietary supplement product database was updated to adapt to the .NET environment. This resulted in the new structure of dietary supplement identifiers. Variables DSDPID, DSDIID, DSDBID were added to indicate the updated supplement ID, ingredient ID, and blend component ID, respectively. The variables

DSDSUPID, DSDINGID, and DSDBCID now indicate the old versions for supplement ID, ingredient ID, and blend component ID, respectively. In addition, six new variables were included to provide information on supplement reformulations: DSDPRDT, DSDPREID, DSDORGID, DSDSGPF, DSDSEQF, DSDLINRF to indicate the product type, previous product ID, original product ID, sequential group formulation, sequential formulation, and linear formulation, respectively.

Data Processing and Editing

The in-house PLD was used for processing and editing of the dietary supplement data since 1999.

Variables pulled directly from the in-house PLD

DSDSUPP: Name of Supplement
This is the name from the front of the product label. The brand name is generally entered first (i.e., Nature's Way) and then the actual product name (i.e., lutein). Information such as the strength (i.e., 60 mg extract) of the product and other qualifiers that help distinguish a product from a similar product (i.e., mega, super, high potency, time release, chewable, extract) are also listed if they are on the front of the product label. Words like "dietary supplement" and "health claims" are not entered as part of the name.

DSDPID: Supplement ID Number
New Supplement ID is a unique number assigned to each product entered in the inhouse PLD.

DSDPRDT: Product Type
If the supplement was entered by NCHS as a regular product, it is equal to 1. If the supplement was created by NCHS as a generic product, it is equal to 2. Otherwise, if the supplement was created by NCHS as a default product, it is equal to 3. Previous Variable DSDGENRC was replaced by code "2" in DSDPRDT.

DSDSUPID: Supplement ID Number-Old version
Old Supplement ID is a 10-digit identifier assigned to each product entered into the in-house PLD. All identifiable products have a supplement ID beginning with the number "1". The next 3 digits (positions 2-4) are "888" or "881" if the supplement was created by NCHS as a generic or default product; otherwise the digits in positions 2-4 are coded "000" or "001". The next 4 digits (positions 5-8) are sequentially assigned by the system for each product. The last 2 digits (positions 9-10) indicate formulations of the same supplement: the first formulation entered into the database = 00, the first reformulation = 01, the next = 02, etc. Note that these are reformulations of the same product. Different versions of products (e.g., liquid vs. tablet, with iron vs. without iron, regular vs. high potency) are considered as different products thus have different 4-digit numbers in positions 5 to 8. When a product name was entered as "no product information available," "refused," or "don't know," its ID number starts with a string of 6s.

DSDSRCE: Supplement Information Source
The source of each product label is recorded into the PLD. Generic and default products do not have a source code.

DSDTYPE: Supplement Type
The type of supplements is recorded into the database. The supplement types are: infant/pediatric, prenatal, mature, or standard . Products are coded as infant/pediatric when the product name states "infant, children, child, or kid/s" or has an indication in the title, label, or the form (e.g., animal shapes) of the supplement that it is intended for children. If this is not the case, but the suggested dose or directions indicate dosage for children only, then the code is infant/pediatric, but if dosages for adults are also included, then the product is coded as standard. Products are coded as "Prenatal" when the product name states prenatal or a derivative of this name or has an indication in the title or label that it is intended for pregnant women. If this is not the case, but the suggested dosage or directions indicate dosage for pregnant women only, the product is coded as a prenatal, but if dosages for non-pregnant adults are also included, the product is coded as standard. Products are coded as "mature"

when the product name or label includes words such as "mature, senior, geriatric, post-menopausal, or silver" or indicates with other words that it is intended for individuals 50 years and over. All other products are coded as "standard".

DSDSERVQ: Serving Size Quantity
This is the “serving size quantity,” which is recorded from the product label’s supplement facts panel for which the nutrient amounts are based on.

DSDSERVU: Serving Size Unit
This is the “serving size unit,” which is recorded from the product label’s supplement facts panel.

DSDPREID: Previous product ID
This represents the parent product from which this current product was derived.

DSDORGID: Original product ID
This represents the original or root product from which this formulation or product was derived.

DSDSGPF: Sequential group formulation
This represents the sequential order of creation from the original product within the group of identical product types.

DSDSEQF: Sequential formulation
This represents the sequential order of creation from the original product.

DSDLINRF: Linear formulation
This represents the direct line of descendancy order of creation from the original product. For example- original product being level 1, a product derived from original level 1 product being level 2 and a product derived from the level 2 product being level 3, and so on.

DSDIID: Ingredient ID
This is the new unique ingredient ID created by the PLD for each ingredient recorded from the product label’s supplement facts panel.

DSDINGID: Ingredient ID-Old version
This is the old unique ingredient ID created by the PLD for each ingredient recorded from the product label’s supplement facts panel.

DSDINGR: Ingredient Name
Ingredient names are recorded from the product label’s supplement facts panel.

DSDOPER: Ingredient Operator
This is a symbol =, <, or > that comes from the product label’s supplement facts panel.

DSDQTY: Ingredient Quantity
Ingredient quantity is recorded for each ingredient listed from the product label’s supplement facts panel. Some nutrient amounts are for a nutrient compound (generally a foreign-made product or an antacid) and these must be converted to an elemental nutrient amount (See Appendix 2 for recommended conversions).

DSDUNIT: Ingredient Unit
Ingredient unit is recorded for each ingredient listed from the product label’s supplement facts panel.

NHANES dietary supplement files have vitamin A and E expressed in different units based on what was reported on the product label’s supplement facts panel. Vitamin E may be reported in International Units (IU) or micrograms (µg). Vitamin A may be reported as International Units

(IU) or as micrograms (µg) of retinol activity equivalents (RAE). Users should be aware of changes in dietary supplement labeling and are advised to use appropriate unit conversions, Unit Conversions.

DSDCAT: Ingredient Category
There are ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

DSDBCNAM: Blend Component Name
These are the ingredient names found within a blend. Blends in products will not give the actual breakdown of ingredient quantities in the blend. The ingredients will usually just be listed, and most of the time a total blend amount is given. The blend ingredients are released in file DSBI.

DSDBID: Blend Component ID
These are new unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCLID: Blend Component ID-Old version
These are old unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCCAT: Blend Component Category
There are blend ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

Variables created for the NHANES-DSD public release files

DSDBLFLG: Blend Flag
This indicator variable denotes whether an ingredient is a blend or not a blend. If the ingredient is a blend, then blend ingredients are contained in file DSBI.

DSDCNTV: Number of Vitamin(s) in Supplement
This variable indicates the number of vitamins in the supplement, including in blends. All ingredients categorized as “vitamins” in variables DSDCAT and DSDBCCAT are added up to get the number of vitamins in the product. If a product has the same vitamin listed as an ingredient as well as a blend, this vitamin was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTM: Number of Mineral(s) in Supplement
This variable indicates the number of minerals in the supplement, including in blends. All ingredients categorized as “minerals” in variables DSDCAT and DSDBCCAT are added up to get the number of minerals in the product. If a product has the same mineral listed as an ingredient as well as a blend, this mineral was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTA: Number of Amino Acid(s) in Supplement
This variable indicates the number of amino acids in the supplement, including in blends. All ingredients categorized as “amino acids” in variables DSDCAT and DSDBCCAT are added up to get the number of amino acids in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTB: Number of Botanical(s) in Supplement
This variable indicates the number of botanicals in the supplement, including in blends. All ingredients categorized as “botanicals” in variables DSDCAT and DSDBCCAT are added up to get the number of botanicals in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTO: Number of Other Ingredients in Supplement
This variable indicates the number of other ingredients in the supplement, including in blends. All ingredients categorized as "other" in variables DSDCAT and DSDBCCAT are added up to get the number of other ingredients in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

The NCHS-DSD file may be linked to both the Household dietary supplement data use files and the 24-hour dietary recall interview dietary supplement files by the supplement ID number (DSDPID).

NCHS-DSD contains information on all DS and antacids reported from 1999-2020. New products will be appended as they are reported in future data releases. There are three files. The "Supplement Information" file (DSPI) and the "Ingredient Information" file (DSII) can be linked by supplement ID number (DSDPID), which is a unique product identifier. The "Ingredient Information" file (DSII) and the "Supplement Blend" file (DSBI) can then be linked by the ingredient ID number (DSDIID).

Layout of DSQ_DSD

(DSPI): Supplement Product Information

Variable Name	Label
DSDPID	Supplement ID number
DSDPRDT	Product Type
DSDSUPP	Supplement Name
DSDSRCE	Supplement Information Source
DSDTYPE	Supplement Type
DSDSERVQ	Serving Size Quantity
DSDSERVU	Serving Size Unit
DSDPREID	Previous product ID
DSDORGID	Original product ID
DSDSGPF	Sequential group formulation
DSDSEQF	Sequential formulation
DSDLINRF	Linear formulation
DSDCNTV	Count of Vitamins in the Supplement
DSDCNTM	Count of Minerals in the Supplement
DSDCNTA	Count of Amino Acids in the Supplement
DSDCNTB	Count of Botanicals in the Supplement
DSDCNTO	Count of Other Ingredients in the Supplement
DSDSUPID	Supplement ID number-Old version

(DSII): Supplement Ingredient Information

Variable Name	Label
DSDPID	Supplement ID Number
DSDSUPP	Supplement Name
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDOPER	Ingredient Operator (<, =, >)
DSDQTY	Ingredient Quantity
DSDUNIT	Ingredient Unit
DSDCAT	Ingredient Category
DSDBLFLG	Blend Flag
DSDINGID	Ingredient ID Number- Old version

(DSBI): Supplement Blend Information

Variable Name	Label
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDBID	Blend Component ID Number
DSDBCNAM	Blend Component Name
DSDBCCAT	Blend Component Category
DSDBCID	Blend Component ID Number-Old version

Analytic Notes

Source of Product Information
The best source of product information is the label itself, but when this cannot be obtained, other sources are used. Information from other sources may not always be an accurate

reflection of what is actually on the supplement label. This is true for the supplement's apparent name as well as for the ingredients. The apparent name on the container is most important, since interviewers see the supplement container and record the name as it appears to them. Differences from what appears on the label are particularly noted for information from the Internet (name and ingredients), the PDR (name), and supplement carton (name). In addition, supplement companies may change the appearance of a label and thus the apparent name without changing the content or may change content with minimal change to the label, or may change both. NCHS attempts to obtain updated labels as they come onto the market, but cannot guarantee complete success. The source of the supplement information is included in the data release.

Using Self-Reported Data
NHANES data are self-reported and recorded by interviewers, and thus may contain inconsistencies or errors. Some inconsistencies have been edited; however, users may notice situations that still need editing. Users are advised to assess the data and edit it as deemed appropriate for the analyses being undertaken.

Codebook and Frequencies

DSDPID - SUPPLEMENT ID NUMBER

Variable Name: DSDPID

SAS Label: SUPPLEMENT ID NUMBER

English Text: Supplement ID number

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28553	Range of Values	188572	188572	
.	Missing	0	188572	

DSDSUPP - SUPPLEMENT NAME

Variable Name: DSDSUPP

SAS Label: SUPPLEMENT NAME

English Text: Supplement name

Code or Value	Value Description	Count	Cumulative	Skip to Item
SUPPLEMENT NAME	Value was recorded	188572	188572	
< blank >	Missing	0	188572	

DSDIID - INGREDIENT ID NUMBER

Variable Name: DSDIID

SAS Label: INGREDIENT ID NUMBER

English Text: Ingredient ID number

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 19815	Range of Values	188572	188572	
.	Missing	0	188572	

DSDINGR - INGREDIENT NAME

Variable Name: DSDINGR

SAS Label: INGREDIENT NAME

English Text: Ingredient Name

Code or Value	Value Description	Count	Cumulative	Skip to Item
INGREDIENT NAME	Value was recorded	188572	188572	
< blank >	Missing	0	188572	

DSDOPER - INGREDIENT OPERATOR

Variable Name: DSDOPER

SAS Label: INGREDIENT OPERATOR

English Text: Ingredient operator

Code or Value	Value Description	Count	Cumulative	Skip to Item
INGREDIENT OPERATOR	Value was recorded	188572	188572	
< blank >	Missing	0	188572	

DSDQTY - INGREDIENT QUANTITY

Variable Name: DSDQTY

SAS Label: INGREDIENT QUANTITY

English Text: Ingredient quantity

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 6000000	Range of Values	188571	188571	
.	Missing	1	188572	

DSDUNIT - INGREDIENT UNIT

Variable Name:	DSDUNIT
SAS Label:	INGREDIENT UNIT
English Text:	Ingredient unit

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	mg	109902	109902	
2	IU	14717	124619	
3	%	2770	127389	
4	mcg	36264	163653	
5	gm	14327	177980	
6	mL	87	178067	
7	kcal	6896	184963	
8	DU	121	185084	
9	HUT	165	185249	
10	LU	85	185334	
11	CU	120	185454	
12	endo-PGO	11	185465	
13	AGU	35	185500	
14	PPM	26	185526	
15	Million	227	185753	
16	Billion	680	186433	
17	LacU	32	186465	
18	X	0	186465	
19	PPB	0	186465	
20	Trace	57	186522	
21	Unknown	668	187190	
22	PU	48	187238	
23	SEU	4	187242	
24	InvU	8	187250	
25	°DP	28	187278	
26	HCU	25	187303	
27	CFU	8	187311	
28	GALU	27	187338	
29	ALU	40	187378	
30	FTU	8	187386	
31	NG	3	187389	
32	mcg DFE	798	188187	
33	mcg RAE	179	188366	
34	mg NE	115	188481	
35	SAPU	10	188491	
36	SU	9	188500	
37	FIP	15	188515	
38	GDU	5	188520	
39	SKB	3	188523	

Code or Value	Value Description	Count	Cumulative	Skip to Item
40	UNIT	20	188543	
41	ENDO-PG	1	188544	
42	BAU	1	188545	
43	HSU	1	188546	
44	BGU	5	188551	
45	AGSU	1	188552	
46	PC	2	188554	
47	SPU	2	188556	
48	XU	4	188560	
49	ENDO-PGU	2	188562	
50	DPPU	1	188563	
51	PGU	1	188564	
52	DPPIV	1	188565	
53	APU	1	188566	
54	USP	3	188569	
55	AG	0	188569	
56	FCC	3	188572	
.	Missing	0	188572	

DSDCAT - INGREDIENT CATEGORY

Variable Name: DSDCAT

SAS Label: INGREDIENT CATEGORY

English Text: Ingredient category

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Vitamin	66877	66877	
2	Mineral	43731	110608	
3	Botanical	14174	124782	
4	Other	54053	178835	
5	Amino acid	4042	182877	
.	Missing	5695	188572	

DSDBLFLG - BLEND FLAG

Variable Name: DSDBLFLG

SAS Label: BLEND FLAG

English Text: Blend flag

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Ingredient is a blend	5695	5695	
2	Ingredient is not a blend	182877	188572	
.	Missing	0	188572	

DSDINGID - INGREDIENT ID NUMBER - OLD VERSION

Variable Name: DSDINGID

SAS Label: INGREDIENT ID NUMBER - OLD VERSION

English Text: Ingredient ID number - old version

Code or Value	Value Description	Count	Cumulative	Skip to Item
INGREDIENT ID NUMBER - OLD VERSION	Value was recorded	142831	142831	
< blank >	Missing	45741	188572	

Appendix 1: Rules for Classifying Ingredients

VITAMINS

An ingredient is classified as a vitamin if it is:

- A single vitamin listed by its name (e.g., vitamin A)
- A standard chemical form of the vitamin (retinol, retinal, retinoic acid) in synthetic or natural form

A vitamin will be classified as "other" when it exists as:

- A precursor or provitamin to the active form of the vitamin (e.g., 7-dehydrocholesterol, a precursor to Vitamin D)
- A complex, since the ingredient content is unclear (e.g., B-complex)

The following appear in supplements as a source of vitamins and are therefore classified as a vitamin:

- Vitamin A: palmitate, vitamin A acetate, vitamin A palmitate
- Vitamin B-1/Thiamin: thiamin monophosphate or TMP, thiamin mononitrate, thiamin hydrochloride
- Vitamin B-2/Riboflavin: riboflavin mononitrate, riboflavin-5-phosphate sodium
- Vitamin B-3/Niacin
- Vitamin B-5/Pantothenic Acid: pantothenate, calcium pantothenate
- Vitamin B-6: pyridoxine hydrochloride, vitamin B6 hydrochloride
- Vitamin B-12/Cobalamin: cyanocobalamin
- Vitamin C/Ascorbic Acid: ascorbyl palmitate, sodium ascorbate
- Vitamin D/Calciferol: cholecalciferol, ergocalciferol, calcitriol
- Vitamin E/Tocopherol: d-alpha tocopheryl acid succinate, dl-alpha tocopheryl acetate, d-alpha tocopheryl acetate, d-alpha tocopherol, d-alpha tocopheryl, tocopherols, mixed tocopherols, vitamin E acetate, tocotrienol
- Vitamin K/Menadione: phytonadione
- Biotin: Choline, choline bitartrate
- Folic Acid/Folate

MINERALS

An ingredient is classified as a mineral if it is a macro or micromineral (trace element):

- in its elemental form (e.g., iron)
- is the source of the mineral in a supplement (e.g., ferrous gluconate, potassium iodide, nickel chloride)

An ingredient containing a mineral is classified as "other" when it is:

- an enzyme (e.g., boron protease)
- a complex, since the ingredient content is unclear (e.g., Trace Mineral Complex)
  used as an electrolyte (chloride, potassium, sodium)

The following are classified as minerals:

- Arsenic
- Copper
- Phosphorus

- Barium
- Fluoride
- Selenium
- Boron
- Iodine
- Silicon
- Bromine
- Iron
- Strontium
- Cadmium
- Magnesium
- Sulfur
- Calcium
- Manganese
- Tin
- Chromium
- Molybdenum
- Vanadium
- Cobalt
- Nickel
- Zinc

BOTANICALS

An ingredient is classified as a botanical if it is:

- part of a plant, tree, shrub, herb, etc.

Botanicals may include the following words:

- Extract, Powder
- Leaf, Root, Flower, Stem, Peel, Fruit
- Component of a botanical that specifically mentions it is from the plant (e.g., soy isoflavones, citrus bioflavonoids)

An ingredient containing a botanical is classified as "other" if it is:

- listed only as an unspecified blend
- a chemical structure derived originally from a botanical (e.g., bromelain, the enzyme found in pineapple; Alliin, a phytochemical in garlic; apple cider vinegar)

AMINO ACIDS

An ingredient is classified as an amino acid if it is an essential or nonessential amino acid. It can exist in:

- it's free form (e.g., lycine, glutamine)
- its post-translational form with one or two added groups (e.g., Cystine, Hydroxylysine, Hydroxyproline, Dimethylglycine, and 3-methylhistidine)
- one of its isomeric forms (e.g., l-tyrosine)
- the source of an amino acid in a supplement (e.g., l-lysine monohydrochloride, glutamic acid hydrochloride)

An amino acid would be classified as "other" if it is:

- in its post-translational form with three or more added groups (Trimethylglycine, Tetramethylglycine, etc.)
- an alpha-keto acid (an amino acid with its amino group, NH3, replaced by a keto group) (e.g., -ketoglutarate)
- a residue of an amino acid (-carboxyglutamic acid also known as GLA)
- as a complex of amino acids (e.g., natural amino acid complex), since the ingredient content is unclear

The following are classified as amino acids:

- Alanine
- Glycine
- Proline
- Arginine
- Histidine
- Serine
- Asparagine
- Isoleucine
- Taurine
- Aspartic
- Acid
- Leucine
- Threonine
- Cysteine
- Lysine
- Tryptophan
- Glutamic
- Acid
- Methionine
- Tyrosine
- Glutamine
- Phenylalanine
- Valine

OTHER

The following are examples of ingredients that would be classified as "other":

- an electrolyte (e.g., chloride, potassium, sodium)
- a hormone (e.g., DHEA, cholesterol), unless if it is the active form of a vitamin (calcitriol)
- an enzyme (e.g., cellulase, glucoamylase)
- Complexes and blends (unless all components are of the same type ex. amino acid blend)
- Bioflavonoids and Isoflavones (if not associated with a plant, in which case it would be classified as a Botanical)
- Vinegars
- Phytochemicals (e.g., lutein, allin)
- Vitamin precursors (e.g., some carotenoids)

Appendix 2: Conversion Factors for Supplement Nutrient Units to Food Units and for Nutrient Compounds to Elemental Nutrients

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
Vitamin A Conversion Factors		
ALPHA CAROTENE	587	1 IU alpha carotene = 7.2 mcg vitamin A
ALPHA CAROTENE	587	1 RAE = 24 mcg alpha carotene
BETA CAROTENE	392	1 IU beta carotene = 0.6 mcg vitamin A
BETA CAROTENE	392	1667 IU beta carotene = 1 mg beta carotene
BETA CAROTENE	392	1 RAE = 12 mcg beta carotene
VITAMIN A*	360	1 IU = 0.3 mcg vitamin A
VITAMIN A*	360	1 RAE = 1 mcg vitamin A
CRYPTOXANTHIN	614	1 RAE = 24 mcg cryptoxanthin
Vitamin D Conversion Factor		
VITAMIN D†	364	40 IU vitamin D = 1 mcg
Vitamin E Conversion Factor		
VITAMIN E‡	365	1 IU = 0.67 mg vitamin E
Calcium Conversion Factors		
CALCIUM CARBONATE	543	40% elemental calcium
CALCIUM L-THREONATE	4003	12.9 % elemental calcium
CALCIUM PANTOTHENATE	395	91.6% pantothenate
Iron Conversion Factors		
FERROUS FUMARATE	782	32.9% elemental iron
IRON FERROCHEL	13380	27.65% elemental iron
Glucosamine Conversion Factors		
GLUCOSAMINE HYDROCHLORIDE	410	83.0% glucosamine
GLUCOSAMINE SULFATE	143	65% glucosamine
GLUCOSAMINE SULFATE .2 KCL	850	29.6% glucosamine
D-GLUCOSAMINE SULFATE.2 NACL	1017	31.3% glucosamine
Magnesium Conversion Factors		
MAGNESIUM ASPARTATE	13643	8.42% elemental magnesium
MAGNESIUM CARBONATE	556	28.9% elemental magnesium
MAGNESIUM HYDROXIDE	544	41.7% elemental magnesium
MAGNESIUM PHOSPHATE TRIBASIC	616	27.7% elemental magnesium
MAGNESIUM TRISILICATE	2054	18.3 % elemental magnesium
Vitamin B-6 Conversion Factor		
PYRIDOXINE HYDROCHLORIDE	470	82% vitamin B-6
Other		
ALUMINUM HYDROXIDE GEL	14221	34.59% elemental aluminum

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
CHROMIUM NICOTINATE	14270	12.43% elemental chromium
CHROMIUM PICOLINATE	487	12.4% elemental chromium
CHOLINE BITARATE	82	41% choline
CHOLINE CITRATE	2248	41% choline
CREATINE MONOHYDRATE	480	88% creatine
CYSTEINE HCL	776	76.9% cysteine
DOCUSATE SODIUM	109	5.1% sodium
GLUTAMIC ACID HYDROCLORIDE	653	80.1% glutamic acid
L-CYSTEINE HCL	488	69.0% cysteine
L-GLUTAMIC ACID HCL	611	80.1% glutamic acid
L-LYSINE HCL	743	80.03% lysine
LYSINE HYDROCHLORIDE	2088	80.03% lysine
POTASSIUM CHOLRIDE	287	52.5% elemental potassium
POTASSIUM PHOSPHATE	575	28.7% elemental potassium
POTASSIUM PHOSPHATE MONOBASIC	615	28.7% elemental potassium
THIAMIN MONONITRATE	468	92% thiamin
ZINC PICOLINATE	2629	21.1% elemental zinc
Basic Unit Conversion		
1 gm = 1000 mg		
1 mg = 1000 mcg		

* Conversion factor used for Vitamin A is Retinol, most common form

† Conversion factor for Calciferol

‡ Conversion factor for Alpha Tocopherol, most common form

National Health and Nutrition Examination Survey

1999-2023 Data Documentation, Codebook, and Frequencies

Dietary Supplement Database - Product Information (DSPI)

Data File: DSPI.xpt

First Published: February 2025

Last Revised: NA

Component Description

The NHANES Dietary Supplement Database (NHANES-DSD) contains detailed information on the dietary supplements (DS) and non-prescription antacids containing calcium and/or magnesium (antacids) reported by survey participants since NHANES 1999. The NHANES-DSD release consists of three datasets containing information on the following: Dietary Supplement Product Information (DSPI), Dietary Supplement Ingredient Information (DSII), and Dietary Supplement Blend Information (DSBI).

Dietary supplement information from the in-house NCHS Product Label Database (PLD) is publicly released in three files that make up the NHANES-DSD. These files incorporate all products that have been reported by respondents since 1999 from the PLD. With subsequent releases, new products reported will be appended to the NHANES-DSD files.

The in-house PLD database is maintained by NCHS nutritionists. NCHS attempts to obtain a product label for all dietary supplements or antacids reported by participants from sources such as the manufacturer or retailer, the Internet, company catalogs, and the Physician's Desk Reference (PDR). Label information has also been obtained from the Dietary Supplement Label Database (DSLD), which is a joint project of the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) and National Library of Medicine (NLM). The DSLD contains the full label contents from a sample of dietary supplement products marketed in the U.S. Selected label information is then entered into the PLD including, but not limited to: supplement name; manufacturer and/or distributor; serving size; form of serving size; and ingredients and amounts. The ingredient information entered into the database is taken directly from the supplement facts box on the label or carton. The PLD is used to assist with data editing and then publicly released in the data files (DSPI, DSII, and DSBI).

In addition to entering labels into the PLD database, NCHS created generic and default dietary supplements and antacids, which are also maintained in the database. Generics were created in the database when we had information about a reported supplement, such as the strength of all ingredients, but no brand name. These were generally single ingredient supplements, which included a strength (e.g., vitamin C 500 mg) or multiple vitamins and/or mineral supplements made by a private label manufacturer that was known to us and for which we had a label with identical ingredients and strengths (e.g., brand X all-purpose multivitamin was reported, and we had a label for brand Y, made by the same manufacturer). When all ingredient strengths were unknown, a default supplement was created in the database. Defaults were created for single ingredient and multiple ingredient supplements based on our own data of most frequently reported supplements of that type, based on the survey cycle in which the data was collected. Created default products and the actual products or strengths that were assigned to these defaults are listed in the documentation for the files associated with participants' use of dietary supplements, located on the NHANES website under the dietary data links.

In 2019, the application used to manage and access NHANES dietary supplement product database was updated to adapt to the .NET environment. This resulted in the new structure of dietary supplement identifiers. Variables DSDPID, DSDIID, DSDBID were added to indicate the updated supplement ID, ingredient ID, and blend component ID, respectively. The variables

DSDSUPID, DSDINGID, and DSDBCID now indicate the old versions for supplement ID, ingredient ID, and blend component ID, respectively. In addition, six new variables were included to provide information on supplement reformulations: DSDPRDT, DSDPREID, DSDORGID, DSDSGPF, DSDSEQF, DSDLINRF to indicate the product type, previous product ID, original product ID, sequential group formulation, sequential formulation, and linear formulation, respectively.

Data Processing and Editing

The in-house PLD was used for processing and editing of the dietary supplement data since 1999.

Variables pulled directly from the in-house PLD

DSDSUPP: Name of Supplement
This is the name from the front of the product label. The brand name is generally entered first (i.e., Nature's Way) and then the actual product name (i.e., lutein). Information such as the strength (i.e., 60 mg extract) of the product and other qualifiers that help distinguish a product from a similar product (i.e., mega, super, high potency, time release, chewable, extract) are also listed if they are on the front of the product label. Words like "dietary supplement" and "health claims" are not entered as part of the name.

DSDPID: Supplement ID Number
New Supplement ID is a unique number assigned to each product entered in the inhouse PLD.

DSDPRDT: Product Type
If the supplement was entered by NCHS as a regular product, it is equal to 1. If the supplement was created by NCHS as a generic product, it is equal to 2. Otherwise, if the supplement was created by NCHS as a default product, it is equal to 3. Previous Variable DSDGENRC was replaced by code "2" in DSDPRDT.

DSDSUPID: Supplement ID Number-Old version
Old Supplement ID is a 10-digit identifier assigned to each product entered into the in-house PLD. All identifiable products have a supplement ID beginning with the number "1". The next 3 digits (positions 2-4) are "888" or "881" if the supplement was created by NCHS as a generic or default product; otherwise the digits in positions 2-4 are coded "000" or "001". The next 4 digits (positions 5-8) are sequentially assigned by the system for each product. The last 2 digits (positions 9-10) indicate formulations of the same supplement: the first formulation entered into the database = 00, the first reformulation = 01, the next = 02, etc. Note that these are reformulations of the same product. Different versions of products (e.g., liquid vs. tablet, with iron vs. without iron, regular vs. high potency) are considered as different products thus have different 4-digit numbers in positions 5 to 8. When a product name was entered as "no product information available," "refused," or "don't know," its ID number starts with a string of 6s.

DSDSRCE: Supplement Information Source
The source of each product label is recorded into the PLD. Generic and default products do not have a source code.

DSDTYPE: Supplement Type
The type of supplements is recorded into the database. The supplement types are: infant/pediatric, prenatal, mature, or standard . Products are coded as infant/pediatric when the product name states "infant, children, child, or kid/s" or has an indication in the title, label, or the form (e.g., animal shapes) of the supplement that it is intended for children. If this is not the case, but the suggested dose or directions indicate dosage for children only, then the code is infant/pediatric, but if dosages for adults are also included, then the product is coded as standard. Products are coded as "Prenatal" when the product name states prenatal or a derivative of this name or has an indication in the title or label that it is intended for pregnant women. If this is not the case, but the suggested dosage or directions indicate dosage for pregnant women only, the product is coded as a prenatal, but if dosages for non-pregnant adults are also included, the product is coded as standard. Products are coded as "mature"

when the product name or label includes words such as "mature, senior, geriatric, post-menopausal, or silver" or indicates with other words that it is intended for individuals 50 years and over. All other products are coded as "standard".

DSDSERVQ: Serving Size Quantity
This is the “serving size quantity,” which is recorded from the product label’s supplement facts panel for which the nutrient amounts are based on.

DSDSERVU: Serving Size Unit
This is the “serving size unit,” which is recorded from the product label’s supplement facts panel.

DSDPREID: Previous product ID
This represents the parent product from which this current product was derived.

DSDORGID: Original product ID
This represents the original or root product from which this formulation or product was derived.

DSDSGPF: Sequential group formulation
This represents the sequential order of creation from the original product within the group of identical product types.

DSDSEQF: Sequential formulation
This represents the sequential order of creation from the original product.

DSDLINRF: Linear formulation
This represents the direct line of descendancy order of creation from the original product. For example- original product being level 1, a product derived from original level 1 product being level 2 and a product derived from the level 2 product being level 3, and so on.

DSDIID: Ingredient ID
This is the new unique ingredient ID created by the PLD for each ingredient recorded from the product label’s supplement facts panel.

DSDINGID: Ingredient ID-Old version
This is the old unique ingredient ID created by the PLD for each ingredient recorded from the product label’s supplement facts panel.

DSDINGR: Ingredient Name
Ingredient names are recorded from the product label’s supplement facts panel.

DSDOPER: Ingredient Operator
This is a symbol =, <, or > that comes from the product label’s supplement facts panel.

DSDQTY: Ingredient Quantity
Ingredient quantity is recorded for each ingredient listed from the product label’s supplement facts panel. Some nutrient amounts are for a nutrient compound (generally a foreign-made product or an antacid) and these must be converted to an elemental nutrient amount (See Appendix 2 for recommended conversions).

DSDUNIT: Ingredient Unit
Ingredient unit is recorded for each ingredient listed from the product label’s supplement facts panel.

NHANES dietary supplement files have vitamin A and E expressed in different units based on what was reported on the product label’s supplement facts panel. Vitamin E may be reported in International Units (IU) or micrograms (μg). Vitamin A may be reported as International Units

(IU) or as micrograms (μg) of retinol activity equivalents (RAE). Users should be aware of changes in dietary supplement labeling and are advised to use appropriate unit conversions, Unit Conversions.

DSDCAT: Ingredient Category
There are ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

DSDBCNAM: Blend Component Name
These are the ingredient names found within a blend. Blends in products will not give the actual breakdown of ingredient quantities in the blend. The ingredients will usually just be listed, and most of the time a total blend amount is given. The blend ingredients are released in file DSBI.

DSDBID: Blend Component ID
These are new unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCLID: Blend Component ID-Old version
These are old unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCCAT: Blend Component Category
There are blend ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

Variables created for the NHANES-DSD public release files

DSDBLFLG: Blend Flag
This indicator variable denotes whether an ingredient is a blend or not a blend. If the ingredient is a blend, then blend ingredients are contained in file DSBI.

DSDCNTV: Number of Vitamin(s) in Supplement
This variable indicates the number of vitamins in the supplement, including in blends. All ingredients categorized as "vitamins" in variables DSDCAT and DSDBCCAT are added up to get the number of vitamins in the product. If a product has the same vitamin listed as an ingredient as well as a blend, this vitamin was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTM: Number of Mineral(s) in Supplement
This variable indicates the number of minerals in the supplement, including in blends. All ingredients categorized as "minerals" in variables DSDCAT and DSDBCCAT are added up to get the number of minerals in the product. If a product has the same mineral listed as an ingredient as well as a blend, this mineral was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTA: Number of Amino Acid(s) in Supplement
This variable indicates the number of amino acids in the supplement, including in blends. All ingredients categorized as "amino acids" in variables DSDCAT and DSDBCCAT are added up to get the number of amino acids in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTB: Number of Botanical(s) in Supplement
This variable indicates the number of botanicals in the supplement, including in blends. All ingredients categorized as "botanicals" in variables DSDCAT and DSDBCCAT are added up to get the number of botanicals in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTO: Number of Other Ingredients in Supplement
This variable indicates the number of other ingredients in the supplement, including in blends. All ingredients categorized as “other” in variables DSDCAT and DSDBCCAT are added up to get the number of other ingredients in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

The NCHS-DSD file may be linked to both the Household dietary supplement data use files and the 24-hour dietary recall interview dietary supplement files by the supplement ID number (DSDPID).

NCHS-DSD contains information on all DS and antacids reported from 1999-2020. New products will be appended as they are reported in future data releases. There are three files. The “Supplement Information” file (DSPI) and the “Ingredient Information” file (DSII) can be linked by supplement ID number (DSDPID), which is a unique product identifier. The “Ingredient Information” file (DSII) and the “Supplement Blend” file (DSBI) can then be linked by the ingredient ID number (DSDIID).

Layout of DSQ_DSD

(DSPI): Supplement Product Information

Variable Name	Label
DSDPID	Supplement ID number
DSDPRDT	Product Type
DSDSUPP	Supplement Name
DSDSRCE	Supplement Information Source
DSDTYPE	Supplement Type
DSDSERVQ	Serving Size Quantity
DSDSERVU	Serving Size Unit
DSDPREID	Previous product ID
DSDORGID	Original product ID
DSDSGPF	Sequential group formulation
DSDSEQF	Sequential formulation
DSDLINRF	Linear formulation
DSDCNTV	Count of Vitamins in the Supplement
DSDCNTM	Count of Minerals in the Supplement
DSDCNTA	Count of Amino Acids in the Supplement
DSDCNTB	Count of Botanicals in the Supplement
DSDCNTO	Count of Other Ingredients in the Supplement
DSDSUPID	Supplement ID number-Old version

(DSII): Supplement Ingredient Information

Variable Name	Label
DSDPID	Supplement ID Number
DSDSUPP	Supplement Name
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDOPER	Ingredient Operator (<, =, >)
DSDQTY	Ingredient Quantity
DSDUNIT	Ingredient Unit
DSDCAT	Ingredient Category
DSDBLFLG	Blend Flag
DSDINGID	Ingredient ID Number- Old version

(DSBI): Supplement Blend Information

Variable Name	Label
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDBID	Blend Component ID Number
DSDBCNAM	Blend Component Name
DSDBCCAT	Blend Component Category
DSDBCID	Blend Component ID Number-Old version

Analytic Notes

Source of Product Information
The best source of product information is the label itself, but when this cannot be obtained, other sources are used. Information from other sources may not always be an accurate

reflection of what is actually on the supplement label. This is true for the supplement's apparent name as well as for the ingredients. The apparent name on the container is most important, since interviewers see the supplement container and record the name as it appears to them. Differences from what appears on the label are particularly noted for information from the Internet (name and ingredients), the PDR (name), and supplement carton (name). In addition, supplement companies may change the appearance of a label and thus the apparent name without changing the content or may change content with minimal change to the label, or may change both. NCHS attempts to obtain updated labels as they come onto the market, but cannot guarantee complete success. The source of the supplement information is included in the data release.

Using Self-Reported Data
NHANES data are self-reported and recorded by interviewers, and thus may contain inconsistencies or errors. Some inconsistencies have been edited; however, users may notice situations that still need editing. Users are advised to assess the data and edit it as deemed appropriate for the analyses being undertaken.

Codebook and Frequencies

DSDPID - SUPPLEMENT ID NUMBER

Variable Name: DSDPID

SAS Label: SUPPLEMENT ID NUMBER

English Text: Supplement ID number

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28553	Range of Values	20235	20235	
.	Missing	0	20235	

DSDPRDT - PRODUCT TYPE

Variable Name: DSDPRDT

SAS Label: PRODUCT TYPE

English Text: Product type

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Regular	17885	17885	
2	Generic	1513	19398	
3	Default	837	20235	
.	Missing	0	20235	

DSDSUPP - SUPPLEMENT NAME

Variable Name: DSDSUPP

SAS Label: SUPPLEMENT NAME

English Text: Supplement Name

Code or Value	Value Description	Count	Cumulative	Skip to Item
SUPPLEMENT NAME	Value was recorded	20235	20235	
< blank >	Missing	0	20235	

DSDSRCE - SUPPLEMENT INFORMATION SOURCE

Variable Name: DSDSRCE

SAS Label: SUPPLEMENT INFORMATION SOURCE

English Text: Supplement Information Source

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Directly from manufacturer	23	23	
2	Directly from distributor	7	30	
4	Inferred from supplement name	1	31	
5	Physician's Desk Reference (PDR)	31	62	
7	Product Catalog	149	211	
8	Internet Listing	4812	5023	
9	Supplement label or carton	11750	16773	
10	Supplement from same manufacturer	143	16916	
11	Dietary Supplement Label Database	1013	17929	
.	Missing	2306	20235	

DSDTYPE - SUPPLEMENT TYPE

Variable Name: DSDTYPE

SAS Label: SUPPLEMENT TYPE

English Text: Supplement Type (prenatal, infant\pediatric, standard, or mature)

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Infant/pediatric	1172	1172	
2	Prenatal	407	1579	
3	Mature	509	2088	
4	Standard	18147	20235	
.	Missing	0	20235	

DSDSERVQ - SERVING SIZE QUANTITY

Variable Name: DSDSERVQ

SAS Label: SERVING SIZE QUANTITY

English Text: Serving size quantity

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 60	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSERVU - SERVING SIZE UNIT

Variable Name:	DSDSERVU
SAS Label:	SERVING SIZE UNIT
English Text:	Serving size unit

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Caplet	1033	1033	
2	Capsule	4995	6028	
3	Dropper	59	6087	
4	Drop	132	6219	
5	Fluid Ounce	99	6318	
6	Gel Cap	25	6343	
9	Lozenge	49	6392	
10	Milliliter	314	6706	
12	Package/Packet	412	7118	
13	Pill	51	7169	
14	Tablespoon/Powder	106	7275	
16	Softgel	2889	10164	
17	Tablespoon/Liquid	203	10367	
18	Tablet	7096	17463	
19	Teaspoon/Liquid	267	17730	
20	Wafer	47	17777	
21	Ounce/Powder	4	17781	
22	Spray/Squirt	23	17804	
24	Scoop/Powder	410	18214	
25	Cup/Powder	8	18222	
27	Chew	1135	19357	
29	Vegicap	477	19834	
30	Can/Liquid	2	19836	
31	Capful	39	19875	
32	Gumball	3	19878	
33	Gram/Powder	94	19972	
34	Teaspoon/Powder	225	20197	
37	Cup/Liquid	1	20198	
38	Gram/Liquid	1	20199	
39	Drop/Lozenge	9	20208	
42	Strip	2	20210	
43	Bottle/Liquid	12	20222	
44	Pieces	6	20228	
99	Unknown	7	20235	
.	Missing	0	20235	

DSDPREID - PREVIOUS PRODUCT ID

Variable Name: DSDPREID

SAS Label: PREVIOUS PRODUCT ID

English Text: Previous product ID

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28550	Range of Values	5842	5842	
.	Missing	14393	20235	

DSDORGID - ORIGINAL PRODUCT ID

Variable Name: DSDORGID

SAS Label: ORIGINAL PRODUCT ID

English Text: Original product ID

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28550	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSGPF - SEQUENTIAL GROUP FORMULATION

Variable Name: DSDSGPF

SAS Label: SEQUENTIAL GROUP FORMULATION

English Text: Sequential group formulation

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 19	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSEQF - SEQUENTIAL FORMULATION

Variable Name: DSDSEQF

SAS Label: SEQUENTIAL FORMULATION

English Text: Sequential formulation

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 29	Range of Values	20235	20235	
.	Missing	0	20235	

DSDLINRF - LINEAR FORMULATION

Variable Name: DSDLINRF

SAS Label: LINEAR FORMULATION

English Text: Linear formulation

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 12	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTV - COUNT OF VITAMINS IN THE SUPPLEMENT

Variable Name: DSDCNTV

SAS Label: COUNT OF VITAMINS IN THE SUPPLEMENT

English Text: Count of vitamins in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 18	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTM - COUNT OF MINERALS IN THE SUPPLEMENT

Variable Name: DSDCNTM

SAS Label: COUNT OF MINERALS IN THE SUPPLEMENT

English Text: Count of minerals in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 72	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTA - COUNT OF AMINO ACIDS IN THE SUPPLEMENT

Variable Name: DSDCNTA

SAS Label: COUNT OF AMINO ACIDS IN THE SUPPLEMENT

English Text: Count of amino acids in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 24	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTB - COUNT OF BOTANICALS IN THE SUPPLEMENT

Variable Name: DSDCNTB

SAS Label: COUNT OF BOTANICALS IN THE SUPPLEMENT

English Text: Count of botanicals in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 125	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTO - COUNT OF OTHER INGREDIENTS IN SUPPLEMENT

Variable Name: DSDCNTO

SAS Label: COUNT OF OTHER INGREDIENTS IN SUPPLEMENT

English Text: Count of other ingredients in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 53	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSUPID - SUPPLEMENT ID NUMBER - OLD VERSION

Variable Name: DSDSUPID

SAS Label: SUPPLEMENT ID NUMBER - OLD VERSION

English Text: Supplement ID Number - old version

Code or Value	Value Description	Count	Cumulative	Skip to Item
SUPPLEMENT ID NUMBER - OLD VERSION	Value was recorded	14629	14629	
< blank >	Missing	5606	20235	

Appendix 1: Rules for Classifying Ingredients

VITAMINS

An ingredient is classified as a vitamin if it is:

- A single vitamin listed by its name (e.g., vitamin A)
- A standard chemical form of the vitamin (retinol, retinal, retinoic acid) in synthetic or natural form

A vitamin will be classified as "other" when it exists as:

- A precursor or provitamin to the active form of the vitamin (e.g., 7-dehydrocholesterol, a precursor to Vitamin D)
- A complex, since the ingredient content is unclear (e.g., B-complex)

The following appear in supplements as a source of vitamins and are therefore classified as a vitamin:

- Vitamin A: palmitate, vitamin A acetate, vitamin A palmitate
- Vitamin B-1/Thiamin: thiamin monophosphate or TMP, thiamin mononitrate, thiamin hydrochloride
- Vitamin B-2/Riboflavin: riboflavin mononitrate, riboflavin-5-phosphate sodium
- Vitamin B-3/Niacin
- Vitamin B-5/Pantothenic Acid: pantothenate, calcium pantothenate
- Vitamin B-6: pyridoxine hydrochloride, vitamin B6 hydrochloride
- Vitamin B-12/Cobalamin: cyanocobalamin
- Vitamin C/Ascorbic Acid: ascorbyl palmitate, sodium ascorbate
- Vitamin D/Calciferol: cholecalciferol, ergocalciferol, calcitriol
- Vitamin E/Tocopherol: d-alpha tocopheryl acid succinate, dl-alpha tocopheryl acetate, d-alpha tocopheryl acetate, d-alpha tocopherol, d-alpha tocopheryl, tocopherols, mixed tocopherols, vitamin E acetate, tocotrienol
- Vitamin K/Menadione: phytonadione
- Biotin: Choline, choline bitartrate
- Folic Acid/Folate

MINERALS

An ingredient is classified as a mineral if it is a macro or micromineral (trace element):

- in its elemental form (e.g., iron)
- is the source of the mineral in a supplement (e.g., ferrous gluconate, potassium iodide, nickel chloride)

An ingredient containing a mineral is classified as "other" when it is:

- an enzyme (e.g., boron protease)
- a complex, since the ingredient content is unclear (e.g., Trace Mineral Complex)
  used as an electrolyte (chloride, potassium, sodium)

The following are classified as minerals:

- Arsenic
- Copper
- Phosphorus

- Barium
- Fluoride
- Selenium
- Boron
- Iodine
- Silicon
- Bromine
- Iron
- Strontium
- Cadmium
- Magnesium
- Sulfur
- Calcium
- Manganese
- Tin
- Chromium
- Molybdenum
- Vanadium
- Cobalt
- Nickel
- Zinc

BOTANICALS

An ingredient is classified as a botanical if it is:

- part of a plant, tree, shrub, herb, etc.

Botanicals may include the following words:

- Extract, Powder
- Leaf, Root, Flower, Stem, Peel, Fruit
- Component of a botanical that specifically mentions it is from the plant (e.g., soy isoflavones, citrus bioflavonoids)

An ingredient containing a botanical is classified as "other" if it is:

- listed only as an unspecified blend
- a chemical structure derived originally from a botanical (e.g., bromelain, the enzyme found in pineapple; Alliin, a phytochemical in garlic; apple cider vinegar)

AMINO ACIDS

An ingredient is classified as an amino acid if it is an essential or nonessential amino acid. It can exist in:

- it’s free form (e.g., lycine, glutamine)
- its post-translational form with one or two added groups (e.g., Cystine, Hydroxylysine, Hydroxyproline, Dimethylglycine, and 3-methylhistidine)
- one of its isomeric forms (e.g., l-tyrosine)
- the source of an amino acid in a supplement (e.g., l-lysine monohydrochloride, glutamic acid hydrochloride)

An amino acid would be classified as "other" if it is:

- in its post-translational form with three or more added groups (Trimethylglycine, Tetramethylglycine, etc.)
- an alpha-keto acid (an amino acid with its amino group, NH3, replaced by a keto group) (e.g., -ketoglutarate)
- a residue of an amino acid (-carboxyglutamic acid also known as GLA)
- as a complex of amino acids (e.g., natural amino acid complex), since the ingredient content is unclear

The following are classified as amino acids:

- Alanine
- Glycine
- Proline
- Arginine
- Histidine
- Serine
- Asparagine
- Isoleucine
- Taurine
- Aspartic
- Acid
- Leucine
- Threonine
- Cysteine
- Lysine
- Tryptophan
- Glutamic
- Acid
- Methionine
- Tyrosine
- Glutamine
- Phenylalanine
- Valine

OTHER

The following are examples of ingredients that would be classified as "other":

- an electrolyte (e.g., chloride, potassium, sodium)
- a hormone (e.g., DHEA, cholesterol), unless if it is the active form of a vitamin (calcitriol)
- an enzyme (e.g., cellulase, glucoamylase)
- Complexes and blends (unless all components are of the same type ex. amino acid blend)
- Bioflavonoids and Isoflavones (if not associated with a plant, in which case it would be classified as a Botanical)
- Vinegars
- Phytochemicals (e.g., lutein, allin)
- Vitamin precursors (e.g., some carotenoids)

Appendix 2: Conversion Factors for Supplement Nutrient Units to Food Units and for Nutrient Compounds to Elemental Nutrients

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
Vitamin A Conversion Factors		
ALPHA CAROTENE	587	1 IU alpha carotene = 7.2 mcg vitamin A
ALPHA CAROTENE	587	1 RAE = 24 mcg alpha carotene
BETA CAROTENE	392	1 IU beta carotene = 0.6 mcg vitamin A
BETA CAROTENE	392	1667 IU beta carotene = 1 mg beta carotene
BETA CAROTENE	392	1 RAE = 12 mcg beta carotene
VITAMIN A*	360	1 IU = 0.3 mcg vitamin A
VITAMIN A*	360	1 RAE = 1 mcg vitamin A
CRYPTOXANTHIN	614	1 RAE = 24 mcg cryptoxanthin
Vitamin D Conversion Factor		
VITAMIN D†	364	40 IU vitamin D = 1 mcg
Vitamin E Conversion Factor		
VITAMIN E‡	365	1 IU = 0.67 mg vitamin E
Calcium Conversion Factors		
CALCIUM CARBONATE	543	40% elemental calcium
CALCIUM L-THREONATE	4003	12.9 % elemental calcium
CALCIUM PANTOTHENATE	395	91.6% pantothenate
Iron Conversion Factors		
FERROUS FUMARATE	782	32.9% elemental iron
IRON FERROCHEL	13380	27.65% elemental iron
Glucosamine Conversion Factors		
GLUCOSAMINE HYDROCHLORIDE	410	83.0% glucosamine
GLUCOSAMINE SULFATE	143	65% glucosamine
GLUCOSAMINE SULFATE .2 KCL	850	29.6% glucosamine
D-GLUCOSAMINE SULFATE.2 NACL	1017	31.3% glucosamine
Magnesium Conversion Factors		
MAGNESIUM ASPARTATE	13643	8.42% elemental magnesium
MAGNESIUM CARBONATE	556	28.9% elemental magnesium
MAGNESIUM HYDROXIDE	544	41.7% elemental magnesium
MAGNESIUM PHOSPHATE TRIBASIC	616	27.7% elemental magnesium
MAGNESIUM TRISILICATE	2054	18.3 % elemental magnesium
Vitamin B-6 Conversion Factor		
PYRIDOXINE HYDROCHLORIDE	470	82% vitamin B-6
Other		
ALUMINUM HYDROXIDE GEL	14221	34.59% elemental aluminum

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
CHROMIUM NICOTINATE	14270	12.43% elemental chromium
CHROMIUM PICOLINATE	487	12.4% elemental chromium
CHOLINE BITARATE	82	41% choline
CHOLINE CITRATE	2248	41% choline
CREATINE MONOHYDRATE	480	88% creatine
CYSTEINE HCL	776	76.9% cysteine
DOCUSATE SODIUM	109	5.1% sodium
GLUTAMIC ACID HYDROCLORIDE	653	80.1% glutamic acid
L-CYSTEINE HCL	488	69.0% cysteine
L-GLUTAMIC ACID HCL	611	80.1% glutamic acid
L-LYSINE HCL	743	80.03% lysine
LYSINE HYDROCHLORIDE	2088	80.03% lysine
POTASSIUM CHOLRIDE	287	52.5% elemental potassium
POTASSIUM PHOSPHATE	575	28.7% elemental potassium
POTASSIUM PHOSPHATE MONOBASIC	615	28.7% elemental potassium
THIAMIN MONONITRATE	468	92% thiamin
ZINC PICOLINATE	2629	21.1% elemental zinc
Basic Unit Conversion		
1 gm = 1000 mg		
1 mg = 1000 mcg		

* Conversion factor used for Vitamin A is Retinol, most common form

† Conversion factor for Calciferol

‡ Conversion factor for Alpha Tocopherol, most common form

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Supplement Use 30-Day - Individual Dietary Supplements (DSQIDS_L)

Data File: DSQIDS_L.xpt

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Last Revised: April 2025

Component Description

The Dietary Supplement and Antacid (DSQ) Section collects information on: 1) prescription and non-prescription dietary supplements (DS), and 2) non-prescription antacids.

In response to the COVID-19 pandemic, the mode of the 30-day dietary supplement data collection changed from personal interviews conducted in participants' homes to telephone interviews conducted after the first 24-hour dietary recall in August 2021-August 2023. Because the 30-day supplement intake was collected by telephone, the interviewer asked survey participants to read to them all the text on the supplement container label. Additionally, the 24-hour supplement intake data collection was discontinued from the 24-hour Day 1 and Day 2 dietary recall. The NHANES August 2021-August 2023 dietary supplement questions are updated to adapt to the telephone mode. Questions on the duration of dietary supplement intake (how long the supplement has been taken) and the reasons for taking the supplement were dropped. Analysts are encouraged to read the documentation to understand the names and structure of the files as well as the variables.

Questions about non-prescription antacids, which contained calcium and/or magnesium (antacids), are also included in this data file.

The DS reported in NHANES August 2021-August 2023 are detailed in the NHANES Dietary Supplement Database (NHANES-DSD) 1999-2023.

30-Day Dietary Supplements Data Files: Two data files were produced from the 30-day dietary supplement and non-prescription antacid sub-sections of the DSQ interview: Total Dietary Supplements file and Individual Dietary Supplements file.

File Name	
Individual Dietary Supplements	DSQIDS_L
Total Dietary Supplements	DSQTOT_L

Individual Dietary Supplements File (DSQIDS_L): Contains detailed information about the types and amounts of individual DS and antacids reported by each participant. The names of the variables included in this file are listed in **Appendix 1**.

The Individual Dietary Supplements File includes one record for each dietary supplement or antacid reported by a survey participant. Only participants that reported taking a dietary supplement or an antacid are included in these files. Each dietary supplement or antacid is identified by a supplement ID number (DSDPID) and each record contains, but is not limited to, the information listed below:

- Name and ID number of the supplement

- Amount of a dietary supplement consumed, which is calculated as the reported amount consumed divided by the serving size from the product label; and
- Amounts of 34 nutrients/dietary components (listed in **Appendix 2**) from each dietary supplement and antacid, as calculated using the NHANES-DSD. If data is missing for any of the variables needed to calculate the amounts consumed for the selected nutrients, this information will be missing for individuals, even though they reported a supplement(s).

This file only includes intake for a select group of nutrients. These records can be linked to the NHANES-DSD, using supplement ID numbers (DSDPID), to obtain more detailed information on reported products. The NHANES-DSD datasets provide information on nutrients in the dietary supplement as reported on the product label. Botanical ingredients would be an example of nutrients not released in the Individual Dietary Supplements files but can be obtained from the NHANES-DSD files.

Total Dietary Supplements Files (DSQTOT_L): Contains, for each participant, average daily total nutrient intakes from DS and antacids. The names for the variables are listed in **Appendix 3**.

The Total Dietary Supplement File provides a summary record of total nutrient intakes from DS and antacids for each individual. This includes users and non-users of DS and antacids. Each total intake record contains, but is not limited to, the following information:

- Whether a dietary supplement was consumed in the past 30 days
- Whether an antacid was consumed in the past 30 days
- Total number of supplements and antacids reported for that participant
- Mean daily intake aggregates of 34 nutrients/dietary components (listed in **Appendix 2**) from all supplements and antacids, as calculated using the NHANES-DSD. If data is missing for any of the variables needed to calculate mean daily intakes, this information will be missing for individuals, even though they may have reported a supplement(s).
- Starting in the 2017-2018 survey cycle, the new variable DSDPID, which indicates supplement ID, was added. The variable DSDSUPID now indicates the old version for supplement ID.

Eligible Sample

All survey participants were eligible for the DSQ.

Interview Setting and Mode of Administration

The DSQ was asked by trained dietary interviewers, using the Computer-Assisted Telephone Interview (CATI) system during the day 1 dietary recall interview. Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant’s intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child’s intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish. The August 2021-August 2022 survey questionnaire can be found on the NHANES website.

Quality Assurance & Quality Control

Data were routinely examined for discrepancies and erroneous entries. Trained nutritionists reviewed incoming data and matched reported dietary supplement entries to known supplements from the in-house NCHS Product Label Database (PLD), where possible; sought additional product labels if feasible; assigned generic or default supplements as appropriate;

transferred or removed products that were not considered DS (i.e., prescription medication); and assigned matching codes (confidence codes). All coding was reviewed by NCHS.

Data Processing and Editing

Data Collection Methods

Dietary Supplements (prescription and non-prescription)

During the interview, survey participants were asked if they had taken a dietary supplement in the past 30 days. Participants used a hand card that listed examples of DS (**Appendix 4**). Those who answered "yes" were asked by the interviewer to read to them all the text on the supplement container label of all the products used. For DS listed on the "Strength Only" List, only the nutrient was selected, and the strength was recorded (see **Appendix 5**). For all other DS reported, the interviewer entered the product's complete name. Interviewers could record up to 30 DS. If the container(s) was not available, the interviewer asked the participant to report verbally the name of the dietary supplement.

Non-prescription antacids

During the telephone interview, survey participants were asked if they had taken non-prescription antacids in the past 30 days. Participants used a hand card that listed examples of non-prescription antacids. Those who answered "yes" were asked by the interviewer to read to them all the text on the antacid container label of all the products used. For each antacid reported, the interviewer entered the product's complete name. If the containers were not available, the interviewer asked the participant to verbally report the name of the antacid.

When the interviewer entered the antacids name into the computer, the name was automatically matched to a prescription drug database (which also contained non-prescription antacid products) on the computer to identify an exact match or similar text matches. The interviewer then selected the best match from a list of possible matches. The original entry of the interviewer and the product selected from the computer list were saved under separate variables for quality control purposes. If an exact match of the product was not found, the interviewer was instructed to select "drug not found on list" from the list. Interviewers recorded up to 30 non-prescription antacids. Nutritionist at NCHS then processed the data collected to determine which products contained calcium and/or magnesium. The non-prescription antacids that contained calcium and/or magnesium were released with the dietary supplement data files.

Participants were also asked how many days each DS was taken in the past 30 days and the amount that was taken on those days. For a non-prescription antacid, participants were asked whether they were taking it as an antacid, as a calcium supplement, or both.

Matching reported dietary supplements to known products

Trained nutritionists, at NCHS, matched the product names entered by the interviewer (including prescription supplement, non-prescription supplement, and antacids) to a known product when possible. NCHS nutritionists first determined if the product is in our in-house PLD. If it is, then the nutritionist verified whether the product label was entered into the system within the 2-year cycle of data collection. If the product label was not entered into the PLD, or the product label was entered before the beginning of the 2-year data collection cycle, NCHS nutritionist attempted to obtain a new product label. NCHS obtained labels for each dietary supplement or non-prescription antacid containing calcium and/or magnesium reported by participants from sources such as the manufacturer, distributor or retailer, the Internet, company catalogs, and the Physician's Desk Reference (PDR). Label information was also obtained from The Dietary Supplement Label Database (DSLD), which is a joint project of the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) and National Library of Medicine (NLM). The DSLD contains the full label contents from a sample of dietary supplement products marketed in the U.S.

NCHS communicates with many major manufacturing company representatives to determine when various product re-formulations become available. Based upon manufacturer advice, we have used a lag time of 5 months after the new re-formulated product hit the market and matched products to participants' accordingly. Despite these precautions, there is no guarantee that the product taken by participants' is matched to the correct formulation in our release files.

NCHS nutritionist attempted to find the exact product(s) reported by participants'; however, this was done with varying degrees of precision. A "matching code or confidence code" (DSDMTCH located in file P_DSQIDS) accompanied each record to indicate the degree of matching certainty. The matches were: 1) Exact or near exact match; 2) Probable match; 3) Generic match; 4) Reasonable match; or 5) Default match. In some cases, no match was made with any certainty. These products were coded 6) No match. Products whose names were reported as "Refused" (DSDSUPP=7777) or "Don't know" (DSDSUPP=9999) have matching codes (DSDMTCH) of 7 and 9, respectively.

NCHS created generic and default DS and antacids, which were also maintained in the database. Generics were created in the database and used when we had collected information about a reported supplement, such as strength of all ingredients, but no brand name. These were generally single ingredient supplements, which included a strength (e.g., vitamin C 500 mg) or multiple vitamins and/or mineral supplements made by a private label manufacturer that was known to us and for which we had a label with identical ingredients and strengths (e.g., brand X all-purpose multivitamin was reported, and we had a label for brand Y, made by the same manufacturer). When all ingredient strengths were not known, a default supplement in the database was used to code the reported supplement. Defaults were created for single ingredient and multiple ingredient supplements based on our own data of most frequently reported supplements of that type, based on the survey cycle in which the data was collected. Created default and generic products and the actual products or strengths that were assigned to these defaults are listed in **Appendix 6**.

After the dietary supplement data was coded and matched to a product in our in-house database, various types of reviews were conducted to ensure the quality of the data. Interviewers' comments are reviewed to ensure that they have been accounted for in coding. Decisions were made about how to code new or unusual dietary supplement products or unusual quantities or units reported by survey participants. Dietary supplements that could not be matched to items in the database were resolved by NCHS nutritionist.

Product information is released from the in-house PLD as the NHANES-DSD, which contains detailed information on the DS and antacids reported by survey participants since NHANES 1999. The NHANES-DSD release consists of three datasets, which contain information on products:

Dietary Supplement Product Information (DSPI)
Dietary Supplement Ingredient Information (DSII)
Dietary Supplement Blend Information (DSBI)

The supplement ID numbers (DSDPID) located in the Individual Dietary Supplements File can be used to merge with the NHANES-DSD files. For more information on the NHANES-DSD, please refer to the documentation and release files located on the NHANES dietary website.

Data Editing

Non-prescription antacids containing calcium and/or magnesium that were reported in the non-prescription antacid section of the DSQ are included in the dietary supplement files.

Prescription products, the following prescription products are included in the dietary supplement files:

- All calcium products except calcium acetate;
- All fluoride products except those in topical gel or cream forms (e.g., stannous

fluoride);
• Over-the-counter niacin, niacinamide, and nicotinic acid;
• All vitamin D products, with the exception of records for Paricalcitol (D2), were retained in the August 2021-August 2023 prescription medication data file; and
• All prenatal products, most prescription vitamins taken by mouth.

All prescription niacin, potassium, and sodium products are retained in the prescription medication file. Combination products that include prescription drug(s) and dietary supplement ingredients are also retained in the prescription medication file (e.g., combination drug alendronate with cholecalciferol or Fosamax with vitamin D3). In August 2021-August 2023, prescription medication section was reduced to two questions (taking prescription medicine and number of prescription medicines). Therefore, none of the above-mentioned prescription products are included in the DS section.

Specific variables and edits:

DSD010: Have you used or taken any vitamins, minerals or other dietary supplements in the past 30 days? Include prescription and non-prescription supplements

Participants with a record of having taken a product determined to be a dietary supplement in the last 30 days are coded "1." This variable was edited and considers DS reported in any of the sub-sections of the DSQ. Some products were mistakenly recorded in the prescription medication sub-section. These data were moved to the dietary supplement files and are counted as supplements for DSD010 and DSDCOUNT. Participants who reported an antacid containing calcium or magnesium as part of the dietary supplement sub-section are also coded "1"; this product was considered to be taken as a calcium supplement. Participants who reported taking an antacid containing calcium or magnesium in the last 30 days that was recorded only in the non-prescription antacid sub-section but did not take any dietary supplement are coded "2." Participants who did not take any product that was determined to be a dietary supplement in the last 30 days are also coded "2," even if they originally reported having taken a supplement. Examples of products that were determined not to be supplements included foods (garlic cloves, raisin bran cereal, PowerBar®), drinks (Ensure®, Gatorade®, tea), over the counter drugs (aspirin, laxatives, electrolyte replacement fluids), homeopathic medicines, and prescription medicines. Prescription medicines were moved from the dietary supplement files to the appropriate prescription medication files. A small number of persons refused to answer this question (coded 7) or did not know whether they used a dietary supplement in the 30 days (coded 9).

DSDCOUNT: The number of DS taken

This variable was computed at NCHS and represents the total number of DS reported by the participant including those DS identified as unknown (DSDPID = 6666666XXX). Antacids that were reported in the dietary supplement sub-section were assumed to be taken as a dietary supplement and also included in the count. Antacids reported in the non-prescription antacid or the prescription medication sub-sections of the DSQ do not contribute to this count. There were also participants who reported the use of a dietary supplement in the past 30 days but did not know the name of the dietary supplement (DSDSUPP = 99999) or refused to report the name of the dietary supplement (DSDSUPP = 77777). Each product reported as refused or don't know is still included in the total count of DS.

DSD010AN: Any non-prescription antacids taken?

This variable was created at NCHS to indicate whether or not an antacid was reported. This variable only considers these types of antacids reported in the non-prescription antacid sub-section of the DSQ. In previous data releases, there were a few antacids reported in the dietary supplement sub-section of the DSQ and these were considered to be taken as DS and included in the DSD010 and DSDCOUNT. There was 2 antacids reported in the dietary supplement sub-section in August 2021-August 2023.

DSDANCNT: The number of non-prescription calcium and/or magnesium - containing antacids taken

This variable was computed at NCHS and represents the total number of antacids reported by the participant. Only these antacids reported in the non-prescription antacid sub-section of the DSQ contributed to this count. Antacids that were reported in the dietary supplement sub-section were assumed to be taken as a supplement and included in the dietary supplement count.

DSDANTA: Created variable that indicates whether an antacid was reported in the DS sub-section or the non-prescription antacid sub-section of the DSQ.

Information on use of non-prescription antacids was sometimes recorded in the dietary supplement sub-section of the DSQ; other times in the non-prescription antacid sub-section. Due to their nutrient content, antacids that contain calcium and/or magnesium are released with the dietary supplement data, irrespective of where they were reported. Only non-prescription antacids that contain calcium and/or magnesium are released in these files; this is not a complete accounting of all non-prescription antacids. Thus, users are cautioned that analyses of these data to estimate the percentage of non-prescription antacids used would not be appropriate.

DSDPID: Supplement ID Number

New Supplement ID is a unique number assigned to each product entered in the in-house PLD.

DSDSUPP: Name of Supplement

This is the name from the front of the product label. The brand name was generally entered first (i.e., Nature’s Way) and then the actual product name (i.e., lutein). Information such as the strength (i.e., 60 mg extract) of the product and other qualifiers that help distinguish a product from a similar product (i.e., mega, super, high potency, time release, chewable, extract) were also listed if they were on the front of the product label. Words like “dietary supplement” and health claims were not entered as part of the name.

DSDMTCH: Matching code confidence codes

Supplements were recorded during the DSQ of the SP questionnaire with varying degrees of accuracy and completeness. NCHS had created a system to specify how certain we were with matching a supplement recorded during the interview with the actual product label.

Exact or near exact match (DSDMTCH=1) indicated that this was the only product that could match this entry.

Probable match (DSDMTCH=2) indicated that the match was not exact, but knowledge of the company’s products strongly suggested that this was the only possible match choice. For example, the entry may not have specified strength or include words such as timed release, but no other options were available for this brand according to manufacturer or retailer information.

Generic match (DSDMTCH=3) indicated we had information on the strength for all ingredients, either: a) as part of the name (e.g., vitamin C 500 mg) or b) because the manufacturer was known and NCHS had an identical product made by this manufacturer for a different distributor or retailer. Thus, the ingredients and amounts were considered to be accurate despite an exact brand match.

Reasonable match (DSDMTCH=4) indicated that the product name may have been incomplete or could have been complete, but other products of this brand also started with these same words so this could not be assured. In these cases, the entered name was matched to either: a) the most frequently reported of these products in the NHANES 2017-2018 and 2019-2020 data, if this could be determined; b) the best-selling product by this company that matched the entered name; or c) the most basic product by this company, as assessed by label wording.

Default match (DSDMTCH=5) indicated that the exact product could not be obtained because the name was imprecise, or the exact brand product could not be located, and no generic was assigned. In these cases, the entered product was matched to a created default product based upon: a) the most commonly reported strengths for single ingredients; b) the most commonly reported brands for major multiple ingredient products such as multivitamins and multivitamin/multi-minerals for children, seniors, or adults, if available; or c) products manufactured by a large, private-label manufacturer.

Finally, a match code of No match (DSDMTCH=6) indicated that no product was found and there was not enough detail in the name to assign a generic or default match with any confidence. The words "no product information available" were listed as the product name (DSDSUPP).

Analysts should be aware that for default matches and matches that chose between several similarly named supplements, there was less certainty that the ingredients and ingredient amounts in the supplement assigned exactly matched those in the supplement actually taken. Additionally, NCHS cannot guarantee in any case that the matched product was the exact product taken or even that any product actually was taken, as these data were self-reported.

DSD070: Dietary supplement container seen by interviewer?

This variable indicated whether the product container was seen by the interviewer. Containers were seen for approximately 87% of the records. A more precise name for a supplement could be recorded by the interviewer, and thus a more precise match to a known supplement can be made, when the interviewer saw the supplement container rather than recording the participant's report of the supplement name (for example, multivitamin/multiminerals were often reported as multivitamins). In general, this was reflected in the matching code, but analysts should be aware that precision was greater when a container was seen.

This variable was mostly unedited. Interviewers asked to see the containers in all three sub-sections of the DSQ. Therefore, most records included in the dietary supplement file contained this information.

DSD090: For how long have you been taking this product or a similar type of product?

This variable has been discontinued in survey cycle August 2021-August 2023.

DSD103: In the past 30 days, on how many days did you take the product?

This variable was unedited. Information was missing for dietary supplement data that was recorded in the prescription medication sub-section, since this question was not asked in that sub-section of the DSQ.

DSD122Q and DSD122U: On the days that you took the product how much did you usually take on a single day?

The data was edited to standardize the amount the participant reported taking and the amount according to the product label serving size. For example, if a participant reported taking 1 tablespoon of a supplement and the label serving size was 3 teaspoons, then the variable was edited to 3 teaspoons (1 tablespoon = 3 teaspoons). There were 581 records in which data was edited.

In most of the cases, using the alternative serving size (information available on some product labels), using simple conversions (i.e., teaspoons to tablespoons) or contacting the manufacturer for information on label serving size (i.e., actual amount for a "capful") provided enough information to make clear edits. However, in some cases the reported amount taken was very different from the product label serving size (n=23). For example, the participant may have reported 1 tablet, but the label serving size is 1 tablespoon. This was assumed to be an error in reporting or an interviewer data entry error. In these cases, the reported amount taken was assumed to be the label serving size. Additionally, all records that were assigned a default product were edited and the label serving size was assigned for DSD122Q and DSD122U.

Information was missing for dietary supplement data that was recorded in the prescription medication sub-section, since this question was not asked in that sub-section of the DSQ.

DSDACTSS: Reported serving size/label serving size

This variable was derived using the information from DSD122Q (reported quantity taken) and information from variable DSDSERVQ (serving size quantity from product label) from the "Dietary Supplement Product Information" file (DSPI) of the NHANES-DSD. File DSPI provided information on the serving size from the product label's supplement facts panel for which the nutrient amounts were based on. The reported amount taken (DSD122Q) was divided by the serving size amount from DSPI (variable DSDSERVQ). Values were set to missing if any information was missing for DSD122Q or DSDSERVQ.

DSD124: Took product on own or doctor advised

This variable has been discontinued in survey cycle August 2021-August 2023.

DSQ128A- DSD128NN: Reason(s) for taking dietary supplement

This variable has been discontinued in survey cycle August 2021-August 2023.

Nutrient variables in the Individual Dietary Supplements file, DSQIDS_L

These variables were created by using files from the NHANES_DSD that contain information on the serving size and the quantity of each nutrient from the product label's supplement facts panel. The participants reported usual amount taken was divided by the serving size from the label in order to determine the actual amount of nutrient consumed. For example, a participant may have reported taking one tablet, but the serving size amount was 2 tablets. Therefore, only half of the nutrients on the label were being consumed. The variable DSDACTSS (Reported serving size/label serving size) indicated the actual amount of product that was consumed. The actual amount of product consumed was then multiplied by the ingredient amount for each dietary supplement or antacid. In the example above, 0.5 would be multiplied by each ingredient/nutrient to estimate the nutrient intake.

Analysts do need to be aware that in some cases data is missing for the amount consumed. In these cases, nutrients could not be calculated and are missing. Analyst should be aware of this, especially when estimating the prevalence of use of certain nutrients by using only the nutrient variables.

DSQIFDFE: Folate, DFE (mcg)

Dietary Folate Equivalents (DFE) were calculated by using a conversion factor of 1.7. This conversion is based on recommendations set by the Institute of Medicines Dietary Reference Intakes (National Academies Press, 2006).

Nutrient variables in the Total Dietary Supplements file, DSQTOT_L

These variables were created by calculating a mean daily intake. The calculation was based on the number of days the participant reported taking each dietary supplement (variable DSD103). For example, if participant X took calcium 600 mg 15 days out of the month, the calculation would be 600mg*15 reported days/30 days per month. If participant X took more than one supplement or antacid containing calcium, they would then be added up for the total mean daily calcium intake from DS.

Analysts do need to be aware that in some cases data is missing for the amount consumed and the number of days the supplement was taken. In these cases, nutrients could not be calculated or mean daily intake could not be estimated and therefore values are missing. Analyst should be aware of this, especially when estimating the prevalence of use of certain nutrients by using only the nutrient variables.

Analytic Notes

Sample weights for 30-day dietary supplement data

The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary supplement data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, the appropriate sample weights must be used.

As mentioned previously, in August 2021-August 2023, the 30-day dietary supplement use was collected following the conclusion of the first 24-hour dietary recall, therefore the same sample weights created for the day 1 recall (i.e., WTDRD1) is provided in the present dataset, and should be used with analysis uses the 30-day dietary supplement data (either alone or when Dietary Supplement data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with 30-day dietary supplement data. These weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using 30-day dietary supplement data and these weights might be larger than standard errors for similar estimates based on MEC weights. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

In NHANES August 2021-August 2023, a total of 11,375 dietary supplements or antacids were reported by survey participants as taken in the last 30 days prior to the interview. Approximately 0.1% of these reported records are missing information on the number of days the supplement was taken and/or the amount usually consumed. This is because no product information was available. Therefore, even though the use of particular supplements was reported, no nutrient content or mean daily intake data for selected nutrients was calculated because of the missing information on the amount usually consumed. Analysts should be cautious when estimating the prevalence of supplement use for specific nutrients. Nutrient variables released in the Individual or Total Dietary Supplements file (DSQIDS_L or DSQTOT_L) should not be used alone to produce such prevalence estimation. It is recommended that analysts merge participant datasets with the NHANES Dietary Supplement Database files to identify the reported supplements containing the nutrients of interest.

In some cases, it may appear as though a participant reported the same dietary supplement more than once. There are a several reasons for these duplications. Participants' may have reported DS with the same generic ingredient but different brand names (i.e., 2 different brands of calcium, but calcium is collected generically with only the strength) or the supplement may have been different forms or dosages of the same product. There were a few cases where the participant did report the exact same dietary supplement, with the same length of use, frequency of use or amount used. In these cases, since the interviewer recorded that a separate dietary supplement container was seen for each reported dietary supplement both mentions of the dietary supplement were retained in the file.

During the data editing process, outlier values were examined. When there was insufficient information to conclude that values were invalid, they were left in the data set. Analysts should examine the distribution of the data and consider whether or not it is appropriate to include or exclude extreme values in a given analysis.

NCHS collects brand name information on supplements whenever feasible, to ensure as much accuracy as possible in finding the label information for the exact product taken and providing exact ingredient information for this product to data users. Brand names are not collected for DS listed on the "Strength Only" List (see **Appendix 5**: Vitamins/Minerals). Only the nutrient is selected, and the strength is recorded. Analyst should be aware that for these specific nutrients, NCHS releases generic information (i.e., calcium 600 mg). Thus, analyses of consumer usage by brand name using NHANES data may not be accurate and is not recommended. Brand names that are available in a limited geographic area of the U.S. are not released; generic names are used for these products.

Deriving nutrient estimates from dietary supplement data

Thirty-four nutrients have already been computed for this release. In order to compute average daily intakes for nutrients or bioactives other than these 34, analyst will need to combine the Individual Dietary Supplements File (DSQIDS_L) with the NHANES-DSD files (DSPI, DSII and DSBI), which contain information on each product and ingredients and amounts in those products.

Intake of multiple supplements with the same nutrient(s), multiple use of the same supplement on the same day, and nutrients in blends should be taken into account in nutrient calculations. Nutrient names and the quantity units need to be harmonized and nutrient amounts from all these calculations must then be summed. Some nutrient amounts are for a nutrient compound (generally a foreign-made product or an antacid) and these must be converted to a nutrient elemental amount.
Analysts need to be aware of question changes over the cycles when looking at trends. Dietary supplement data was reported as times per month in 1999-2000, days in the past month in 2001-2002, and days in the past 30 days in 2003-2023.

Please refer to the on-line NHANES Dietary Tutorial, which is available on the NHANES website, for further details on how to use these datasets.

Estimating total nutrient intakes from all sources (foods, beverages and DS)

To estimate total dietary nutrient intake, nutrients from diet, supplements, and antacids should be combined. Although in August 2021-August 2023, the 30-day DS data was collected with the first 24-hour dietary recall using similar data collection methods, the reference intake period is different. Therefore, combining nutrient intake from these different variables to estimate total nutrient intake requires thoughtful consideration of the analytic goal and methods, and deserves accurate description of methods, assumptions, and weaknesses in any presentation of results.

All the key concepts and caveats regarding estimating nutrient intakes from dietary (foods and beverages) intake apply when estimating total nutrient intake from both foods and supplements. For these analyses, the sample of participants with reliable data for both supplement intake and the First Day recall are selected. Then, for each participant the average daily nutrient intake of supplements from the 30-day supplements use files are determined and added to the nutrient intake from the 24-hour recall. The 30-day supplement files can be linked to the dietary files by the respondent sequence number (SEQN). For more information on the data, refer to the documentations accompanying the 24-hour dietary intakes and dietary supplement datasets.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

WTDRD1 - Dietary day one sample weight

Variable Name: WTDRD1

SAS Label: Dietary day one sample weight

English Text: Dietary day one sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5708.632624 to 394679.4226	Range of Values	11369	11369	
0	Day 1 dietary recall not done/incomplete	6	11375	
.	Missing	0	11375	

DSDPID - Supplement ID Number

Variable Name: DSDPID

SAS Label: Supplement ID Number

English Text: Supplement ID Number

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
275 to 6666667314	Range of Values	11375	11375	
.	Missing	0	11375	

DSDSUPP - Supplement Name

Variable Name: DSDSUPP

SAS Label: Supplement Name

English Text:

Target:

Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
Supplement Name	Value was recorded	11375	11375	
< blank >	Missing	0	11375	

DSDANTA - Antacid reported as a dietary supplement

Variable Name: DSDANTA

SAS Label: Antacid reported as a dietary supplement

English Text: Antacid reported as a dietary supplement

English Instructions: Antacids that are reported with dietary supplements (DSQ) are coded 1, antacids reported with medications in the antacid section are coded 2, dietary supplements that are not antacids are coded 0.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	Non-antacid supplement	10461	10461	
1	Antacid reported with dietary supplement (DSQ)	5	10466	
2	Antacid reported with medication (RXQ)	909	11375	
.	Missing	0	11375	

DSD070 - Was container seen?

Variable Name: DSD070

SAS Label: Was container seen?

English Text: Was container seen?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	10083	10083	
2	No	1287	11370	
.	Missing	5	11375	

DSDMTCH - Matching code

Variable Name: DSDMTCH

SAS Label: Matching code

English Text: Matching code

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Exact or near exact match	6165	6165	
2	Probable match	557	6722	
3	Generic match	2877	9599	
4	Reasonable match	687	10286	
5	Default match	924	11210	
6	No match	158	11368	
7	Refused	0	11368	
9	Don't know	7	11375	
.	Missing	0	11375	

DSD103 - Days supplement taken, past 30 days

Variable Name: DSD103

SAS Label: Days supplement taken, past 30 days

English Text: In the past 30 days, on how many days did you take {PRODUCT NAME}?

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 30	Range of Values	11341	11341	
7777	Refused	0	11341	
9999	Don't know	28	11369	
.	Missing	6	11375	

DSD122Q - Quantity of supplement taken daily

Variable Name: DSD122Q

SAS Label: Quantity of supplement taken daily

English Text: On the days that (you/SP) took (PRODUCT NAME), how much did (you/SP), usually take on a single day?

English Instructions: CAPI INSTRUCTIONS: RESPONSE FIELD SHOULD ALLOW FOR 3 NUMERIC ENTRIES AND INCLUDE A DECIMAL. ALLOW 0 OR 1 ENTRIES TO THE LEFT OF THE DECIMAL AND 0,1 OR 2 ENTRIES TO THE RIGHT OF THE DECIMAL. ENTER NUMBER

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.027 to 120	Range of Values	11368	11368	
777777	Refused	0	11368	
999999	Don't know	2	11370	
.	Missing	5	11375	

DSD122U - Dosage form

Variable Name:	DSD122U
SAS Label:	Dosage form
English Text:	On the days that {you/SP} took {PRODUCT NAME}, how much did {you/SP} usually take on a single day?
English Instructions:	CAPI INSTRUCTION: RESPONSE FIELD SHOULD ALLOW FOR 3 NUMERIC ENTRIES AND INCLUDE A DECIMAL. ALLOW 0 OR 1 ENTRIES TO THE LEFT OF THE DECIMAL AND 0, 1 OR 2 ENTRIES TO THE RIGHT OF THE DECIMAL. IF FORM CODE 1 THROUGH 8 IN DSQ.077, AUTOMATICALLY CODE THE UNIT CODE 1 AND SKIP TO BOX NEW 4A. IF FORM CODE 12 IN DSQ.077, AUTOMATICALLY CODE THE UNIT CODE 13 FOR DSQ.123U AND SKIP TO BOX NEW 4A. IF FORM CODE 13 IN DSQ.077, AUTOMATICALLY CODE THE UNIT CODE 20 FOR DSQ.123U AND SKIP TO BOX NEW 4A. IF FORM CODE 14 IN DSQ.077, AUTOMATICALLY CODE THE UNIT CODE 17 FOR DSQ.123U AND SKIP TO BOX NEW 4A. IF FORM CODE 16 IN DSQ.077, AUTOMATICALLY CODE THE UNIT CODE 6 FOR DSQ.123U AND SKIP TO BOX NEW 4A. IF FORM CODE 9 IN DSQ.077, DISPLAY THE UNIT CODES 1, 6, 7, 11, 12, 13, 15, 16, 17, 18, 20, 21, 23, 27, 28, 30, 91, 77, 99 FOR DSQ.123U. IF FORM CODE 10, 17 IN DSQ.077, DISPLAY THE UNIT CODES 2, 3, 5, 7, 11, 12, 15, 18, 19, 23, 27, 29, 91, 77, 99 FOR DSQ.123U. IF FORM CODE 11, 15 IN DSQ.077, DISPLAY THE UNIT CODES 11, 12, 15, 16, 18, 21, 23, 27, 28, 91, 77, 99 FOR DSQ.123U. IF FORM CODE 91, 77, 99 IN DSQ.077, DISPLAY ENTIRE PICK LIST FOR DSQ.123U. IF CONTAINER NOT SEEN (CODE 2 IN DSQ.071), DISPLAY ENTIRE PICK LIST FOR DSQ.123U. ENTER UNIT/FORM.
Target:	Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	TABLETS, CAPSULES, PILLS, CAPLETS, SOFTGELS, GEL CAPS, VEGICAPS, CHEWABLE TABLETS	9217	9217	
2	Droppers	5	9222	
3	Drops	54	9276	
5	Injections/shots	0	9276	
6	LOZENGES/COUGH DROPS	12	9288	
7	Milliliters	117	9405	
11	Tablespoons	42	9447	
12	Teaspoons	172	9619	
13	Wafers	5	9624	
15	Cans	0	9624	
16	Grams	33	9657	
17	Dots	0	9657	
18	Cups	0	9657	
19	Sprays/Squirts	3	9660	
20	CHEWS/GUMMIES	1428	11088	
21	Scoop	134	11222	
22	CC	0	11222	
23	Capful	4	11226	
27	Ounces	13	11239	
28	Packages/Packets	110	11349	
29	Vial	0	11349	
30	Gumball	0	11349	
31	Strips	0	11349	
32	Bottle	4	11353	
41	Pieces	14	11367	
77	Refused	0	11367	
99	Don't know	3	11370	
.	Missing	5	11375	

DSDACTSS - Reported serving size/label serving size

Variable Name: DSDACTSS

SAS Label: Reported serving size/label serving size

English Text: Reported serving size/label serving size

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.05 to 16	Range of Values	11209	11209	
.	Missing	166	11375	

DSQIKCAL - Energy (kcal)

Variable Name: DSQIKCAL

SAS Label: Energy (kcal)

English Text: Energy (kcal)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 2010	Range of Values	3094	3094	
.	Missing	8281	11375	

DSQIPROT - Protein (gm)

Variable Name: DSQIPROT

SAS Label: Protein (gm)

English Text: Protein (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 75	Range of Values	387	387	
.	Missing	10988	11375	

DSQICARB - Carbohydrate (gm)

Variable Name: DSQICARB

SAS Label: Carbohydrate (gm)

English Text: Carbohydrate (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 415.5	Range of Values	2406	2406	
.	Missing	8969	11375	

DSQISUGR - Total sugars (gm)

Variable Name: DSQISUGR

SAS Label: Total sugars (gm)

English Text: Total sugars (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 64	Range of Values	1681	1681	
.	Missing	9694	11375	

DSQIFIBE - Dietary fiber (gm)

Variable Name: DSQIFIBE

SAS Label: Dietary fiber (gm)

English Text: Dietary fiber (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 45	Range of Values	307	307	
.	Missing	11068	11375	

DSQITFAT - Total fat (gm)

Variable Name: DSQITFAT

SAS Label: Total fat (gm)

English Text: Total fat (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.3 to 28	Range of Values	915	915	
.	Missing	10460	11375	

DSQISFAT - Total saturated fatty acids (gm)

Variable Name: DSQISFAT

SAS Label: Total saturated fatty acids (gm)

English Text: Total saturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 4.6	Range of Values	207	207	
.	Missing	11168	11375	

DSQIMFAT - Total monounsaturated fatty acids (gm)

Variable Name: DSQIMFAT

SAS Label: Total monounsaturated fatty acids (gm)

English Text: Total monounsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 6	Range of Values	91	91	
.	Missing	11284	11375	

DSQIPFAT - Total polyunsaturated fatty acids (gm)

Variable Name: DSQIPFAT

SAS Label: Total polyunsaturated fatty acids (gm)

English Text: Total polyunsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.3 to 18	Range of Values	230	230	
.	Missing	11145	11375	

DSQICHOL - Cholesterol (mg)

Variable Name: DSQICHOL

SAS Label: Cholesterol (mg)

English Text: Cholesterol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 98	Range of Values	420	420	
.	Missing	10955	11375	

DSQILYCO - Lycopene (mcg)

Variable Name: DSQILYCO

SAS Label: Lycopene (mcg)

English Text: Lycopene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
99 to 20000	Range of Values	535	535	
.	Missing	10840	11375	

DSQILZ - Lutein + zeaxanthin (mcg)

Variable Name: DSQILZ

SAS Label: Lutein + zeaxanthin (mcg)

English Text: Lutein + zeaxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
50 to 275000	Range of Values	780	780	
.	Missing	10595	11375	

DSQIVB1 - Thiamin (Vitamin B1) (mg)

Variable Name: DSQIVB1

SAS Label: Thiamin (Vitamin B1) (mg)

English Text: Thiamin (Vitamin B1) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.005 to 400	Range of Values	1805	1805	
.	Missing	9570	11375	

DSQIVB2 - Riboflavin (Vitamin B2) (mg)

Variable Name: DSQIVB2

SAS Label: Riboflavin (Vitamin B2) (mg)

English Text: Riboflavin (Vitamin B2) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 400	Range of Values	1842	1842	
.	Missing	9533	11375	

DSQINIAC - Niacin (mg)

Variable Name: DSQINIAC

SAS Label: Niacin (mg)

English Text: Niacin (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 2000	Range of Values	2038	2038	
.	Missing	9337	11375	

DSQIVB6 - Vitamin B6 (mg)

Variable Name: DSQIVB6

SAS Label: Vitamin B6 (mg)

English Text: Vitamin B6 (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 500	Range of Values	2526	2526	
.	Missing	8849	11375	

DSQIFA - Folic acid (mcg)

Variable Name: DSQIFA

SAS Label: Folic acid (mcg)

English Text: Folic acid (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 7200	Range of Values	2431	2431	
.	Missing	8944	11375	

DSQIFDFE - Folate, DFE (mcg)

Variable Name: DSQIFDFE

SAS Label: Folate, DFE (mcg)

English Text: Folate, DFE (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
9 to 12240	Range of Values	2431	2431	
.	Missing	8944	11375	

DSQICHL - Total choline (mg)

Variable Name: DSQICHL

SAS Label: Total choline (mg)

English Text: Total choline (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 1175	Range of Values	297	297	
.	Missing	11078	11375	

DSQIVB12 - Vitamin B12 (mcg)

Variable Name: DSQIVB12

SAS Label: Vitamin B12 (mcg)

English Text: Vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.3 to 10000	Range of Values	2823	2823	
.	Missing	8552	11375	

DSQIVC - Vitamin C (mg)

Variable Name: DSQIVC

SAS Label: Vitamin C (mg)

English Text: Vitamin C (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 8000	Range of Values	3415	3415	
.	Missing	7960	11375	

DSQIVK - Vitamin K (mcg)

Variable Name: DSQIVK

SAS Label: Vitamin K (mcg)

English Text: Vitamin K (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5 to 45000	Range of Values	1253	1253	
.	Missing	10122	11375	

DSQIVD - Vitamin D (D2 + D3) (mcg)

Variable Name: DSQIVD

SAS Label: Vitamin D (D2 + D3) (mcg)

English Text: Vitamin D (D2 + D3) (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 2500	Range of Values	4213	4213	
.	Missing	7162	11375	

DSQICALC - Calcium (mg)

Variable Name: DSQICALC

SAS Label: Calcium (mg)

English Text: Calcium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.8 to 3000	Range of Values	3401	3401	
.	Missing	7974	11375	

DSQIPHOS - Phosphorus (mg)

Variable Name: DSQIPHOS

SAS Label: Phosphorus (mg)

English Text: Phosphorus (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 1035	Range of Values	862	862	
.	Missing	10513	11375	

DSQIMAGN - Magnesium (mg)

Variable Name: DSQIMAGN

SAS Label: Magnesium (mg)

English Text: Magnesium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 4080	Range of Values	1985	1985	
.	Missing	9390	11375	

DSQIIRON - Iron (mg)

Variable Name: DSQIIRON

SAS Label: Iron (mg)

English Text: Iron (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.07 to 195	Range of Values	1152	1152	
.	Missing	10223	11375	

DSQIZINC - Zinc (mg)

Variable Name: DSQIZINC

SAS Label: Zinc (mg)

English Text: Zinc (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 307.69	Range of Values	2769	2769	
.	Missing	8606	11375	

DSQICOPP - Copper (mg)

Variable Name: DSQICOPP

SAS Label: Copper (mg)

English Text: Copper (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.0004 to 6	Range of Values	1439	1439	
.	Missing	9936	11375	

DSQISODI - Sodium (mg)

Variable Name: DSQISODI

SAS Label: Sodium (mg)

English Text: Sodium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.3 to 1820	Range of Values	1540	1540	
.	Missing	9835	11375	

DSQIPOTA - Potassium (mg)

Variable Name: DSQIPOTA

SAS Label: Potassium (mg)

English Text: Potassium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 3120	Range of Values	1064	1064	
.	Missing	10311	11375	

DSQISELE - Selenium (mcg)

Variable Name: DSQISELE

SAS Label: Selenium (mcg)

English Text: Selenium (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 400	Range of Values	1413	1413	
.	Missing	9962	11375	

DSQICAFF - Caffeine (mg)

Variable Name: DSQICAFF

SAS Label: Caffeine (mg)

English Text: Caffeine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
13 to 500	Range of Values	11	11	
.	Missing	11364	11375	

DSQIIODI - Iodine (mcg)

Variable Name: DSQIIODI

SAS Label: Iodine (mcg)

English Text:

Target:

Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4 to 1500	Range of Values	1863	1863	
.	Missing	9512	11375	

Appendix 1: Variables in the Individual Dietary Supplements File (DSQIDS_L)

Variables	Variable Label
SEQN	Respondent sequence number
DSDPID	NCHS Supplement ID
DSDANTA	Antacid reported as a dietary supplement
DSD070	Was container seen?
DSDMTCH	Matching code
DSD103	Days supplement taken, past 30 days
DSD122Q	Quantity of supplement taken per day
DSD122U	Dosage form
DSDACTSS	Reported serving size/label serving size
DSQIKCAL	Energy (kcal)
DSQIPROT	Protein (gm)
DSQICARB	Carbohydrate (gm)
DSQISUGR	Total sugars (gm)
DSQIFIBE	Dietary fiber (gm)
DSQITFAT	Total fat (gm)
DSQISFAT	Total saturated fatty acids (gm)
DSQIMFAT	Total monounsaturated fatty acids (gm)
DSQIPFAT	Total polyunsaturated fatty acids (gm)
DSQICHOL	Cholesterol (mg)
DSQILYCO	Lycopene (mcg)
DSQILZ	Lutein + zeaxanthin (mcg)
DSQIVB1	Thiamin (Vitamin B1) (mg)
DSQIVB2	Riboflavin (Vitamin B2) (mg)
DSQINIAC	Niacin (mg)
DSQIVB6	Vitamin B6 (mg)
DSQIFA	Folic acid (mcg)
DSQIFDFE	Folate, DFE (mcg)
DSQICHL	Total choline (mg)
DSQIVB12	Vitamin B12 (mcg)
DSQIVC	Vitamin C (mg)
DSQIVK	Vitamin K (mcg)
DSQIVD	Vitamin D (D2 + D3) (mcg)
DSQICALC	Calcium (mg)
DSQIPHOS	Phosphorus (mg)
DSQIMAGN	Magnesium (mg)
DSQIIRON	Iron (mg)
DSQIZINC	Zinc (mg)
DSQICOPP	Copper (mg)
DSQISODI	Sodium (mg)

Variables	Variable Label
DSQIPOTA	Potassium (mg)
DSQISELE	Selenium (mcg)
DSQICAFF	Caffeine (mg)
DSQIIODI	Iodine (mcg)

Appendix 2: List of Nutrients/Dietary Components (Unit)

Energy and Macronutrients
Food energy (kcal) €
Protein (g) €
Carbohydrate (g) €
Fat, total (g) €
Alcohol (g)
Sugars, total (g) €
Dietary fiber, total (g) €
Water (moisture) (g)*
Saturated fatty acids, total (g) €
Monounsaturated fatty acids, total (g) €
Polyunsaturated fatty acids, total (g) €
Cholesterol (mg) €

Individual fatty acids:
4:0 (g)
6:0 (g)
8:0 (g)
10:0 (g)
12:0 (g)
14:0 (g)
16:0 (g)
18:0 (g)
16:1 (g)
18:1 (g)
20:1 (g)
22:1 (g)
18:2 (g)
18:3 (g)
18:4 (g)
20:4 (g)
20:5 n-3 (g)
22:5 n-3 (g)
22:6 n-3 (g)

Vitamins, Minerals, and Other Components
Vitamin A as retinol activity equivalents (µg) ¥
Retinol (µg) ¥
Carotenoids:
  Carotene, alpha (µg) ¥
  Carotene, beta (µg) ¥
  Cryptoxanthin, beta (µg) ¥
  Lycopene (µg) €
  Lutein + zeaxanthin (µg) €
Vitamin E as alpha-tocopherol (mg) ¥
  Added vitamin E as alpha-tocopherol (mg)
Vitamin D (D2 + D3) (µg) €
Vitamin K as phylloquinone (µg) €
Vitamin C (mg) €
Thiamin (mg) €
Riboflavin (mg) €
Niacin (mg) €
Vitamin B-6 (mg) €
Folate, total (µg)
  Folate as dietary folate equivalents (µg) €
  Folic acid (µg) €
  Food folate (µg)
Choline, total (mg) €
Vitamin B-12 (µg) €
  Added vitamin B-12 (µg)
Calcium (mg) €

Iron (mg) €
Magnesium (mg) €
Phosphorus (mg) €
Potassium (mg) €
Sodium (mg) €
Zinc (mg) €
Copper (mg) €
Selenium (µg) €
Caffeine (mg) €
Theobromine (mg)

Iodine (mcg)†

* Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)

€Indicates nutrients in which data is available for dietary supplements and non-prescription antacids containing calcium and/or magnesium

¥Indicates that data will be available in a later release cycle

† Only included in the dietary supplement files

Appendix 3: Variables in the Total Dietary Supplement File (DSQTOT_L)

Variables	Variable Label
SEQN	Respondent sequence number
DSD010	Any Dietary Supplements Taken?
DSDCOUNT	Total # of Dietary Supplements Taken
DSD010AN	Any Antacids Taken?
DSDANCNT	Total # of Antacids Taken
DSQTKCAL	Energy (kcal)
DSQTPROT	Protein (gm)
DSQTCARB	Carbohydrate (gm)
DSQTSUGR	Total sugars (gm)
DSQTFIBE	Dietary fiber (gm)
DSQTTFAT	Total fat (gm)
DSQTSFAT	Total saturated fatty acids (gm)
DSQTMFAT	Total monounsaturated fatty acids (gm)
DSQTPFAT	Total polyunsaturated fatty acids (gm)
DSQTCHOL	Cholesterol (mg)
DSQTLYCO	Lycopene (mcg)
DSQTLZ	Lutein + zeaxanthin (mcg)
DSQTVB1	Thiamin (Vitamin B1) (mg)
DSQTVB2	Riboflavin (Vitamin B2) (mg)
DSQTNIAC	Niacin (mg)
DSQTVB6	Vitamin B6 (mg)
DSQTFA	Folic acid (mcg)
DSQTFDFE	Folate, DFE (mcg)
DSQTCHL	Total choline (mg)
DSQTVB12	Vitamin B12 (mcg)
DSQTVC	Vitamin C (mg)
DSQTVK	Vitamin K (mcg)
DSQTVD	Vitamin D (D2 + D3) (mcg)
DSQTCALC	Calcium (mg)
DSQTPHOS	Phosphorus (mg)
DSQTMAGN	Magnesium (mg)
DSQTIRON	Iron (mg)
DSQTZINC	Zinc (mg)
DSQTCOPP	Copper (mg)
DSQTSODI	Sodium (mg)
DSQTPOTA	Potassium (mg)
DSQTSELE	Selenium (mcg)
DSQTCAFF	Caffeine (mg)
DSQTIODI	Iodine (mcg)

Appendix 4: Dietary Supplement Hand card

VITAMINS **MINERALS**	Calcium, Vitamin C, Calcium and Iron, Vitamin E, Magnesium, Zinc, Calcium plus Vitamin D
MULTI-VITAMIN--MULTI-MINERALS	Flintstones, One a Day, Prenatals, Tri-Vi-Flor, B-Complex, Centrum
HERBALS AND BOTANICLAS	Echinacea, Garlic, Saw Palmetto, Ginkgo, Ginseng
FIBER	Metamucil, Fibercon, Benefiber
AMINO ACIDS	Lysine, Methionine, Tryptophan
OTHERS	Fish Oil, Chondrotin, Glucosamine

Appendix 5: Vitamins/Minerals on the "Strength Only" List

- Vitamin A
- Vitamin B6
- Vitamin B12
- Vitamin C
- Vitamin D (D3)
- Vitamin E
- Calcium
- Chromium (Chromium Picolinate)
- Folate (Folic Acid)
- Iron (Ferrous Xxxate)
- Magnesium
- Potassium
- Selenium
- Zinc (Zinc Gluconate)

Appendix 6: Created Default Supplements and Antacids

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
5-HTP (5-Hydroxytryptophan)	100 mg	Most Commonly Reported Strength
Acetyl L-Carnitine	500 mg	Most Commonly Reported Strength
Alpha Lipoic Acid	300 mg	Commonly Available Strength
Amino Acid	ON Optimum Nutrition Superior Amino 2222 Tablets	Commonly Available Product
Apple Cider Vinegar Tablets	Nature's Bounty Apple Cider Vinegar 480 mg Per Serving	Commonly Available Product
Arginine Powder	3000 mg	Commonly Available Strength
Artichoke	500 mg	Most Commonly Reported Strength
Ashwagandha	500 mg	Most Commonly Reported Strength
B 50 B-Complex	GNC B-Complex 50	Commonly Available Product
Balanced B 100 B-Complex	Nature Made Time Release B-100 Complex	Commonly Available Product
BCAA Powder	ON Optimum Nutrition Instantized BCAA 5000 Powder	Commonly Available Product
Beet Powder	7 grams	Commonly Available Strength
Biotin	5000 mcg	Most Commonly Reported Strength
Black Cumin Seed Oil	500 Mg	Commonly Available Strength
Borage Oil	1000 mg	Commonly Available Strength
Brewer's Yeast Powder	Solgar Brewer's Yeast Powder	Commonly Available Product
Calcium	600 mg	Most Commonly Reported Strength
Calcium & Magnesium	Calcium 500 mg, Magnesium 250 mg	Most Commonly Reported Strengths
Calcium + D + K	Viactiv Calcium+ Max Formula Bone Strengthening 650 mg Calcium	Commonly Available Product
Calcium + Magnesium Liquid	Calcium 1000 mg, Magnesium 500 mg	Commonly Available Strengths
Calcium + Vitamin D 20 mcg	Calcium 600 mg, Vitamin D 20 mcg	Most Commonly Reported Strengths
Calcium + Vitamin D 2000 IU	Calcium 600 mg, Vitamin D 2000 IU	Commonly Available Strengths
Calcium 1200 mg with Vitamin D	Calcium 600 mg, Vitamin D 40 mcg	Commonly Available Strengths

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Calcium 250 mg With Vitamin D	Calcium 250 mg, Vitamin D 3.1 mcg	Commonly Available Strengths
Calcium 500 mg With Vitamin D	Calcium 500 mg, Vitamin D 10 mcg	Most Commonly Reported Strengths
Calcium 600 mg With Vitamin D	Calcium 600 mg, Vitamin D 20 mcg	Most Commonly Reported Strengths
Calcium Magnesium & Zinc	Calcium 333 mg, Magnesium 133 mg & Zinc 5 mg	Commonly Available Strengths
Calcium With Vitamin D	Calcium 600 mg, Vitamin D 20 mcg	Most Commonly Reported Strengths
Calcium With Vitamin D & Minerals	Caltrate Bone Health Advanced Calcium 600+D3 Plus Minerals	Commonly Available Product
CBD Gummies	Wyld CBD Gummies 500 mg CBD 25 mg CBD Per Gummy	Commonly Available Product
CBD Oil	Sunsoil CBD Oil 10 mg CBD Servings	Commonly Available Product
Children’s Multivitamin/Multimineral	Flintstones Complete Children’s Multivitamin	Most Commonly Reported Product
Chromium	200 mcg	Most Commonly Reported Strength
Cinnamon	500 mg	Most Commonly Reported Strength
Citrus Bergamot	500 mg	Commonly Available Strength
Cod Liver Oil Softgels	Spring Valley Cod Liver Oil Plus Vitamins A & D3	Commonly Available Product
Coenzyme Q-10	100 mg	Most Commonly Reported Strength
Collagen	1000 mg	Commonly Available Strength
Collagen & Vitamin C	Neocell Super Collagen + Vitamin C Collagen Type 1 & 3	Commonly Available Product
Collagen Powder	Neocell Super Collagen Peptides Collagen Type 1&3	Commonly Available Product
Collinsonia Root	500 mg	Commonly Available Strength
CoQ10 + Red Yeast Rice	CoQ10 60 mg, Red Yeast Rice 600 mg	Commonly Available Strengths
Cranberry	500 mg	Most Commonly Reported Strength
Creatine Capsules	750 mg	Commonly Available Strength
Creatine Monohydrate	5000 mg	Most Commonly Reported Strength
Curcumin	500 mg	Commonly Available Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Dairy Digestive Caplets	Lactaid Original Lactase Enzyme	Commonly Available Product
DHA	200 mg	Commonly Available Strength
DHEA	25 mg	Most Commonly Reported Strength
Echinacea	400 mg	Most Commonly Reported Strength
Echinacea Liquid	1.0 ml	Commonly Available Strength
Elderberry	575 mg	Commonly Available Strength
Elderberry Liquid	Nature's Way Sambucus Standardized Elderberry Original Syrup	Commonly Available Product
Enzymes	Genuine N-Zimes Dr. Howell's Original Formula Digestive Enzyme	Commonly Available Product
Evening Primrose Oil	1000 mg	Most Commonly Reported Strength
Eye Health Complex	Equate Advanced Eye Health Complex	Commonly Available Product
Eye Multi-Vitamin and Mineral	Bausch + Lomb Preservision Eye Vitamin And Mineral AREDS Formula Soft Gels	Commonly Available Product
Fiber Capsules	Metamucil Psyllium Fiber 3-In-1 Fiber Multiple Health Benefits	Most Commonly Reported Product
Fiber Powder	Metamucil Psyllium Fiber 4-In-1 Fiber Made With Real Sugar Unflavored Stone Ground Texture	Commonly Available Product
Fish Oil	Nature's Bounty Fish Oil 1000 Mg | 300 mg Of Omega-3	Commonly Available Product
Fish, Flax and Borage Oils	Puritan's Pride Maximum Strength Triple Omega 3-6-9 Fish, Flax & Borage Oils	Commonly Available Product
Flax Seed Oil	1000 mg	Most Commonly Reported Strength
Folate	1333 mcg DFE	Most Commonly Reported Strength
GABA	500 mg	Commonly Available Strength
Garcinia Cambogia	500 mg	Commonly Available Strength
Garlic	1000 mg	Most Commonly Reported Strength
Ginger	550 mg	Most Commonly Reported Strength
Ginkgo Biloba	120 mg	Most Commonly Reported Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Ginseng	250 mg	Most Commonly Reported Strength
Glucosamine	1000 mg	Most Commonly Reported Strength
Glucosamine & MSM	750 mg & 750 mg	Most Commonly Reported Strengths
Glucosamine 1500 mg with MSM	1500 mg & 1500 mg	Most Commonly Reported Strengths
Glucosamine Chondroitin	750 mg & 600 mg	Most Commonly Reported Strengths
Glucosamine Chondroitin & MSM	Puritan's Pride Glucosamine Chondroitin MSM 3 Per Day Formula	Commonly Available Product
Glutamine Powder	4500 mg	Commonly Available Strength
Green Tea	500 mg	Commonly Available Strength
Greens Powder	Newvitality Royal Greens Original Blend	Commonly Available Product
Gummy Adult Apple Cider Vinegar	Vitafusion Gummy Vitamins Apple Cider Vinegar	Commonly Available Product
Gummy Adult Calcium + Vitamin D	Vitafusion Gummy Vitamins Calcium 500 mg Per Serving with Vitamin D3 & Calcium	Commonly Available Product
Gummy Adult Coq10	Vitafusion Gummy Supplements Coq10 200 mg Per Serving	Commonly Available Product
Gummy Adult Cranberry	Spring Valley Adult Gummy Cranberry 500 mg Per Serving	Commonly Available Product
Gummy Adult Elderberry with Vitamin C & Zinc	Sambucol Black Elderberry Black Elderberry Gummies with Vitamin C & Zinc	Commonly Available Product
Gummy Adult Fiber	Vitafusion Fiber Well Gummies Sugar Free 5g Fiber Per Serving	Commonly Available Product
Gummy Adult Hair, Skin & Nails	Nature's Bounty Optimal Solutions Hair, Skin & Nails Gummies 2,500 mcg Biotin Per Serving	Commonly Available Product
Gummy Adult Lion's Mane Mushroom	500 mg	Commonly Available Strength
Gummy Adult Melatonin	Vitafusion Melatonin Sugarfree Gummy 3 mg Of Melatonin Per Serving	Commonly Available Product
Gummy Adult Melatonin Extra Strength	Vitafusion Gummy Supplements Melatonin Extra Strength 5 mg Per Serving	Commonly Available Product
Gummy Adult Melatonin Max Strength	Vitafusion Gummy Supplements Melatonin Max Strength 10 mg Per Serving	Commonly Available Product
Gummy Adult Men's Multivitamin / Multimineral	Vitafusion Adult Gummy Vitamins Men's Multi Daily Multivitamin	Commonly Available Product

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Gummy Adult Multivitamin / Multimineral	Vitafusion Adult Gummy Vitamins Multivites Multivitamin Gummies	Commonly Available Product
Gummy Adult Omega-3	Vitafusion Gummy Vitamins Omega-3 EPA/DHA	Commonly Available Product
Gummy Adult Prenatal Multivitamin	Vitafusion Gummy Vitamins Prenatal Multivitamin with Folate & 50 mg DHA Per Serving	Commonly Available Product
Gummy Adult Probiotic	Schiff Digestive Advantage Daily Probiotic Gummies	Commonly Available Product
Gummy Adult Vitamin C	Vitafusion Gummy Vitamins Power C High Potency Vitamin C	Commonly Available Product
Gummy Adult Vitamin C Extra Strength	Vitafusion Gummy Vitamins Power C 500 mg Per Serving Extra Strength	Commonly Available Product
Gummy Adult Vitamin D	Vitafusion Gummy Vitamins Vitamin D3 High Potency	Commonly Available Product
Gummy Adult Vitamin D Extra Strength	Vitafusion Gummy Vitamins Vitamin D3 Extra Strength 75 mcg Per Serving	Commonly Available Product
Gummy Adult Women's Multivitamin / Multimineral	Vitafusion Adult Gummy Vitamins Women's Multi Daily Multivitamin	Commonly Available Product
Gummy Adult Zinc	Vitafusion Power Zinc Gummy Vitamins	Commonly Available Product
Gummy Bear Elderberry	Flintstones Supplements Elderberry Gummies With Immunity Support From Vitamin C + Zinc	Commonly Available Product
Gummy Bear Melatonin	Natrol Kids Melatonin Gummies	Commonly Available Product
Gummy Bear Multivitamin	L'il Critters Gummy Vites Daily Multivitamin Plus Vitamins C & D	Most Commonly Reported Product
Gummy Bear Multivitamin + Probiotic	Olly Kids Multi + Probiotic	Commonly Available Product
Gummy Bear Probiotic	Schiff Digestive Advantage Daily Probiotics Kids Gummies	Commonly Available Product
Gummy Bear Vitamin D	Nature Made Kids First Vitamin D3	Commonly Available Product
Hair, Skin and Nails	Nature's Bounty Optimal Solutions Hair, Skin & Nails 3,000 mcg Biotin Per Serving	Commonly Available Product
Horny Goat Weed	500 mg	Commonly Available Strength
Iodine Drops	150 mcg	Commonly Available Strength
Iron	65 mg	Most Commonly Reported Strength
Krill Oil	Schiff MegaRed Superior Omega-3 Krill Oil 350 mg	Commonly Available Product
Lactobacillus Acidophilus	Nature's Bounty Acidophilus Probiotic 100 Million Organisms	Commonly Available Product

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Lion’s Mane Mushroom	500 mg	Commonly Available Strength
Lion’s Mane Mushroom Powder	1000 mg	Commonly Available Strength
Liquid B Complex	Nature's Bounty Sublingual Liquid B Complex	Commonly Available Product
Liquid Coenzyme Q-10	100 mg	Commonly Available Strength
Liquid Fish Oil	Carlson Norwegian The Very Finest Fish Oil 1,600 mg Omega-3s	Commonly Available Product
Liquid Flaxseed Oil	Barlean's Fresh Flax Oil Organic Pure & Unrefined Fresh Cold Pressed	Commonly Available Product
Liquid Vitamin B-12	5000 mcg	Commonly Available Strength
Liquid Vitamin C	1000 mg	Commonly Available Strength
L-Tyrosine	500 mg	Commonly Available Strength
Lutein	20 mg	Most Commonly Reported Strength
Lutein & Zeaxanthin	Lutein 25 mg, Zeaxanthin 5 mg	Most Commonly Reported Strengths
Lysine	500 mg	Most Commonly Reported Strength
Maca	500 mg	Commonly Available Strength
Magnesium	250 mg	Most Commonly Reported Strength
Magnesium Powder	325 mg	Commonly Available Strength
Melatonin	5 mg	Most Commonly Reported Strength
Men’s 50+ Multivitamin / Multimineral	Centrum Silver Men 50+ Multivitamin/Multimineral	Most Commonly Reported Product
Men’s Multivitamin/Multimineral	One A Day Men's Complete Multivitamin Multivitamin/Multimineral Bayer	Most Commonly Reported Product
Moringa	400 mg	Commonly Available Strength
Moringa Powder	6 grams	Commonly Available Strength
Multivitamin / Multimineral	Centrum Adults Multivitamin/Multimineral GSK	Most Commonly Reported Product
Multivitamin / Multimineral Liquid	Centrum Liquid Adults Multivitamin/Multimineral	Commonly Available Product
Multivitamin Multimineral Pack	Nature Made Daily Diabetes Health Pack	Commonly Available Product

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Myo-Inositol & D-Chiro Inositol Powder	Theralogix Ovasitol Inositol Powder 40:1 Inositol Blend	Commonly Available Product
Neem	475 mg	Commonly Available Strength
Oil of Oregano Liquid	34 mg	Commonly Available Strength
Omega 3	Carlson Wild Caught Super Omega-3 Gems 1,200 mg Omega-3s	Commonly Available Product
Pediatric Iron Drops	Enfamil Fer-In-Sol Essential Iron For Infants & Toddlers Iron Drops	Commonly Available Product
Pediatric Vitamin D Drops	Enfamil D-Vi-Sol Vitamin D Drops	Commonly Available Product
Poly-Vitamin With Iron Drops	Enfamil Multivitamin & Iron Infant & Toddler Poly-Vi-Sol With Iron	Commonly Available Product
Potassium	99 mg	Most Commonly Reported Strength
Prenatal Vitamins	Spring Valley Prenatal Multivitamin/Multimineral For Pregnant Or Nursing Women	Commonly Available Product
Probiotic	Shaklee +Probiotic Optiflora Pearl Bifidus & Acidophilus	Commonly Available Product
Protein Powder	Pure Protein 100% Whey Protein	Commonly Available Product
Psyllium Fiber	Metamucil Psyllium Fiber 4-In-1 Fiber Made With Real Sugar Unflavored Stone Ground Texture	Commonly Available Product
Red Yeast Rice	600 mg	Most Commonly Reported Strength
Reishi Mushroom	600 mg	Commonly Available Strength
Saw Palmetto	450 mg	Most Commonly Reported Strength
Selenium	200 mcg	Most Commonly Reported Strength
Senior Multivitamin / Multimineral	Centrum Silver Adults 50+ Multivitamin/Multimineral GSK	Most Commonly Reported Product
Silica	35 mg	Commonly Available Strength
Sodium Fluoride Drops	0.5 mg	Commonly Available Strength
Spirulina Powder	BareOrganics Raw Organic Spirulina Powder	Commonly Available Product
Stress B-Complex	Nature Made Stress B-Complex With Key B Vitamins +Vitamin C & Zinc	Commonly Available Product
Super B-Complex	Nature Made Super B-Complex With C	Commonly Available Product
Tart Cherry Extract	1200 mg	Commonly Available Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Tribulus Terrestris	500 mg	Commonly Available Strength
Turkey Tail Mushroom Mycelium Powder	1000 mg	Commonly Available Strength
Turmeric	500 mg	Most Commonly Reported Strength
Ubiquinol	100 mg	Commonly Available Strength
Valerian	500 mg	Commonly Available Strength
Vitamin A	2400 mcg	Most Commonly Reported Strength
Vitamin B-1 (Thiamin)	100 mg	Most Commonly Reported Strength
Vitamin B-12	1000 mcg	Most Commonly Reported Strength
Vitamin B-6	100 mg	Most Commonly Reported Strength
Vitamin B-Complex	GNC B-Complex 50	Commonly Available Product
Vitamin C	500 mg	Most Commonly Reported Strength
Vitamin C Packet	Emergen-C 1,000 Mg Vitamin C Antioxidants, B Vitamins & Electrolytes Fizzy Drink Mix	Most Commonly Reported Product
Vitamin C Powder	1000 mg	Commonly Available Strength
Vitamin D	50 mcg	Most Commonly Reported Strength
Vitamin E	180 mg	Most Commonly Reported Strength
Vitamin K	100 mcg	Most Commonly Reported Strength
Vitamins D & K	Vitamin D 125 mcg, Vitamin K 90 mcg	Most Commonly Reported Strengths
Vitamins D & K Liquid	Vitamin D 25 mcg, Vitamin K 10 mcg	Commonly Available Strengths
Whey Protein	Pure Protein 100% Whey Protein	Commonly Available Product
Women's 50+ Multivitamin / Multimineral	Centrum Silver Women 50+ Multivitamin/Multimineral GSK	Most Commonly Reported Product
Women's Multivitamin / Multimineral	One A Day Women's Complete Multivitamin Multivitamin/Multimineral Bayer	Most Commonly Reported Product
Zinc	50 mg	Most Commonly Reported Strength
Zinc Liquid	15 mg	Most Commonly Reported Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
ZMA	Magnesium 450 mg, Zinc 30 mg, Vitamin B-6 10.5 mg	Commonly Available Strengths

Default Antacid	Antacid Assigned	Selection of Assigned Antacid Based On:
Default Antacid & Antigas Liquid	Equate Regular Strength Antacid Antacid & Antigas	Commonly Available Product
Default Antacid Aluminum & Magnesium	Gaviscon Regular Strength Alumina & Magnesium Trisilicate Tablets / Antacid	Commonly Available Product
Default Antacid Calcium & Magnesium	Rolaids Regular Strength Antacid / Calcium & Magnesium	Most Commonly Reported Product
Default Antacid Plus Tablets	Rolaids Advanced Antacid Plus Anti-Gas Multi-Symptom	Commonly Available Product
Default Calcium Antacid	Tums Calcium Carbonate Antacid Regular Strength 500 GSK	Most Commonly Reported Product

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Dietary Supplement Use 30-Day - Total Dietary Supplements (DSQTOT_L)

Data File: DSQTOT_L.xpt

First Published: February 2025

Last Revised: April 2025

Component Description

The Dietary Supplement and Antacid (DSQ) Section collects information on: 1) prescription and non-prescription dietary supplements (DS), and 2) non-prescription antacids.

In response to the COVID-19 pandemic, the mode of the 30-day dietary supplement data collection changed from personal interviews conducted in participants' homes to telephone interviews conducted after the first 24-hour dietary recall in August 2021-August 2023. Because the 30-day supplement intake was collected by telephone, the interviewer asked survey participants to read to them all the text on the supplement container label. Additionally, the 24-hour supplement intake data collection was discontinued from the 24-hour Day 1 and Day 2 dietary recall. The NHANES August 2021-August 2023 dietary supplement questions are updated to adapt to the telephone mode. Questions on the duration of dietary supplement intake (how long the supplement has been taken) and the reasons for taking the supplement were dropped. Analysts are encouraged to read the documentation to understand the names and structure of the files as well as the variables.

Questions about non-prescription antacids, which contained calcium and/or magnesium (antacids), are also included in this data file.

The DS reported in NHANES August 2021-August 2023 are detailed in the NHANES Dietary Supplement Database (NHANES-DSD) 1999-2023.

30-Day Dietary Supplements Data Files: Two data files were produced from the 30-day dietary supplement and non-prescription antacid sub-sections of the DSQ interview: Total Dietary Supplements file and Individual Dietary Supplements file.

File Name	
Individual Dietary Supplements	DSQIDS_L
Total Dietary Supplements	DSQTOT_L

Individual Dietary Supplements File (DSQIDS_L): Contains detailed information about the types and amounts of individual DS and antacids reported by each participant. The names of the variables included in this file are listed in **Appendix 1**.

The Individual Dietary Supplements File includes one record for each dietary supplement or antacid reported by a survey participant. Only participants that reported taking a dietary supplement or an antacid are included in these files. Each dietary supplement or antacid is identified by a supplement ID number (DSDPID) and each record contains, but is not limited to, the information listed below:

- Name and ID number of the supplement

- Amount of a dietary supplement consumed, which is calculated as the reported amount consumed divided by the serving size from the product label; and
- Amounts of 34 nutrients/dietary components (listed in **Appendix 2**) from each dietary supplement and antacid, as calculated using the NHANES-DSD. If data is missing for any of the variables needed to calculate the amounts consumed for the selected nutrients, this information will be missing for individuals, even though they reported a supplement(s).

This file only includes intake for a select group of nutrients. These records can be linked to the NHANES-DSD, using supplement ID numbers (DSDPID), to obtain more detailed information on reported products. The NHANES-DSD datasets provide information on nutrients in the dietary supplement as reported on the product label. Botanical ingredients would be an example of nutrients not released in the Individual Dietary Supplements files but can be obtained from the NHANES-DSD files.

Total Dietary Supplements Files (DSQTOT_L): Contains, for each participant, average daily total nutrient intakes from DS and antacids. The names for the variables are listed in **Appendix 3**.

The Total Dietary Supplement File provides a summary record of total nutrient intakes from DS and antacids for each individual. This includes users and non-users of DS and antacids. Each total intake record contains, but is not limited to, the following information:

- Whether a dietary supplement was consumed in the past 30 days
- Whether an antacid was consumed in the past 30 days
- Total number of supplements and antacids reported for that participant
- Mean daily intake aggregates of 34 nutrients/dietary components (listed in **Appendix 2**) from all supplements and antacids, as calculated using the NHANES-DSD. If data is missing for any of the variables needed to calculate mean daily intakes, this information will be missing for individuals, even though they may have reported a supplement(s).
- Starting in the 2017-2018 survey cycle, the new variable DSDPID, which indicates supplement ID, was added. The variable DSDSUPID now indicates the old version for supplement ID.

Eligible Sample

All survey participants were eligible for the DSQ.

Interview Setting and Mode of Administration

The DSQ was asked by trained dietary interviewers, using the Computer-Assisted Telephone Interview (CATI) system during the day 1 dietary recall interview. Interviews were conducted with a proxy for participants less than six years of age (who was generally the person most knowledgeable about the participant’s intake). Interviews of children aged 6 to 8 were conducted with a proxy and the child was present to assist in reporting intake information. Interviews of children aged 9-11, were conducted with the child and the assistance of a proxy familiar with the child’s intake. Participants 12 years or older answered for themselves. Dietary interviewers conducted telephone interviews in English and Spanish. The August 2021-August 2022 survey questionnaire can be found on the NHANES website.

Quality Assurance & Quality Control

Data were routinely examined for discrepancies and erroneous entries. Trained nutritionists reviewed incoming data and matched reported dietary supplement entries to known supplements from the in-house NCHS Product Label Database (PLD), where possible; sought additional product labels if feasible; assigned generic or default supplements as appropriate;

transferred or removed products that were not considered DS (i.e., prescription medication); and assigned matching codes (confidence codes). All coding was reviewed by NCHS.

Data Processing and Editing

Data Collection Methods

Dietary Supplements (prescription and non-prescription)

During the interview, survey participants were asked if they had taken a dietary supplement in the past 30 days. Participants used a hand card that listed examples of DS (**Appendix 4**). Those who answered "yes" were asked by the interviewer to read to them all the text on the supplement container label of all the products used. For DS listed on the "Strength Only" List, only the nutrient was selected, and the strength was recorded (see **Appendix 5**). For all other DS reported, the interviewer entered the product's complete name. Interviewers could record up to 30 DS. If the container(s) was not available, the interviewer asked the participant to report verbally the name of the dietary supplement.

Non-prescription antacids

During the telephone interview, survey participants were asked if they had taken non-prescription antacids in the past 30 days. Participants used a hand card that listed examples of non-prescription antacids. Those who answered "yes" were asked by the interviewer to read to them all the text on the antacid container label of all the products used. For each antacid reported, the interviewer entered the product's complete name. If the containers were not available, the interviewer asked the participant to verbally report the name of the antacid.

When the interviewer entered the antacids name into the computer, the name was automatically matched to a prescription drug database (which also contained non-prescription antacid products) on the computer to identify an exact match or similar text matches. The interviewer then selected the best match from a list of possible matches. The original entry of the interviewer and the product selected from the computer list were saved under separate variables for quality control purposes. If an exact match of the product was not found, the interviewer was instructed to select "drug not found on list" from the list. Interviewers recorded up to 30 non-prescription antacids. Nutritionist at NCHS then processed the data collected to determine which products contained calcium and/or magnesium. The non-prescription antacids that contained calcium and/or magnesium were released with the dietary supplement data files.

Participants were also asked how many days each DS was taken in the past 30 days and the amount that was taken on those days. For a non-prescription antacid, participants were asked whether they were taking it as an antacid, as a calcium supplement, or both.

Matching reported dietary supplements to known products

Trained nutritionists, at NCHS, matched the product names entered by the interviewer (including prescription supplement, non-prescription supplement, and antacids) to a known product when possible. NCHS nutritionists first determined if the product is in our in-house PLD. If it is, then the nutritionist verified whether the product label was entered into the system within the 2-year cycle of data collection. If the product label was not entered into the PLD, or the product label was entered before the beginning of the 2-year data collection cycle, NCHS nutritionist attempted to obtain a new product label. NCHS obtained labels for each dietary supplement or non-prescription antacid containing calcium and/or magnesium reported by participants from sources such as the manufacturer, distributor or retailer, the Internet, company catalogs, and the Physician's Desk Reference (PDR). Label information was also obtained from The Dietary Supplement Label Database (DSLD), which is a joint project of the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) and National Library of Medicine (NLM). The DSLD contains the full label contents from a sample of dietary supplement products marketed in the U.S.

NCHS communicates with many major manufacturing company representatives to determine when various product re-formulations become available. Based upon manufacturer advice, we have used a lag time of 5 months after the new re-formulated product hit the market and matched products to participants' accordingly. Despite these precautions, there is no guarantee that the product taken by participants' is matched to the correct formulation in our release files.

NCHS nutritionist attempted to find the exact product(s) reported by participants'; however, this was done with varying degrees of precision. A "matching code or confidence code" (DSDMTCH located in file P_DSQIDS) accompanied each record to indicate the degree of matching certainty. The matches were: 1) Exact or near exact match; 2) Probable match; 3) Generic match; 4) Reasonable match; or 5) Default match. In some cases, no match was made with any certainty. These products were coded 6) No match. Products whose names were reported as "Refused" (DSDSUPP=7777) or "Don't know" (DSDSUPP=9999) have matching codes (DSDMTCH) of 7 and 9, respectively.

NCHS created generic and default DS and antacids, which were also maintained in the database. Generics were created in the database and used when we had collected information about a reported supplement, such as strength of all ingredients, but no brand name. These were generally single ingredient supplements, which included a strength (e.g., vitamin C 500 mg) or multiple vitamins and/or mineral supplements made by a private label manufacturer that was known to us and for which we had a label with identical ingredients and strengths (e.g., brand X all-purpose multivitamin was reported, and we had a label for brand Y, made by the same manufacturer). When all ingredient strengths were not known, a default supplement in the database was used to code the reported supplement. Defaults were created for single ingredient and multiple ingredient supplements based on our own data of most frequently reported supplements of that type, based on the survey cycle in which the data was collected. Created default and generic products and the actual products or strengths that were assigned to these defaults are listed in **Appendix 6**.

After the dietary supplement data was coded and matched to a product in our in-house database, various types of reviews were conducted to ensure the quality of the data. Interviewers' comments are reviewed to ensure that they have been accounted for in coding. Decisions were made about how to code new or unusual dietary supplement products or unusual quantities or units reported by survey participants. Dietary supplements that could not be matched to items in the database were resolved by NCHS nutritionist.

Product information is released from the in-house PLD as the NHANES-DSD, which contains detailed information on the DS and antacids reported by survey participants since NHANES 1999. The NHANES-DSD release consists of three datasets, which contain information on products:

Dietary Supplement Product Information (DSPI)
Dietary Supplement Ingredient Information (DSII)
Dietary Supplement Blend Information (DSBI)

The supplement ID numbers (DSDPID) located in the Individual Dietary Supplements File can be used to merge with the NHANES-DSD files. For more information on the NHANES-DSD, please refer to the documentation and release files located on the NHANES dietary website.

Data Editing

Non-prescription antacids containing calcium and/or magnesium that were reported in the non-prescription antacid section of the DSQ are included in the dietary supplement files.

Prescription products, the following prescription products are included in the dietary supplement files:

- All calcium products except calcium acetate;
- All fluoride products except those in topical gel or cream forms (e.g., stannous

fluoride);
• Over-the-counter niacin, niacinamide, and nicotinic acid;
• All vitamin D products, with the exception of records for Paricalcitol (D2), were retained in the August 2021-August 2023 prescription medication data file; and
• All prenatal products, most prescription vitamins taken by mouth.

All prescription niacin, potassium, and sodium products are retained in the prescription medication file. Combination products that include prescription drug(s) and dietary supplement ingredients are also retained in the prescription medication file (e.g., combination drug alendronate with cholecalciferol or Fosamax with vitamin D3). In August 2021-August 2023, prescription medication section was reduced to two questions (taking prescription medicine and number of prescription medicines). Therefore, none of the above-mentioned prescription products are included in the DS section.

Specific variables and edits:

DSD010: Have you used or taken any vitamins, minerals or other dietary supplements in the past 30 days? Include prescription and non-prescription supplements

Participants with a record of having taken a product determined to be a dietary supplement in the last 30 days are coded "1." This variable was edited and considers DS reported in any of the sub-sections of the DSQ. Some products were mistakenly recorded in the prescription medication sub-section. These data were moved to the dietary supplement files and are counted as supplements for DSD010 and DSDCOUNT. Participants who reported an antacid containing calcium or magnesium as part of the dietary supplement sub-section are also coded "1"; this product was considered to be taken as a calcium supplement. Participants who reported taking an antacid containing calcium or magnesium in the last 30 days that was recorded only in the non-prescription antacid sub-section but did not take any dietary supplement are coded "2." Participants who did not take any product that was determined to be a dietary supplement in the last 30 days are also coded "2," even if they originally reported having taken a supplement. Examples of products that were determined not to be supplements included foods (garlic cloves, raisin bran cereal, PowerBar®), drinks (Ensure®, Gatorade®, tea), over the counter drugs (aspirin, laxatives, electrolyte replacement fluids), homeopathic medicines, and prescription medicines. Prescription medicines were moved from the dietary supplement files to the appropriate prescription medication files. A small number of persons refused to answer this question (coded 7) or did not know whether they used a dietary supplement in the 30 days (coded 9).

DSDCOUNT: The number of DS taken

This variable was computed at NCHS and represents the total number of DS reported by the participant including those DS identified as unknown (DSDPID = 6666666XXX). Antacids that were reported in the dietary supplement sub-section were assumed to be taken as a dietary supplement and also included in the count. Antacids reported in the non-prescription antacid or the prescription medication sub-sections of the DSQ do not contribute to this count. There were also participants who reported the use of a dietary supplement in the past 30 days but did not know the name of the dietary supplement (DSDSUPP = 99999) or refused to report the name of the dietary supplement (DSDSUPP = 77777). Each product reported as refused or don't know is still included in the total count of DS.

DSD010AN: Any non-prescription antacids taken?

This variable was created at NCHS to indicate whether or not an antacid was reported. This variable only considers these types of antacids reported in the non-prescription antacid sub-section of the DSQ. In previous data releases, there were a few antacids reported in the dietary supplement sub-section of the DSQ and these were considered to be taken as DS and included in the DSD010 and DSDCOUNT. There was 2 antacids reported in the dietary supplement sub-section in August 2021-August 2023.

DSDANCNT: The number of non-prescription calcium and/or magnesium - containing antacids taken

This variable was computed at NCHS and represents the total number of antacids reported by the participant. Only these antacids reported in the non-prescription antacid sub-section of the DSQ contributed to this count. Antacids that were reported in the dietary supplement sub-section were assumed to be taken as a supplement and included in the dietary supplement count.

DSDANTA: Created variable that indicates whether an antacid was reported in the DS sub-section or the non-prescription antacid sub-section of the DSQ.

Information on use of non-prescription antacids was sometimes recorded in the dietary supplement sub-section of the DSQ; other times in the non-prescription antacid sub-section. Due to their nutrient content, antacids that contain calcium and/or magnesium are released with the dietary supplement data, irrespective of where they were reported. Only non-prescription antacids that contain calcium and/or magnesium are released in these files; this is not a complete accounting of all non-prescription antacids. Thus, users are cautioned that analyses of these data to estimate the percentage of non-prescription antacids used would not be appropriate.

DSDPID: Supplement ID Number

New Supplement ID is a unique number assigned to each product entered in the in-house PLD.

DSDSUPP: Name of Supplement

This is the name from the front of the product label. The brand name was generally entered first (i.e., Nature's Way) and then the actual product name (i.e., lutein). Information such as the strength (i.e., 60 mg extract) of the product and other qualifiers that help distinguish a product from a similar product (i.e., mega, super, high potency, time release, chewable, extract) were also listed if they were on the front of the product label. Words like "dietary supplement" and health claims were not entered as part of the name.

DSDMTCH: Matching code confidence codes

Supplements were recorded during the DSQ of the SP questionnaire with varying degrees of accuracy and completeness. NCHS had created a system to specify how certain we were with matching a supplement recorded during the interview with the actual product label.

Exact or near exact match (DSDMTCH=1) indicated that this was the only product that could match this entry.

Probable match (DSDMTCH=2) indicated that the match was not exact, but knowledge of the company's products strongly suggested that this was the only possible match choice. For example, the entry may not have specified strength or include words such as timed release, but no other options were available for this brand according to manufacturer or retailer information.

Generic match (DSDMTCH=3) indicated we had information on the strength for all ingredients, either: a) as part of the name (e.g., vitamin C 500 mg) or b) because the manufacturer was known and NCHS had an identical product made by this manufacturer for a different distributor or retailer. Thus, the ingredients and amounts were considered to be accurate despite an exact brand match.

Reasonable match (DSDMTCH=4) indicated that the product name may have been incomplete or could have been complete, but other products of this brand also started with these same words so this could not be assured. In these cases, the entered name was matched to either: a) the most frequently reported of these products in the NHANES 2017-2018 and 2019-2020 data, if this could be determined; b) the best-selling product by this company that matched the entered name; or c) the most basic product by this company, as assessed by label wording.

Default match (DSDMTCH=5) indicated that the exact product could not be obtained because the name was imprecise, or the exact brand product could not be located, and no generic was assigned. In these cases, the entered product was matched to a created default product based upon: a) the most commonly reported strengths for single ingredients; b) the most commonly reported brands for major multiple ingredient products such as multivitamins and multivitamin/multi-minerals for children, seniors, or adults, if available; or c) products manufactured by a large, private-label manufacturer.

Finally, a match code of No match (DSDMTCH=6) indicated that no product was found and there was not enough detail in the name to assign a generic or default match with any confidence. The words "no product information available" were listed as the product name (DSDSUPP).

Analysts should be aware that for default matches and matches that chose between several similarly named supplements, there was less certainty that the ingredients and ingredient amounts in the supplement assigned exactly matched those in the supplement actually taken. Additionally, NCHS cannot guarantee in any case that the matched product was the exact product taken or even that any product actually was taken, as these data were self-reported.

DSD070: Dietary supplement container seen by interviewer?

This variable indicated whether the product container was seen by the interviewer. Containers were seen for approximately 87% of the records. A more precise name for a supplement could be recorded by the interviewer, and thus a more precise match to a known supplement can be made, when the interviewer saw the supplement container rather than recording the participant's report of the supplement name (for example, multivitamin/multiminerals were often reported as multivitamins). In general, this was reflected in the matching code, but analysts should be aware that precision was greater when a container was seen.

This variable was mostly unedited. Interviewers asked to see the containers in all three sub-sections of the DSQ. Therefore, most records included in the dietary supplement file contained this information.

DSD090: For how long have you been taking this product or a similar type of product?

This variable has been discontinued in survey cycle August 2021-August 2023.

DSD103: In the past 30 days, on how many days did you take the product?

This variable was unedited. Information was missing for dietary supplement data that was recorded in the prescription medication sub-section, since this question was not asked in that sub-section of the DSQ.

DSD122Q and DSD122U: On the days that you took the product how much did you usually take on a single day?

The data was edited to standardize the amount the participant reported taking and the amount according to the product label serving size. For example, if a participant reported taking 1 tablespoon of a supplement and the label serving size was 3 teaspoons, then the variable was edited to 3 teaspoons (1 tablespoon = 3 teaspoons). There were 581 records in which data was edited.

In most of the cases, using the alternative serving size (information available on some product labels), using simple conversions (i.e., teaspoons to tablespoons) or contacting the manufacturer for information on label serving size (i.e., actual amount for a "capful") provided enough information to make clear edits. However, in some cases the reported amount taken was very different from the product label serving size (n=23). For example, the participant may have reported 1 tablet, but the label serving size is 1 tablespoon. This was assumed to be an error in reporting or an interviewer data entry error. In these cases, the reported amount taken was assumed to be the label serving size. Additionally, all records that were assigned a default product were edited and the label serving size was assigned for DSD122Q and DSD122U.

Information was missing for dietary supplement data that was recorded in the prescription medication sub-section, since this question was not asked in that sub-section of the DSQ.

DSDACTSS: Reported serving size/label serving size

This variable was derived using the information from DSD122Q (reported quantity taken) and information from variable DSDSERVQ (serving size quantity from product label) from the "Dietary Supplement Product Information" file (DSPI) of the NHANES-DSD. File DSPI provided information on the serving size from the product label's supplement facts panel for which the nutrient amounts were based on. The reported amount taken (DSD122Q) was divided by the serving size amount from DSPI (variable DSDSERVQ). Values were set to missing if any information was missing for DSD122Q or DSDSERVQ.

DSD124: Took product on own or doctor advised

This variable has been discontinued in survey cycle August 2021-August 2023.

DSQ128A- DSD128NN: Reason(s) for taking dietary supplement

This variable has been discontinued in survey cycle August 2021-August 2023.

Nutrient variables in the Individual Dietary Supplements file, DSQIDS_L

These variables were created by using files from the NHANES_DSD that contain information on the serving size and the quantity of each nutrient from the product label's supplement facts panel. The participants reported usual amount taken was divided by the serving size from the label in order to determine the actual amount of nutrient consumed. For example, a participant may have reported taking one tablet, but the serving size amount was 2 tablets. Therefore, only half of the nutrients on the label were being consumed. The variable DSDACTSS (Reported serving size/label serving size) indicated the actual amount of product that was consumed. The actual amount of product consumed was then multiplied by the ingredient amount for each dietary supplement or antacid. In the example above, 0.5 would be multiplied by each ingredient/nutrient to estimate the nutrient intake.

Analysts do need to be aware that in some cases data is missing for the amount consumed. In these cases, nutrients could not be calculated and are missing. Analyst should be aware of this, especially when estimating the prevalence of use of certain nutrients by using only the nutrient variables.

DSQIFDFE: Folate, DFE (mcg)

Dietary Folate Equivalents (DFE) were calculated by using a conversion factor of 1.7. This conversion is based on recommendations set by the Institute of Medicines Dietary Reference Intakes (National Academies Press, 2006).

Nutrient variables in the Total Dietary Supplements file, DSQTOT_L

These variables were created by calculating a mean daily intake. The calculation was based on the number of days the participant reported taking each dietary supplement (variable DSD103). For example, if participant X took calcium 600 mg 15 days out of the month, the calculation would be 600mg*15 reported days/30 days per month. If participant X took more than one supplement or antacid containing calcium, they would then be added up for the total mean daily calcium intake from DS.

Analysts do need to be aware that in some cases data is missing for the amount consumed and the number of days the supplement was taken. In these cases, nutrients could not be calculated or mean daily intake could not be estimated and therefore values are missing. Analyst should be aware of this, especially when estimating the prevalence of use of certain nutrients by using only the nutrient variables.

Analytic Notes

Sample weights for 30-day dietary supplement data

The August 2021–August 2023 NHANES sample design is similar to past cycles that drew a multi-year, stratified, clustered four-stage sample of the U.S. civilian noninstitutionalized population to provide national health estimates. To adapt to the continuing pandemic environment, this cycle also included an updated sample design to minimize contact between interviewers and survey participants. An overview of sample design, nonresponse bias assessment, and analytic guides for NHANES August 2021-August 2023 is provided on the NHANES website. In addition, a document is also available online to provide additional information on the changes that pertain to the dietary supplement data collection in this survey cycle.

Similar to prior cycles, sample weights are constructed to encompass the unequal probabilities of selection, as well as adjustments for non-participation by selected sample persons. In order to produce national, representative estimates, the appropriate sample weights must be used.

As mentioned previously, in August 2021-August 2023, the 30-day dietary supplement use was collected following the conclusion of the first 24-hour dietary recall, therefore the same sample weights created for the day 1 recall (i.e., WTDRD1) is provided in the present dataset, and should be used with analysis uses the 30-day dietary supplement data (either alone or when Dietary Supplement data are used in conjunction with MEC data). The set of weights (WTDRD1) is applicable to the 6,754 participants with 30-day dietary supplement data. These weights were constructed by taking the MEC sample weights (WTMEC2YR) and further adjusting for: (a) the additional non-response; and (b) the differential allocation by weekdays (Monday through Thursday) and weekends (Friday through Sunday) for the dietary intake data collection. These weights are more variable than the MEC weights, and the sample size is smaller, so estimated standard errors using 30-day dietary supplement data and these weights might be larger than standard errors for similar estimates based on MEC weights. Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

In NHANES August 2021-August 2023, a total of 11,375 dietary supplements or antacids were reported by survey participants as taken in the last 30 days prior to the interview. Approximately 0.1% of these reported records are missing information on the number of days the supplement was taken and/or the amount usually consumed. This is because no product information was available. Therefore, even though the use of particular supplements was reported, no nutrient content or mean daily intake data for selected nutrients was calculated because of the missing information on the amount usually consumed. Analysts should be cautious when estimating the prevalence of supplement use for specific nutrients. Nutrient variables released in the Individual or Total Dietary Supplements file (DSQIDS_L or DSQTOT_L) should not be used alone to produce such prevalence estimation. It is recommended that analysts merge participant datasets with the NHANES Dietary Supplement Database files to identify the reported supplements containing the nutrients of interest.

In some cases, it may appear as though a participant reported the same dietary supplement more than once. There are a several reasons for these duplications. Participants' may have reported DS with the same generic ingredient but different brand names (i.e., 2 different brands of calcium, but calcium is collected generically with only the strength) or the supplement may have been different forms or dosages of the same product. There were a few cases where the participant did report the exact same dietary supplement, with the same length of use, frequency of use or amount used. In these cases, since the interviewer recorded that a separate dietary supplement container was seen for each reported dietary supplement both mentions of the dietary supplement were retained in the file.

During the data editing process, outlier values were examined. When there was insufficient information to conclude that values were invalid, they were left in the data set. Analysts should examine the distribution of the data and consider whether or not it is appropriate to include or exclude extreme values in a given analysis.

NCHS collects brand name information on supplements whenever feasible, to ensure as much accuracy as possible in finding the label information for the exact product taken and providing exact ingredient information for this product to data users. Brand names are not collected for DS listed on the "Strength Only" List (see **Appendix 5**: Vitamins/Minerals). Only the nutrient is selected, and the strength is recorded. Analyst should be aware that for these specific nutrients, NCHS releases generic information (i.e., calcium 600 mg). Thus, analyses of consumer usage by brand name using NHANES data may not be accurate and is not recommended. Brand names that are available in a limited geographic area of the U.S. are not released; generic names are used for these products.

Deriving nutrient estimates from dietary supplement data

Thirty-four nutrients have already been computed for this release. In order to compute average daily intakes for nutrients or bioactives other than these 34, analyst will need to combine the Individual Dietary Supplements File (DSQIDS_L) with the NHANES-DSD files (DSPI, DSII and DSBI), which contain information on each product and ingredients and amounts in those products.

Intake of multiple supplements with the same nutrient(s), multiple use of the same supplement on the same day, and nutrients in blends should be taken into account in nutrient calculations. Nutrient names and the quantity units need to be harmonized and nutrient amounts from all these calculations must then be summed. Some nutrient amounts are for a nutrient compound (generally a foreign-made product or an antacid) and these must be converted to a nutrient elemental amount.
Analysts need to be aware of question changes over the cycles when looking at trends. Dietary supplement data was reported as times per month in 1999-2000, days in the past month in 2001-2002, and days in the past 30 days in 2003-2023.

Please refer to the on-line NHANES Dietary Tutorial, which is available on the NHANES website, for further details on how to use these datasets.

Estimating total nutrient intakes from all sources (foods, beverages and DS)

To estimate total dietary nutrient intake, nutrients from diet, supplements, and antacids should be combined. Although in August 2021-August 2023, the 30-day DS data was collected with the first 24-hour dietary recall using similar data collection methods, the reference intake period is different. Therefore, combining nutrient intake from these different variables to estimate total nutrient intake requires thoughtful consideration of the analytic goal and methods, and deserves accurate description of methods, assumptions, and weaknesses in any presentation of results.

All the key concepts and caveats regarding estimating nutrient intakes from dietary (foods and beverages) intake apply when estimating total nutrient intake from both foods and supplements. For these analyses, the sample of participants with reliable data for both supplement intake and the First Day recall are selected. Then, for each participant the average daily nutrient intake of supplements from the 30-day supplements use files are determined and added to the nutrient intake from the 24-hour recall. The 30-day supplement files can be linked to the dietary files by the respondent sequence number (SEQN). For more information on the data, refer to the documentations accompanying the 24-hour dietary intakes and dietary supplement datasets.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

WTDRD1 - Dietary day one sample weight

Variable Name: WTDRD1

SAS Label: Dietary day one sample weight

English Text: Dietary day one sample weight

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
5708.632624 to 408505.81364	Range of Values	6754	6754	
0	Day 1 dietary recall not done/incomplete	2106	8860	
.	Missing	0	8860	

DSDCOUNT - Total # of Dietary Supplements Taken

Variable Name: DSDCOUNT

SAS Label: Total # of Dietary Supplements Taken

English Text: Includes all supplements and the antacids reported with supplements, but not antacids reported with medications.

English Instructions: < blank >

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 30	Range of Values	6731	6731	
77	Refused	13	6744	
99	Don't know	10	6754	
.	Missing	2106	8860	

DSDANCNT - Total # of Antacids Taken

Variable Name: DSDANCNT

SAS Label: Total # of Antacids Taken

English Text: Includes all antacids reported with medications.

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 3	Range of Values	6720	6720	
77	Refused	3	6723	
99	Don't know	5	6728	
.	Missing	2132	8860	

DSD010 - Any Dietary Supplements Taken?

Variable Name: DSD010

SAS Label: Any Dietary Supplements Taken?

English Text: The next questions are about {your/SP's} use of dietary supplements and medications during the past month. {Have you/Has SP} used or taken any vitamins, minerals or other dietary supplements in the past month? Include those products prescribed by a health professional such as a doctor or dentist, and those that do not require a prescription. This card lists some examples of different types of dietary supplements.

English Instructions: HAND CARD DSQ1

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	3762	3762	
2	No	2969	6731	
7	Refused	13	6744	
9	Don't know	10	6754	
.	Missing	2106	8860	

DSD010AN - Any Antacids Taken?

Variable Name: DSD010AN

SAS Label: Any Antacids Taken?

English Text:

Target:

Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	876	876	
2	No	5844	6720	
7	Refused	3	6723	
9	Don't know	5	6728	
.	Missing	2132	8860	

DSQTKCAL - Energy (kcal)

Variable Name: DSQTKCAL

SAS Label: Energy (kcal)

English Text: Energy (kcal)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.03 to 340	Range of Values	2100	2100	
.	Missing	6760	8860	

DSQTPROT - Protein (gm)

Variable Name: DSQTPROT

SAS Label: Protein (gm)

English Text: Protein (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 37	Range of Values	343	343	
.	Missing	8517	8860	

DSQTCARB - Carbohydrate (gm)

Variable Name: DSQTCARB

SAS Label: Carbohydrate (gm)

English Text: Carbohydrate (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 113	Range of Values	1750	1750	
.	Missing	7110	8860	

DSQTSUGR - Total sugars (gm)

Variable Name: DSQTSUGR

SAS Label: Total sugars (gm)

English Text: Total sugars (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 66	Range of Values	1284	1284	
.	Missing	7576	8860	

DSQTFIBE - Dietary fiber (gm)

Variable Name: DSQTFIBE

SAS Label: Dietary fiber (gm)

English Text: Dietary fiber (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 45	Range of Values	282	282	
.	Missing	8578	8860	

DSQTTFAT - Total fat (gm)

Variable Name: DSQTTFAT

SAS Label: Total fat (gm)

English Text: Total fat (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.03 to 30	Range of Values	763	763	
.	Missing	8097	8860	

DSQTSFAT - Total saturated fatty acids (gm)

Variable Name: DSQTSFAT

SAS Label: Total saturated fatty acids (gm)

English Text: Total saturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.03 to 5.3	Range of Values	197	197	
.	Missing	8663	8860	

DSQTMFAT - Total monounsaturated fatty acids (gm)

Variable Name: DSQTMFAT

SAS Label: Total monounsaturated fatty acids (gm)

English Text: Total monosaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 5	Range of Values	91	91	
.	Missing	8769	8860	

DSQTPFAT - Total polyunsaturated fatty acids (gm)

Variable Name: DSQTPFAT

SAS Label: Total polyunsaturated fatty acids (gm)

English Text: Total polyunsaturated fatty acids (gm)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 18	Range of Values	224	224	
.	Missing	8636	8860	

DSQTCHOL - Cholesterol (mg)

Variable Name: DSQTCHOL

SAS Label: Cholesterol (mg)

English Text: Cholesterol (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 56.7	Range of Values	417	417	
.	Missing	8443	8860	

DSQTLYCO - Lycopene (mcg)

Variable Name: DSQTLYCO

SAS Label: Lycopene (mcg)

English Text: Lycopene (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10 to 20000	Range of Values	531	531	
.	Missing	8329	8860	

DSQTLZ - Lutein + zeaxanthin (mcg)

Variable Name: DSQTLZ

SAS Label: Lutein + zeaxanthin (mcg)

English Text: Lutein + zeaxanthin (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2 to 275000	Range of Values	741	741	
.	Missing	8119	8860	

DSQTVB1 - Thiamin (Vitamin B1) (mg)

Variable Name: DSQTVB1

SAS Label: Thiamin (Vitamin B1) (mg)

English Text: Thiamin (Vitamin B1) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.003 to 401.5	Range of Values	1632	1632	
.	Missing	7228	8860	

DSQTVB2 - Riboflavin (Vitamin B2) (mg)

Variable Name: DSQTVB2

SAS Label: Riboflavin (Vitamin B2) (mg)

English Text: Riboflavin (Vitamin B2) (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.004 to 401.7	Range of Values	1663	1663	
.	Missing	7197	8860	

DSQTNIAC - Niacin (mg)

Variable Name: DSQTNIAC

SAS Label: Niacin (mg)

English Text: Niacin (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.1 to 2013.3	Range of Values	1829	1829	
.	Missing	7031	8860	

DSQTVB6 - Vitamin B6 (mg)

Variable Name: DSQTVB6

SAS Label: Vitamin B6 (mg)

English Text: Vitamin B6 (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.01 to 500	Range of Values	2237	2237	
.	Missing	6623	8860	

DSQTFA - Folic acid (mcg)

Variable Name: DSQTFA

SAS Label: Folic acid (mcg)

English Text: Folic acid (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 9032	Range of Values	2183	2183	
.	Missing	6677	8860	

DSQTFDFE - Folate, DFE (mcg)

Variable Name: DSQTFDFE

SAS Label: Folate, DFE (mcg)

English Text: Folate, DFE (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 15354	Range of Values	2183	2183	
.	Missing	6677	8860	

DSQTCHL - Total choline (mg)

Variable Name: DSQTCHL

SAS Label: Total choline (mg)

English Text: Total choline (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.001 to 1175	Range of Values	288	288	
.	Missing	8572	8860	

DSQTVB12 - Vitamin B12 (mcg)

Variable Name: DSQTVB12

SAS Label: Vitamin B12 (mcg)

English Text: Vitamin B12 (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 9333.33	Range of Values	2411	2411	
.	Missing	6449	8860	

DSQTVC - Vitamin C (mg)

Variable Name: DSQTVC

SAS Label: Vitamin C (mg)

English Text: Vitamin C (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.001 to 8000	Range of Values	2622	2622	
.	Missing	6238	8860	

DSQTVK - Vitamin K (mcg)

Variable Name: DSQTVK

SAS Label: Vitamin K (mcg)

English Text: Vitamin K (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.4 to 22500	Range of Values	1208	1208	
.	Missing	7652	8860	

DSQTVD - Vitamin D (D2 + D3) (mcg)

Variable Name: DSQTVD

SAS Label: Vitamin D (D2 + D3) (mcg)

English Text: Vitamin D (D2 + D3) (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.04 to 1262.08	Range of Values	3016	3016	
.	Missing	5844	8860	

DSQTCALC - Calcium (mg)

Variable Name: DSQTCALC

SAS Label: Calcium (mg)

English Text: Calcium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.5 to 3383.3	Range of Values	2426	2426	
.	Missing	6434	8860	

DSQTPHOS - Phosphorus (mg)

Variable Name: DSQTPHOS

SAS Label: Phosphorus (mg)

English Text: Phosphorus (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 583.3	Range of Values	834	834	
.	Missing	8026	8860	

DSQTMAGN - Magnesium (mg)

Variable Name: DSQTMAGN

SAS Label: Magnesium (mg)

English Text: Magnesium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 1691.7	Range of Values	1613	1613	
.	Missing	7247	8860	

DSQTIRON - Iron (mg)

Variable Name: DSQTIRON

SAS Label: Iron (mg)

English Text: Iron (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.003 to 150	Range of Values	1051	1051	
.	Missing	7809	8860	

DSQTZINC - Zinc (mg)

Variable Name: DSQTZINC

SAS Label: Zinc (mg)

English Text: Zinc (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 200	Range of Values	2314	2314	
.	Missing	6546	8860	

DSQTCOPP - Copper (mg)

Variable Name: DSQTCOPP

SAS Label: Copper (mg)

English Text: Copper (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.00001 to 6	Range of Values	1311	1311	
.	Missing	7549	8860	

DSQTSODI - Sodium (mg)

Variable Name: DSQTSODI

SAS Label: Sodium (mg)

English Text: Sodium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.03 to 1092	Range of Values	1245	1245	
.	Missing	7615	8860	

DSQTPOTA - Potassium (mg)

Variable Name: DSQTPOTA

SAS Label: Potassium (mg)

English Text: Potassium (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.2 to 3120	Range of Values	960	960	
.	Missing	7900	8860	

DSQTSELE - Selenium (mcg)

Variable Name: DSQTSELE

SAS Label: Selenium (mcg)

English Text: Selenium (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.7 to 400	Range of Values	1351	1351	
.	Missing	7509	8860	

DSQTCAFF - Caffeine (mg)

Variable Name: DSQTCAFF

SAS Label: Caffeine (mg)

English Text: Caffeine (mg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
10 to 133	Range of Values	9	9	
.	Missing	8851	8860	

DSQTIODI - Iodine (mcg)

Variable Name: DSQTIODI

SAS Label: Iodine (mcg)

English Text: Iodine (mcg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.5 to 1500	Range of Values	1816	1816	
.	Missing	7044	8860	

Appendix 1: Variables in the Individual Dietary Supplements File (DSQIDS_L)

Variables	Variable Label
SEQN	Respondent sequence number
DSDPID	NCHS Supplement ID
DSDANTA	Antacid reported as a dietary supplement
DSD070	Was container seen?
DSDMTCH	Matching code
DSD103	Days supplement taken, past 30 days
DSD122Q	Quantity of supplement taken per day
DSD122U	Dosage form
DSDACTSS	Reported serving size/label serving size
DSQIKCAL	Energy (kcal)
DSQIPROT	Protein (gm)
DSQICARB	Carbohydrate (gm)
DSQISUGR	Total sugars (gm)
DSQIFIBE	Dietary fiber (gm)
DSQITFAT	Total fat (gm)
DSQISFAT	Total saturated fatty acids (gm)
DSQIMFAT	Total monounsaturated fatty acids (gm)
DSQIPFAT	Total polyunsaturated fatty acids (gm)
DSQICHOL	Cholesterol (mg)
DSQILYCO	Lycopene (mcg)
DSQILZ	Lutein + zeaxanthin (mcg)
DSQIVB1	Thiamin (Vitamin B1) (mg)
DSQIVB2	Riboflavin (Vitamin B2) (mg)
DSQINIAC	Niacin (mg)
DSQIVB6	Vitamin B6 (mg)
DSQIFA	Folic acid (mcg)
DSQIFDFE	Folate, DFE (mcg)
DSQICHL	Total choline (mg)
DSQIVB12	Vitamin B12 (mcg)
DSQIVC	Vitamin C (mg)
DSQIVK	Vitamin K (mcg)
DSQIVD	Vitamin D (D2 + D3) (mcg)
DSQICALC	Calcium (mg)
DSQIPHOS	Phosphorus (mg)
DSQIMAGN	Magnesium (mg)
DSQIIRON	Iron (mg)
DSQIZINC	Zinc (mg)
DSQICOPP	Copper (mg)
DSQISODI	Sodium (mg)

Variables	Variable Label
DSQIPOTA	Potassium (mg)
DSQISELE	Selenium (mcg)
DSQICAFF	Caffeine (mg)
DSQIIODI	Iodine (mcg)

Appendix 2: List of Nutrients/Dietary Components (Unit)

Energy and Macronutrients
Food energy (kcal) €
Protein (g) €
Carbohydrate (g) €
Fat, total (g) €
Alcohol (g)
Sugars, total (g) €
Dietary fiber, total (g) €
Water (moisture) (g)*
Saturated fatty acids, total (g) €
Monounsaturated fatty acids, total (g) €
Polyunsaturated fatty acids, total (g) €
Cholesterol (mg) €

Individual fatty acids:
4:0 (g)
6:0 (g)
8:0 (g)
10:0 (g)
12:0 (g)
14:0 (g)
16:0 (g)
18:0 (g)
16:1 (g)
18:1 (g)
20:1 (g)
22:1 (g)
18:2 (g)
18:3 (g)
18:4 (g)
20:4 (g)
20:5 n-3 (g)
22:5 n-3 (g)
22:6 n-3 (g)

Vitamins, Minerals, and Other Components
Vitamin A as retinol activity equivalents (µg) ¥
Retinol (µg) ¥
Carotenoids:
  Carotene, alpha (µg) ¥
  Carotene, beta (µg) ¥
  Cryptoxanthin, beta (µg) ¥
  Lycopene (µg) €
  Lutein + zeaxanthin (µg) €
Vitamin E as alpha-tocopherol (mg) ¥
  Added vitamin E as alpha-tocopherol (mg)
Vitamin D (D2 + D3) (µg) €
Vitamin K as phylloquinone (µg) €
Vitamin C (mg) €
Thiamin (mg) €
Riboflavin (mg) €
Niacin (mg) €
Vitamin B-6 (mg) €
Folate, total (µg)
  Folate as dietary folate equivalents (µg) €
  Folic acid (µg) €
  Food folate (µg)
Choline, total (mg) €
Vitamin B-12 (µg) €
  Added vitamin B-12 (µg)
Calcium (mg) €

Iron (mg) €
Magnesium (mg) €
Phosphorus (mg) €
Potassium (mg) €
Sodium (mg) €
Zinc (mg) €
Copper (mg) €
Selenium (µg) €
Caffeine (mg) €
Theobromine (mg)

Iodine (mcg)†

* Value reflects moisture present in all foods, beverages, and water consumed as a beverage (variables DR1IMOIS, DR2IMOIS, DR1TMOIS, DR2TMOIS)

€Indicates nutrients in which data is available for dietary supplements and non-prescription antacids containing calcium and/or magnesium

¥Indicates that data will be available in a later release cycle

† Only included in the dietary supplement files

Appendix 3: Variables in the Total Dietary Supplement File (DSQTOT_L)

Variables	Variable Label
SEQN	Respondent sequence number
DSD010	Any Dietary Supplements Taken?
DSDCOUNT	Total # of Dietary Supplements Taken
DSD010AN	Any Antacids Taken?
DSDANCNT	Total # of Antacids Taken
DSQTKCAL	Energy (kcal)
DSQTPROT	Protein (gm)
DSQTCARB	Carbohydrate (gm)
DSQTSUGR	Total sugars (gm)
DSQTFIBE	Dietary fiber (gm)
DSQTTFAT	Total fat (gm)
DSQTSFAT	Total saturated fatty acids (gm)
DSQTMFAT	Total monounsaturated fatty acids (gm)
DSQTPFAT	Total polyunsaturated fatty acids (gm)
DSQTCHOL	Cholesterol (mg)
DSQTLYCO	Lycopene (mcg)
DSQTLZ	Lutein + zeaxanthin (mcg)
DSQTVB1	Thiamin (Vitamin B1) (mg)
DSQTVB2	Riboflavin (Vitamin B2) (mg)
DSQTNIAC	Niacin (mg)
DSQTVB6	Vitamin B6 (mg)
DSQTFA	Folic acid (mcg)
DSQTFDFE	Folate, DFE (mcg)
DSQTCHL	Total choline (mg)
DSQTVB12	Vitamin B12 (mcg)
DSQTVC	Vitamin C (mg)
DSQTVK	Vitamin K (mcg)
DSQTVD	Vitamin D (D2 + D3) (mcg)
DSQTCALC	Calcium (mg)
DSQTPHOS	Phosphorus (mg)
DSQTMAGN	Magnesium (mg)
DSQTIRON	Iron (mg)
DSQTZINC	Zinc (mg)
DSQTCOPP	Copper (mg)
DSQTSODI	Sodium (mg)
DSQTPOTA	Potassium (mg)
DSQTSELE	Selenium (mcg)
DSQTCAFF	Caffeine (mg)
DSQTIODI	Iodine (mcg)

Appendix 4: Dietary Supplement Hand card

VITAMINS **MINERALS**	Calcium, Vitamin C, Calcium and Iron, Vitamin E, Magnesium, Zinc, Calcium plus Vitamin D
MULTI-VITAMIN--MULTI-MINERALS	Flintstones, One a Day, Prenatals, Tri-Vi-Flor, B-Complex, Centrum
HERBALS AND BOTANICLAS	Echinacea, Garlic, Saw Palmetto, Ginkgo, Ginseng
FIBER	Metamucil, Fibercon, Benefiber
AMINO ACIDS	Lysine, Methionine, Tryptophan
OTHERS	Fish Oil, Chondrotin, Glucosamine

Appendix 5: Vitamins/Minerals on the "Strength Only" List

- Vitamin A
- Vitamin B6
- Vitamin B12
- Vitamin C
- Vitamin D (D3)
- Vitamin E
- Calcium
- Chromium (Chromium Picolinate)
- Folate (Folic Acid)
- Iron (Ferrous Xxxate)
- Magnesium
- Potassium
- Selenium
- Zinc (Zinc Gluconate)

Appendix 6: Created Default Supplements and Antacids

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
5-HTP (5-Hydroxytryptophan)	100 mg	Most Commonly Reported Strength
Acetyl L-Carnitine	500 mg	Most Commonly Reported Strength
Alpha Lipoic Acid	300 mg	Commonly Available Strength
Amino Acid	ON Optimum Nutrition Superior Amino 2222 Tablets	Commonly Available Product
Apple Cider Vinegar Tablets	Nature's Bounty Apple Cider Vinegar 480 mg Per Serving	Commonly Available Product
Arginine Powder	3000 mg	Commonly Available Strength
Artichoke	500 mg	Most Commonly Reported Strength
Ashwagandha	500 mg	Most Commonly Reported Strength
B 50 B-Complex	GNC B-Complex 50	Commonly Available Product
Balanced B 100 B-Complex	Nature Made Time Release B-100 Complex	Commonly Available Product
BCAA Powder	ON Optimum Nutrition Instantized BCAA 5000 Powder	Commonly Available Product
Beet Powder	7 grams	Commonly Available Strength
Biotin	5000 mcg	Most Commonly Reported Strength
Black Cumin Seed Oil	500 Mg	Commonly Available Strength
Borage Oil	1000 mg	Commonly Available Strength
Brewer's Yeast Powder	Solgar Brewer's Yeast Powder	Commonly Available Product
Calcium	600 mg	Most Commonly Reported Strength
Calcium & Magnesium	Calcium 500 mg, Magnesium 250 mg	Most Commonly Reported Strengths
Calcium + D + K	Viactiv Calcium+ Max Formula Bone Strengthening 650 mg Calcium	Commonly Available Product
Calcium + Magnesium Liquid	Calcium 1000 mg, Magnesium 500 mg	Commonly Available Strengths
Calcium + Vitamin D 20 mcg	Calcium 600 mg, Vitamin D 20 mcg	Most Commonly Reported Strengths
Calcium + Vitamin D 2000 IU	Calcium 600 mg, Vitamin D 2000 IU	Commonly Available Strengths
Calcium 1200 mg with Vitamin D	Calcium 600 mg, Vitamin D 40 mcg	Commonly Available Strengths

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Calcium 250 mg With Vitamin D	Calcium 250 mg, Vitamin D 3.1 mcg	Commonly Available Strengths
Calcium 500 mg With Vitamin D	Calcium 500 mg, Vitamin D 10 mcg	Most Commonly Reported Strengths
Calcium 600 mg With Vitamin D	Calcium 600 mg, Vitamin D 20 mcg	Most Commonly Reported Strengths
Calcium Magnesium & Zinc	Calcium 333 mg, Magnesium 133 mg & Zinc 5 mg	Commonly Available Strengths
Calcium With Vitamin D	Calcium 600 mg, Vitamin D 20 mcg	Most Commonly Reported Strengths
Calcium With Vitamin D & Minerals	Caltrate Bone Health Advanced Calcium 600+D3 Plus Minerals	Commonly Available Product
CBD Gummies	Wyld CBD Gummies 500 mg CBD 25 mg CBD Per Gummy	Commonly Available Product
CBD Oil	Sunsoil CBD Oil 10 mg CBD Servings	Commonly Available Product
Children’s Multivitamin/Multimineral	Flintstones Complete Children’s Multivitamin	Most Commonly Reported Product
Chromium	200 mcg	Most Commonly Reported Strength
Cinnamon	500 mg	Most Commonly Reported Strength
Citrus Bergamot	500 mg	Commonly Available Strength
Cod Liver Oil Softgels	Spring Valley Cod Liver Oil Plus Vitamins A & D3	Commonly Available Product
Coenzyme Q-10	100 mg	Most Commonly Reported Strength
Collagen	1000 mg	Commonly Available Strength
Collagen & Vitamin C	Neocell Super Collagen + Vitamin C Collagen Type 1 & 3	Commonly Available Product
Collagen Powder	Neocell Super Collagen Peptides Collagen Type 1&3	Commonly Available Product
Collinsonia Root	500 mg	Commonly Available Strength
CoQ10 + Red Yeast Rice	CoQ10 60 mg, Red Yeast Rice 600 mg	Commonly Available Strengths
Cranberry	500 mg	Most Commonly Reported Strength
Creatine Capsules	750 mg	Commonly Available Strength
Creatine Monohydrate	5000 mg	Most Commonly Reported Strength
Curcumin	500 mg	Commonly Available Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Dairy Digestive Caplets	Lactaid Original Lactase Enzyme	Commonly Available Product
DHA	200 mg	Commonly Available Strength
DHEA	25 mg	Most Commonly Reported Strength
Echinacea	400 mg	Most Commonly Reported Strength
Echinacea Liquid	1.0 ml	Commonly Available Strength
Elderberry	575 mg	Commonly Available Strength
Elderberry Liquid	Nature's Way Sambucus Standardized Elderberry Original Syrup	Commonly Available Product
Enzymes	Genuine N-Zimes Dr. Howell's Original Formula Digestive Enzyme	Commonly Available Product
Evening Primrose Oil	1000 mg	Most Commonly Reported Strength
Eye Health Complex	Equate Advanced Eye Health Complex	Commonly Available Product
Eye Multi-Vitamin and Mineral	Bausch + Lomb Preservision Eye Vitamin And Mineral AREDS Formula Soft Gels	Commonly Available Product
Fiber Capsules	Metamucil Psyllium Fiber 3-In-1 Fiber Multiple Health Benefits	Most Commonly Reported Product
Fiber Powder	Metamucil Psyllium Fiber 4-In-1 Fiber Made With Real Sugar Unflavored Stone Ground Texture	Commonly Available Product
Fish Oil	Nature's Bounty Fish Oil 1000 Mg | 300 mg Of Omega-3	Commonly Available Product
Fish, Flax and Borage Oils	Puritan's Pride Maximum Strength Triple Omega 3-6-9 Fish, Flax & Borage Oils	Commonly Available Product
Flax Seed Oil	1000 mg	Most Commonly Reported Strength
Folate	1333 mcg DFE	Most Commonly Reported Strength
GABA	500 mg	Commonly Available Strength
Garcinia Cambogia	500 mg	Commonly Available Strength
Garlic	1000 mg	Most Commonly Reported Strength
Ginger	550 mg	Most Commonly Reported Strength
Ginkgo Biloba	120 mg	Most Commonly Reported Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Ginseng	250 mg	Most Commonly Reported Strength
Glucosamine	1000 mg	Most Commonly Reported Strength
Glucosamine & MSM	750 mg & 750 mg	Most Commonly Reported Strengths
Glucosamine 1500 mg with MSM	1500 mg & 1500 mg	Most Commonly Reported Strengths
Glucosamine Chondroitin	750 mg & 600 mg	Most Commonly Reported Strengths
Glucosamine Chondroitin & MSM	Puritan's Pride Glucosamine Chondroitin MSM 3 Per Day Formula	Commonly Available Product
Glutamine Powder	4500 mg	Commonly Available Strength
Green Tea	500 mg	Commonly Available Strength
Greens Powder	Newvitality Royal Greens Original Blend	Commonly Available Product
Gummy Adult Apple Cider Vinegar	Vitafusion Gummy Vitamins Apple Cider Vinegar	Commonly Available Product
Gummy Adult Calcium + Vitamin D	Vitafusion Gummy Vitamins Calcium 500 mg Per Serving with Vitamin D3 & Calcium	Commonly Available Product
Gummy Adult Coq10	Vitafusion Gummy Supplements Coq10 200 mg Per Serving	Commonly Available Product
Gummy Adult Cranberry	Spring Valley Adult Gummy Cranberry 500 mg Per Serving	Commonly Available Product
Gummy Adult Elderberry with Vitamin C & Zinc	Sambucol Black Elderberry Black Elderberry Gummies with Vitamin C & Zinc	Commonly Available Product
Gummy Adult Fiber	Vitafusion Fiber Well Gummies Sugar Free 5g Fiber Per Serving	Commonly Available Product
Gummy Adult Hair, Skin & Nails	Nature's Bounty Optimal Solutions Hair, Skin & Nails Gummies 2,500 mcg Biotin Per Serving	Commonly Available Product
Gummy Adult Lion's Mane Mushroom	500 mg	Commonly Available Strength
Gummy Adult Melatonin	Vitafusion Melatonin Sugarfree Gummy 3 mg Of Melatonin Per Serving	Commonly Available Product
Gummy Adult Melatonin Extra Strength	Vitafusion Gummy Supplements Melatonin Extra Strength 5 mg Per Serving	Commonly Available Product
Gummy Adult Melatonin Max Strength	Vitafusion Gummy Supplements Melatonin Max Strength 10 mg Per Serving	Commonly Available Product
Gummy Adult Men's Multivitamin / Multimineral	Vitafusion Adult Gummy Vitamins Men's Multi Daily Multivitamin	Commonly Available Product

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Gummy Adult Multivitamin / Multimineral	Vitafusion Adult Gummy Vitamins Multivites Multivitamin Gummies	Commonly Available Product
Gummy Adult Omega-3	Vitafusion Gummy Vitamins Omega-3 EPA/DHA	Commonly Available Product
Gummy Adult Prenatal Multivitamin	Vitafusion Gummy Vitamins Prenatal Multivitamin with Folate & 50 mg DHA Per Serving	Commonly Available Product
Gummy Adult Probiotic	Schiff Digestive Advantage Daily Probiotic Gummies	Commonly Available Product
Gummy Adult Vitamin C	Vitafusion Gummy Vitamins Power C High Potency Vitamin C	Commonly Available Product
Gummy Adult Vitamin C Extra Strength	Vitafusion Gummy Vitamins Power C 500 mg Per Serving Extra Strength	Commonly Available Product
Gummy Adult Vitamin D	Vitafusion Gummy Vitamins Vitamin D3 High Potency	Commonly Available Product
Gummy Adult Vitamin D Extra Strength	Vitafusion Gummy Vitamins Vitamin D3 Extra Strength 75 mcg Per Serving	Commonly Available Product
Gummy Adult Women's Multivitamin / Multimineral	Vitafusion Adult Gummy Vitamins Women's Multi Daily Multivitamin	Commonly Available Product
Gummy Adult Zinc	Vitafusion Power Zinc Gummy Vitamins	Commonly Available Product
Gummy Bear Elderberry	Flintstones Supplements Elderberry Gummies With Immunity Support From Vitamin C + Zinc	Commonly Available Product
Gummy Bear Melatonin	Natrol Kids Melatonin Gummies	Commonly Available Product
Gummy Bear Multivitamin	L'il Critters Gummy Vites Daily Multivitamin Plus Vitamins C & D	Most Commonly Reported Product
Gummy Bear Multivitamin + Probiotic	Olly Kids Multi + Probiotic	Commonly Available Product
Gummy Bear Probiotic	Schiff Digestive Advantage Daily Probiotics Kids Gummies	Commonly Available Product
Gummy Bear Vitamin D	Nature Made Kids First Vitamin D3	Commonly Available Product
Hair, Skin and Nails	Nature's Bounty Optimal Solutions Hair, Skin & Nails 3,000 mcg Biotin Per Serving	Commonly Available Product
Horny Goat Weed	500 mg	Commonly Available Strength
Iodine Drops	150 mcg	Commonly Available Strength
Iron	65 mg	Most Commonly Reported Strength
Krill Oil	Schiff MegaRed Superior Omega-3 Krill Oil 350 mg	Commonly Available Product
Lactobacillus Acidophilus	Nature's Bounty Acidophilus Probiotic 100 Million Organisms	Commonly Available Product

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Lion’s Mane Mushroom	500 mg	Commonly Available Strength
Lion’s Mane Mushroom Powder	1000 mg	Commonly Available Strength
Liquid B Complex	Nature's Bounty Sublingual Liquid B Complex	Commonly Available Product
Liquid Coenzyme Q-10	100 mg	Commonly Available Strength
Liquid Fish Oil	Carlson Norwegian The Very Finest Fish Oil 1,600 mg Omega-3s	Commonly Available Product
Liquid Flaxseed Oil	Barlean's Fresh Flax Oil Organic Pure & Unrefined Fresh Cold Pressed	Commonly Available Product
Liquid Vitamin B-12	5000 mcg	Commonly Available Strength
Liquid Vitamin C	1000 mg	Commonly Available Strength
L-Tyrosine	500 mg	Commonly Available Strength
Lutein	20 mg	Most Commonly Reported Strength
Lutein & Zeaxanthin	Lutein 25 mg, Zeaxanthin 5 mg	Most Commonly Reported Strengths
Lysine	500 mg	Most Commonly Reported Strength
Maca	500 mg	Commonly Available Strength
Magnesium	250 mg	Most Commonly Reported Strength
Magnesium Powder	325 mg	Commonly Available Strength
Melatonin	5 mg	Most Commonly Reported Strength
Men’s 50+ Multivitamin / Multimineral	Centrum Silver Men 50+ Multivitamin/Multimineral	Most Commonly Reported Product
Men’s Multivitamin/Multimineral	One A Day Men's Complete Multivitamin Multivitamin/Multimineral Bayer	Most Commonly Reported Product
Moringa	400 mg	Commonly Available Strength
Moringa Powder	6 grams	Commonly Available Strength
Multivitamin / Multimineral	Centrum Adults Multivitamin/Multimineral GSK	Most Commonly Reported Product
Multivitamin / Multimineral Liquid	Centrum Liquid Adults Multivitamin/Multimineral	Commonly Available Product
Multivitamin Multimineral Pack	Nature Made Daily Diabetes Health Pack	Commonly Available Product

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Myo-Inositol & D-Chiro Inositol Powder	Theralogix Ovasitol Inositol Powder 40:1 Inositol Blend	Commonly Available Product
Neem	475 mg	Commonly Available Strength
Oil of Oregano Liquid	34 mg	Commonly Available Strength
Omega 3	Carlson Wild Caught Super Omega-3 Gems 1,200 mg Omega-3s	Commonly Available Product
Pediatric Iron Drops	Enfamil Fer-In-Sol Essential Iron For Infants & Toddlers Iron Drops	Commonly Available Product
Pediatric Vitamin D Drops	Enfamil D-Vi-Sol Vitamin D Drops	Commonly Available Product
Poly-Vitamin With Iron Drops	Enfamil Multivitamin & Iron Infant & Toddler Poly-Vi-Sol With Iron	Commonly Available Product
Potassium	99 mg	Most Commonly Reported Strength
Prenatal Vitamins	Spring Valley Prenatal Multivitamin/Multimineral For Pregnant Or Nursing Women	Commonly Available Product
Probiotic	Shaklee +Probiotic Optiflora Pearl Bifidus & Acidophilus	Commonly Available Product
Protein Powder	Pure Protein 100% Whey Protein	Commonly Available Product
Psyllium Fiber	Metamucil Psyllium Fiber 4-In-1 Fiber Made With Real Sugar Unflavored Stone Ground Texture	Commonly Available Product
Red Yeast Rice	600 mg	Most Commonly Reported Strength
Reishi Mushroom	600 mg	Commonly Available Strength
Saw Palmetto	450 mg	Most Commonly Reported Strength
Selenium	200 mcg	Most Commonly Reported Strength
Senior Multivitamin / Multimineral	Centrum Silver Adults 50+ Multivitamin/Multimineral GSK	Most Commonly Reported Product
Silica	35 mg	Commonly Available Strength
Sodium Fluoride Drops	0.5 mg	Commonly Available Strength
Spirulina Powder	BareOrganics Raw Organic Spirulina Powder	Commonly Available Product
Stress B-Complex	Nature Made Stress B-Complex With Key B Vitamins +Vitamin C & Zinc	Commonly Available Product
Super B-Complex	Nature Made Super B-Complex With C	Commonly Available Product
Tart Cherry Extract	1200 mg	Commonly Available Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
Tribulus Terrestris	500 mg	Commonly Available Strength
Turkey Tail Mushroom Mycelium Powder	1000 mg	Commonly Available Strength
Turmeric	500 mg	Most Commonly Reported Strength
Ubiquinol	100 mg	Commonly Available Strength
Valerian	500 mg	Commonly Available Strength
Vitamin A	2400 mcg	Most Commonly Reported Strength
Vitamin B-1 (Thiamin)	100 mg	Most Commonly Reported Strength
Vitamin B-12	1000 mcg	Most Commonly Reported Strength
Vitamin B-6	100 mg	Most Commonly Reported Strength
Vitamin B-Complex	GNC B-Complex 50	Commonly Available Product
Vitamin C	500 mg	Most Commonly Reported Strength
Vitamin C Packet	Emergen-C 1,000 Mg Vitamin C Antioxidants, B Vitamins & Electrolytes Fizzy Drink Mix	Most Commonly Reported Product
Vitamin C Powder	1000 mg	Commonly Available Strength
Vitamin D	50 mcg	Most Commonly Reported Strength
Vitamin E	180 mg	Most Commonly Reported Strength
Vitamin K	100 mcg	Most Commonly Reported Strength
Vitamins D & K	Vitamin D 125 mcg, Vitamin K 90 mcg	Most Commonly Reported Strengths
Vitamins D & K Liquid	Vitamin D 25 mcg, Vitamin K 10 mcg	Commonly Available Strengths
Whey Protein	Pure Protein 100% Whey Protein	Commonly Available Product
Women's 50+ Multivitamin / Multimineral	Centrum Silver Women 50+ Multivitamin/Multimineral GSK	Most Commonly Reported Product
Women's Multivitamin / Multimineral	One A Day Women's Complete Multivitamin Multivitamin/Multimineral Bayer	Most Commonly Reported Product
Zinc	50 mg	Most Commonly Reported Strength
Zinc Liquid	15 mg	Most Commonly Reported Strength

Default Supplement	Assigned Strength or Supplement	Selection of Assigned Strength or Supplement Based On:
ZMA	Magnesium 450 mg, Zinc 30 mg, Vitamin B-6 10.5 mg	Commonly Available Strengths

Default Antacid	Antacid Assigned	Selection of Assigned Antacid Based On:
Default Antacid & Antigas Liquid	Equate Regular Strength Antacid Antacid & Antigas	Commonly Available Product
Default Antacid Aluminum & Magnesium	Gaviscon Regular Strength Alumina & Magnesium Trisilicate Tablets / Antacid	Commonly Available Product
Default Antacid Calcium & Magnesium	Rolaids Regular Strength Antacid / Calcium & Magnesium	Most Commonly Reported Product
Default Antacid Plus Tablets	Rolaids Advanced Antacid Plus Anti-Gas Multi-Symptom	Commonly Available Product
Default Calcium Antacid	Tums Calcium Carbonate Antacid Regular Strength 500 GSK	Most Commonly Reported Product